

Two-Sleeve Portfolio: 80% Optimised Core / 20% Moonshot Sleeve

A GBP 200,000 model portfolio for a UK investor on interactive investor

Analysis date 2026-09-17. All prices and statistics as at the close of **2026-09-16** (the last full trading day). Base currency GBP.

This is the **moonshot rebuild**. The core sleeve, the as-of date, the five-year window and every market input are unchanged from the 16 September build; Sleeve A has been rebuilt from scratch under a new mandate, and 34 of its 40 names are different. The 16 September quality-sleeve build is archived under `archive/2026-09-16-quality-sleeve/` and the 11 September build under `archive/2026-09-11/`. Every difference between this build and the one it replaces is itemised in `data/REFRESH_DIFF.md` and summarised at the end of section 4; section 6 sets out the new mandate in full.

This is educational analysis, not personalised investment advice. Nobody who worked on this project is a licensed adviser, this is not a personal recommendation, and no number in this report is a forecast. Everything here is a model output built on five years of Yahoo Finance price history, and the single most important thing to understand about it is in section 9. **Most positions in the moonshot sleeve are expected to lose money and some are expected to go to zero; the sleeve is sized so that this is survivable, not so that it is unlikely.**

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1. Executive summary

1.1 The portfolio at a glance

GBP 200,000 is split into two sleeves. The **core sleeve** takes 80% (GBP 160,000) and is sized by mean-variance optimisation at a 12% target volatility. The **moonshot sleeve** takes 20% (GBP 40,000) as exactly 40 equal GBP 1,000 positions in listed companies chosen for a defended 10x-100x case over five to ten years, accepting that most of them may go to zero. One core line, SPCX, is not sized by the optimiser at all: it is a fixed 2.5% policy weight, for the reason given in 1.3.

Line	Instrument	Bucket	Weight of core	GBP	Price (GBP)	Shares	Invested GBP
IITU.L	iShares S&P 500 Information Technology Sector UCITS ETF USD (Acc)	etf_equity	5.4%	8,631	37.9900	227	8,624
ISF.L	iShares Core FTSE 100 UCITS ETF GBP (Dist)	etf_equity	13.5%	21,656	10.4620	2,070	21,656
SMGB.L	VanEck Semiconductor UCITS ETF	etf_equity	16.6%	26,568	76.7800	346	26,566
SGLN.L	iShares Physical Gold ETC	etc_commodity	14.8%	23,726	62.7000	378	23,701
BRNT.L	WisdomTree Brent Crude Oil	etc_commodity	5.2%	8,274	73.3813	112	8,219
RR.L	Rolls-Royce Holdings plc	stock	5.0%	8,000	14.4060	555	7,995
PLTR	Palantir Technologies Inc.	stock	5.0%	8,000	129.3951	61	7,893
SPCX *(policy weight)*	Space Exploration Technologies Corp. (SpaceX)	stock	2.5%	4,000	111.9831	35	3,919
AZN.L	AstraZeneca PLC	stock	1.1%	1,724	121.3600	14	1,699
SHEL.L	Shell plc	stock	5.0%	8,000	35.7650	223	7,976
HSBA.L	HSBC Holdings plc	stock	5.0%	8,000	15.0780	530	7,991
BRK-B	Berkshire Hathaway Inc. Class B	stock	1.4%	2,276	385.7956	5	1,929
IBTM.L	iShares \$ Treasury Bond 7-10yr UCITS ETF USD (Dist)	etf_bond	10.0%	16,000	123.9200	129	15,986
DBMG.L	iMGP DBi Managed Futures UCITS ETF R USD (Acc)	trend	9.5%	15,144	100.2200	151	15,133
Core total	14 lines		100.0%	160,000			159,287
Moonshot sleeve	40 names x GBP 1,000, equal weight at cost	moonshot	(20% of total)	40,000			39,962
Portfolio total	54 lines			200,000			199,249

Source: data/combined_positions.csv; data/analysis_results.json -> portfolios.Moderate12.gbp/shares | shares = floor(GBP allocated / GBP price at the 2026-09-16 close); the two gilts are priced per GBP 100 nominal

Whole shares only, so GBP 751 of the GBP 200,000 stays in cash (GBP 713 of it inside the core sleeve). The weighted ongoing charge on the core funds is 0.20% a year; the single stocks and the moonshot sleeve carry no fund fee.

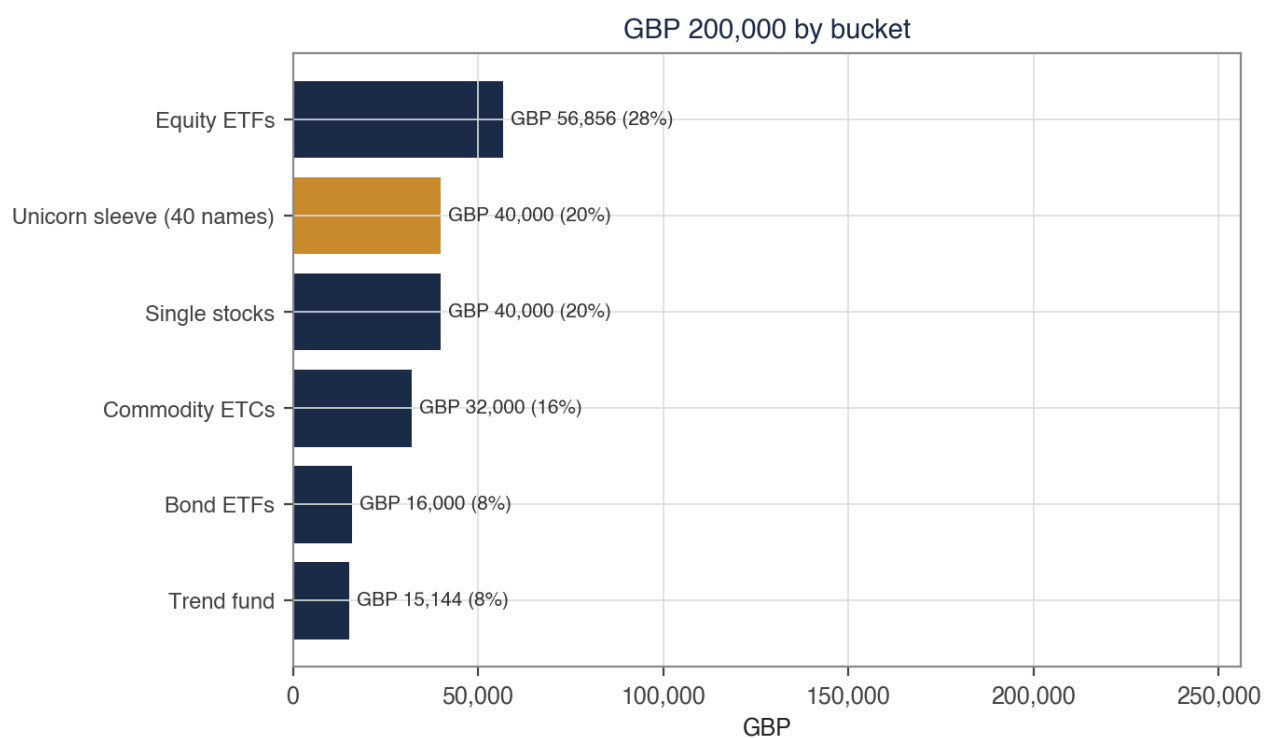


Figure 1. The GBP 200,000 split by sleeve and by bucket. Source: data/combined_positions.csv, as at 2026-09-16.

1.2 Headline statistics

Statistic	Core sleeve (Moderate12)	Moonshot sleeve	Combined GBP 200,000
Expected return, 50/50 blend (the optimiser input)	15.4%	10.0%	14.3%
Expected return, historical 5y only	23.6%	12.1%	n/a
Expected return, CAPM / yield only	7.2%	7.9%	n/a
Ex-ante volatility (Ledoit-Wolf, optimised 97.5%)	12.0%	n/a	n/a
Ex-ante volatility (all lines incl. SPCX)	12.8%	n/a (equal weight, not optimised)	14.2%
Ex-ante Sharpe	0.91	n/a	0.74
Realised 5y volatility (daily)	12.1%	26.5%	13.0%
Realised 5y Sharpe (daily)	1.67	0.32	1.37
Realised 5y Sortino (daily, MAR = Bank Rate)	2.45	0.46	1.97
Beta vs VWRP.L (as held)	0.79	0.87	0.83
Upside beta (beta+)	0.85	0.77	0.85
Downside beta (beta-)	0.77	0.91	0.82
Beta asymmetry (beta+ minus beta-)	0.08	-0.13	0.03
Beta asymmetry excluding the SPCX policy line	-0.09	n/a	n/a
Daily return skew, 5y	-0.49	0.11	-0.49
Excess kurtosis, 5y	2.71	1.32	2.23
Backtest 5y CAGR (monthly rebalanced, no costs)	26.0%	12.9%	23.9%
Backtest 5y max drawdown (month-end)	-3.8%	-43.0%	-7.4%
Realised 5y maximum drawdown (daily)	-13.4%	-45.5%	-16.8%
95% daily VaR / CVaR	1.13% / 1.68%	2.59% / 3.44%	1.23% / 1.82%
99% daily VaR / CVaR	1.96% / 2.82%	3.85% / 4.89%	2.13% / 2.99%

Source: data/analysis_results.json -> portfolios.Moderate12, unicorn_sleeve.stats, combined_portfolio; data/backtest_monthly.csv for the moonshot sleeve's monthly row | 5y daily window 2021-09-16 to 2026-09-16 (1,261 return days); rf = BoE Bank Rate 3.75%

For scale: the benchmark VWRP.L (FTSE All-World in GBP) returned 71.2% over the same five years, with 13.0% annualised volatility and a -17.6% maximum drawdown.

Read the moonshot column with care. Those are five-year statistics on a set of names selected in September 2026 for a five- to ten-year payoff whose entire design is "most go to zero, one or two go up 10x-30x". They are reported because the combined-book covariance needs a single return series for the sleeve, not because they forecast anything. Section 6.6 sets out the payoff arithmetic that actually describes the sleeve, and section 6.8 explains why every one of these numbers is worse than the sleeve it replaced.

1.3 Verdict

What the model does well. It is built on total-return series that were rebuilt from scratch in GBP rather than trusting Yahoo's adjusted close, and three separate price-data faults were found and repaired along the way - a 100x quote-scale glitch and two placeholder opening prints in DBMG.L, a pounds-to-pence quote-convention break in SMEA.L, and, new in this refresh, a single fabricated flat-OHLC bar in WLDS.L (section 2.2). Left alone the DBMG.L fault would have put a 7,795% volatility into the covariance matrix and corrupted every weight; the WLDS.L bar alone would have overstated that fund's five-year volatility by 8.9 percentage points. The covariance is shrunk, not raw. The optimiser is solved from 96 random starts per objective with cross-seeding and a dominance check, so the reported portfolios are not local accidents. Every constraint in the brief binds or is satisfied to 1e-06. The result - a core sleeve with a blended expected return of 15.4% at a 12% ex-ante volatility, beta 0.79 to world equity, and a realised 5-year Sharpe of 1.67 - is internally coherent and reproducible from the shipped files.

What it cannot claim. It cannot claim to be a forecast. The five years 2021-2026 contained a gold bull market, an AI capital-expenditure boom and an oil spike, and the sample means inherit all three: the historical-only expected return for the core is 23.6% while the CAPM/yield-only estimate is 7.2%. The headline 15.4% sits between them and is far more likely to be too high than too low. It cannot claim a track record: the backtest applies today's weights to the past, charges no dealing commission, no 0.5% stamp duty, no FX conversion charge and no spread, and the moonshot sleeve's history is survivorship-biased because the 40 names were picked in 2026 from companies that still exist. It cannot claim to have found positive skew against the market: on daily data no long-only portfolio in this universe achieved a positive beta asymmetry on the optimised part alone (section 2.5), and the new sleeve's own beta asymmetry is now clearly negative at -0.13. And it cannot claim precision in the weights themselves - a mean-variance optimiser is a maximiser of estimation error, and swapping one reasonable expected-return vector for another moves 83% of the portfolio. Above all, **it cannot claim that the moonshot sleeve will work**: nothing in this report establishes that a dossier-defended 40 is more likely to contain a 10x than any other 40, and a five-year backtest cannot test that claim even in principle.

The moonshot sleeve is now a separate decision from the core. The core sleeve in this build is bit-for-bit the one published on 16 September: same weights, same expected returns, same risk. What changed is Sleeve A, and the change costs the combined book 1.2 percentage points of ex-ante expected return (15.5% to 14.3%) and 0.07 of ex-ante Sharpe, and takes 3.0 points off the realised five-year backtest CAGR. Whether that is a price worth paying cannot be settled by any number in this report: it is the difference between owning forty good small companies and owning forty lottery tickets with written-down theses, and the evidence for the second only arrives over five to ten years. Section 6.8 puts the whole cost in one table.

The three judgement calls that drive the core result. First, **the expected-return estimator**. Using a 50/50 blend of the 5-year sample mean and a structural CAPM/yield estimate is a defensible middle course (Michaud 1989), but it is a choice, and it is the choice that puts Rolls-Royce, Palantir and a semiconductor ETF at or near their caps: their 5-year sample means are 61%, 59% and 33% a year. Feed the optimiser the structural estimate alone and 83% of the portfolio changes - the stocks and commodities collapse and bonds take a third of the book. Second, **the 12% volatility target**. 'Moderate risk' is defined here as 12% annualised ex-ante volatility and nothing else; that single number is what makes this portfolio hold 25% in individual stocks and 20% in commodities rather than the safer max-Sharpe mix at 9.6%. Moving the target moves everything. Third, **the SPCX policy weight**. SpaceX listed on 2026-06-12 and has 67 daily bars in the window. It cannot be optimised, so it was given a flat 2.5% (GBP 4,000) of the core by decision, not by the model - and because beta is linear in the weights, that one 2.5% line is the difference between the core's headline beta asymmetry reading +0.08 and reading -0.09. A judgement call that flips the sign of a headline statistic deserves to be named as one.

1.4 Recent developments, 11-16 September 2026

This build moves the as-of date from the 11 September close to the 16 September close. Three trading days is not long, but it contained two central-bank decisions, a UK inflation print and an oil-supply scare, and it is worth setting out what the model does and does not absorb from them.

The Fed hiked. On 2026-09-16 the FOMC raised the target range 25bp to **3.75%-4.00%**, unanimously 12-0 - its first increase since July 2023 - describing inflation as "remaining elevated". The Summary of Economic Projections put the median funds rate at 4.1% at end-2026, implying one further hike this year. Chair Kevin Warsh then

declined to give forward guidance and distanced himself from the Committee's own dot plot; equities sold off through the press conference and money markets moved to roughly a 50% chance of another hike in October. Sources: [FOMC statement](<https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm>), [SEP](<https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm>), [CNBC](<https://www.cnbc.com/2026/09/16/treasury-yield-bond-market-fed-decision.html>). The ECB separately raised its deposit rate 25bp to 2.50%, effective 16 September.

UK CPI reaccelerated, but not in the core. August CPI rose to **3.1%** year on year from 2.9% in July, a five-month high, on a 0.5% monthly rise driven almost entirely by transport - petrol up 9.1p a litre, motor-fuel inflation 23.0%. **Core CPI was unchanged at 2.6% and services inflation unchanged at 3.4%** ([ONS](<https://www.ons.gov.uk/economy/inflationandpriceindices/bulletins/consumerpriceinflation/august2026>)). The acceleration is energy-led rather than broad, which is why the market read it as dovish. The MPC did not meet inside this window: its decision fell on **17 September, the day after the as-of date**, with roughly 96-97% priced for a hold at 3.75% and the July 6-3 split (Greene, Mann and Pill each preferring 4.00%) expected to repeat. **Bank Rate is therefore still 3.75% in this model, unchanged from the 11 September build**, and so is every CAPM intercept and every Sharpe denominator built on it.

Yields: the long end is the story. The US 10-year Treasury went through 5% for the first time since 2007, touching 5.04% before the meeting and settling at **5.01%**, with commentary attributing the move to term premium - large deficits, heavy Treasury issuance and heavy AI-related corporate issuance competing for the same capital. UK 30-year gilts sit at **5.89%**, around the highest since March 1998, but on the 16th itself the 10-year **fell** about 8bp to **5.30%** and the 30-year eased about 3bp as the in-line CPI print landed and the oil rally paused. Separately, the Bank was reported to be about to stop actively selling 20- and 30-year gilts from the Asset Purchase Facility, and the DMO has already cut long conventional gilt sales to roughly 9% of its GBP 252.1bn remit.

Conflict and energy. Drones launched from Iraq struck pumping stations on Aramco's East-West pipeline and Riyadh shut the line, removing the main workaround for an effectively closed Strait of Hormuz. Brent settled at \$108.75 on 15 September, a four-month high, then fell 2.7% to **\$105.83** on the 16th as Saudi Arabia signalled half of pipeline capacity back within days. The US Strategic Petroleum Reserve is down to 285.0 million barrels, the lowest since November 1982. Italian Eurofighters shot down a suspected Russian drone over Lithuania on 15 September, and Houthi forces captured the Greater and Lesser Hanish islands in the Red Sea.

Sectors and holdings. Semiconductors had a hard window - SMGB.L -3.41%, ASML.AS -5.56% - after AI executives led by Anthropic's Dario Amodei, publicly backed over the preceding weekend by OpenAI, Microsoft and xAI, called for slower frontier-model development; chip-equipment names read the call across to future fab capacity when markets opened on Monday the 14th and ASML.AS fell 7.2% that day ([CIO Dive](<https://www.ciodive.com/news/AI-slowdown-CIO-enterprise-adoption/830340/>), [Motley Fool](<https://www.fool.com/investing/2026/09/14/why-asml-stock-is-falling-today/>)). China extended rare-earth licensing and export controls on the 15th; against that, TSMC and Samsung both committed to ASML's High-NA EUV. The BIS and an ECB research team both warned in the same fortnight about debt-financed AI capital expenditure. Oil lines gained (BRNT.L +1.60%, SHEL.L +0.72%). Gold was flat in sterling (SGLN.L -0.02%) despite a 1.9% fall in dollars, because sterling weakened - a clean illustration of why the brief insists on GBP terms. Among core single names: AZN.L +3.66% on positive COPD and lung-cancer readouts against a failed first-line breast-cancer endpoint; PLTR +4.25%; HSBA.L -2.89% after a broker downgrade; RR.L -0.96% despite a Rolls-Royce SMR early-works contract with CEZ for Temelin; SPCX -0.22% with its 24 September lock-up still ahead of it.

Four of the six names carried over into the new moonshot sleeve moved in the window, and they are the only sleeve-level news that is still about something this portfolio holds: **IMSR -10.43%**, the largest fall in either sleeve, a multi-day de-rating of a pre-revenue SMR developer as financing costs rose rather than an event (it filed a Graphite Topical Report with the NRC on the 10th); **ARA.TO +10.57%**, the largest gain, on a **US EXIM Bank letter of interest for up to USD 750m** for the Louisiana rare-earth project - a non-binding letter, but one larger than the whole company; **BIOA -8.72%**, traced to the Morgan Stanley healthcare conference on the 16th where the company acknowledged a setback in the wider cardiovascular-inflammation field and a pause in one planned late-stage study; and **FTC.L -4.90%** despite winning an **USD 8m follow-on US satellite mmWave contract** on the 16th itself. SLX.AX (-5.59%) and VUL.AX (-5.49%) fell with the small-cap complex on no company-specific catalyst. The 11-16 September moves of the 34 names that were in the sleeve then and are not in it now are documented in [archive/2026-09-16-quality-sleeve/](#) and are no longer relevant to this portfolio. Full

detail, with sources, is in `universe/NEWS_AND_IMPACT_2026-09-16.md`.

How any of this reaches the model.

Channel	Does the model see it?	Where it lands
Bank Rate (the CAPM risk-free rate)	Yes, as an input	Unchanged at 3.75%; the MPC met on 17 September, after the as-of date
Fed funds target range	Yes, as an input to one line	3.50-3.75% to 3.75-4.00%; enters IBTM.L's interest-parity estimate, cutting its expected GBP return 20bp
US 10y Treasury yield	Yes, as an input to one line	4.96% to 5.01%; the other half of IBTM.L's estimate
Gilt clean prices and redemption yields	Yes, as inputs	T36N 96.75 / 5.30% / modified duration 7.46; T56 93.155 / 5.87% / 14.06
Equity risk premium	Yes, as an input	Unchanged at 4.17% (Damodaran mature-market); contributes exactly 0.00 pp of change this refresh
The 16 September market reaction	Yes, but only as three days of price data	Already inside the closes, and inside the 1,261-day return sample where it is invisible
Dot plot, withdrawn forward guidance, October FOMC odds	No	Forward assumptions the model does not hold
24 September SPCX lock-up (328.4m shares)	No	Calendared supply on a line with 67 bars; the reason it is sized by policy, not optimised
The GUTS Nasdaq delisting hearing, stayed 11 September	No	A tradeability risk on one moonshot line, carried openly in section 6.8 rather than modelled
28 October UK Budget, halved fiscal headroom	No	The largest known UK risk in the next six weeks
US high-yield spreads in their richest decile (~265-270bp)	No	Five-year HY history contains both tight and wide spreads, so it flatters the forward risk of starting here
BIS and ECB warnings on AI-capex concentration	No	The covariance treats the AI complex's correlation as a statistic, not as a fragility

Source: `universe/NEWS_AND_IMPACT_2026-09-16.md`; `universe/news_sectors-holdings_2026-09-16.json`; `universe/news_rates-inflation_2026-09-16.json`; `universe/news_geopolitics-energy_2026-09-16.json`; `universe/market_params.json`; `data/REFRESH_DIFF.md`

The point that matters most. The optimiser's expected returns are **five-year estimates**. They react to news only through the risk-free rate, the equity risk premium and the yield inputs - and in this refresh the first two did not move at all. Everything else the news did is in the model twice over: once properly, as the starting prices at which the GBP 200,000 would be deployed and as the gilt and Treasury yields that price three bond lines, and once ineffectually, as three trading days inside a 1,261-day sample, where a 25bp Fed hike moves an annualised five-year mean by an amount that rounds to nothing. That is not a flaw to be fixed by adding news to the model; it is the honest limit of a five-year statistical estimate, and it is why this section exists separately from the optimisation. Nothing in the 11-16 September flow triggered a core replacement: all 37 core instruments were still listed, solvent and tradeable on ii at the as-of close. Sleeve A was replaced the following day, but for a change of mandate rather than anything in this news flow - and the two names that did leave the **new** sleeve were removed by its own fact-check, not by the market (section 6.2).

2. Objective and method

2.1 The two sleeves

	Core sleeve (B)	Moonshot sleeve (A)
Capital	GBP 160,000 (80%)	GBP 40,000 (20%)
Construction	mean-variance optimisation, long only	40 equal positions of GBP 1,000 at cost
Universe	37 instruments: 12 equity UCITS ETFs, 6 commodity ETCs, 11 single stocks, 5 bond ETFs, 2 individual gilts, 1 trend-following fund	40 listed companies, median market cap USD 220m, chosen for a defended 10x-100x mechanism
Objective	maximise expected return subject to 12% ex-ante volatility	a small number of 10x-100x winners over 5-10 years, accepting that most positions may go to zero; each name sized so that a total loss costs 0.5% of the portfolio
Sizing input	Ledoit-Wolf shrunk covariance, 50/50 blended expected returns	none - equal weight by construction
Maintenance	drift back to the optimised weights, in tolerance bands	let winners run; trim half only above 3% of the total portfolio; annual thesis review

Source: PROJECT_BRIEF.md MOONSHOT SLEEVE MANDATE; SPEC_ANALYSIS.md section 8; universe/core_universe.json; universe/unicom_final.json

The moonshot sleeve is deliberately **not** optimised. Forty pre-revenue and early-revenue companies have no reliable covariance structure, and any optimiser handed them would concentrate the sleeve into whichever three names happened to have the best recent five years. Equal weight is the honest answer when you do not believe your own estimates - and in this sleeve, where thirteen of the forty have under three years of usable history or no revenue at all, the estimates deserve even less belief than usual.

2.2 Data, window and GBP total-return construction

Item	Choice
Source	Yahoo Finance via yfinance, fetched 2026-09-17 for the 2026-09-16 close
Full history window	2016-09-16 to 2026-09-16
Core statistics and optimisation window	2021-09-16 to 2026-09-16 (5 years, daily; 1,261 return days)
Trading calendar	VWRP.L trading days, unioned with the ^FTSE/^FTAS sessions VWRP.L is missing inside its own operating range; gaps forward-filled up to 3 days
Benchmark for betas	VWRP.L (FTSE All-World, GBP). ^FTAS (FTSE All-Share) reported alongside
Risk-free rate	BoE Bank Rate 3.75% , in force at 2026-09-16 (held since 2026-07-30)
Equity risk premium	Damodaran mature-market implied ERP 4.17% (2026-07-01)
Annualisation	252 trading days; volatility x sqrt(252)

Source: data/analysis_results.json -> config, market_params_used; universe/market_params.json

Nothing in this section changed in the moonshot rebuild. The price panel, the repairs, the calendar and the patched 2026-09-16 bars are the same objects the 16 September build used; the only change is that 34 of the 40 Sleeve A columns are different tickers. The panel was re-fetched and re-verified from scratch, and one housekeeping trap was found in doing so: the stale *_raw.csv panels left on disk by the previous build still carried the OLD 40 names, and because analysis.py reads the raw panel in preference to the deliverable, leaving them in place would have silently re-analysed the old sleeve. They were cleared before the moonshot fetch and rebuilt from it; the check is recorded in data/REFRESH_DIFF.md note 10.1.

Total return was rebuilt, not borrowed. For every instrument the closing price was split-adjusted from the split events, then a total-return index was compounded as $TR_t = TR_{(t-1)} \times (P_t + D_t) / P_{(t-1)}$ with dividends in the same unit as the price. Yahoo's Adj Close column is discarded entirely for LSE lines. The reason is a

documented Yahoo fault: for tickers quoted in pence, Yahoo's adjusted close applies only one hundredth of each dividend. This was re-checked empirically on three LSE names in this refresh before relying on it.

Ticker	Ex-div date	Prior close	Yahoo prior Adj Close	Adj Close if the FULL dividend were applied	Adj Close if 1/100 of it were applied
RR.L	2026-08-06	1,570.40	1,570.34	1,564.26	1,570.34
SHEL.L	2026-08-13	3,322.50	3,322.21	3,293.28	3,322.21
ISF.L	2026-06-18	1,034.00	1,033.87	1,020.83	1,033.87

Source: data/fetch_log.json -> gbp_dividend_check | Yahoo's Adj Close matches the 1/100 column in all three cases

The raw Dividends column itself is fine - it is already in pence for pence-quoted lines, with dividend/price ratios in a plausible 0.4%-1.4% band. It is only Yahoo's own adjustment that is wrong. Pence lines are divided by 100 to reach pounds; other currencies are converted at the daily GBP cross rate, forward-filled. Three instruments have no usable own history and are spliced onto a proxy (section 3.2).

A new Yahoo fault specific to this as-of date. Yahoo's daily history() omits the **2026-09-16 bar for most LSE-listed ETFs and ETCs** - the series runs 2026-09-15 straight to 2026-09-17 - while LSE single stocks and every non-UK line carry the bar. The LSE was open; the closes exist. Left alone, a pipeline forward-filling the 15 September close would have silently mis-stated fourteen core instruments (SGLN.L would have read 6,161 instead of 6,270; SMGB.L 75.33 instead of 76.78). **Twenty-five tickers had their 2026-09-16 bar rebuilt** from 60-minute intraday data, with the close taken from the official figure recorded in universe/core_universe.json (captured from info['previousClose'] during the live 17 September session and cross-checked against eighteen LSE lines that did carry a direct 16 September print - all eighteen matched exactly). Two of the twenty-five - the gilt proxies IGLT.L and GLTL.L - had no recorded official close and fall back to the last intraday 16:00 bar, which is flagged per ticker in data/fetch_log.json. This is now on the project's standing Yahoo-trap list.

Three price-data faults were found and repaired.

- **DBMG.L quote-scale glitches.** 33 bars in DBMG.L's own-history segment (2025-04-23 to 2025-08-11) were quoted 100x too low, producing fabricated -99% falls and +7,500% to +9,900% recoveries; two further opening prints (2025-04-22/23) are flat, zero-volume placeholders and are masked, with the DBMF proxy segment re-anchored to the first clean bar (2025-04-24). *Effect:* DBMG.L 5y annualised volatility falls from 7,795% to 15.3%; its beta to VWRP.L from 2.04 to 0.13. Because it feeds the shared Ledoit-Wolf covariance, this changes every optimised weight.
- **SMEA.L quote-convention break.** The feed is quoted in pounds to 2023-11-24 and in pence from 2024-04-25; the earlier segment is rescaled onto the current convention and the mis-quoted stretch in between (which is mostly a Yahoo data hole) is masked. *Effect:* SMEA.L 5y annualised volatility falls from 5,245% to 14.0%.
- **WLDS.L placeholder bar, new in this refresh.** The recovered 2025-10-24 session (see below) brought with it a bar for WLDS.L with Open = High = Low = Close = 8.9258 and Volume = 0, against settled closes of 6.6480 on the 23rd and 6.7285 on the 27th: +34.26% then -24.62%, a one-day round trip. An MSCI World Small Cap index did not move 34% in a day, and VWRP.L and ISF.L are forward-filled flat on that date, so nothing in the market explains it. *Effect of leaving it in:* WLDS.L 5y volatility 24.48% instead of 15.61%, skew +5.335 instead of -0.386, maximum drawdown -27.57% instead of -21.55%. The bar is masked and the spanning return dropped, not synthesised. A panel-wide scan of all 90 raw files over the full ten years for the same flat-OHLC, zero-volume, reversed-move signature returns exactly two instances - this one and the SMEA.L bar on the same date. Three Sleeve A lines in the build where this repair was first made also moved hard on 2025-10-24 (ARA.TO +17.6%, AMPX +10.8%, SLX.AX +10.2%); their bars carry real ranges and heavy volume, so those are genuine and were left alone. ARA.TO and SLX.AX are still in the sleeve; AMPX is not.

Why 2025-10-24 is new. The common calendar is built as VWRP.L's own trading days unioned with the ^FTSE/^FTAS sessions VWRP.L is missing inside its own operating range. That union is what makes the 2026-09-16 bar available at all, and it also recovered two genuine LSE sessions the previous build had silently dropped - 2025-10-24 and 2026-03-06 - which is the entire reason the return count rises from 1,259 to 1,261 (the three-day window roll is net zero: three days in, three days out). One consequence to keep in mind: VWRP.L has no own bar on either recovered date, so the benchmark return there is a forward-filled 0.00%, and every beta, beta+ and beta- regression now contains those two manufactured observations. Harmless for well-behaved lines; it is precisely what made the two placeholder bars damaging.

The repaired panels were written over the deliverable price files; the untouched fetch output is preserved as *_raw.csv. A third data decision is a judgement rather than a fault: IMSR (Terrestrial Energy) uses bars from 2025-11-01 onward only, because everything earlier is pre-de-SPAC shell data.

Source: data/analysis_results.json -> log, data_repairs, sources_assumptions_corrections; data/fetch_log.json -> patched_bars_2026-09-16, scale_glitch_repairs; data/REFRESH_DIFF.md

2.3 Estimators

Expected returns are reported three ways. The optimiser is fed the third.

1. **Historical.** The arithmetic mean of daily returns over the 5-year window, times 252. Arithmetic rather than geometric, because the expectation of the one-period return is what a mean-variance optimiser needs.
2. **Structural.** For equities, equity ETFs, commodity ETCs and the trend fund: CAPM, $r_f + \beta \times ERP$, using the instrument's own GBP beta to VWRP.L. For the two individual gilts: the redemption yield at the as-of clean price. For the high-yield and EM credit ETFs: the distribution yield minus a 1% expected default loss. For IBTM.L (US Treasuries, GBP-quoted but unhedged): the US 10-year yield plus the GBP-USD policy-rate differential, which is what interest parity implies the expected GBP return should be.
3. **Blend.** 50% of each. This is the optimiser input.

The blend is not a fudge; it is the response to a known problem. Sample means are by a wide margin the noisiest input to mean-variance optimisation, and the optimiser is maximally sensitive to exactly that input (Michaud 1989). An equal blend with a structural estimate halves the weight on the noise without pretending the sample contains no information at all. Section 4.6 shows what the portfolio looks like under each of the three vectors on its own, and how much trading it would take to move between them.

A note on which ERP. CAPM here pairs a UK risk-free rate (Bank Rate 3.75%) with Damodaran's **mature-market (global) implied ERP, 4.17%** as at 2026-07-01 - both mature-market constructs, which is the internally consistent pairing the brief asks for. An interim pass of this refresh had substituted a different, legitimate Damodaran series - his monthly implied ERP for the S&P 500, 4.09% at 2026-09-01, computed off the index's own price level and a US Treasury yield - which answers a different question. That was reverted; the US series is retained in market_params.json as a labelled sensitivity only. Swapping it in moves the core sleeve's blended expected return by at most 0.035 pp (MinVariance 6.31% to 6.29%, Moderate12 15.41% to 15.37%). **The ERP is therefore unchanged from the 11 September build, and contributes exactly 0.00 pp of this refresh's expected-return change.**

Covariance is shrunk. The sample covariance of a 36-asset problem estimated on 1,135 usable daily observations is invertible but badly conditioned, and the optimiser will happily exploit any spurious near-arbitrage in it. Ledoit and Wolf's (2004) constant-correlation shrinkage was implemented from the paper (not from a library) and applied: $\Sigma = \delta \times F + (1 - \delta) \times S$, where F is the constant-correlation target built from the average sample correlation and S is the MLE sample covariance. The optimal intensity came out at **delta = 0.035840**, with the components recorded in the log.

Ledoit-Wolf component	Value
delta	0.035840
r_bar	0.232380
pi	0.000191
rho	0.000070
gamma	0.000003
kappa	40.678742
T	1,135.000000
N	36.000000
reference	Ledoit & Wolf (2004), Journal of Portfolio Management 30(4)

Source: data/analysis_results.json -> covariance.ledoit_wolf

A delta of about 0.036 is small, which is what one should expect with 1,135 observations for 36 assets: the sample matrix is already reasonably well estimated and shrinkage only has to do a little work. It is not zero, and it matters

most for the pairs with the least overlapping history. The complete-case sample is 1,135 of the 1,261 window days, unchanged from the previous build - the masked SMEA.L stretch (2023-11-27 to 2024-04-24) and the GHYS.L hole (2026-01-02 to 2026-01-23) are what the optimiser drops.

2.4 The optimiser and its constraints

Sequential least-squares quadratic programming (SLSQP), long only, weights summing to 97.5% of the core (SPCX takes the other 2.5% by policy). Each non-convex objective is solved from 96 random feasible starts plus every other solved portfolio as a start, repeated for up to 4 cross-seeding rounds, and then checked against every other feasible portfolio on its own objective; the run aborts if any portfolio is dominated. None was.

Constraint	Limit
Single etf equity line	max 20%
Single etf bond line	max 20%
Single etc commodity line	max 20%
Single gilt line	max 20%
Single trend line	max 15%
Single stock line	max 5%
Group: equity etf (etf_equity)	min 30%, max -
Group: commodity (etc_commodity)	min 5%, max 20%
Group: bonds (etf_bond, gilt)	min 10%, max 40%
Group: stocks (stock)	min -, max 25%
Group: trend (trend)	min -, max 15%
Volatility (headline portfolio)	ex-ante Ledoit-Wolf vol <= 12%
Long only	no short positions, no leverage

Source: data/analysis_results.json -> config.constraints

'Moderate risk' is defined as a 12% annualised ex-ante volatility, and nothing else. That is the brief's definition and it is worth being explicit about what it does and does not capture. It is a forward-looking number from the shrunk covariance matrix, not a realised one. It says nothing about drawdown, about tail shape, or about how the portfolio behaves when correlations go to one. For reference, the realised 5-year daily volatility of the resulting portfolio was 12.1% - close to target, but that is a coincidence of the sample, not a property of the method. Four portfolios are reported so the target can be seen in context: minimum variance, maximum Sharpe, this 12%-target portfolio, and a skew-tilted maximum-Sortino portfolio.

The 12% constraint binds on the ex-ante volatility of the **optimised 97.5%**, and it binds at exactly 12.00% in this build as it did in the last. Adding the 2.5% SPCX line at its 67-bar pairwise covariance raises the full-core figure to 12.8%. Both numbers are reported throughout; SPCX's row and column of that matrix are not reliable.

2.5 The positive-skew measures, and what they found

The brief asks for a portfolio with an asymmetric payoff - more participation on the way up than on the way down. Four measures were computed for every asset and every portfolio:

- **Upside beta (beta+)**: the OLS slope of the asset's excess return on the benchmark's, computed only on days when the benchmark's excess return was positive.
- **Downside beta (beta-)**: the same slope on days when it was negative.
- **Beta asymmetry**: beta+ minus beta-. Positive means the payoff is convex against the market.
- **Skewness** of the return distribution itself, and **Sortino** (excess return divided by downside deviation, with the minimum acceptable return set to Bank Rate).

The honest finding is that no long-only portfolio in this universe achieved positive daily beta asymmetry on the optimised part alone. Measured on the 97.5% the optimiser actually controls, every one of the six portfolios has beta+ below beta-:

Portfolio	beta+ (excl. SPCX)	beta- (excl. SPCX)	Asymmetry (excl. SPCX)	Asymmetry as held (incl. SPCX)	Realised 5y Sortino	Realised 5y skew
MinVariance	0.26	0.29	-0.03	0.15	0.74	-0.49
MaxSharpe	0.51	0.59	-0.08	0.09	2.55	-0.56
Moderate12	0.66	0.75	-0.09	0.08	2.45	-0.49
MaxSortino	0.50	0.57	-0.07	0.11	2.56	-0.47
EqualWeight	0.64	0.69	-0.04	0.13	1.54	-0.40
NaiveReference	0.73	0.75	-0.02	0.15	1.14	-0.34

Source: data/analysis_results.json -> portfolios.* | daily, 5y window 2021-09-16 to 2026-09-16, benchmark VWRP.L

MaxSortino is the closest to the goal. It has the highest realised Sortino of the six (2.564, against MaxSharpe's 2.547 and Moderate12's 2.451), the least negative daily skew among the optimised portfolios, and the least negative beta asymmetry on the optimised part (-0.069, against Moderate12's -0.092). It is the portfolio the brief's skew objective actually points at. It is not the headline because it sits at a 9.7% ex-ante volatility rather than the 12% the brief specifies as 'moderate'.

Three warnings about the asymmetry number. It changes sign with the measurement frequency: on monthly data the same weights give beta+ above beta- for the optimised part too. It changes sign with the treatment of SPCX: because beta is linear in the weights, the 2.5% SPCX line at its 67-bar beta+ of 8.34 moves the core's beta+ by about 0.21 and flips the headline reading from -0.09 to +0.08. And SPCX's 67-bar betas are not estimates in any statistical sense - the three extra bars this refresh added moved its beta from 2.83 to 2.95, a 0.12 move from a 4.7% increase in sample size, which is itself the clearest demonstration that the number is not converged. The correct reading is that **the positive-skew claim is unproven in either direction** on this data.

The moonshot sleeve is where the positive skew is - and it is weaker than it was. The equal-weight sleeve index has a daily return skew of **+0.11** over the five years, still right-tailed unlike every core portfolio, with excess kurtosis of 1.32. That is the expected shape for a basket of 40 small companies where the downside per name is bounded at GBP 1,000 and the upside is not, and it is the shape most vulnerable to survivorship bias, because the names were chosen in 2026.

But the sleeve's beta asymmetry is now clearly negative, at -0.13, against 0.003 for the quality sleeve it replaced: it captured 0.77 of the benchmark's up days and 0.91 of its down days. Taken at face value that is the opposite of convexity. Taken honestly it is a statement about 2021-2026 - a window in which frontier microcaps fell with the market and did not rise with it - rather than about the payoff the sleeve is designed to produce, which is a five- to ten-year distribution that a daily regression cannot see. Both readings are in section 6.7, and neither is settled here. **The combined GBP 200,000 book's beta asymmetry falls with it, from +0.07 to +0.03.**

3. The core universe

3.1 The 37 instruments

Every line below is purchasable by a UK retail client on interactive investor: LSE-listed UCITS ETFs and ETCs with a UK KID, ordinary shares on ii's dealt exchanges, and two individual gilts. No US-domiciled ETF appears, because UK retail clients cannot buy them - the issuers do not produce a PRIIPs Key Information Document, and without one a broker cannot legally sell the product to a retail investor. That is industry-wide regulation, not an ii policy. All 37 were re-verified as listed, solvent and tradeable at the 2026-09-16 close; none was added, removed or substituted, and the moonshot rebuild did not touch this sleeve at all.

Bucket	Ticker	ii EPIC	Name	Ccy	TER	YTM	Domicile	ii evidence	History proxy
etf_equity	IITU.L	IITU	iShares S&P 500 Information Technology Sector UCITS ETF USD (Acc)	GBP	0.15%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	VMID.L	VMID	Vanguard FTSE 250 UCITS ETF (GBP) Distributing	GBP	0.10%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	IWQU.L	IWQU	iShares Edge MSCI World Quality Factor UCITS ETF USD (Acc)	USD	0.25%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	VWRP.L	VWRP	Vanguard FTSE All-World UCITS ETF (USD) Accumulating	GBP	0.14%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	CSP1.L	CSP1	iShares Core S&P 500 UCITS ETF USD (Acc)	GBP	0.07%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	ISF.L	ISF	iShares Core FTSE 100 UCITS ETF GBP (Dist)	GBP	0.07%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	EMIM.L	EMIM	iShares Core MSCI EM IMI UCITS ETF USD (Acc)	GBP	0.18%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	WLDS.L	WLDS	iShares MSCI World Small Cap UCITS ETF USD (Acc)	GBP	0.35%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	SMEA.L	SMEA	iShares Core MSCI Europe UCITS ETF EUR (Acc)	GBP	0.12%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	IJPA.L	IJPA	iShares Core MSCI Japan IMI UCITS ETF USD (Acc)	USD	0.12%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	SMGB.L	SMGB	VanEck Semiconductor UCITS ETF	GBP	0.35%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_equity	XDWH.L	XDWH	Xtrackers MSCI World Health Care UCITS ETF 1C	USD	0.25%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etc_commodity	SGLN.L	SGLN	iShares Physical Gold ETC	GBP	0.12%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etc_commodity	SSLN.L	SSLN	iShares Physical Silver ETC	GBP	0.20%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etc_commodity	BRNT.L	BRNT	WisdomTree Brent Crude Oil	USD	0.49%	n/a	Jersey	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etc_commodity	COPA.L	COPA	WisdomTree Copper	USD	0.49%	n/a	Jersey	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etc_commodity	ICOM.L	ICOM	iShares Diversified Commodity Swap UCITS ETF	USD	0.19%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-

Bucket	Ticker	ii EPIC	Name	Ccy	TER	YTM	Domicile	ii evidence	History proxy
etc_commodity	YCA.L	YCA	Yellow Cake plc	GBP	n/a	n/a	Jersey	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	RR.L	RR.	Rolls-Royce Holdings plc	GBP	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	PLTR	PLTR	Palantir Technologies Inc.	USD	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	SPCX	SPCX	Space Exploration Technologies Corp. (SpaceX)	USD	n/a	n/a	-	ii international markets list (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	AZN.L	AZN	AstraZeneca PLC	GBP	n/a	n/a	-	ii international markets list (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	SHEL.L	SHEL	Shell plc	GBP	n/a	n/a	-	ii international markets list (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	HSBA.L	HSBA	HSBC Holdings plc	GBP	n/a	n/a	-	ii international markets list (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	ULVR.L	ULVR	Unilever PLC	GBP	n/a	n/a	-	ii international markets list (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	MSFT	MSFT	Microsoft Corporation	USD	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	ASML.AS	ASML	ASML Holding N.V.	EUR	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	SIE.DE	SIE	Siemens Aktiengesellschaft	EUR	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
stock	BRK-B	BRK.B	Berkshire Hathaway Inc. Class B	USD	n/a	n/a	-	ii instrument page (accessed 2026-09-14, re-verified 2026-09-16)	-
etf_bond	IHYU.L	IHYU	iShares \$ High Yield Corp Bond UCITS ETF USD (Dist)	USD	0.50%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_bond	IHYG.L	IHYG	iShares EUR High Yield Corp Bond UCITS ETF EUR (Dist)	EUR	0.50%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_bond	GHYS.L	GHYS	iShares Global High Yield Corp Bond GBP Hedged UCITS ETF (Dist)	GBP	0.55%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-

Bucket	Ticker	ii EPIC	Name	Ccy	TER	YTM	Domicile	ii evidence	History proxy
etf_bond	IEMB.L	IEMB	iShares J.P. Morgan \$ EM Bond UCITS ETF USD (Dist)	USD	0.45%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
etf_bond	IBTM.L	IBTM	iShares \$ Treasury Bond 7-10yr UCITS ETF USD (Dist)	GBP	0.07%	n/a	Ireland	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	-
gilt	T36N	T36N	4 7/8% Treasury Gilt 2036	GBP	n/a	5.300%	United Kingdom	ii bonds & gilts guide (online dealing, GBP 3.99)	IGLT.L (whole history)
gilt	T56	T56	5 3/8% Treasury Gilt 2056	GBP	n/a	5.870%	United Kingdom	ii bonds & gilts guide (online dealing, GBP 3.99)	GLTL.L (whole history)
trend	DBMG.L	DBMG	iMGP DBi Managed Futures UCITS ETF R USD (Acc)	GBP	0.75%	n/a	Luxembourg	LSE UCITS ETF/ETC with UK KID; ii deals all LSE ETFs online	DBMF (spliced before 2025-04-24)

Source: universe/core_universe.json -> instruments | TER and ISIN from justETF / issuer factsheets; gilt clean price, YTM and duration from Hargreaves Lansdown as at 2026-09-16, recorded in universe/market_params.json

Bucket counts: 12 equity ETFs, 6 commodity ETCs, 11 individual stocks, 5 bond ETFs, 2 individual gilts, 1 trend-following fund. Three instruments carry a proxy history flag everywhere they appear.

3.2 Resolutions and corrections made while building the universe

SPCX is SpaceX, and it is now a real listed security. Space Exploration Technologies Corp. completed its IPO on 2026-06-12 on NASDAQ at an offer price of USD 135, closing its first day near USD 161. The ticker resolves on Yahoo with 67 daily bars inside the five-year window and a 2026-09-16 close of USD 150.88. No substitution was made and no proxy was used. But 67 bars cannot support a covariance, a beta, a skew or a drawdown estimate, so SPCX is excluded from the optimiser universe entirely and carries a fixed 2.5% policy weight (GBP 4,000) instead, as an editable lever. Its statistics are reported throughout with an explicit insufficient-history warning. Rocket Lab (RKL.B) was identified as the nearest listed space pure-play if a proxy were ever wanted, and rejected as a weak one: Starlink connectivity, not launch, is the majority of SpaceX's revenue. One dating correction was made in this refresh: a news snapshot recorded a USD 143.51 close for 16 September against a USD 148.15 previous close; the yfinance series gives 14 September USD 148.15, 15 September USD 143.49 and 16 September USD 150.88, and the re-verification batch agrees on USD 150.88, so that snapshot appears to have been taken during the 15 September session and mislabelled. USD 150.88 is used; both figures are recorded in core_universe.json.

DBMG is what it says it is, and needs a proxy. DBMG.L is the LSE pence line of the iMGP DBi Managed Futures UCITS ETF (LU2951555585, Luxembourg, accumulating, TER 0.75%), listed 2025-04-22. At about 1.4 years it is below the brief's three-year minimum, so its history before 2025-04-24 is the US-listed sister fund DBMF converted to GBP and re-anchored to the first clean own bar. DBMF itself is US-domiciled and **not** purchasable on ii - it is a history proxy only, never a holding.

VMID was chosen over MIDD for the FTSE 250. Both track the identical index, so the 30 basis points a year of fee difference (VMID 0.10%, MIDD 0.40%) is the dominant long-horizon difference and compounds directly into return. MIDD's only advantage is Yahoo history back to 2009 against VMID's 2014 - but VMID already covers the brief's full ten-year window, so the longer history buys nothing this analysis needs. MIDD is recorded as the alternative if a pre-2014 crisis-inclusive history is ever required.

There is no helium ETC, anywhere. Helium is not an exchange-traded futures commodity: it is sold under bilateral long-term offtake contracts with no liquid futures curve, and an ETC needs exactly such a curve (or physical storage) to collateralise. The only route is listed equities. Two AIM candidates were examined - Pulsar Helium (PLSR.L) and Helium One Global (HE1.L) - and both were deliberately excluded from the core: they are sub-scale pre-revenue explorers, PLSR has under two years of history, and HE1 trades below one penny, where the bid-offer spread alone would swamp any weight an optimiser could sensibly assign. That is moonshot-sleeve territory, not core.

The hedged-bond correction. The brief treated SHYU and SEMB as GBP-hedged alternatives to IHYU and IEMB. They are not. An FX regression over the five-year daily window gives $\text{SHYU} \times \text{GBPUSD} / \text{IHYU} = 0.9998$ (sd 0.005) and $\text{SEMB}/100 \times \text{GBPUSD} / \text{IEMB} = 0.9999$ (sd 0.005): both are simply the GBP-quoted lines of the

same **unhedged** USD funds. Listing currency is not a hedging decision. The same test confirmed CSP1/CSPX, SMGB/SMH and IBTM/IDTM are matched pairs. The only genuinely share-class-hedged high-yield fund found was GHYS.L, and it is in the universe. No GBP-hedged EM sovereign ETF with usable history exists: the only candidate, EMBG.L, launched 2026-07-24.

The gilt benchmark choice, re-priced to 2026-09-16. The two individual gilts have no Yahoo ticker at all, so their column labels are LSE EPICs, not symbols, and their return history is entirely the proxy ETF. The 10-year line is **T36N** - 4 7/8% Treasury Gilt 2036, GB00BWB1N39, syndicated April 2026, clean price **96.75**, redemption yield **5.30%**, modified duration **7.46** - proxied by IGLT.L, whose 10.36-year weighted average maturity is a good duration match. An earlier draft used T4Q (4 1/4% Treasury Gilt 2036); T36N is the newer on-the-run issue and is the one carried through the model, with T4Q retained in `market_params.json` as the alternative. The 30-year line is **T56** - 5 3/8% Treasury Gilt 2056, clean price **93.155**, redemption yield **5.87%**, modified duration **14.06** - proxied by GLTL.L. Both the clean price and the redemption yield for each gilt now come from the same `market_params.json` record at the same date; the previous build paired a 2026-09-16 yield with a 2026-09-14 clean price, and that has been corrected. It matters only where a gilt is actually held: at 96.75 rather than 96.06, MinVariance's 7.235% T36N position buys **119** units rather than 120, and Variant A's 16.419% position buys **271** rather than 273. The duration match on the 30-year line remains **imperfect and understated in the model**: GLTL tracks a 15-year-plus index whose duration is longer than the individual gilt's, so the proxy overstates the gilt's rate sensitivity. More fundamentally, a gilt held to maturity and a perpetually-rolling gilt ETF are different instruments; the proxy's risk should not be read as the gilt's hold-to-maturity risk.

One further data decision. For pence-quoted tickers, Yahoo's `info['marketCap']` is already in pounds, not pence, and was treated accordingly. The previous build had one line, SGL.DE, for which yfinance returned neither `marketCap` nor `sharesOutstanding`, so its market cap, enterprise value and P/E were left blank rather than guessed; SGL.DE was a Sleeve A name and is not in this build, and **every one of the 51 stocks in this build has a computable market cap**.

4. Optimisation results

4.1 The six portfolios side by side

	Min variance	Max Sharpe	Moderate12 (headline)	Max Sortino	Equal weight (ref)	Naive (ref)
Holdings	14	14	14	15	37	37
Expected return, blend	6.3%	13.4%	15.4%	13.4%	11.1%	9.8%
Expected return, historical only	6.9%	20.2%	23.6%	20.3%	15.1%	12.4%
Expected return, CAPM/yield only	5.8%	6.6%	7.2%	6.5%	7.0%	7.3%
Ex-ante vol (LW, optimised 97.5%)	5.5%	9.6%	12.0%	9.7%	10.6%	10.8%
Ex-ante vol (all 37 lines)	6.1%	10.3%	12.8%	10.4%	11.3%	11.5%
Ex-ante Sharpe	0.42	0.93	0.91	0.92	0.65	0.53
Beta vs VWRP.L (as held)	0.36	0.63	0.79	0.61	0.74	0.81
Beta vs ^FTAS (as held)	0.27	0.48	0.57	0.49	0.61	0.62
beta+ / beta- (as held)	0.47 / 0.32	0.71 / 0.61	0.85 / 0.77	0.70 / 0.59	0.84 / 0.71	0.92 / 0.77
Asymmetry (as held / excl. SPCX)	0.15 / -0.03	0.09 / -0.08	0.08 / -0.09	0.11 / -0.07	0.13 / -0.04	0.15 / -0.02
Realised 5y return (total)	39%	170%	218%	171%	107%	81%
Realised 5y CAGR (daily)	6.8%	22.0%	26.1%	22.1%	15.7%	12.6%
Realised 5y vol (daily)	5.5%	9.6%	12.1%	9.7%	10.6%	10.8%
Realised 5y Sharpe	0.53	1.73	1.67	1.73	1.07	0.80
Realised 5y Sortino	0.74	2.55	2.45	2.56	1.54	1.14
Realised 5y skew	-0.49	-0.56	-0.49	-0.47	-0.40	-0.34
Realised 5y excess kurtosis	2.63	3.20	2.71	3.13	2.79	2.65
Realised 5y max drawdown	-6.7%	-10.0%	-13.4%	-9.7%	-11.9%	-13.7%
Backtest 5y CAGR (monthly)	6.7%	21.7%	26.0%	22.0%	15.8%	12.8%
Backtest 5y max DD (month-end)	-3.9%	-2.4%	-3.8%	-2.7%	-7.4%	-7.4%
Weighted TER	0.23%	0.16%	0.20%	0.16%	0.17%	0.21%
Invested GBP	159,265	159,575	159,287	159,253	157,709	158,586
Residual cash GBP	735	425	713	747	2,291	1,414

Source: data/analysis_results.json -> portfolios.*; data/weights.csv | EqualWeight and NaiveReference are unconstrained comparison references, not optimiser output. Every column in this table is unchanged from the 16 September build except the realised and backtest total-return and CAGR rows, which move by up to 0.2 pp because a level-base defect was fixed on 2026-09-17 (see the end of section 4)

EqualWeight and NaiveReference are references, not candidates. EqualWeight in particular **breaches** the 25% all-stocks cap (10 stocks at 2.71% each plus SPCX at 2.50% = 29.6%) and is kept only for comparison. The four optimised portfolios satisfy every bound, the budget and all five group constraints to within 1e-06.

4.2 Weights

Ticker	Name	Bucket	Min variance	Max Sharpe	Moderate12	Max Sortino	Equal weight	Naive
IITU.L	iShares S&P 500 Information Technology Sector UCITS ETF USD (Acc)	etf_equity	-	7.7%	5.4%	2.8%	2.7%	5.0%
VMID.L	Vanguard FTSE 250 UCITS ETF (GBP) Distributing	etf_equity	4.9%	-	-	-	2.7%	5.0%
IWQU.L	iShares Edge MSCI World Quality Factor UCITS ETF USD (Acc)	etf_equity	-	-	-	-	2.7%	5.0%
VWRP.L	Vanguard FTSE All-World UCITS ETF (USD) Accumulating	etf_equity	-	-	-	-	2.7%	5.0%
CSP1.L	iShares Core S&P 500 UCITS ETF USD (Acc)	etf_equity	4.2%	-	-	-	2.7%	5.0%
ISF.L	iShares Core FTSE 100 UCITS ETF GBP (Dist)	etf_equity	14.7%	15.9%	13.5%	19.6%	2.7%	5.0%
EMIM.L	iShares Core MSCI EM IMI UCITS ETF USD (Acc)	etf_equity	1.9%	-	-	-	2.7%	5.0%
WLDS.L	iShares MSCI World Small Cap UCITS ETF USD (Acc)	etf_equity	-	-	-	-	2.7%	5.0%
SMEA.L	iShares Core MSCI Europe UCITS ETF EUR (Acc)	etf_equity	-	-	-	-	2.7%	5.0%
IJPA.L	iShares Core MSCI Japan IMI UCITS ETF USD (Acc)	etf_equity	-	-	-	-	2.7%	5.0%
SMGB.L	VanEck Semiconductor UCITS ETF	etf_equity	-	6.4%	16.6%	7.5%	2.7%	5.0%
XDWH.L	Xtrackers MSCI World Health Care UCITS ETF 1C	etf_equity	4.3%	-	-	-	2.7%	5.0%
SGLN.L	iShares Physical Gold ETC	etc_commodity	5.6%	16.4%	14.8%	16.5%	2.7%	1.7%
SSLN.L	iShares Physical Silver ETC	etc_commodity	-	-	-	-	2.7%	1.7%
BRNT.L	WisdomTree Brent Crude Oil	etc_commodity	-	3.6%	5.2%	3.5%	2.7%	1.7%
COPA.L	WisdomTree Copper	etc_commodity	-	-	-	-	2.7%	1.7%
ICOM.L	iShares Diversified Commodity Swap UCITS ETF	etc_commodity	4.9%	-	-	-	2.7%	1.7%
YCA.L	Yellow Cake plc	etc_commodity	-	-	-	-	2.7%	1.7%
RR.L	Rolls-Royce Holdings plc	stock	-	5.0%	5.0%	5.0%	2.7%	0.8%
PLTR	Palantir Technologies Inc.	stock	-	3.7%	5.0%	5.0%	2.7%	0.8%
SPCX	Space Exploration Technologies Corp. (SpaceX)	stock	2.5%	2.5%	2.5%	2.5%	2.5%	2.5%
AZN.L	AstraZeneca PLC	stock	-	0.5%	1.1%	0.5%	2.7%	0.8%
SHEL.L	Shell plc	stock	-	4.8%	5.0%	3.6%	2.7%	0.8%
HSBA.L	HSBC Holdings plc	stock	-	5.0%	5.0%	5.0%	2.7%	0.8%
ULVR.L	Unilever PLC	stock	3.3%	-	-	1.3%	2.7%	0.8%
MSFT	Microsoft Corporation	stock	-	-	-	-	2.7%	0.8%
ASML.AS	ASML Holding N.V.	stock	-	-	-	-	2.7%	0.8%
SIE.DE	Siemens Aktiengesellschaft	stock	-	-	-	-	2.7%	0.8%
BRK-B	Berkshire Hathaway Inc. Class B	stock	0.8%	3.5%	1.4%	2.1%	2.7%	0.8%
IHYU.L	iShares \$ High Yield Corp Bond UCITS ETF USD (Dist)	etf_bond	-	-	-	-	2.7%	2.1%
IHYG.L	iShares EUR High Yield Corp Bond UCITS ETF EUR (Dist)	etf_bond	-	-	-	-	2.7%	2.1%
GHYS.L	iShares Global High Yield Corp Bond GBP Hedged UCITS ETF (Dist)	etf_bond	12.8%	-	-	-	2.7%	2.1%
IEMB.L	iShares J.P. Morgan \$ EM Bond UCITS ETF USD (Dist)	etf_bond	-	-	-	-	2.7%	2.1%
IBTM.L	iShares \$ Treasury Bond 7-10yr UCITS ETF USD (Dist)	etf_bond	20.0%	16.3%	10.0%	16.2%	2.7%	2.1%
T36N	4 7/8% Treasury Gilt 2036	gilt	7.2%	-	-	-	2.7%	2.1%
T56	5 3/8% Treasury Gilt 2056	gilt	-	-	-	-	2.7%	2.1%

Ticker	Name	Bucket	Min variance	Max Sharpe	Moderate12	Max Sortino	Equal weight	Naive
DBMG.L	iMGP DBi Managed Futures UCITS ETF R USD (Acc)	trend	12.9%	8.7%	9.5%	8.8%	2.7%	5.0%
	Total		100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

Source: data/weights.csv | fractions of the GBP 160,000 core sleeve; SPCX fixed at 2.5% by policy in every column

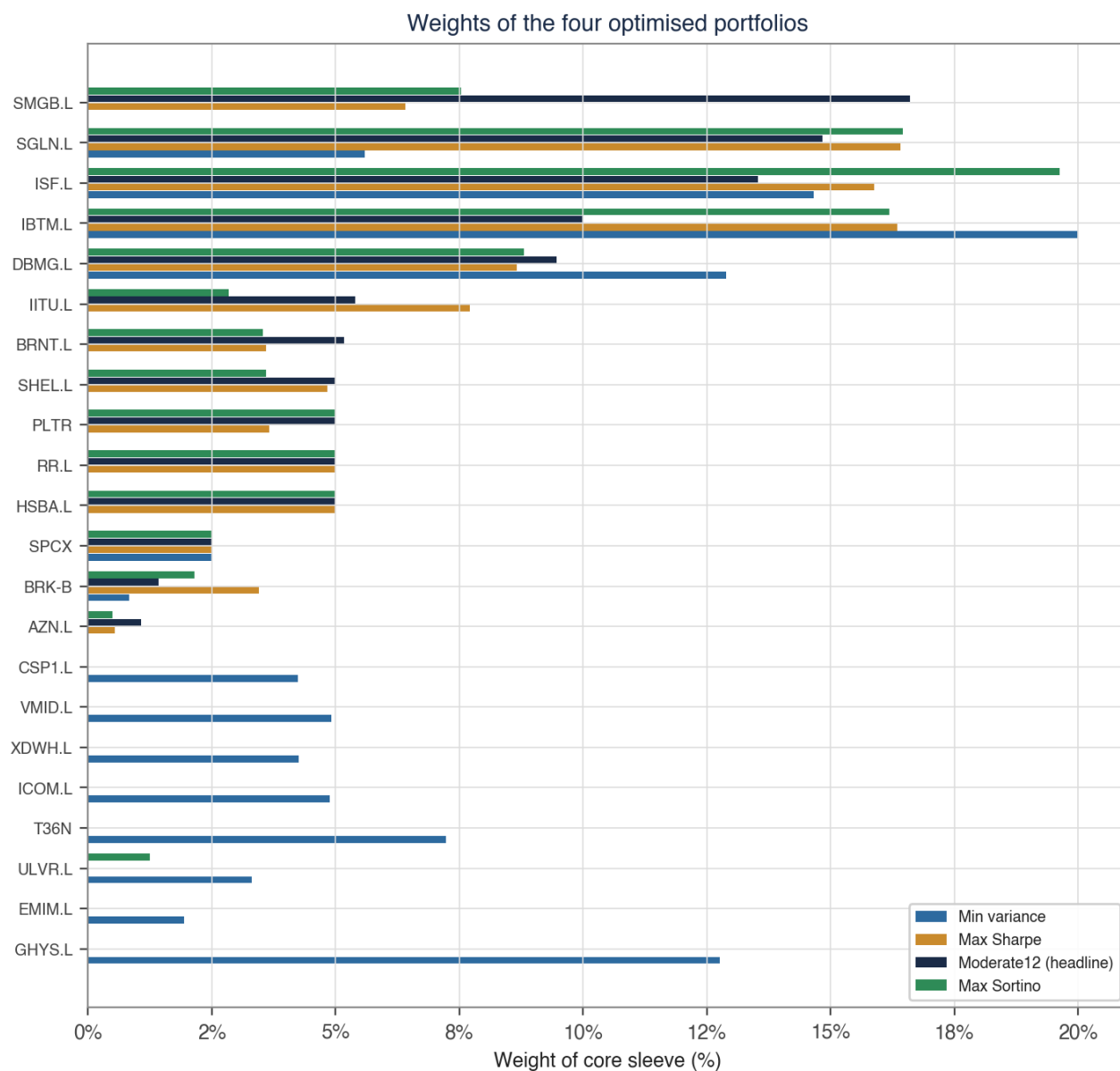


Figure 3. Weights of the four optimised portfolios and the two references. Source: data/weights.csv.

4.3 The efficient frontier

The concentration is real and should not be glossed over. SMGB.L, the VanEck Semiconductor UCITS ETF, takes 16.6% of the core weight and contributes 35.8% of its ex-ante variance. What stops it going further is the 20% single-fund cap, not diversification. Its marginal risk contribution (0.258) is the highest of any held line except Palantir's. A semiconductor ETF with a 1.91 beta to world equity and a 32% annualised volatility is doing a great deal of the work in a portfolio labelled 'moderate' - and it was also the line that lost the most over 11-16 September (-3.41%), which is why its weight came down 0.47 pp in the 16 September refresh rather than up. Section 4.6 prices what it would cost to force that concentration down.

Other features worth naming:

- The **commodity group sits at its 20% ceiling** (20.0%), almost all of it gold (14.8% SGLN.L) plus a 5.2% Brent line. Gold's five-year sample return was 20.0% a year with a beta of only 0.05, which is exactly the combination a mean-variance optimiser loves and exactly the combination least likely to repeat.
- The **single-stock group sits at its 25% ceiling** (25.0%), with Rolls-Royce, Palantir, Shell and HSBC all pinned at the 5% per-stock cap and SPCX taking 2.5% by policy. AstraZeneca and Berkshire take the residual. Four of six stock lines at their cap means the cap, not the model, is choosing the size.
- The **bond group sits at its 10% floor** (10.0%), entirely in IBTM.L. Both gilts and all four credit ETFs get zero. Section 4.6 explains why, and it is the most consequential zero in the model.
- **Equity ETFs take 35.5%**, above their 30% floor, split between the semiconductor line, the FTSE 100 and a US technology line.
- The **trend fund takes 9.5%** of a possible 15%. DBMG.L's repaired history shows a 0.13 beta and a 15.3% volatility - a genuine diversifier, and the line most dependent on a data repair.

4.5 Risk contributions

A weight is not a risk. The Euler decomposition splits the portfolio's ex-ante volatility into per-asset contributions that sum exactly to it: $MCR_i = (\Sigma w)_i / \sigma_p$, and $RC_i = w_i \times MCR_i$.

Ticker	Name	Weight	Marginal risk contribution	Risk contribution	% of total risk
SMGB.L	VanEck Semiconductor UCITS ETF	16.6%	0.258	0.0429	35.8%
PLTR	Palantir Technologies Inc.	5.0%	0.335	0.0167	14.0%
ISF.L	iShares Core FTSE 100 UCITS ETF GBP (Dist)	13.5%	0.080	0.0109	9.0%
RR.L	Rolls-Royce Holdings plc	5.0%	0.212	0.0106	8.8%
IITU.L	iShares S&P 500 Information Technology Sector UCITS ETF USD (Acc)	5.4%	0.170	0.0091	7.6%
SGLN.L	iShares Physical Gold ETC	14.8%	0.049	0.0073	6.1%
BRNT.L	WisdomTree Brent Crude Oil	5.2%	0.121	0.0062	5.2%
HSBA.L	HSBC Holdings plc	5.0%	0.120	0.0060	5.0%
SHEL.L	Shell plc	5.0%	0.106	0.0053	4.4%
DBMG.L	iMGP DBi Managed Futures UCITS ETF R USD (Acc)	9.5%	0.049	0.0047	3.9%
BRK-B	Berkshire Hathaway Inc. Class B	1.4%	0.051	0.0007	0.6%
AZN.L	AstraZeneca PLC	1.1%	0.042	0.0005	0.4%
IBTM.L	iShares \$ Treasury Bond 7-10yr UCITS ETF USD (Dist)	10.0%	-0.009	-0.0009	-0.7%
	Total (optimised 97.5%)	97.5%		0.1200	100.0%

Source: data/analysis_results.json -> portfolios.Moderate12.risk_contribution; data/risk_contribution.csv | ex-ante, Ledoit-Wolf covariance; SPCX is outside this matrix

Weight vs. risk contribution, Moderate12

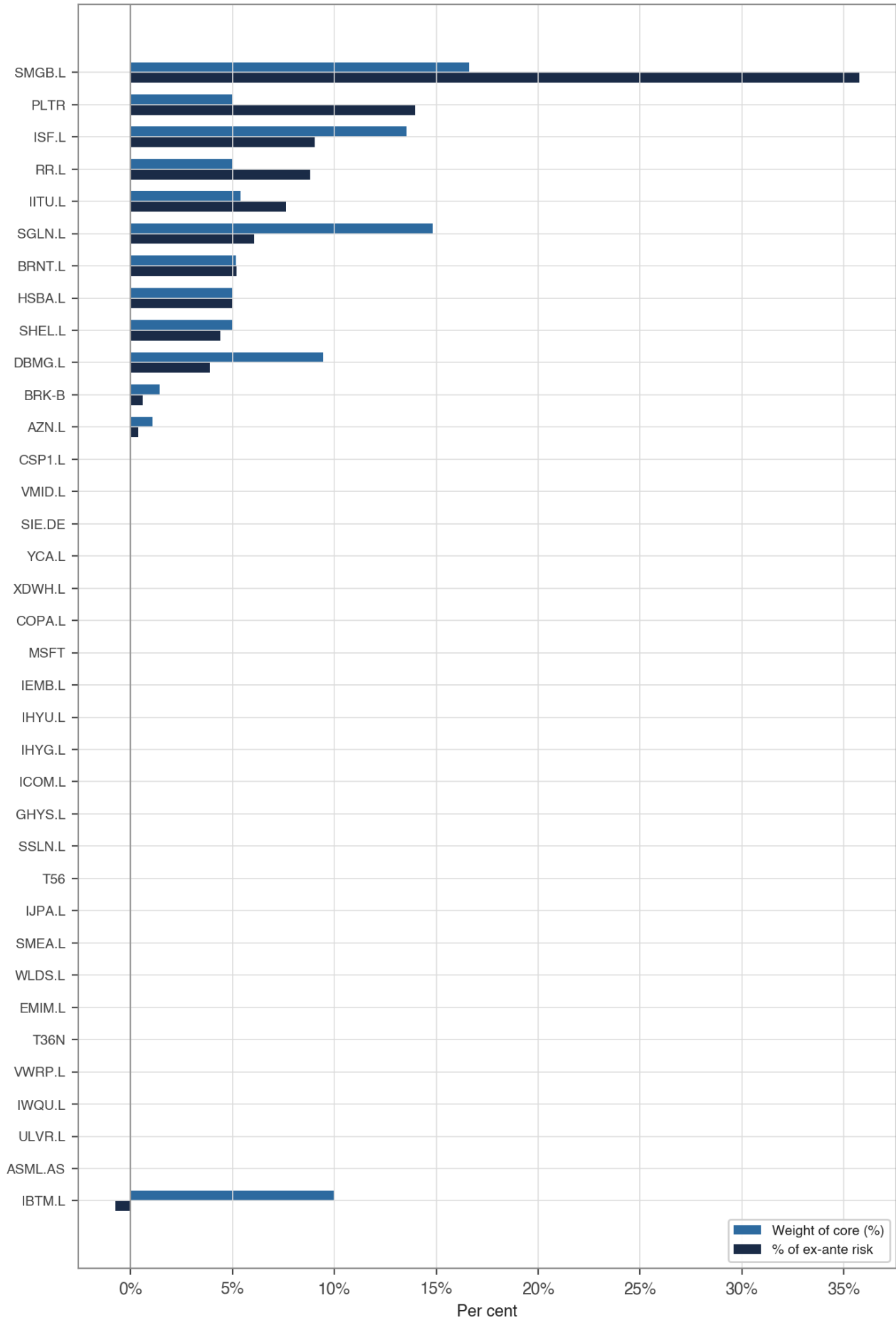


Figure 5. Weight against percentage of ex-ante risk for each held line. Source: data/risk_contribution.csv.

Three lines - SMGB.L, PLTR and RR.L - carry 58.6% of the risk on 26.6% of the money. Gold does the opposite: 14.8% of the weight for 6.1% of the risk. That is the diversification the optimiser is buying, and it is why the commodity group is pinned at its ceiling.

4.6 Sensitivity to the expected-return estimate

This is the single most important sensitivity in the report. The optimiser input is the 50/50 blend. Re-solving the identical problem with each estimator on its own gives:

Expected-return vector fed to the optimiser	Moderate12 expected return (optimised part)	Ex-ante vol	One-way turnover vs the headline	Top holdings
Historical 5y sample mean only	23.3%	12.0%	7.8%	SMGB.L 19.4%, SGLN.L 14.0%, DBMG.L 13.5%, ISF.L 12.1%, IBTM.L 10.0%
CAPM / yield only	7.8%	12.0%	82.7%	IHYU.L 20.0%, CSP1.L 19.4%, GHYS.L 16.3%, IITU.L 13.8%, SMGB.L 8.5%
50/50 blend (the headline)	15.0%	12.0%	0.0%	SMGB.L 16.6%, SGLN.L 14.8%, ISF.L 13.5%, IBTM.L 10.0%, DBMG.L 9.5%

Source: data/analysis_results.json -> expected_return_sensitivity

Read the turnover column first. Moving from the blend to the historical-only vector would require trading 7.8% of the portfolio - the two vectors largely agree. Moving to the CAPM/yield-only vector would require trading 82.7% of it. That is not a sensitivity; that is a different portfolio. Under the structural vector the single-stock group collapses from 25% to 4.8% including the SPCX policy line, commodities fall from 20% to their 5% floor, bonds rise to 36.3% plus a 3.7% gilt line, and equity ETFs take just over half the book. The same constraints, the same covariance, the same optimiser - and an almost completely different answer.

The same picture holds for the max-Sharpe portfolio: 7.5% turnover to the historical vector, 78.7% to the structural one.

Two structured variants were solved as well, each changing exactly one thing about the headline problem.

	Headline Moderate12	Variant A: gilts at redemption yield	Variant B: tighter concentration caps
What changed	-	T36N and T56 priced at their redemption yield instead of the 50/50 blend	single fund line 20% -> 10%, single stock 5% -> 4%
Expected return (blend basis)	15.4%	14.9%	14.9%
Ex-ante vol (optimised part)	12.0%	12.0%	12.0%
Ex-ante Sharpe	0.91	0.87	0.88
Holdings	14	14	17
Largest single line	SMGB.L 16.6%	SMGB.L 17.7%	10.0% (four lines at the 10% cap)
Gilt weight	0.0%	T36N 16.4%	0.0%
Bond ETF weight	10.0%	0.0%	10.0%
Weighted TER	0.20%	0.19%	0.22%
Backtest 5y CAGR	26.0%	24.7%	24.1%
Backtest 5y max DD	-3.8%	-7.6%	-5.6%
Realised 5y Sharpe	1.67	1.58	1.57
One-way turnover vs the headline	-	19.1%	32.1%

Source: data/analysis_variants.json -> variants.*.Moderate12; data/variants_summary.csv; data/weights_variants.csv

Why the gilts get zero weight, and what changes under variant A.

The two individual gilts and the long-duration bond ETF get no weight in the headline portfolio for a specific and arguable reason. The 50/50 blend averages a forward-looking redemption yield with a backward-looking five-year sample return, and for gilts those two numbers point in opposite directions:

Gilt	5y sample return (a year)	Redemption yield	50/50 blend used in the headline
T36N (4 7/8% Treasury Gilt 2036)	-4.0%	5.3%	0.6%
T56 (5 3/8% Treasury Gilt 2056)	-11.1%	5.9%	-2.6%

Source: data/expected_returns.csv; universe/market_params.json benchmark_gilt_10y / benchmark_gilt_30y

The five-year sample return is the once-in-a-generation repricing from roughly 1% yields to 5.3% - a capital loss that **has already happened and cannot happen again from today's level**. Blending it into a forward estimate double-counts the past. For a gilt held to maturity the redemption yield is the better estimate, full stop.

Variant A prices this. Feeding the optimiser the redemption yield for the two gilts and leaving everything else unchanged moves **19.1%** of the portfolio: the 10-year gilt T36N takes 16.4% of the core, the entire bond-ETF allocation (IBTM.L's 10.0%) disappears into it, the trend fund falls from 9.5% to 8.6%, and the FTSE 100 line is roughly halved (13.5% to 6.9%). The cost is 0.5% of blended expected return and 0.04 of ex-ante Sharpe; the backtest CAGR falls from 26.0% to 24.7% and the month-end drawdown roughly doubles to -7.6%.

This is a judgement call the model exposes rather than settles. Variant A is a single-change sensitivity, not a coherent all-forward-looking vector - it leaves the credit ETFs and IBTM.L on the blend even though their estimate #2 is already yield-based. And a redemption yield is the forward return only if the gilt is actually held to maturity with coupons reinvested at that yield, while the covariance matrix still uses the rolling-ETF proxy's risk. The Inputs tab of the Excel model is where this should be switched and re-run, not a settled answer here.

Variant B prices the concentration. Tightening the single-fund cap to 10% and the single-stock cap to 4% pushes SMGB.L from 16.6% to 10.0%, spreads the book across 17 lines instead of 14, and brings the top risk contributor down. It costs 0.5% of blended expected return at the same 12% volatility - in other words, forcing diversification here is close to free on this data. It requires trading 32.1% of the portfolio, and it raises the weighted TER slightly (to 0.22%) because the cheap concentrated lines are replaced by more of the expensive ones. One caution specific to this variant: with four lines pinned at the 10% cap the residual allocation is a near-degenerate choice between two near-substitute high-yield ETFs whose blended estimates are within a few basis points of each other, and a tiny input move flips it (GHYS.L 5.41% to 4.44%, IHYU.L 4.59% to 5.56% between the two builds). That is a property of the tighter cap set, not new information.

What changed since the 11 September build, and what the moonshot rebuild changed

Two things have changed since 11 September, and they are of completely different kinds. The 16 September refresh moved the as-of date and re-priced everything. The 17 September moonshot rebuild moved nothing except Sleeve A, and moved all of it. Both are set out below, because the core sleeve's numbers are inherited unchanged from the first and the sleeve's numbers come entirely from the second.

Part one: the 11-to-16 September refresh (inherited unchanged). The as-of date moved from the 2026-09-11 close to the 2026-09-16 close. The estimators, the constraints, the 12% target, the benchmark and the 2.5% SPCX policy weight were identical, and the universe files carried no swaps: all 37 core instruments and all 40 names of the then-current Sleeve A survived. Only the CONFIG dates, the rate and yield inputs, two masked bad bars and the gilt clean-price source changed.

No allocation of consequence moved. The largest one-way turnover across the eight portfolios is 1.9% of the core sleeve. The headline Moderate-12% portfolio turns over **1.48%** and its expected return falls **-0.09 pp**, 15.50% to 15.41%; its ex-ante volatility is unchanged at 12.00% because the constraint binds in both builds, so the optimiser keeps taking the same risk and gets paid slightly less for it. Its beta rises +0.006 to 0.794: replacing the risk it lost in SMGB.L (-0.47 pp of weight after a -3.41% three days) with cheaper beta in ISF.L (+0.78 pp) and IITU.L (+0.48 pp). MaxSharpe and MaxSortino, free of a binding vol target, de-risked instead - both cut SMGB.L by about a point and lost more expected return (-0.17 pp and -0.18 pp) while losing almost no Sharpe. GMV is the only portfolio whose expected return rose (+0.04 pp).

Two published statistics did move in that refresh, and not because of the market. Two placeholder bars on 2025-10-24 (WLDS.L and SMEA.L) were masked, changing those funds' reported volatility, skew and drawdown materially - WLDS.L's five-year volatility fell 8.87 pp - while moving no weight by more than 0.1 pp. The calendar separately recovered two genuine LSE sessions inside the old span, which is the whole of the 1,259 to 1,261 change in return days; the window roll itself was net zero. "Nothing material moved" was true of the portfolio and false of the per-asset statistics.

Inputs. Bank Rate unchanged at 3.75%; ERP unchanged at 4.17%. Gilt redemption yields fell (T36N 5.466% to 5.30%, T56 6.003% to 5.87%) as both prices rose into the CPI print. IBTM.L is the one line whose structural estimate fell for a policy reason: the US 10-year rose 5bp but the Fed hiked 25bp, so the interest-parity carry term fell and the net estimate is -20bp. Line-by-line detail is in archive/2026-09-16-quality-sleeve/data/REFRESH_DIFF.md.

Part two: the 17 September moonshot rebuild. This one has exactly one driver. The as-of date, the 2,526-row calendar, the 1,261 return days, Bank Rate, the ERP, the gilt and Treasury yields, the SPCX policy weight, the constraints and the core universe are all **identical** between the two builds, and both read the same `universe/market_params.json`. There is no window roll and no market move to attribute anything to. The only input that changed is the 40-name Sleeve A list, and it changed by **34 of 40 names**.

The core sleeve did not move, and that was verified rather than assumed. The optimiser never sees Sleeve A: its universe is the 36 core lines with at least three years of history, its covariance is the core-only Ledoit-Wolf matrix and its expected returns are the core-only blend. The inputs reproduce to 3e-16 (`cov_lw.csv`, `cov_annual.csv`, `corr.csv`, `expected_returns.csv`), the Ledoit-Wolf intensity is bit-identical at $\delta = 0.035840$, and MinVariance, **Moderate12 (the headline)**, EqualWeight and NaiveReference reproduce to 3e-15 or better on every weight. MaxSharpe and MaxSortino miss a literal 1e-9 bar by 2.1e-08 and 9.0e-09 respectively - not a data difference but a flat objective surface, where last-bit float noise between two runs of threaded BLAS moves an SLSQP solution stopping at $\text{ftol } 1\text{e-}12$. The money impact is **GBP 0.0034 on a GBP 160,000 sleeve**, and no reported statistic moves in its fourth decimal.

One published defect was found and fixed, and it touches every total return and CAGR in this report. Blind verification found that a return series' total-return level was being compounded from 1.0 *on* the date of its first return rather than on the day before it, so that first return was counted in every distributional statistic but lost from the compounded level. `total_return` was therefore taken over 1,260 of the 1,261 returns, and `cagr` annualised it over the wrong span. It was reproduced to ten decimal places and fixed in `scripts/analysis.py`. Mean, volatility, Sharpe, Sortino, skew, kurtosis, VaR, CVaR, the betas and the correlations are all unaffected. What moves is small but real: the headline core portfolio's realised five-year total return reads 218% rather than 219% and its CAGR 26.07% rather than 26.13%, the monthly backtest CAGRs of EqualWeight and NaiveReference move up 0.2 pp, and the archived quality sleeve restates from 182.4% to 183.7%. **Every old-versus-new comparison in section 6.8 restates the archived figures onto the corrected basis first**, so the two sides stay like-for-like.

What the sleeve change cost. The sleeve's blended expected return falls from 15.77% to **9.99%**, its realised five-year CAGR from 23.2% to **9.1%** and its Sharpe from 0.77 to **0.32**, while its volatility barely moves (26.8% to 26.5%) and its maximum drawdown deepens from -40.2% to **-45.5%**. The combined GBP 200,000 book inherits all of that at a fifth of the size: ex-ante expected return 15.48% to **14.33%**, ex-ante volatility 14.42% to **14.23%**, ex-ante Sharpe 0.813 to **0.743**, realised backtest CAGR 26.0% to **23.0%**. The sleeve's share of total portfolio risk is almost unchanged at 31.9%. Section 6.8 carries the full table and the reasons.

One honest wrinkle in the combined covariance. The 77x77 pairwise-complete matrix is made positive semi-definite by eigenvalue clipping applied to the *whole* matrix. Replacing 34 of the 40 sleeve rows changes the negative eigenvalues that have to be clipped (-0.1463 to -0.1032), so the repair redistributes slightly differently across the core block too - largest cell change 0.0115, on the SPCX row, and the largest implied core volatility change is SGLN.L at 0.85 pp. The **optimiser's** covariance is untouched (max abs change 3.1e-16), so no core weight moves; but the combined book's ex-ante volatility, betas and risk split are computed from a matrix whose core block has shifted marginally. That is a property of the PSD repair, not a market event, and it is disclosed rather than smoothed over.

Source: data/REFRESH_DIFF.md; data/refresh_diff.json; universe/refresh_changelog.json; archive/2026-09-16-quality-sleeve/data/REFRESH_DIFF.md

5. Risk

5.1 Per-instrument risk statistics, core universe

All figures are 5-year daily, GBP total return, benchmark VWRP.L, over 2021-09-16 to 2026-09-16. VaR and CVaR are historical (not parametric) and are daily losses expressed as positive numbers.

Ticker	Beta	beta+	beta-	Asym.	Skew	Exc. kurt.	Sortino	Vol	Max DD
IITU.L	1.48	1.47	1.35	0.12	-0.16	1.46	1.17	22.6%	-28.0%
VMID.L	0.82	0.88	0.84	0.04	0.07	2.47	0.08	15.2%	-28.8%
IWQU.L	0.98	0.98	0.91	0.07	0.06	2.39	0.63	16.7%	-18.7%
VWRP.L	1.00	1.00	1.00	0.00	-0.38	3.05	0.85	13.0%	-17.6%
CSP1.L	1.06	1.08	1.04	0.04	-0.28	2.72	0.95	14.5%	-20.8%
ISF.L	0.65	0.71	0.74	-0.03	-0.60	4.70	0.99	12.5%	-12.7%
EMIM.L	0.93	0.89	0.93	-0.04	-0.21	2.79	0.52	16.5%	-21.0%
WLDS.L	1.07	1.09	1.13	-0.04	-0.38	2.29	0.39	15.6%	-21.6%
SMEA.L	0.84	0.92	0.86	0.06	-0.11	3.48	0.56	14.0%	-15.8%
IJPA.L	0.80	0.78	0.76	0.02	0.09	1.96	0.52	19.1%	-19.5%
SMGB.L	1.91	1.80	1.71	0.09	-0.16	1.21	1.31	32.4%	-36.2%
XDWH.L	0.55	0.58	0.53	0.05	-0.03	2.04	0.23	15.7%	-18.3%
SGLN.L	0.05	-0.03	0.07	-0.10	-0.32	4.84	1.37	17.1%	-24.9%
SSLN.L	0.42	0.29	0.53	-0.23	-0.69	18.10	0.64	39.4%	-64.9%
BRNT.L	0.11	-0.41	0.58	-0.99	-0.32	3.01	1.01	37.0%	-33.4%
COPA.L	0.48	0.32	0.61	-0.29	-1.37	21.07	0.41	26.6%	-26.4%
ICOM.L	0.09	-0.21	0.27	-0.48	-0.28	2.25	0.77	18.0%	-28.5%
YCA.L	0.99	0.95	1.06	-0.11	0.25	2.28	0.48	36.4%	-48.2%
RR.L	1.31	1.41	1.45	-0.04	1.19	13.13	2.20	41.6%	-55.1%
PLTR	1.44	1.42	1.00	0.42	1.17	8.49	1.32	67.2%	-76.1%
SPCX *	2.95	8.34	1.40	6.93	0.51	2.52	0.17	90.3%	-48.9%
AZN.L	0.43	0.53	0.49	0.03	-0.24	5.75	0.53	24.7%	-26.7%
SHEL.L	0.47	0.20	0.87	-0.67	-0.47	3.39	1.21	24.4%	-20.6%
HSBA.L	0.85	0.89	1.09	-0.20	-0.51	4.35	1.93	24.7%	-22.0%
ULVR.L	0.11	0.27	0.01	0.26	-0.15	13.75	0.10	19.7%	-28.4%
MSFT	0.67	0.65	0.47	0.18	0.52	6.20	0.58	29.2%	-34.1%
ASML.AS	1.92	1.82	1.65	0.17	-0.18	2.84	0.64	40.2%	-46.4%
SIE.DE	1.41	1.43	1.58	-0.15	0.06	3.20	0.69	30.8%	-37.0%
BRK-B	0.34	0.34	0.34	-0.00	-0.10	3.27	0.85	18.8%	-20.6%
IHYU.L	0.28	0.27	0.23	0.04	0.33	2.90	0.12	10.0%	-10.5%
IHYG.L	0.24	0.27	0.21	0.06	0.46	3.38	-0.22	7.4%	-15.5%
GHYS.L	0.27	0.29	0.28	0.01	-0.11	3.81	-0.11	6.1%	-14.7%
IEMB.L	0.30	0.34	0.21	0.13	0.36	3.81	-0.23	11.5%	-15.2%
IBTM.L	-0.03	-0.03	-0.14	0.11	0.28	2.38	-0.70	9.4%	-16.3%
T36N	0.04	0.05	-0.05	0.10	1.25	16.12	-1.10	10.3%	-33.5%
T56	0.07	0.02	-0.06	0.08	2.16	31.97	-1.12	19.7%	-53.0%
DBMG.L	0.13	-0.10	0.20	-0.30	-0.16	2.76	0.61	15.3%	-31.6%
VWRP.L (benchmark)	1.00	1.00	1.00	0.00	-0.38	3.05	0.85	13.0%	-17.6%

Ticker	95% VaR	95% CVaR	99% VaR	99% CVaR
IITU.L	2.25%	3.11%	3.57%	4.52%
VMID.L	1.49%	2.14%	2.68%	3.14%
IWQU.L	1.61%	2.29%	2.77%	3.52%
VWRP.L	1.27%	1.90%	2.11%	3.15%

Ticker	95% VaR	95% CVaR	99% VaR	99% CVaR
CSP1.L	1.44%	2.11%	2.37%	3.31%
ISF.L	1.23%	1.90%	2.37%	3.28%
EMIM.L	1.65%	2.43%	2.99%	3.83%
WLDS.L	1.52%	2.28%	2.95%	3.77%
SMEA.L	1.32%	2.05%	2.60%	3.28%
IJPA.L	1.88%	2.60%	2.96%	3.77%
SMGB.L	3.21%	4.46%	5.21%	6.38%
XDWH.L	1.48%	2.17%	2.59%	3.27%
SGLN.L	1.59%	2.50%	3.05%	4.26%
SSLN.L	3.41%	5.70%	6.27%	11.74%
BRNT.L	3.63%	5.40%	6.40%	8.77%
COPA.L	2.50%	3.66%	3.78%	6.04%
ICOM.L	1.68%	2.60%	3.15%	4.07%
YCA.L	3.62%	4.92%	5.79%	7.17%
RR.L	3.54%	5.28%	5.95%	9.10%
PLTR	6.05%	8.33%	9.82%	12.44%
SPCX *	7.58%	11.68%	14.47%	15.84%
AZN.L	2.18%	3.59%	4.16%	6.44%
SHEL.L	2.34%	3.68%	4.53%	5.85%
HSBA.L	2.32%	3.81%	4.78%	6.27%
ULVR.L	1.63%	2.77%	3.04%	5.08%
MSFT	2.94%	3.95%	4.18%	5.77%
ASML.AS	3.84%	5.63%	6.51%	8.86%
SIE.DE	2.95%	4.38%	5.17%	6.59%
BRK-B	1.76%	2.59%	3.04%	4.15%
IHYU.L	0.96%	1.35%	1.60%	1.95%
IHYG.L	0.71%	0.99%	1.15%	1.45%
GHYS.L	0.62%	0.89%	1.02%	1.38%
IEMB.L	1.11%	1.57%	1.94%	2.40%
IBTM.L	0.95%	1.26%	1.43%	1.87%
T36N	0.98%	1.36%	1.45%	2.14%
T56	1.79%	2.54%	2.75%	4.01%
DBMG.L	1.49%	2.23%	2.80%	3.41%
VWRP.L (benchmark)	1.27%	1.90%	2.11%	3.15%

* = fewer than three years of daily history in the window; every statistic on that row is reported for completeness and is **not** an estimate. Only SPCX carries the flag in the core universe (67 bars from 2026-06-12).

Source: data/stats_daily.csv | 5y daily 2021-09-16 to 2026-09-16; rf = 3.75%; benchmark VWRP.L in GBP

A few readings worth pulling out:

- **Gold (SGLN.L) has a beta of 0.05 to world equity** on a 17.1% volatility and a 20.0% sample return. Near-zero beta with a large positive sample mean is exactly what a mean-variance optimiser reaches for, and it is why the commodity group sits at its ceiling. Note that its return distribution is left-tailed (skew -0.32, excess kurtosis 4.84) - low beta is not the same thing as a safe asset.
- **Brent (BRNT.L) is the one genuinely counter-cyclical line:** beta+ -0.41 against beta- 0.58, an asymmetry of -0.99 - it rose on the days world equity fell, over this window. That is a 2022-2026 sample property, not a law, and it comes with a 37.0% volatility. The three days added in this refresh sharpened it: the oil complex gained 1.87% into a soft global tape.

- **The trend fund (DBMG.L) has a beta of 0.13** - the lowest equity beta of any line except gold and the gilt proxies - and it is the line whose statistics depended entirely on the data repair described in 2.2.
- **The two gilt proxies have essentially no equity beta** (0.04 and 0.07, with negative downside betas) but five-year drawdowns of -33.5% and -53.0% and Sortino ratios below -1. Long gilts were not a safe haven over this window; they were the single largest source of loss in the universe, and the 30-year line lost more than half its value.
- **The largest single-name tail is Rolls-Royce**: a 1.19 skew with 13.13 excess kurtosis and a -55.1% drawdown inside the window - a right-tailed distribution that still lost more than half its value along the way.
- **SPCX's numbers are meaningless as estimates**: beta 2.95, beta+ 8.34, volatility 90.3%, drawdown -48.9%, all from 67 days. They are printed so nobody has to wonder, not so anybody can use them. Three extra bars this refresh moved the beta from 2.83 to 2.95.

5.2 Correlations

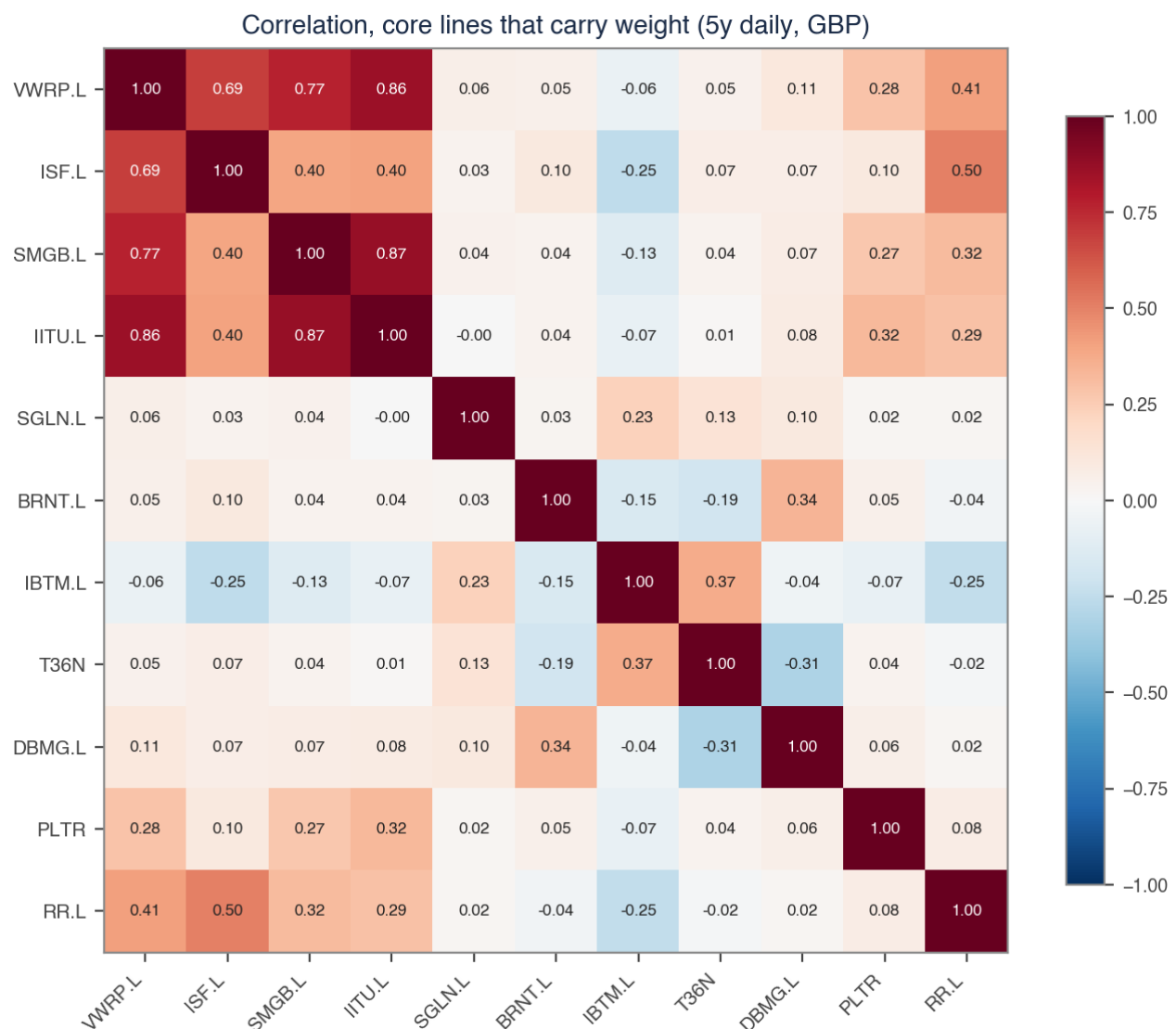


Figure 6. Correlation matrix of the 36-instrument optimiser universe, 5y daily GBP total returns on the 1,135-day complete-case sample. Source: data/corr.csv.

A readable slice of it - the lines that actually carry weight, plus the benchmark:

	VWRP.L	ISF.L	SMGB.L	IITU.L	SGLN.L	BRNT.L	IBTM.L	T36N	DBMG.L
VWRP.L	1.00	0.69	0.77	0.86	0.06	0.05	-0.06	0.05	0.11
ISF.L	0.69	1.00	0.40	0.40	0.03	0.10	-0.25	0.07	0.07

	VWRP.L	ISF.L	SMGB.L	IITU.L	SGLN.L	BRNT.L	IBTM.L	T36N	DBMG.L
SMGB.L	0.77	0.40	1.00	0.87	0.04	0.04	-0.13	0.04	0.07
IITU.L	0.86	0.40	0.87	1.00	-0.00	0.04	-0.07	0.01	0.08
SGLN.L	0.06	0.03	0.04	-0.00	1.00	0.03	0.23	0.13	0.10
BRNT.L	0.05	0.10	0.04	0.04	0.03	1.00	-0.15	-0.19	0.34
IBTM.L	-0.06	-0.25	-0.13	-0.07	0.23	-0.15	1.00	0.37	-0.04
T36N	0.05	0.07	0.04	0.01	0.13	-0.19	0.37	1.00	-0.31
DBMG.L	0.11	0.07	0.07	0.08	0.10	0.34	-0.04	-0.31	1.00
PLTR	0.28	0.10	0.27	0.32	0.02	0.05	-0.07	0.04	0.06
RR.L	0.41	0.50	0.32	0.29	0.02	-0.04	-0.25	-0.02	0.02

	PLTR	RR.L
VWRP.L	0.28	0.41
ISF.L	0.10	0.50
SMGB.L	0.27	0.32
IITU.L	0.32	0.29
SGLN.L	0.02	0.02
BRNT.L	0.05	-0.04
IBTM.L	-0.07	-0.25
T36N	0.04	-0.02
DBMG.L	0.06	0.02
PLTR	1.00	0.08
RR.L	0.08	1.00

Source: data/corr.csv

The average pairwise correlation across the whole optimiser universe is 0.23 (0.2324 exactly) - which is the constant-correlation target the Ledoit-Wolf shrinkage pulls towards. It rose slightly, from 0.2208, when the two placeholder bars were masked. Gold, the trend fund and the gilt proxies are the genuinely low-correlation lines; everything in the equity complex moves together, and the semiconductor ETF is the most correlated of the lot with the benchmark (0.77).

5.3 Drawdowns

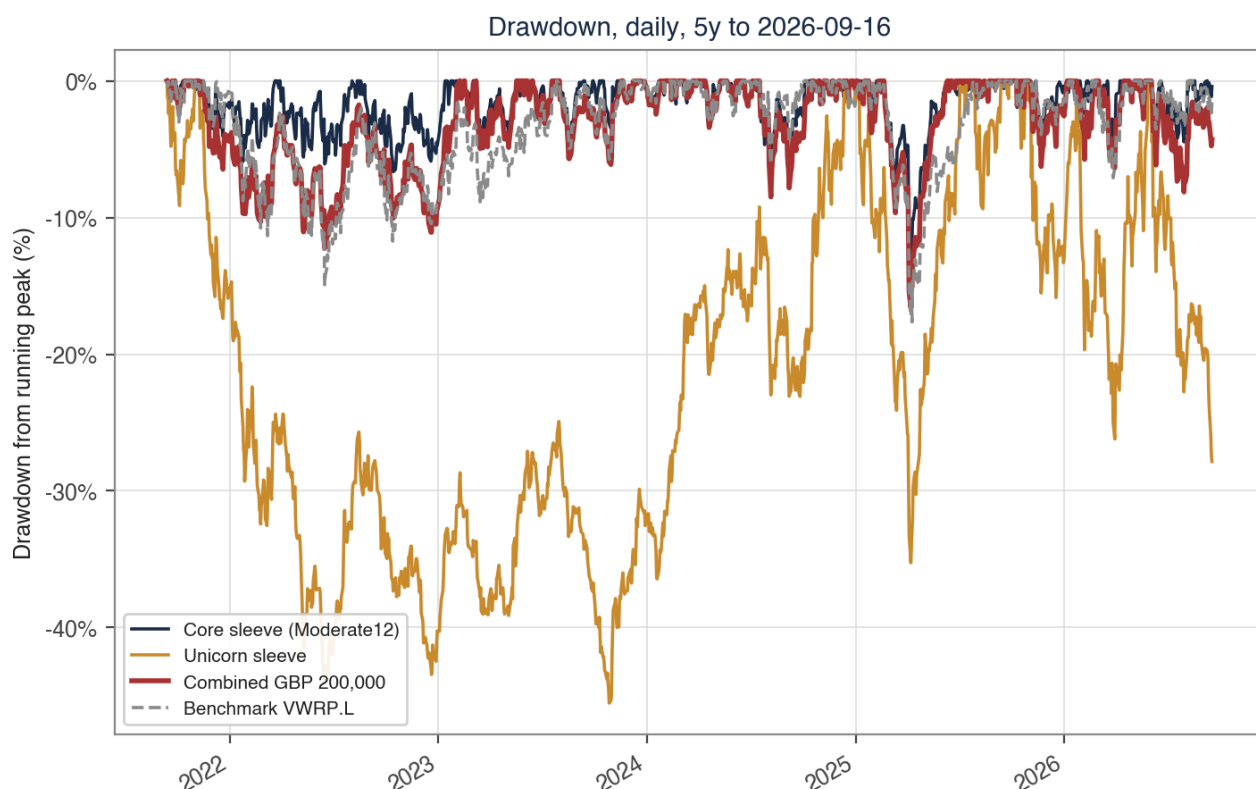


Figure 7. Drawdown from running peak, daily, for the headline core sleeve, the moonshot sleeve, the combined GBP 200,000 portfolio and the benchmark. 5y to 2026-09-16. Source: `data/prices_gbp_daily.csv`, `data/analysis_results.json`.

Series	Max drawdown (daily)	Peak	Trough
Core sleeve (Moderate12)	-13.4%	2025-02-18	2025-04-07
Moonshot sleeve	-45.5%	2021-11-05	2023-10-27
Combined GBP 200,000	-16.8%	2025-02-19	2025-04-07
Benchmark VWRP.L	-17.6%	2025-01-22	2025-04-09

Source: `data/analysis_results.json` -> `portfolios.Moderate12.realised_daily_5y`, `unicorn_sleeve.stats`, `combined_portfolio.realised_daily_5y`; `data/stats_daily.csv`

The core sleeve's worst episode was the February-April 2025 equity fall, where it lost 13.4% against the benchmark's deeper fall. The moonshot sleeve's worst was much earlier, much larger and much longer: a 45.5% fall running from 2021-11-05 all the way to 2023-10-27, two years of small-cap growth de-rating rather than a single shock. That is 5.4 points deeper and nearly four times longer than the sleeve it replaces, and it is the single figure a prospective holder should look at hardest. Because the two sleeves' worst episodes are different events years apart, the combined portfolio's drawdown (16.8%) is well below the weighted average of the two - but note that the combined figure did deepen, from 15.9% to 16.8%, purely from the sleeve change.

5.4 Backtest

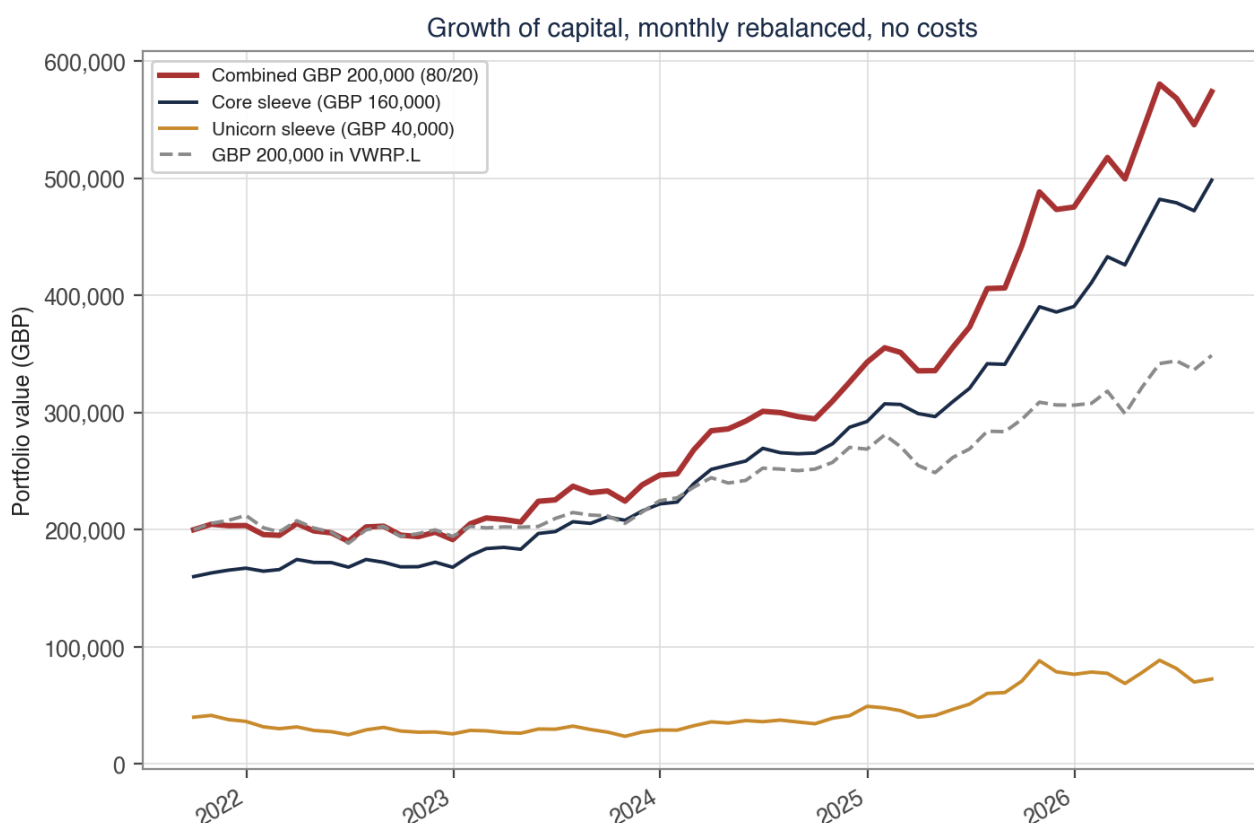


Figure 8. Growth of GBP 160,000 in the core sleeve, GBP 40,000 in the moonshot sleeve and GBP 200,000 combined, against the benchmark, at fixed target weights rebalanced every month end. 2021-10-31 to 2026-08-31. No dealing costs, taxes, spreads or FX charges. Source: `data/backtest_monthly.csv`.

This is not a track record. It applies weights chosen in 2026 to returns from 2021 onwards. The names were selected knowing what happened. Read it as a consistency check on the optimiser's output, not as evidence about the future.

Series	Start	Months	CAGR	Volatility	Max DD (month-end)	Sharpe	Sortino	Total return
MinVariance	2021-10-31	59	6.7%	5.4%	-3.9%	0.54	0.94	38%
MaxSharpe	2021-10-31	59	21.7%	7.9%	-2.4%	2.06	6.48	163%
Moderate12 (headline)	2021-10-31	59	26.0%	10.3%	-3.8%	1.95	5.82	211%
MaxSortino	2021-10-31	59	22.0%	8.2%	-2.7%	2.03	6.47	166%
EqualWeight	2021-10-31	59	15.8%	8.7%	-7.4%	1.31	2.46	105%
NaïveReference	2021-10-31	59	12.8%	9.2%	-7.4%	0.96	1.65	81%
Moonshot sleeve	2021-10-31	59	12.9%	33.3%	-43.0%	0.41	0.71	81%
Combined 80/20	2021-10-31	59	23.9%	14.0%	-7.4%	1.34	3.10	187%
Benchmark VWRP.L	2021-10-31	59	12.0%	11.4%	-11.5%	0.73	1.20	74%
Benchmark ^FTAS	2021-10-31	58	7.3%	10.7%	-10.6%	0.36	0.52	40%

Source: `data/backtest_monthly.csv`; `data/analysis_results.json` -> `portfolios.*.backtest_5y` | fixed target weights, rebalanced at every month end on month-end GBP total-return levels; no costs. The moonshot-sleeve and benchmark rows are computed here on the same convention from `data/backtest_monthly.csv`, with the level based at the month end BEFORE the first return (the corrected convention described at the end of section 4)

The moonshot row is the one that moved. On the same monthly convention the sleeve compounds at 12.9% a year against the quality sleeve's 26.3%, with a 33.3% annualised volatility and a **-43.0% month-end drawdown**. That drags the combined book from 26.6% to 23.9% and its month-end drawdown from -6.8% to -7.4%. It is also worth noticing what the sleeve does to the *shape* of the combined series rather than its level: the combined book's monthly Sortino falls from 3.67 to 3.10 while its Sharpe falls from 1.55 to 1.34, i.e. it lost slightly more of its upside than of its downside protection.

The 2021-onward all-live variant. A true ten-year backtest of this universe is impossible: the last core line to start is SMGB.L on 2020-12-10, with VWRP.L (2019-07-25), DBMF/DBMG.L (2019-05-08) and PLTR (2020-09-30) the

other late starters. The 'all assets live' variant therefore begins on 2021-01-31, the first month end after every constituent exists, and runs 68 months:

Portfolio	All-live backtest (from 2021-01-31, 68 months)				Max-available (from 2016-10-31, 119 months)			
	CAGR	Vol	Max DD	Sharpe	CAGR	Vol	Max DD	Sharpe
MinVariance	6.9%	5.2%	-3.9%	0.58	6.0%	5.7%	-6.9%	0.39
MaxSharpe	20.9%	7.5%	-2.4%	2.09	14.7%	9.1%	-13.4%	1.15
Moderate12	25.2%	9.7%	-3.8%	1.99	17.0%	11.2%	-17.3%	1.13
MaxSortino	21.1%	7.8%	-2.7%	2.03	14.4%	9.5%	-15.0%	1.07
EqualWeight	16.2%	8.2%	-7.4%	1.42	13.4%	9.3%	-13.6%	1.00
NaiveReference	13.3%	8.7%	-7.4%	1.05	11.4%	9.3%	-13.2%	0.81

Source: data/analysis_results.json -> portfolios.*.backtest_all_assets_live / .backtest_max_available

The 'max available' column runs the full ten years but renormalises the weights over whichever names exist on each date, so its early years are a **different portfolio**. It is included for completeness and should not be compared like-for-like: the headline portfolio's ten-year CAGR of 17.0% with a 17.3% drawdown is a materially different object from the 26.0% / 3.8% of the five-year run.

5.5 Calendar-year returns

Series	2021	2022	2023	2024	2025	2026
MinVariance	2.3%	2.6%	1.8%	8.5%	10.9%	7.0%
MaxSharpe	3.6%	3.3%	24.8%	27.7%	29.5%	19.0%
Moderate12	4.5%	0.5%	32.2%	31.7%	33.6%	27.6%
MaxSortino	2.7%	2.6%	25.0%	29.3%	31.3%	19.0%
EqualWeight	3.2%	-2.8%	20.8%	18.4%	23.8%	15.7%
NaiveReference	4.4%	-4.6%	15.4%	14.3%	18.6%	16.0%
MoonshotSleeve	-8.9%	-29.3%	12.9%	69.3%	55.5%	-15.6%
Combined_80_20	1.8%	-5.9%	28.7%	39.1%	38.6%	20.7%
Benchmark_VWRP	4.4%	-8.4%	15.6%	19.6%	13.9%	13.9%

Source: data/analysis_results.json -> calendar_year_returns | 2021 is a part year from 2021-10-31; 2026 runs to 2026-08-31

2022 is the year that matters, and the moonshot sleeve makes it matter more. The benchmark fell 8.4%, the headline core sleeve made **+0.5%** - the gold, oil and trend allocations doing exactly what they are there for - and the moonshot sleeve fell **29.3%**, against 12.3% for the quality sleeve. The combined portfolio fell 5.9% rather than 1.5%. That is the honest price of the mandate in the one genuinely bad year inside the window: a fifth of the book in frontier microcaps roughly quadrupled the sleeve's contribution to a drawdown year.

The sleeve's calendar pattern is also much lumpier than the one it replaced - -8.9%, -29.3%, +12.9%, **+69.3%**, +55.5%, then **-15.6%** in the eight months to August 2026. Two of six periods carry almost all of the return and two carry almost all of the loss, which is exactly what a basket of binary outcomes looks like and exactly why an annualised mean is a poor description of it. Nothing in 2023-2026 is a stress test for the core; those were four strong years for almost everything it holds. Note that the 2026 column ends on 2026-08-31, so none of the September news discussed in section 1.4 is in it.

5.6 VaR and CVaR of the combined portfolio

Horizon / confidence	Core sleeve	Moonshot sleeve	Combined GBP 200,000	Combined, in GBP
95% daily VaR	1.13%	2.59%	1.23%	GBP 2,469
95% daily CVaR	1.68%	3.44%	1.82%	GBP 3,648
99% daily VaR	1.96%	3.85%	2.13%	GBP 4,265
99% daily CVaR	2.82%	4.89%	2.99%	GBP 5,972

Source: data/analysis_results.json -> portfolios.Moderate12.realised_daily_5y, unicorn_sleeve.stats, combined_portfolio.realised_daily_5y | historical VaR/CVaR on 1,261 daily observations

Read those as: on the worst one day in twenty over the last five years, the combined portfolio lost about GBP 2,469; averaged across all of those worst days, about GBP 3,648. On the worst one day in a hundred, about GBP 4,265, averaging GBP 5,972. These are historical, in-sample figures on a five-year window that contained no banking crisis and no 2008-style dislocation.

The combined portfolio's risk splits between the sleeves as **68.1% core / 31.9% moonshot**, almost exactly where the quality sleeve left it (67.8 / 32.2). That stability is not a coincidence and it is worth understanding: the new names are individually far more volatile (mean annualised name volatility 93.1% against 80.3%) but much less correlated with each other (mean pairwise correlation inside the sleeve 0.064 against 0.093), and at sleeve level the two effects very nearly cancel. The moonshot sleeve is 20% of the money and roughly a third of the risk - which is the intended shape, but is worth knowing before it is described as a 'satellite'.

5.7 What these numbers do NOT include

Every return, Sharpe, drawdown and VaR figure in this report is gross. Specifically, none of them include:

- **Dealing commission.** Building this portfolio is 54 trades. On ii's now-confirmed published rates that is GBP 283 on the Plus plan (37 UK and US lines at GBP 3.99, 17 other international lines at GBP 7.99), about 0.14% of capital - plus a GBP 49 telephone fee on each of the two lines ii deals by phone only, and every rebalance costs more.
- **Stamp duty.** 0.5% SDRT on purchases of UK main-market company shares. On this portfolio that is RR.L, AZN.L, SHEL.L and HSBA.L only: all three UK moonshot lines (GELN.L, FTC.L, BGO.L) are on AIM and therefore exempt, as are ETFs, ETCs and gilts.
- **The FX conversion charge.** 0.75% on the first GBP 50,000 converted and 0.25% above that on the Plus plan, applied to the USD, EUR, CAD, AUD and HKD lines - and again on the way out.
- **Bid-offer spread.** Not modelled at all. On the smallest moonshot names this is by a wide margin the largest single cost, and it has grown in this build: the sleeve's median name now trades about USD 2.4m a day against USD 6.0m before, nine names trade under USD 250k a day, and one (HYT.AX) is quoted in fractions of an Australian cent.
- **The ii platform fee,** which is a fixed monthly charge rather than a percentage - and, newly relevant, the Core plan is capped at a GBP 100,000 portfolio, so a GBP 200,000 book needs the Plus plan at GBP 14.99 a month or a split across accounts.
- **Tax,** other than the stamp duty noted. Inside an ISA or SIPP there is none; outside one there is dividend tax and capital gains tax.
- **Survivorship in the moonshot sleeve.** The 40 names were selected in 2026 from companies that still exist and are still listed. A sleeve chosen on the same mandate in 2021 would contain names that have since failed or delisted, and the sleeve's 9.1% five-year CAGR is inflated by that fact in a way that cannot be quantified from this data. The bias is larger here than in the sleeve this replaces, because the new names are smaller, earlier-stage and more likely to have been the kind of company that does not survive five years.
- **Exit liquidity in the moonshot sleeve.** A GBP 1,000 ticket is fillable in every one of the 40; on the thinnest two it is more than 30% of a typical day's turnover on roughly a quarter to a third of sessions. Nothing in this model assumes, or prices, the ability to sell.

Section 8 puts numbers on the first four.

6. The moonshot sleeve

6.1 The mandate changed, so the sleeve changed

Sleeve A is no longer a basket of good small companies. On 2026-09-17 the owner's instruction replaced the screen the previous build used with a mandate, recorded verbatim in PROJECT_BRIEF.md:

""build the unicorn sleeve out for proper moonshot companies with smaller caps and liquidity now, with proper fundamental analysis, market trends, key backers etc. - you can with the most confidence explain why this company will be the next big thing; no restrictions or guidelines on what you pick as long as you can defend the selection.""

That is a different instrument from the one it replaces. **The target outcome is a small number of 10x-100x winners over a five- to ten-year hold, accepting that most positions may go to zero.** It is explicitly not a quality-small-cap basket, and the sleeve should not be judged on its median holding.

Only two hard constraints survive. First, the line must be purchasable by a UK retail client on interactive investor - a listed equity on an ii-dealt exchange (LSE/AIM, NYSE/NASDAQ/NYSE American, TSX, Euronext, Xetra, Milan, Madrid, SIX, ASX, HKEX, SGX), and **not** TSX-Venture, OTC or pink sheets. Second, it must be a listed operating company, not a shell, not a SPAC without a target, and not in administration.

Everything else became a judgement to be defended rather than a filter. Market cap, revenue, cash runway, dilution history and liquidity are inputs to the argument, not gates. Pre-revenue is allowed. Negative book equity is allowed. Illiquidity is allowed, provided a GBP 1,000 ticket is fillable. The previous sleeve's screen - market cap USD 100m-5bn, an operating business with a revenue path, at least twelve months of cash - has been withdrawn in full.

Each selected name therefore carries a dossier written to be attacked: what it does, the market trend and why now, the technology and why it is hard to copy, fundamentals, **key backers with a URL and a date for every claim**, the 12-24 month catalysts, the explicit path to 10x and the end-state market capitalisation it implies, whether 100x is even arithmetically possible, what kills it, and a confidence score with a named falsifier. The forty dossiers are in universe/dossiers/; the design note is universe/MOONSHOT_SLEEVE_DESIGN.md; the audit trail is universe/moonshot/.

6.2 How the 40 were chosen

Every count below is re-derived from the files themselves rather than asserted.

Stage	Count	Where
Ideation rows across six moonshot clusters	150	universe/moonshot/ideas_*.json
Distinct candidates after de-duplication	128	same
Independently verified (Yahoo data, exchange, ii-dealability, still operating, backer claims)	114, in 10 batches	universe/moonshot/verify_batch_1..10.json
Not reached by a verification batch, therefore excluded rather than judged on unverified data	14	audit trail
Scored on asymmetry / backing / timing	128 / 128 / 126	universe/moonshot/judge_*.json
Carrying a full three-lens weighted score	126	same
Selected by weighted score under a sector cap of 4	40	universe/unicorn_final.json
Dossiers written	42 (the 40 selected plus 2 later removed)	universe/dossiers/
Adversarial fact-check groups, 6-8 load-bearing claims re-derived per dossier	5 groups, about 250 claims	universe/moonshot/factcheck_group_1..5.json

Stage	Count	Where
Theses falsified or asymmetry invalidated at fact-check, and replaced	2	below

Source: universe/unicorn_final.json -> method, changes; universe/MOONSHOT_SLEEVE_DESIGN.md sections 2 and 7; universe/moonshot/

The judging weights are the mandate's, not a choice made here.

Lens	Weight	What it asks
Asymmetry / next-big-thing potential	50%	Is there a mechanism by which this becomes 10x larger, and is the starting price small enough for that to matter?
Execution and backing	30%	Who has done real diligence and stayed - strategic investors, named funds, government contracts, board quality, insider ownership?
Timing and trend	20%	Is the catalyst inside a 12-24 month window, and is the market moving toward this now?

No minimum score was applied, the sector cap is 4 of 40, and ties break toward the smaller market cap. The selected weighted scores run from **8.24 (IPWR, Ideal Power)** down to **5.97 (BGO.L, Bango)**.

Two names were removed after their dossiers were written, which is the only real evidence that the process can say no to its own work.

- **HEX.L, Helix Exploration - thesis falsified.** Its only quantified route to 10x rested on the company's own "NPV8 of US\$304m on US\$19m of initial capex" against "P50 gross unrisks prospective resources of 2.3 Bcf". Re-fetched on 2026-09-17, the source page does not state whether that 2.3 Bcf is recoverable helium or raw gas and publishes no price, flow rate or opex. Own arithmetic: under the raw-gas reading, 2.3 Bcf x 1.5% grade is 34,500 Mcf of helium, worth roughly USD 10-21m gross and undiscounted at USD 300-600/Mcf against USD 19m of capex - short of the stated NPV by 15-30x. The two readings differ by about 100x and the source supports neither, so the valuation anchor was withdrawn. Replaced by **Archer Materials (AXE.AX)**.
- **NAUT, Nautilus Biotechnology - asymmetry invalidated.** The technology case survived the fact-check; the asymmetry did not. Nautilus was selected on "technology priced below net cash", and the fact-check surfaced an at-the-market equity facility of up to **USD 125,000,000** signed with TD Securities (USA) LLC on 2026-09-11 (8-K, CIK 1808805, accession 0001808805-26-000041) - about 101% of the entire market capitalisation, with a full draw at USD 0.97 issuing roughly 129m shares against 127.2m outstanding. A negative enterprise value is not underwritable when the denominator can double at the seller's discretion. Replaced by **Prime Medicine (PRME)**.

Four further dossiers were corrected but kept - GFUZ (market cap corrected from a yfinance trap of USD 631m to USD 498m and the 10x bar restated upward to USD 9.4-13.6bn fully diluted), 2498.HK (the Cainiao stake softened from "largest shareholder at 10.46%" to "top-two, percentage withdrawn"), VUL.AX (BASF permits covered 2D seismic only) and ALCLS.PA (AstraZeneca's November 2023 tranche was USD 105m *including* the USD 25m upfront, not on top of it). One judging adjustment is recorded rather than hidden: both incoming names had been marked down on backing purely because their backer claims were unverified at judging time. PRME's Bristol Myers Squibb partnership, Cystic Fibrosis Foundation support and PM359 RMAF designation were all confirmed, so its backing score rose 4.6 to 6.4 (weighted 5.93 to 6.47). AXE.AX's IonQ agreement was confirmed - but confirmation showed that **Archer pays IonQ USD 1.5m over three years**, so it is commercial intent rather than financial backing, and backing rose only 3.2 to 3.8 (weighted 5.96 to 6.14). Archer therefore enters with the weakest backer set in the sleeve, and its own dossier says so.

Six dossiers carry claims explicitly labelled UNVERIFIED and were left labelled rather than quietly deleted: Caribou's 2023 Pfizer investment, Filtronic's SpaceX warrant mechanics, Clarity's Fast Track designation, Median Technologies' sensitivity and specificity figures, BTQ's QPerfect consideration, and RoboSense's Cainiao stake. A sceptic can see exactly which load-bearing bricks are still untested.

Judge scores are an ordering of panel conviction, not a sizing input, and not probabilities. Every position is GBP 1,000 regardless of rank.

6.3 The sleeve rules changed too

This is the part most likely to be skipped and most likely to decide the outcome.

- **GBP 1,000 per name at cost.** Equal weight at entry only.
- **Let winners run.** There is no annual rebalance back to GBP 1,000. Rebalancing a moonshot sleeve is the mechanism by which you sell the one position that was supposed to pay for the other thirty-nine.
- **Trim only on size.** When a position exceeds **3% of the total GBP 200,000 portfolio** - GBP 6,000, i.e. about 6x from cost - sell half.
- **Annual thesis review.** Replace any name whose thesis is falsified. Each dossier names its own falsifier, so this is a test rather than an opinion, and the falsifier is reproduced in every per-name block in section 7.2.
- **No stop losses.** A 70% drawdown is an expected event in this sleeve, not a signal.

The previous sleeve was equal-weighted and rebalanced annually. That rule has been removed, and the removal is deliberate: a moonshot sleeve that rebalances is a quality sleeve with worse companies in it. Note the consequence for the statistics in this report - the sleeve index used everywhere below is an equal-weight, **daily-rebalanced** index, which is the opposite of the rule actually adopted. It is used because the combined-book covariance needs a single return series, and it flatters the sleeve relative to the buy-and-hold basket that would actually be held. That mismatch is named again in section 9.

6.4 The profile of the 40

Measure	Value
Median market cap	USD 220.3m
Smallest	USD 7.6m (HYT.AX, HyTerra)
Largest	USD 945.5m (2498.HK, RoboSense)
Under USD 100m	10 of 40
Under USD 250m	24 of 40
Over USD 500m	9 of 40
Median average daily traded value	USD 2,008k a day
Thinnest	USD 18.1k a day (LIS.AX, Li-S Energy)
Deepest	USD 32,650k a day (LAES, SEALSQ)
Under USD 250k a day - fillable, not exitable	9 of 40
Median dossier confidence score	6.0 / 10
Names where 100x is arithmetically open at all	11 of 40
Names with fewer than three years of daily history in the window	6 of 40 (GUTS, IMSR, 2498.HK, GFUZ, BIOA, BTQ)
Reporting no revenue at all in Yahoo's latest figures	9 of 40; two more report under USD 1m (BTQ at zero, IPWR at USD 30k)
Negative book stockholders' equity	3 of 40 (ALMDT.PA, GFUZ, AISP)
Positive operating margin on Yahoo's latest figures	4 of 40 (IDN, FTC.L, BGO.L, and SLDP whose 2,978% reading is an artefact of a tiny revenue base)
Profitable, net-cash and cash-generative, on the design note's assessment	1 of 40 (FTC.L, Filtronic)

Source: universe/unicorn_final.json -> profile, names[]; data/stats_daily.csv insufficient_history | market cap and traded value in USD at the 2026-09-16 close. The traded-value figures here are the sleeve file's own; data/fetch_log.json -> avg_daily_value_usd_k_3m recomputes the same statistic over the 63 trading rows to 2026-09-16 and agrees within a few per cent on every name (median USD 2,391k, thinnest USD 18.7k), which is the basis used for the old-versus-new comparison in 6.8

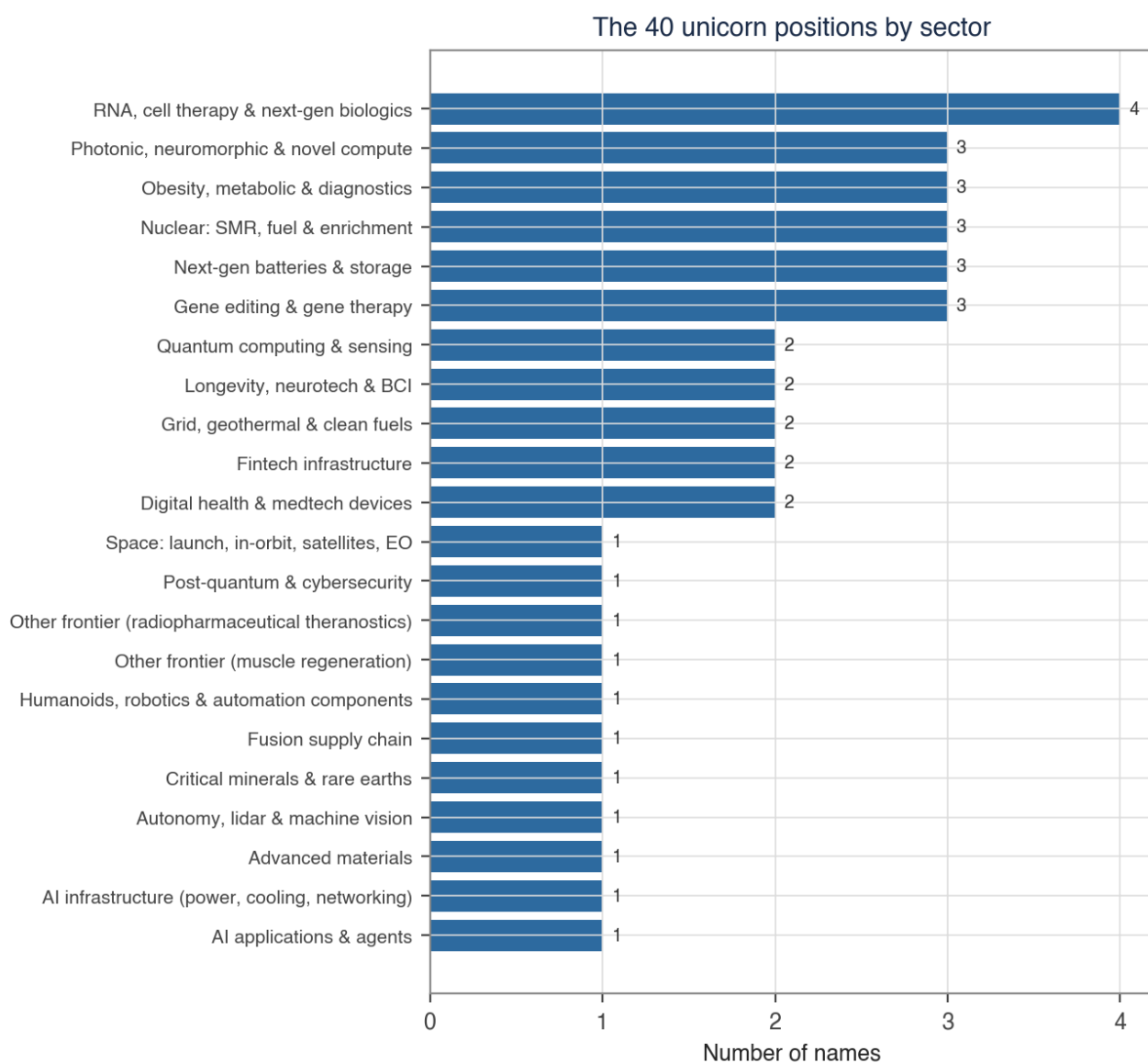


Figure 9. The 40 moonshot positions by sector. The sector cap is 4 of 40 and one sector reaches it. Source: [universe/unicorn_final.json](#) -> [sector_counts](#).

Sector	Names	Tickers
RNA, cell therapy & next-gen biologics	4	CRBU, ALCLS.PA, PRQR, MOLN.SW
Gene editing & gene therapy	3	VYGR, EDIT, PRME
Next-gen batteries & storage	3	GELN.L, SLDP, LIS.AX
Nuclear: SMR, fuel & enrichment	3	LTBR, SLX.AX, IMSR
Obesity, metabolic & diagnostics	3	QSI, GUTS, ALMDT.PA
Photonic, neuromorphic & novel compute	3	ALMU, WBT.AX, GSIT
Digital health & medtech devices	2	NYXH, EMV.AX
Fintech infrastructure	2	IDN, BGO.L
Grid, geothermal & clean fuels	2	VUL.AX, HYT.AX
Longevity, neurotech & BCI	2	ONWD.BR, BIOA
Quantum computing & sensing	2	AXE.AX, BTQ
AI applications & agents	1	AISP
AI infrastructure (power, cooling, networking)	1	IPWR
Advanced materials	1	AL3.AX

Sector	Names	Tickers
Autonomy, lidar & machine vision	1	SEYE.ST
Critical minerals & rare earths	1	ARA.TO
Fusion supply chain	1	GFUZ
Humanoids, robotics & automation components	1	2498.HK
Other frontier (muscle regeneration)	1	MSCL.TO
Other frontier (radiopharmaceutical theranostics)	1	CU6.AX
Post-quantum & cybersecurity	1	LAES
Space: launch, in-orbit, satellites, EO	1	FTC.L
Total	40	22 sectors

Source: universe/unicorn_final.json -> sector_counts, names[].sector

Twenty-two sectors for forty names is a deliberate consequence of the cap: the largest single cluster is four RNA, cell-therapy and next-generation biologics names, and thirteen sectors are represented by a single position. Read the sector labels as a description of what the sleeve is exposed to, not as a diversification claim. Add up the labels that are really one risk and the picture changes: four RNA and cell-therapy names, three gene-editing names, three obesity and metabolic diagnostics, two digital-health devices, two longevity and neurotech, one muscle-regeneration and one radiopharmaceutical - **sixteen of the forty are, in practice, clinical-trial and FDA risk**. A second cluster of three - Lightbridge, Silex and Terrestrial Energy - is nuclear licensing risk. Twenty-two sector labels do not mean twenty-two independent bets.

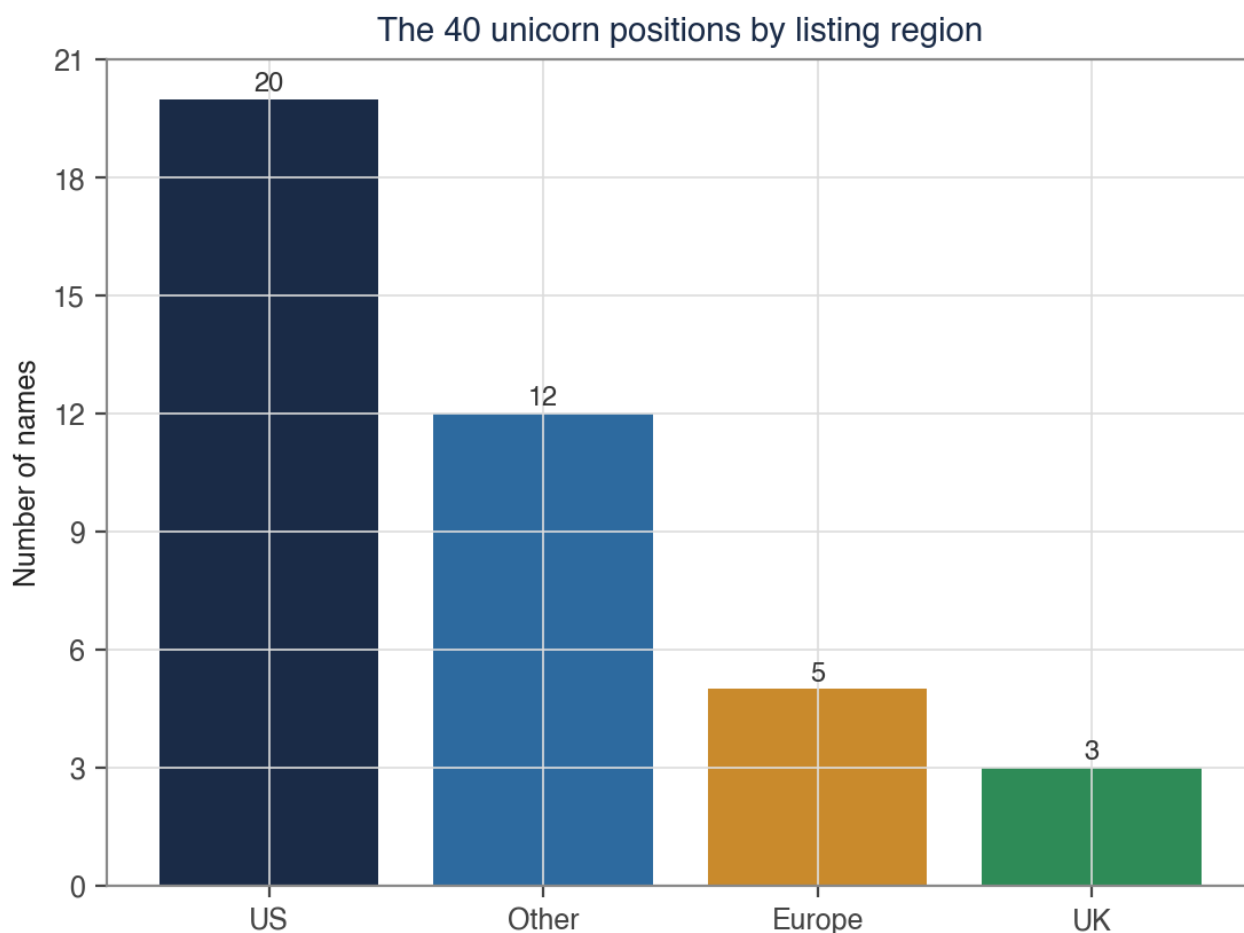


Figure 10. The 40 moonshot positions by listing region. Source: universe/unicorn_final.json -> region_counts.

Listing region	Names	Exchanges
US	20	NASDAQ (Capital Market, Global Market and Global Select tiers; NYXH also carries a Euronext Brussels line)

Listing region	Names	Exchanges
Other	12	ASX 9, Toronto Stock Exchange 2, HKEX 1
Europe	5	Euronext Brussels, Euronext Paris, Euronext Growth Paris, Nasdaq Stockholm, SIX Swiss Exchange
UK	3	LSE AIM (all three)
Total	40	10 venues across 4 regions

Source: universe/unicorn_final.json -> region_counts, names[].exchange

The region field records the listing venue, not the economic exposure, and on this sleeve the gap is wide. ONWD.BR is a Dutch-incorporated company with its research base in Switzerland, listed in Brussels. NYXH is filed under US because its primary line is on NASDAQ, but it is a Belgian company. ALCLS.PA is the Paris ordinary line of a company whose ADR trades in New York and whose largest shareholder is AstraZeneca. 2498.HK is a mainland-Chinese business. PRQR is a Dutch company on NASDAQ. Do not read the region table as a map of where the money is at risk.

It also matters for cost, and more here than in the previous sleeve. **Seventeen of the fifty-four lines in the whole portfolio are outside the UK and US**, and two of them - SEYE.ST on Nasdaq Stockholm and MOLN.SW on SIX - can only be dealt on ii **by telephone**, which carries a GBP 49 fee on top of the normal commission. One further line, ALMDT.PA, sits on Euronext **Growth** Paris, a junior MTF that ii's public disclosure does not mention by name either way; it is carried on a best-evidence basis and flagged for checking before dealing. Section 8.1 prices all of this.

6.5 The 40 positions

#	Ticker	Company	Exchange	Region	Sector	Mkt cap (USD m)	Avg daily value (USD k)	Score	Conf
1	IPWR	Ideal Power Inc.	NasdaqCM	US	AI infrastructure (power, cooling, networking)	64	998	8.24	7
2	VYGR	Voyager Therapeutics, Inc.	NASDAQ	US	Gene editing & gene therapy	188	1,936	7.54	7
3	AL3.AX	AML3D Limited	ASX	Other	Advanced materials	55	229	7.23	6
4	ONWD.BR	ONWARD Medical N.V.	Euronext Brussels	Europe	Longevity, neurotech & BCI	232	361	7.16	7
5	LTBR	Lightbridge Corporation	NASDAQ Capital Market	US	Nuclear: SMR, fuel & enrichment	228	6,691	7.13	7
6	ARA.TO	Aclara Resources Inc.	Toronto Stock Exchange	Other	Critical minerals & rare earths	670	266	7.05	8
7	SLX.AX	Silex Systems Limited	ASX	Other	Nuclear: SMR, fuel & enrichment	871	3,852	7.04	8
8	NYXH	Nyxoah SA	NASDAQ (also Euronext Brussels, NYXH.BR)	US	Digital health & medtech devices	158	457	7.01	7
9	ALMU	Aeluma, Inc.	NasdaqCM	US	Photonic, neuromorphic & novel compute	246	8,973	6.88	5
10	WBT.AX	Weebit Nano Limited	ASX	Other	Photonic, neuromorphic & novel compute	523	6,197	6.78	7
11	VUL.AX	Vulcan Energy Resources Limited	ASX	Other	Grid, geothermal & clean fuels	746	4,072	6.68	7
12	IDN	Intellicheck, Inc.	NASDAQ	US	Fintech infrastructure	48	1,276	6.61	7
13	GELN.L	Gelion plc	LSE (AIM)	UK	Next-gen batteries & storage	44	35	6.59	6
14	EDIT	Editas Medicine, Inc.	NASDAQ	US	Gene editing & gene therapy	416	6,975	6.59	6
15	SEYE.ST	Smart Eye AB (publ)	Nasdaq Stockholm	Europe	Autonomy, lidar & machine vision	213	2,718	6.54	7
16	QSI	Quantum-Si Incorporated	NASDAQ	US	Obesity, metabolic & diagnostics	145	2,732	6.49	5
17	CRBU	Caribou Biosciences, Inc.	NASDAQ	US	RNA, cell therapy & next-gen biologics	134	2,699	6.48	6
18	ALCLS.PA	Cellectis S.A.	Euronext Paris	Europe	RNA, cell therapy & next-gen biologics	142	581	6.48	6
19	GUTS	Fractyl Health, Inc.	NASDAQ	US	Obesity, metabolic & diagnostics	90	1,202	6.47	5
20	PRME	Prime Medicine, Inc.	NasdaqGM	US	Gene editing & gene therapy	544	11,289	6.47	6
21	PRQR	ProQR Therapeutics N.V.	NASDAQ	US	RNA, cell therapy & next-gen biologics	262	2,080	6.46	7
22	ALMDT.PA	Median Technologies SA	Euronext Growth Paris	Europe	Obesity, metabolic & diagnostics	232	868	6.44	6
23	CU6.AX	Clarity Pharmaceuticals Ltd	ASX	Other	Other frontier (radiopharmaceutical theranostics)	558	4,388	6.43	6
24	LAES	SEALSQ Corp	NasdaqGS	US	Post-quantum & cybersecurity	489	32,650	6.42	6
25	IMSR	Terrestrial Energy Inc.	NASDAQ Global Select	US	Nuclear: SMR, fuel & enrichment	352	9,979	6.38	6
26	EMV.AX	EMVision Medical Devices Ltd	ASX	Other	Digital health & medtech devices	109	40	6.36	6
27	SLDP	Solid Power, Inc.	NASDAQ Global Select	US	Next-gen batteries & storage	583	8,607	6.25	6
28	HYT.AX	HyTerra Ltd	ASX	Other	Grid, geothermal & clean fuels	8	19	6.24	3
29	MSCL.TO	Satellos Bioscience Inc.	Toronto Stock Exchange	Other	Other frontier (muscle regeneration)	203	203	6.22	6

#	Ticker	Company	Exchange	Region	Sector	Mkt cap (USD m)	Avg daily value (USD k)	Score	Conf
30	2498.HK	Robosense Technology Co., Ltd	HKEX	Other	Humanoids, robotics & automation components	946	14,143	6.18	6
31	GFUZ	General Fusion Group Ltd.	NASDAQ Global Market	US	Fusion supply chain	498	1,350	6.15	4
32	AXE.AX	Archer Materials Limited	ASX	Other	Quantum computing & sensing	37	111	6.14	4
33	MOLN.SW	Molecular Partners AG	SIX Swiss Exchange	Europe	RNA, cell therapy & next-gen biologics	152	65	6.14	6
34	LIS.AX	Li-S Energy Limited	ASX	Other	Next-gen batteries & storage	68	18	6.12	6
35	GSIT	GSI Technology, Inc.	NasdaqGS	US	Photonic, neuromorphic & novel compute	185	4,856	6.11	6
36	BIOA	BioAge Labs, Inc.	NASDAQ	US	Longevity, neurotech & BCI	356	14,076	6.10	6
37	AISP	Airship AI Holdings, Inc.	NASDAQ	US	AI applications & agents	66	918	6.08	5
38	FTC.L	Filtronic plc	LSE AIM	UK	Space: launch, in-orbit, satellites, EO	690	6,520	6.05	7
39	BTQ	BTQ Technologies Corp.	NasdaqGM	US	Quantum computing & sensing	387	11,963	6.02	5
40	BGO.L	Bango PLC	LSE (AIM)	UK	Fintech infrastructure	64	146	5.97	6
Total		40 names x GBP 1,000							

#	Ticker	Price (local)	Price (GBP)	Shares
1	IPWR	3.890 USD	2.887	346
2	VYGR	3.050 USD	2.264	441
3	AL3.AX	0.135 AUD	0.071	14,004
4	ONWD.BR	2.800 EUR	2.398	417
5	LTBR	6.470 USD	4.802	208
6	ARA.TO	3.660 CAD	1.951	512
7	SLX.AX	4.390 AUD	2.322	430
8	NYXH	1.550 USD	1.150	869
9	ALMU	13.430 USD	9.968	100
10	WBT.AX	3.050 AUD	1.613	619
11	VUL.AX	2.240 AUD	1.185	843
12	IDN	2.380 USD	1.766	566
13	GELN.L	0.145 GBP	0.145	6,896
14	EDIT	2.710 USD	2.011	497
15	SEYE.ST	53.050 SEK	4.027	248
16	QSI	0.730 USD	0.542	1,845
17	CRBU	1.250 USD	0.928	1,077
18	ALCLS.PA	1.151 EUR	0.986	1,014
19	GUTS	0.563 USD	0.418	2,393
20	PRME	3.000 USD	2.227	449
21	PRQR	1.860 USD	1.380	724
22	ALMDT.PA	3.785 EUR	3.242	308
23	CU6.AX	2.090 AUD	1.105	904
24	LAES	2.190 USD	1.625	615
25	IMSR	4.250 USD	3.154	317
26	EMV.AX	1.735 AUD	0.918	1,089

#	Ticker	Price (local)	Price (GBP)	Shares
27	SLDP	2.390 USD	1.774	563
28	HYT.AX	0.006 AUD	0.003	315,091
29	MSCL.TO	13.000 CAD	6.931	144
30	2498.HK	16.090 HKD	1.523	656
31	GFUZ	7.840 USD	5.819	171
32	AXE.AX	0.185 AUD	0.098	10,219
33	MOLN.SW	3.210 CHF	2.909	343
34	LIS.AX	0.150 AUD	0.079	12,603
35	GSIT	4.810 USD	3.570	280
36	BIOA	7.750 USD	5.752	173
37	AISP	1.905 USD	1.414	707
38	FTC.L	2.330 GBP	2.330	429
39	BTQ	2.630 USD	1.952	512
40	BGO.L	0.625 GBP	0.625	1,600
	Total			invested GBP 39,962

Source: universe/unicorn_final.json -> names; data/unicorn_stats.csv; data/combined_positions.csv | prices at the 2026-09-16 close; shares = floor(GBP 1,000 / GBP price). 'Score' is the weighted three-lens judge score (asymmetry 50%, backing 30%, timing 20%); 'Conf' is the dossier author's own confidence, 1-10. Pence-quoted LSE lines have already been divided by 100, so their 'local' price is shown in pounds

Four of these positions buy more than ten thousand shares, all of them on the ASX (AL3.AX 14,004; LIS.AX 12,603; AXE.AX 10,219; and HYT.AX 315,091 at AUD 0.006, i.e. six-tenths of an Australian cent). A sub-cent quote and a GBP 1,000 ticket do not combine comfortably: on HyTerra the minimum price increment is a large fraction of the price, so the bid-offer spread alone can cost several per cent of the position on the way in and again on the way out.

6.6 The arithmetic of the payoff, and the base rate

This sleeve is designed to have a bad median outcome. The most likely single result for any one of these 40 positions is a loss, and for a meaningful number of them a total loss. Several are pre-revenue, several have negative book equity, several have under eighteen months of cash, and at least a dozen have a near-certain equity raise inside the holding period. Being right on the science and diluted out of the return is the single most common way this sleeve loses money; it is named explicitly in the "what kills it" line of roughly half the dossiers, and it is the reason NAUT was removed.

No hit rate has been modelled, because none can be known. What can be set out is the arithmetic of what a sleeve like this needs. Cost is 40 x GBP 1,000 = GBP 40,000, which is 20% of the GBP 200,000 book.

Scenario	0x	~1x	3x	10x	30x	Sleeve value	vs GBP 40,000 cost	Effect on the whole book
Sleeve fails	28	10	2	0	0	GBP 16,000	-60%	-12% of total
One big winner	30	6	2	1	1	GBP 52,000	+30%	+6% of total
Two big winners	30	5	2	2	1	GBP 67,000	+68%	+13.5% of total
Works as intended	25	8	4	2	1	GBP 70,000	+75%	+15% of total

Source: own calculation; universe/MOONSHOT_SLEEVE_DESIGN.md section 6 | positions valued at cost x multiple; no dealing costs, FX, spread or tax. These are illustrations of arithmetic, NOT probability-weighted forecasts

Three things follow, and they are the whole design.

First, the losses are capped and the winners are not. No position can lose more than GBP 1,000, but a single 30x turns GBP 1,000 into GBP 30,000 - three-quarters of the entire sleeve's cost recovered by one line. That is why the sleeve holds 40 names rather than 10: the payoff is driven by the maximum, not the mean, and which name it is cannot be known in advance. Note what the table also shows - the "one big winner" row beats the "sleeve fails" row by GBP 36,000 while still holding thirty zeros.

Second, the trim rule exists because the rebalancing instinct is the enemy. A position that reaches 3% of a GBP 200,000 book is worth GBP 6,000, up 6x. Selling half at that point banks GBP 3,000 - three times the original

stake - while leaving GBP 3,000 to keep compounding. Selling all of it, or rebalancing back to GBP 1,000 annually, would systematically destroy the only outcome that makes the sleeve worth owning.

Third, the sleeve is sized so that total failure is survivable. If every one of these 40 companies went to zero the GBP 200,000 portfolio would fall 20% - a bad year in an equity market, not a catastrophe. The 80% core sleeve is what pays for the mortgage. **Do not size this sleeve larger than you would be willing to write off entirely.**

The published base rate is worse than most people expect. Bessembinder (2018) examined all 25,332 US common stocks in CRSP from 1926 to 2016 and found that **57.8% of them delivered lifetime buy-and-hold returns below one-month Treasury bills**, that the single most frequent outcome for a stock over its life was a **-100% return**, and that **just 4.3% of firms accounted for the entire net wealth creation** of the US market over ninety years, with the top 86 companies - 0.33% of the sample - accounting for about half of it. Source: Hendrik Bessembinder, "Do stocks outperform Treasury bills?", *Journal of Financial Economics* 129(3), 440-457, 2018 ([SSRN](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2900447)).

Two readings of that finding sit side by side here and both are honest. The optimistic reading is that a positively skewed market makes a diversified 40-name basket of asymmetric bets a rational way to buy exposure to the tail, because missing the 4% is what destroys concentrated portfolios. The sceptical reading is that Bessembinder's 4.3% is a *description of the whole market*, including the mega-caps this sleeve deliberately excludes, and that a portfolio drawn exclusively from pre-revenue frontier microcaps is drawn from the part of the distribution where the -100% outcome is most common, not least. **Nothing in this report establishes that this particular 40 is more likely to contain a winner than any other 40.** The sleeve is a bet that a defended, sourced dossier improves the odds relative to a random draw; that bet is untested and unfalsifiable inside a five-year backtest.

6.7 The sleeve as a portfolio

Statistic	Moonshot sleeve (equal weight, daily rebalanced)
Date range used	2021-09-17 to 2026-09-16 (1,261 return days; level based at 2021-09-16)
Live names at the start / end	27 / 40 of 40
Annualised arithmetic mean	12.2%
CAGR	9.1%
Total return over the window	54.8%
Volatility	26.5%
Sharpe / Sortino	0.32 / 0.46
Skew	+0.11
Excess kurtosis	1.32
Beta / beta+ / beta-	0.87 / 0.77 / 0.91
Beta asymmetry	-0.13
Alpha vs VWRP.L (annualised)	1.7%
R-squared vs VWRP.L	0.18
Correlation with benchmark	0.43
Max drawdown	-45.5% (2021-11-05 to 2023-10-27)
Best / worst day	9.38% / -7.09%
95% daily VaR / CVaR	2.59% / 3.44%
99% daily VaR / CVaR	3.85% / 4.89%
Expected return: historical / CAPM / blend	12.1% / 7.9% / 10.0%

Source: data/analysis_results.json -> unicorn_sleeve

The index starts on 2021-09-17 with 27 of the 40 names live and reaches all 40 by the end, so the early index is a smaller and different portfolio from the late one. Its R-squared against world equity is **0.18**, so more than four-fifths of what this sleeve does is idiosyncratic. That is the point of it, and it is also why these statistics should carry the least confidence of anything in this report.

Three readings deserve to be pulled out, and two of them are uncomfortable.

- **The skew is still positive, and it is smaller than the sleeve it replaces.** At +0.11 the daily distribution is mildly right-tailed, against +0.29 for the quality sleeve. The bounded-downside, unbounded-upside shape that section 2.5 identifies is real but faint on daily data - which should not surprise anyone, because a five-year daily return series is the wrong instrument for measuring a payoff whose whole shape is a five- to ten-year lottery.
- **The beta asymmetry is now clearly negative, at -0.13.** The sleeve captures 0.77 of the benchmark's up days and 0.91 of its down days. For a sleeve built for convexity, that is the opposite of the desired reading, and it is not a rounding artefact - the quality sleeve's figure was 0.003. What it records is that the 2021-2026 window is mostly the small-cap growth de-rating, in which frontier microcaps fell with the market and did not rise with it.
- **The maximum drawdown is -45.5% and it lasted two years.** From 2021-11-05 to 2023-10-27 the equal-weight sleeve index lost nearly half its value, and the recovery is inside the 2024-2025 rally. A holder of this sleeve should expect an episode of that size, and the design note is explicit that a 70% drawdown in an individual name is an expected event rather than a signal.

6.8 What changed against the quality sleeve, and what it cost

The 2026-09-16 sleeve was built to the old brief's screen - market cap USD 100m-5bn, an operating business with a revenue path, at least twelve months of cash, a structurally growing sector. That produced a portfolio of **good small companies**. Six of those forty were re-proposed because they genuinely fit the new mandate rather than merely surviving it (ARA.TO, SLX.AX, VUL.AX, IMSR, BIOA, FTC.L); **thirty-four were dropped and thirty-four new names took their place.**

	Old (quality sleeve)	New (moonshot sleeve)
Names	40	40
Kept	-	6
Replaced	-	34
Median market cap (USD m)	818.0	220.3
Market cap range (USD m)	190.9 - 2,864.5	7.6 - 945.5
Median average daily traded value (USD k)	6,043	2,391
Thinnest name (USD k a day)	261	19
Listing regions	US 16, Other 11, Europe 7, UK 6	US 20, Other 12, Europe 5, UK 3
Sectors used	21	22
Live names on the sleeve index's first date	28 of 40	27 of 40

Source: data/REFRESH_DIFF.md section 5; data/refresh_diff.json | traded value on both sides is data/fetch_log.json's avg_daily_value_usd_k_3m method - mean(Close x Volume) over the 63 trading rows to 2026-09-16 - recomputed from the archived build's own raw bars for the old column, so the two are like-for-like

The old sleeve's **smallest** name (Exodus, USD 190.9m) was larger than 15 of the 40 names here, and larger than this sleeve's median. That single comparison is the change in one line.

What it cost, measured on the same five years. Nothing else in the model moved: the as-of date, the 1,261-day window, the calendar, Bank Rate, the ERP, the gilt yields, the core universe and every core weight are identical between the two builds. The only input that changed is the 40-name list, so every difference below is attributable to it and to nothing else.

Statistic	Old (quality)	New (moonshot)	Change
Mean return (annualised, arithmetic)	24.4%	12.2%	-12.19 pp
CAGR	23.2%	9.1%	-14.06 pp
Total return over the window	183.7%	54.8%	-128.95 pp
Volatility	26.8%	26.5%	-0.32 pp
Sharpe	0.771	0.320	-0.451
Sortino	1.170	0.465	-0.706
Skew	0.289	0.108	-0.181

Statistic	Old (quality)	New (moonshot)	Change
Excess kurtosis	1.149	1.320	+0.171
Max drawdown	-40.2%	-45.5%	-5.36 pp
Beta / beta+ / beta-	1.11 / 1.07 / 1.06	0.87 / 0.77 / 0.91	-0.24 / -0.29 / -0.16
Beta asymmetry	+0.003	-0.133	-0.136
R-squared vs VWRP.L	0.292	0.182	-0.109
Expected return, historical / CAPM / blend	22.9% / 8.6% / 15.8%	12.1% / 7.9% / 10.0%	-10.8 / -0.7 / -5.8 pp

Source: data/REFRESH_DIFF.md section 6; data/analysis_results.json -> unicorn_sleeve;
archive/2026-09-16-quality-sleeve/data/analysis_results.json | the old CAGR and total-return figures are *RESTATED* onto the corrected level base (section 9, note on the level-base fix) so both columns are on the same convention; as published the archived figures were 182.37% and 23.09%

Read that table as a cost, not as a verdict. The new sleeve's realised five-year CAGR is 14.1 percentage points lower than the sleeve it replaces, its Sharpe less than half, and its drawdown five points deeper. Three things should be said about that plainly.

First, **this is what the mandate asked for.** A basket of pre-revenue frontier microcaps priced in 2026 has, as its five-year history, the 2022-2023 small-cap de-rating. The quality sleeve's 23.2% CAGR is the record of companies that had already worked. Selecting on realised five-year return is exactly the selection the new mandate forbids.

Second, **the five-year statistics are close to meaningless as a forecast for this sleeve** and are reported because the combined-book covariance needs them, not because they predict anything. A payoff distribution whose entire design is "most go to zero, one or two go up 10x-30x over five to ten years" is not summarised by an annualised mean and a standard deviation.

Third, **the distributional cost is real and should not be explained away.** Volatility barely moved (26.8% to 26.5%) because forty idiosyncratic names diversify each other either way, but the *shape* got worse on two measures that this report has consistently used to judge the sleeve: skew fell from +0.29 to +0.11, and beta asymmetry flipped from about zero to clearly negative. At the covariance level the design is working as intended - mean individual name volatility rose from 80.3% to 93.1% a year while mean pairwise correlation inside the sleeve fell from 0.093 to 0.064, i.e. more idiosyncratic risk and less shared factor risk - but that is a statement about the sleeve's construction, not evidence that the bets are good.

At the whole-portfolio level it is a fifth of the size. The combined GBP 200,000 book is 80% unchanged core plus 20% new sleeve, so it inherits the change at 0.2x: ex-ante expected return falls from 15.5% to **14.3%**, ex-ante volatility from 14.4% to **14.2%**, ex-ante Sharpe from 0.81 to **0.74**, and the realised five-year backtest CAGR from 26.0% to **23.0%**. The sleeve's share of total portfolio risk barely moves, 32.2% to **31.9%**. Section 4's closing subsection and data/REFRESH_DIFF.md carry the rest.

One liquidity consequence is new and is not in any of those numbers. The nine thinnest names trade under USD 250k a day and the thinnest under USD 19k. A GBP 1,000 ticket - about USD 1.34k - is 7.4% of a typical day's turnover in Li-S Energy and 7.1% in HyTerra, and on roughly a quarter to a third of actual sessions it would have been more than 30% of that day's entire turnover on both. Those positions are fillable with limit orders and patience; they are not exitable into bad news at any size. Nothing in this model prices that.

Ticker	Company	Avg daily value (USD k)	Market cap (USD m)	GBP 1,000 as % of a day
LIS.AX	Li-S Energy Limited	18.1	68.3	7.40%
HYT.AX	HyTerra Ltd	18.9	7.6	7.09%
GELN.L	Gelion plc	35.0	44.4	3.83%
EMV.AX	EMVision Medical Devices Ltd	40.1	109.0	3.34%
MOLN.SW	Molecular Partners AG	64.7	152.0	2.07%
AXE.AX	Archer Materials Limited	111.1	37.3	1.21%
BGO.L	Bango PLC	145.7	64.3	0.92%
MSCL.TO	Satellos Bioscience Inc.	202.8	203.1	0.66%
AL3.AX	AML3D Limited	228.6	54.6	0.59%

Ticker	Company	Avg daily value (USD k)	Market cap (USD m)	GBP 1,000 as % of a day
ARA.TO	Aclara Resources Inc.	266.2	670.2	0.50%

Source: universe/unicorn_final.json -> names[.avg_daily_value_usd_k; own calculation converting GBP 1,000 at the as-of GBPUSD rate of 1.3473 | Yahoo volume, not venue-reported ADV

Five tradeability items were raised and four resolved. SEYE.ST (Nasdaq Stockholm) and MOLN.SW (SIX) are **phone-only** on ii and carry a GBP 49 telephone fee each way on top of commission. ONWD.BR (Euronext Brussels) and 2498.HK (HKEX) are dealt online with no VIE-structure exclusion found anywhere on ii.co.uk. **ALMDT.PA remains open:** Euronext Growth Paris is a junior MTF that ii's public disclosure does not distinguish from main-market Paris, and a help-centre search for "Euronext Growth" returns no results - the position is carried on a best-evidence "online" call that should be confirmed by instrument search (ISIN FR0011049824) or by telephone before an order is placed. Separately, **GUTS (Fractyl Health) received a Nasdaq delisting determination on 2026-09-10**, stayed by a Hearings Panel request filed on 2026-09-11; if the panel is not persuaded the line leaves Nasdaq, and OTC is outside the ii mandate entirely. That item is unresolved and is the single largest tradeability risk in the sleeve.

7. Individual holdings

One block per single company: the 11 individual stocks in the core sleeve first, then the 40 moonshot positions in rank order. Every figure comes from `data/fundamentals_table.csv` and `data/stats_daily.csv`; fields Yahoo does not supply are marked rather than left blank or guessed. The moonshot blocks carry six extra fields the core blocks do not - key backers, catalysts, the 10x end-state market cap, whether 100x is arithmetically possible, the dossier's confidence score and its named falsifier - all of them from `universe/unicorn_final.json` and `universe/dossiers/<TICKER>.md`.

How to read the blocks.

- **Prices** are the 2026-09-16 close. Pence-quoted LSE lines are shown in pounds (already divided by 100).
- **52-week high and low** are computed from the intraday High/Low columns over the trailing 365 calendar days, not from Yahoo's `info` fields, which are unreliable (they have been seen returning 0.0). % below 52w high is computed from the same high, so the two always agree.
- **Market cap** is own-calculated as shares outstanding x the GBP price per share. Yahoo's own `marketCap` is in the company's reporting currency and is priced live, not at the as-of date; where the two differ by more than 5% it is because the company has multiple share classes, and the block says so. Thirteen names carry that note in this build (SPCX 40%, BRK-B 36%, ALCLS.PA 23%, HYT.AX 17%, ALMU 14%, AL3.AX 11%, GSIT 9%, ALMDT.PA 8%, WBT.AX 7%, SLDP 6%, IMSR 34%, EMV.AX 5%, LTBR 5%).
- **Valuation multiples** are Yahoo's, rescaled by $\text{close}(2026-09-16) / \text{price-at-fetch}$ so they are priced at the as-of date rather than at the 2026-09-17 fetch. The median rescale factor is 0.9846 and the range 0.9000 to 1.2000 - a wider spread than the previous build's, because the moonshot names are small and moved further between the as-of close and the live fetch. Three lines (QSI, GUTS, GSIT) had a further session trade before `.info` was captured; the rescale uses the raw 2026-09-16 bar directly, so those rows are still priced at the as-of date.
- **Balance-sheet and income figures** are in the company's own reporting currency as supplied by Yahoo, which is not always the quote currency: SEALSQ (LAES) is Swiss and ProQR (PRQR) and Nyxoah (NYXH) are Dutch and Belgian respectively, and all three report in US dollars because their primary listing is on NASDAQ. No line in this build carries a reporting currency that differs from the one shown, but the underlying statement figures are whatever Yahoo held on 2026-09-17 and cross-company comparisons of them deserve care. **For nine of the forty moonshot names Yahoo reports no revenue at all**, and for two more it reports under USD 1m; those rows are pre-revenue companies, not data faults.
- **Beta (own)** is the 5-year daily GBP beta against VWRP.L; **Beta (Yahoo)** is Yahoo's own figure, on its own unstated window and currency, shown for contrast. $\text{Ke} = \text{rf} + \text{beta_own} \times \text{ERP} = 3.75\% + \text{beta} \times 4.17\%$. $\text{WACC} = \text{E}/(\text{D}+\text{E}) \times \text{Ke} + \text{D}/(\text{D}+\text{E}) \times \text{Kd} \times (1 - \text{t})$, with $\text{Kd} = \text{FY interest expense} / \text{total debt}$ (bounded 0-30%, fallback $\text{rf} + 2\%$) and $\text{t} = \text{tax} / \text{pretax income}$ (bounded 0-35%, fallback 25%).
- **Returns** are GBP total returns to 2026-09-16.
- **One rounding trap.** Yahoo reports trailing earnings per share for pence-quoted LSE lines rounded to the nearest penny. For a small company earning about a penny a share that makes the trailing P/E almost meaningless: **FTC.L reads 115.5 purely because the denominator is 0.02**. The forward P/E, which uses an unrounded estimate, is the more informative figure on that row - 59.6 against 115.5. The other two UK moonshot lines (GELN.L, BGO.L) report negative trailing earnings, so they show no trailing P/E at all.
- **Negative multiples are printed, not suppressed.** A loss-making pre-revenue company has a negative forward P/E and a negative EV/EBITDA, and roughly two-thirds of the moonshot sleeve does. They are arithmetic, not signals, and they are shown rather than blanked so that nobody has to wonder whether a blank means "loss-making" or "not supplied".

7.1 Core sleeve single stocks

1. Rolls-Royce Holdings plc (RR.L)

LSE | Industrials / Aerospace & Defense | United Kingdom | quoted in GBp | Core sleeve weight 5.0% = GBP 8,000, 555 shares

Price (local)	14.406 GBP	Price (GBP)	14.406
52w high / low (local)	15.860 / 9.900	% below 52w high	-9.2%
Market cap (GBP)	GBP 118.63bn	Market cap (USD m)	159,831
Enterprise value (GBP)	GBP 116.52bn	Trailing P/E	38.96
Forward P/E	28.48	Price / book	40.89
EV / EBITDA	22.86	Dividend yield	0.85%
Total debt	GBP 4.37bn	Cash	GBP 6.48bn
Net debt	GBP -2.11bn	Debt / equity (book)	1.60
Interest cover (EBITDA / interest)	16.88	Interest cover (EBIT proxy)	23.96
Capex, latest FY	GBP -985.0m	Capex / revenue	4.3%
Free cash flow	GBP 4.60bn	Revenue (latest FY)	GBP 23.16bn
Operating margin	21.1%	Revenue growth	20.6%
Beta (Yahoo)	1.19	Beta (own, 5y daily GBP)	1.31
Upside beta (beta+)	1.41	Downside beta (beta-)	1.45
CAPM cost of equity Ke	9.2%	WACC	9.1%
Cost of debt Kd	6.9% (interest expense (FY) / total debt)	Tax rate	15.8%
Return YTD (GBP)	26.3%	Return 1y (GBP)	29.3%
Volatility 5y	41.6%	Max drawdown 5y	-55.1%
Skew / excess kurtosis	1.19 / 13.13	Sharpe / Sortino 5y	1.37 / 2.20

Source: data/fundamentals_table.csv row RR.L; data/stats_daily.csv row RR.L

Business. Rolls-Royce Holdings plc designs and manages mission-critical power systems in the United Kingdom and internationally. The company operates through three segments: Civil Aerospace, Defence, and Power Systems. The Civil Aerospace segment develops, manufactures, markets, and sells aero engines for large commercial aircraft, regional jet, and business aviation markets, as well as provides aftermarket services.

Why it is in the core. User-named; civil-aero aftermarket annuity plus defence and SMR optionality - the UK's clearest structural-edge industrial.

2. Palantir Technologies Inc. (PLTR)

NasdaqGS | Technology / Software - Infrastructure | United States | quoted in USD | Core sleeve weight 5.0% = GBP 8,000, 61 shares

Price (local)	174.340 USD	Price (GBP)	129.395
52w high / low (local)	207.520 / 106.370	% below 52w high	-16.0%
Market cap (GBP)	GBP 297.70bn	Market cap (USD m)	401,106
Enterprise value (GBP)	GBP 290.87bn	Trailing P/E	147.40
Forward P/E	74.24	Price / book	42.39
EV / EBITDA	147.19	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 211.4m	Cash	USD 9.41bn
Net debt	USD -9.20bn	Debt / equity (book)	0.02
Interest cover (EBITDA / interest)	767.33	Interest cover (EBIT proxy)	478.63
Capex, latest FY	USD -33.9m	Capex / revenue	0.6%
Free cash flow	USD 2.16bn	Revenue (latest FY)	USD 6.16bn
Operating margin	47.1%	Revenue growth	92.8%
Beta (Yahoo)	1.62	Beta (own, 5y daily GBP)	1.44
Upside beta (beta+)	1.42	Downside beta (beta-)	1.00
CAPM cost of equity Ke	9.8%	WACC	9.8%
Cost of debt Kd	1.6% (interest expense (FY) / total debt)	Tax rate	1.4%
Return YTD (GBP)	-2.0%	Return 1y (GBP)	6.6%
Volatility 5y	67.2%	Max drawdown 5y	-76.1%
Skew / excess kurtosis	1.17 / 8.49	Sharpe / Sortino 5y	0.82 / 1.32

Source: data/fundamentals_table.csv row PLTR; data/stats_daily.csv row PLTR

Business. Palantir Technologies Inc. builds and deploys software platforms for the intelligence community to assist in counterterrorism investigations and operations in the United States, the United Kingdom, and internationally.

Why it is in the core. User-named; government and commercial AI deployment platform with entrenched contracts - high beta, extreme valuation multiple.

3. Space Exploration Technologies Corp. (SpaceX) (SPCX)

NasdaqGS | Industrials / Aerospace & Defense | United States | quoted in USD | Core sleeve weight 2.5% = GBP 4,000, 35 shares

Insufficient history. SPCX listed on 2026-06-12 and has 67 daily bars. Its beta, upside and downside betas, volatility, skew and drawdown are arithmetic on that sample, not statistical estimates, and it is excluded from the optimiser covariance. Its 2.5% weight is a policy decision, not a model output. **Share-class note.** Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 40.1%. The company has more than one share class; the headline below is shares x price.

Price (local)	150.880 USD	Price (GBP)	111.983
52w high / low (local)	225.640 / 104.830	% below 52w high	-33.1%
Market cap (GBP)	GBP 861.86bn	Market cap (USD m)	1,161,217
Enterprise value (GBP)	GBP 817.10bn	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	84.36	Price / book	15.23
EV / EBITDA	186.69	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 39.71bn	Cash	USD 100.01bn
Net debt	USD -60.30bn	Debt / equity (book)	0.71
Interest cover (EBITDA / interest)	3.03	Interest cover (EBIT proxy)	-1.17
Capex, latest FY	USD -20.91bn	Capex / revenue	90.7%
Free cash flow	n/a (not reported by Yahoo)	Revenue (latest FY)	USD 23.04bn
Operating margin	-1.8%	Revenue growth	91.9%
Beta (Yahoo)	n/a (not reported by Yahoo)	Beta (own, 5y daily GBP)	2.95
Upside beta (beta+)	8.34	Downside beta (beta-)	1.40
CAPM cost of equity Ke	16.0%	WACC	15.6%
Cost of debt Kd	4.9% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	n/a (listed 2026-06-12; no full year of history)	Return 1y (GBP)	n/a (listed 2026-06-12; no full year of history)
Volatility 5y	90.3%	Max drawdown 5y	-48.9%
Skew / excess kurtosis	0.51 / 2.52	Sharpe / Sortino 5y	0.11 / 0.17

Source: data/fundamentals_table.csv row SPCX; data/stats_daily.csv row SPCX

Business. Space Exploration Technologies Corp. provides satellite-based broadband services in the United States, Ireland, Canada, and internationally. It operates through three operating segments: Space, Connectivity, and AI. The Space segment designs, manufactures, and launches reusable rockets to provide access to space.

Why it is in the core. User's 'SPCX' resolved: SpaceX IPO'd on NASDAQ 2026-06-12, so the real listed security is included rather than a substitute.

4. AstraZeneca PLC (AZN.L)

LSE | Healthcare / Drug Manufacturers - General | United Kingdom | quoted in GBp | Core sleeve weight 1.1% = GBP 1,724, 14 shares

Price (local)	121.360 GBP	Price (GBP)	121.360
52w high / low (local)	157.320 / 98.920	% below 52w high	-22.9%
Market cap (GBP)	GBP 188.22bn	Market cap (USD m)	253,593
Enterprise value (GBP)	GBP 215.60bn	Trailing P/E	23.89
Forward P/E	13.98	Price / book	4.94
EV / EBITDA	10.83	Dividend yield	2.02%
Total debt	GBP 32.35bn	Cash	GBP 4.97bn
Net debt	GBP 27.38bn	Debt / equity (book)	0.66
Interest cover (EBITDA / interest)	12.33	Interest cover (EBIT proxy)	8.68
Capex, latest FY	GBP -5.91bn	Capex / revenue	9.6%
Free cash flow	GBP 4.91bn	Revenue (latest FY)	GBP 61.37bn
Operating margin	23.5%	Revenue growth	6.4%
Beta (Yahoo)	0.20	Beta (own, 5y daily GBP)	0.43
Upside beta (beta+)	0.53	Downside beta (beta-)	0.49
CAPM cost of equity Ke	5.5%	WACC	5.3%
Cost of debt Kd	5.0% (interest expense (FY) / total debt)	Tax rate	17.5%
Return YTD (GBP)	-10.5%	Return 1y (GBP)	3.3%
Volatility 5y	24.7%	Max drawdown 5y	-26.7%
Skew / excess kurtosis	-0.24 / 5.75	Sharpe / Sortino 5y	0.38 / 0.53

Source: data/fundamentals_table.csv row AZN.L; data/stats_daily.csv row AZN.L

Business. AstraZeneca PLC, a biopharmaceutical company, focuses on the discovery, development, manufacture, and commercialization of prescription medicines.

Why it is in the core. UK-listed global pharma with an oncology pipeline - defensive earnings and the largest weight in the FTSE 100.

5. Shell plc (SHEL.L)

LSE | Energy / Oil & Gas Integrated | United Kingdom | quoted in GBp | Core sleeve weight 5.0% = GBP 8,000, 223 shares

Price (local)	35.765 GBP	Price (GBP)	35.765
52w high / low (local)	37.585 / 25.538	% below 52w high	-4.8%
Market cap (GBP)	GBP 204.40bn	Market cap (USD m)	275,401
Enterprise value (GBP)	GBP 246.10bn	Trailing P/E	10.69
Forward P/E	9.68	Price / book	1.51
EV / EBITDA	4.27	Dividend yield	3.23%
Total debt	GBP 73.08bn	Cash	GBP 31.37bn
Net debt	GBP 41.70bn	Debt / equity (book)	0.42
Interest cover (EBITDA / interest)	12.15	Interest cover (EBIT proxy)	7.28
Capex, latest FY	GBP -18.95bn	Capex / revenue	6.4%
Free cash flow	GBP 21.46bn	Revenue (latest FY)	GBP 296.60bn
Operating margin	16.7%	Revenue growth	44.7%
Beta (Yahoo)	-0.22	Beta (own, 5y daily GBP)	0.47
Upside beta (beta+)	0.20	Downside beta (beta-)	0.87
CAPM cost of equity Ke	5.7%	WACC	5.3%
Cost of debt Kd	6.5% (interest expense (FY) / total debt)	Tax rate	35.0%
Return YTD (GBP)	34.1%	Return 1y (GBP)	40.2%
Volatility 5y	24.4%	Max drawdown 5y	-20.6%
Skew / excess kurtosis	-0.47 / 3.39	Sharpe / Sortino 5y	0.86 / 1.21

Source: data/fundamentals_table.csv row SHEL.L; data/stats_daily.csv row SHEL.L

Business. Shell plc operates as an energy and petrochemical company in Europe, Asia, Oceania, Africa, the United States, and other parts of the Americas. It operates through the following segments: Integrated Gas, Upstream, Marketing, Chemicals and Products, and Renewables and Energy Solutions.

Why it is in the core. Energy cash flows and buybacks; an equity-form inflation hedge that pays you to hold it, unlike the oil ETC's roll drag.

6. HSBC Holdings plc (HSBA.L)

LSE | Financial Services / Banks - Diversified | United Kingdom | quoted in GBp | Core sleeve weight 5.0% = GBP 8,000, 530 shares

Price (local)	15.078 GBP	Price (GBP)	15.078
52w high / low (local)	16.100 / 9.593	% below 52w high	-6.3%
Market cap (GBP)	GBP 258.11bn	Market cap (USD m)	347,764
Enterprise value (GBP)	GBP -123.96bn	Trailing P/E	14.22
Forward P/E	10.44	Price / book	1.98
EV / EBITDA	n/a (not reported by Yahoo)	Dividend yield	3.76%
Total debt	GBP 837.58bn	Cash	GBP 1,219.65bn
Net debt	GBP -382.08bn	Debt / equity (book)	4.23
Interest cover (EBITDA / interest)	n/a (not reported by Yahoo)	Interest cover (EBIT proxy)	1.47
Capex, latest FY	GBP -4.66bn	Capex / revenue	6.9%
Free cash flow	n/a (not reported by Yahoo)	Revenue (latest FY)	GBP 67.43bn
Operating margin	58.8%	Revenue growth	25.4%
Beta (Yahoo)	0.57	Beta (own, 5y daily GBP)	0.85
Upside beta (beta+)	0.89	Downside beta (beta-)	1.09
CAPM cost of equity Ke	7.3%	WACC	6.2%
Cost of debt Kd	7.5% (interest expense (FY) / total debt)	Tax rate	22.7%
Return YTD (GBP)	33.4%	Return 1y (GBP)	55.8%
Volatility 5y	24.7%	Max drawdown 5y	-22.0%
Skew / excess kurtosis	-0.51 / 4.35	Sharpe / Sortino 5y	1.35 / 1.93

Source: data/fundamentals_table.csv row HSBA.L; data/stats_daily.csv row HSBA.L

Business. HSBC Holdings plc engages in the provision of banking and financial products and services worldwide. It operates through four segments: Hong Kong, UK, Corporate and Institutional Banking, and International Wealth and Premier Banking. The Hong Kong segment is involved in the retail banking and wealth and commercial banking of HSBC Hong Kong and Hang Seng Bank.

Why it is in the core. Financials exposure with an Asia-weighted balance sheet - rate-sensitive in the opposite direction to the bond sleeve.

7. Unilever PLC (ULVR.L)

LSE | Consumer Defensive / Household & Personal Products | United Kingdom | quoted in GBp | Not held in the headline portfolio (zero weight)

Price (local)	46.400 GBP	Price (GBP)	46.400
52w high / low (local)	57.113 / 36.440	% below 52w high	-18.8%
Market cap (GBP)	GBP 99.91bn	Market cap (USD m)	134,615
Enterprise value (GBP)	GBP 126.08bn	Trailing P/E	21.22
Forward P/E	15.75	Price / book	7.22
EV / EBITDA	11.59	Dividend yield	3.55%
Total debt	GBP 32.11bn	Cash	GBP 5.94bn
Net debt	GBP 26.17bn	Debt / equity (book)	2.07
Interest cover (EBITDA / interest)	9.91	Interest cover (EBIT proxy)	8.92
Capex, latest FY	GBP -1.59bn	Capex / revenue	3.1%
Free cash flow	GBP 6.73bn	Revenue (latest FY)	GBP 50.62bn
Operating margin	20.3%	Revenue growth	0.5%
Beta (Yahoo)	0.45	Beta (own, 5y daily GBP)	0.11
Upside beta (beta+)	0.27	Downside beta (beta-)	0.01
CAPM cost of equity Ke	4.2%	WACC	3.8%
Cost of debt Kd	3.4% (interest expense (FY) / total debt)	Tax rate	28.5%
Return YTD (GBP)	-2.1%	Return 1y (GBP)	-13.9%
Volatility 5y	19.7%	Max drawdown 5y	-28.4%
Skew / excess kurtosis	-0.15 / 13.75	Sharpe / Sortino 5y	0.07 / 0.10

Source: data/fundamentals_table.csv row ULVR.L; data/stats_daily.csv row ULVR.L

Business. Unilever PLC operates as a fast-moving consumer goods company in the Asia Pacific, Africa, the Americas, and Europe. It operates through four segments: Beauty & Wellbeing, Personal Care, Home Care, and Foods. The Beauty & Wellbeing segment offers hair care, such as shampoo, conditioner, and styling; face, hand, and body moisturizer skin care products; and Prestige Beauty and Wellbeing products.

Why it is in the core. Low-beta consumer staples with EM revenue - the lowest-volatility equity line here, which is what makes the 12% vol target reachable.

8. Microsoft Corporation (MSFT)

NasdaqGS | Technology / Software - Infrastructure | United States | quoted in USD | Not held in the headline portfolio (zero weight)

Price (local)	490.300 USD	Price (GBP)	363.901
52w high / low (local)	553.720 / 349.200	% below 52w high	-11.5%
Market cap (GBP)	GBP 2,702.16bn	Market cap (USD m)	3,640,745
Enterprise value (GBP)	GBP 2,740.73bn	Trailing P/E	26.92
Forward P/E	20.47	Price / book	8.11
EV / EBITDA	19.01	Dividend yield	0.81%
Total debt	USD 128.81bn	Cash	USD 76.84bn
Net debt	USD 51.97bn	Debt / equity (book)	0.22
Interest cover (EBITDA / interest)	63.66	Interest cover (EBIT proxy)	55.39
Capex, latest FY	USD -115.95bn	Capex / revenue	34.9%
Free cash flow	USD 16.55bn	Revenue (latest FY)	USD 331.84bn
Operating margin	45.1%	Revenue growth	17.7%
Beta (Yahoo)	1.11	Beta (own, 5y daily GBP)	0.67
Upside beta (beta+)	0.65	Downside beta (beta-)	0.47
CAPM cost of equity Ke	6.5%	WACC	6.4%
Cost of debt Kd	2.4% (interest expense (FY) / total debt)	Tax rate	19.4%
Return YTD (GBP)	2.0%	Return 1y (GBP)	-0.9%
Volatility 5y	29.2%	Max drawdown 5y	-34.1%
Skew / excess kurtosis	0.52 / 6.20	Sharpe / Sortino 5y	0.39 / 0.58

Source: data/fundamentals_table.csv row MSFT; data/stats_daily.csv row MSFT

Business. Microsoft Corporation, a technology company, develops and supports a portfolio of technology solutions for individuals and businesses worldwide. Its products include operating systems, server applications, business solution applications, software development tools, desktop and server management tools, and video games; and devices, such as PCs, tablets, gaming and entertainment consoles, other intelligent devices, and related accessories.

Why it is in the core. Quality mega-cap compounder with the deepest AI-infrastructure moat - the low-volatility way to own the same theme as PLTR.

9. ASML Holding N.V. (ASML.AS)

Euronext Amsterdam | Technology / Semiconductor Equipment & Materials | Netherlands | quoted in EUR | Not held in the headline portfolio (zero weight)

Price (local)	1,396.200 EUR	Price (GBP)	1,195.766
52w high / low (local)	1,741.000 / 729.500	% below 52w high	-19.8%
Market cap (GBP)	GBP 459.29bn	Market cap (USD m)	618,827
Enterprise value (GBP)	GBP 454.50bn	Trailing P/E	54.10
Forward P/E	26.58	Price / book	24.19
EV / EBITDA	39.34	Dividend yield	0.57%
Total debt	EUR 1.98bn	Cash	EUR 7.58bn
Net debt	EUR -5.60bn	Debt / equity (book)	0.09
Interest cover (EBITDA / interest)	114.02	Interest cover (EBIT proxy)	97.42
Capex, latest FY	EUR -1.63bn	Capex / revenue	4.6%
Free cash flow	EUR 8.44bn	Revenue (latest FY)	EUR 35.33bn
Operating margin	37.1%	Revenue growth	21.3%
Beta (Yahoo)	1.36	Beta (own, 5y daily GBP)	1.92
Upside beta (beta+)	1.82	Downside beta (beta-)	1.65
CAPM cost of equity Ke	11.8%	WACC	11.7%
Cost of debt Kd	6.0% (interest expense (FY) / total debt)	Tax rate	17.7%
Return YTD (GBP)	49.5%	Return 1y (GBP)	102.9%
Volatility 5y	40.2%	Max drawdown 5y	-46.4%
Skew / excess kurtosis	-0.18 / 2.84	Sharpe / Sortino 5y	0.44 / 0.64

Source: data/fundamentals_table.csv row ASML.AS; data/stats_daily.csv row ASML.AS

Business. ASML Holding N.V. provides lithography solutions for the development, production, marketing, sales, upgrading, and servicing of advanced semiconductor equipment systems. The company offers lithography, metrology, and inspection systems. It also provides extreme ultraviolet lithography systems; and deep ultraviolet lithography systems comprising immersion and dry lithography systems solutions to manufacture various range of semiconductor nodes and technologies.

Why it is in the core. EUV lithography monopoly - the single hardest-to-replicate chokepoint in the entire semiconductor supply chain.

10. Siemens Aktiengesellschaft (SIE.DE)

XETRA | Industrials / Specialty Industrial Machinery | Germany | quoted in EUR | Not held in the headline portfolio (zero weight)

Price (local)	260.500 EUR	Price (GBP)	223.103
52w high / low (local)	291.700 / 198.000	% below 52w high	-10.7%
Market cap (GBP)	GBP 174.08bn	Market cap (USD m)	234,545
Enterprise value (GBP)	GBP 207.48bn	Trailing P/E	25.45
Forward P/E	19.52	Price / book	2.89
EV / EBITDA	19.35	Dividend yield	2.10%
Total debt	EUR 49.92bn	Cash	EUR 10.93bn
Net debt	EUR 38.99bn	Debt / equity (book)	0.69
Interest cover (EBITDA / interest)	7.63	Interest cover (EBIT proxy)	7.60
Capex, latest FY	EUR -2.44bn	Capex / revenue	3.0%
Free cash flow	EUR 6.50bn	Revenue (latest FY)	EUR 81.11bn
Operating margin	14.9%	Revenue growth	7.3%
Beta (Yahoo)	1.28	Beta (own, 5y daily GBP)	1.41
Upside beta (beta+)	1.43	Downside beta (beta-)	1.58
CAPM cost of equity Ke	9.6%	WACC	8.2%
Cost of debt Kd	3.3% (interest expense (FY) / total debt)	Tax rate	23.1%
Return YTD (GBP)	9.3%	Return 1y (GBP)	15.0%
Volatility 5y	30.8%	Max drawdown 5y	-37.0%
Skew / excess kurtosis	0.06 / 3.20	Sharpe / Sortino 5y	0.48 / 0.69

Source: data/fundamentals_table.csv row SIE.DE; data/stats_daily.csv row SIE.DE

Business. Siemens Aktiengesellschaft, a technology company, focuses in the areas of automation and digitalization across the world. It operates through Digital Industries, Smart Infrastructure, Mobility, Siemens Healthineers, and Siemens Financial Services (SFS) segments.

Why it is in the core. European industrial automation and grid infrastructure - capex-cycle exposure absent from every other line in this universe.

11. Berkshire Hathaway Inc. Class B (BRK-B)

NYSE | Financial Services / Insurance - Diversified | United States | quoted in USD | Core sleeve weight 1.4% = GBP 2,276, 5 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 35.6%. The company has more than one share class; the headline below is shares x price.

Price (local)	519.800 USD	Price (GBP)	385.796
52w high / low (local)	537.740 / 464.010	% below 52w high	-3.3%
Market cap (GBP)	GBP 543.21bn	Market cap (USD m)	731,897
Enterprise value (GBP)	GBP 367.38bn	Trailing P/E	13.35
Forward P/E	23.91	Price / book	0.00
EV / EBITDA	3.79	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 128.60bn	Cash	USD 365.51bn
Net debt	USD -236.91bn	Debt / equity (book)	0.13
Interest cover (EBITDA / interest)	25.78	Interest cover (EBIT proxy)	17.27
Capex, latest FY	USD -20.93bn	Capex / revenue	5.4%
Free cash flow	USD 71.98bn	Revenue (latest FY)	USD 384.69bn
Operating margin	32.6%	Revenue growth	10.0%
Beta (Yahoo)	0.60	Beta (own, 5y daily GBP)	0.34
Upside beta (beta+)	0.34	Downside beta (beta-)	0.34
CAPM cost of equity Ke	5.2%	WACC	4.9%
Cost of debt Kd	3.9% (interest expense (FY) / total debt)	Tax rate	18.4%
Return YTD (GBP)	3.4%	Return 1y (GBP)	5.1%
Volatility 5y	18.8%	Max drawdown 5y	-20.6%
Skew / excess kurtosis	-0.10 / 3.27	Sharpe / Sortino 5y	0.59 / 0.85

Source: data/fundamentals_table.csv row BRK-B; data/stats_daily.csv row BRK-B

Business. Berkshire Hathaway Inc., together with its subsidiaries, engages in the insurance, freight rail transportation, and utility businesses.

Why it is in the core. Diversified low-beta US quality with a large cash reserve - behaves partly like a call on market dislocation, damping the tech concentration.

7.2 Moonshot sleeve, in rank order

1. Ideal Power Inc. (IPWR)

NasdaqCM | Technology / Semiconductors | United States | quoted in USD | Moonshot sleeve rank 1 of 40, GBP 1,000, 346 shares

Price (local)	3.890 USD	Price (GBP)	2.887
52w high / low (local)	9.300 / 2.615	% below 52w high	-58.2%
Market cap (GBP)	GBP 47.4m	Market cap (USD m)	64
Enterprise value (GBP)	GBP 17.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-3.33	Price / book	1.48
EV / EBITDA	-1.97	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 358.4k	Cash	USD 41.3m
Net debt	USD -40.9m	Debt / equity (book)	0.03
Interest cover (EBITDA / interest)	-916.70	Interest cover (EBIT proxy)	-831.88
Capex, latest FY	USD -449.5k	Capex / revenue	1485.9%
Free cash flow	USD -5.3m	Revenue (latest FY)	USD 30.2k
Operating margin	-58758.9%	Revenue growth	354.9%
Beta (Yahoo)	1.84	Beta (own, 5y daily GBP)	1.08
Upside beta (beta+)	0.68	Downside beta (beta-)	1.41
CAPM cost of equity Ke	8.3%	WACC	8.2%
Cost of debt Kd	3.5% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	25.8%	Return 1y (GBP)	-25.1%
Volatility 5y	87.3%	Max drawdown 5y	-84.5%
Skew / excess kurtosis	1.48 / 8.22	Sharpe / Sortino 5y	0.07 / 0.11
Judge score (asymmetry / backing / timing)	8.24 (8.4 / 7.8 / 8.5)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 998k	GBP 1,000 as % of a day	0.13%
10x end-state market cap	USD 0.64bn	100x arithmetically possible?	Arithmetically live (~\$6.4bn) but low probability

Source: data/fundamentals_table.csv row IPWR; data/stats_daily.csv row IPWR; universe/unicorn_final.json; universe/dossiers/IPWR.md

Business. Ideal Power Inc. focuses on the development and commercialization of its bidirectional bipolar junction TRANSistor solid-state switch technology. The company offers SymCool Power Module designed to meet low conduction loss needs of the solid-state circuit protection and EV contactor markets.

Thesis. Ideal Power licenses B-TRAN, the only bidirectional bipolar transistor in existence: one device that conducts and blocks in both directions, halving conduction loss versus the back-to-back MOSFET/IGBT pairs every solid-state circuit breaker uses today. Nvidia's 800VDC Rubin Ultra rack architecture creates a socket that no incumbent device fills well, and a US hyperscaler is co-developing an 800V DC prototype for exactly that architecture. Market cap \$63.9m at the 2026-09-16 close with \$41.3m of cash and essentially no debt, so the market values the whole platform at roughly \$22m.

Key backers. US hyperscaler (unnamed) - Nvidia Rubin Ultra 800V DC prototype, delivery end-Q4 2026; Stellantis - second shipment of custom packaged B-TRAN samples plus joint work on its solid-state contactor system spec; high-volume Asian foundry - long-term supply agreement with functional first silicon; lead Asia customer - low-current SSCB, production-ready units targeted Q4 2026; distribution partner - first stocking order for 800V SSCB reference-design kits. All from the Q2 2026 release (PRNewswire, 2026-08-13).

Catalysts, 12-24 months. Hyperscaler 800V DC prototype delivery end-Q4 2026 (highest value per unit of market cap); production-ready low-current SSCB units at the Asia customer Q4 2026; Stellantis moving from evaluation to design-in; a first licence with disclosed royalty terms; foundry volume ramp.

Why 10x. 10x = \$639m; since \$41m of the \$64m is cash, the operating business need only reach ~\$598m. That requires one pathway - datacentre SSCB or EV contactor - to convert into a production design-in with device or royalty revenue in the tens of millions. An emerging power-semi platform with a named hyperscaler and a named auto OEM at \$600m is unremarkable for the category. 100x = \$6.4bn is arithmetically live (Navitas has traded near that on GaN) if B-TRAN becomes standard content in both DC datacentre protection and EV contactors; low probability, but the ~14-15 quarter runway is long enough to find out.

What kills it. Pre-revenue for over a decade with a prior pivot (it was a solar inverter company) and \$5,800 of quarterly revenue; hyperscaler co-development routinely ends without a purchase order and the partner is unnamed so nothing can be independently verified; Infineon/onsemi/Wolfspeed/Navitas can attack the same sockets with proven SiC and GaN at scale, and double-sided wafer processing is a real yield risk at volume.

What falsifies it. The Q4 2026 hyperscaler prototype slipping materially or being dropped. If that milestone passes without converting to a named commercial agreement during 2027, the thesis is broken regardless of the balance sheet.

Confidence. 7/10. Unusually clean asymmetry: debt-free, ~4-year runway, technology priced at ~\$22m against a named Nvidia-architecture programme.

2. Voyager Therapeutics, Inc. (VYGR)

NASDAQ | Healthcare / Biotechnology | United States | quoted in USD | Moonshot sleeve rank 2 of 40, GBP 1,000, 441 shares

Price (local)	3.050 USD	Price (GBP)	2.264
52w high / low (local)	5.550 / 2.940	% below 52w high	-45.0%
Market cap (GBP)	GBP 139.5m	Market cap (USD m)	188
Enterprise value (GBP)	GBP 77.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-2.39	Price / book	1.06
EV / EBITDA	-0.95	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 32.6m	Cash	USD 116.3m
Net debt	USD -83.8m	Debt / equity (book)	0.12
Interest cover (EBITDA / interest)	-10.35	Interest cover (EBIT proxy)	-10.26
Capex, latest FY	USD -2.6m	Capex / revenue	7.5%
Free cash flow	USD -77.3m	Revenue (latest FY)	USD 34.5m
Operating margin	-833.9%	Revenue growth	-39.1%
Beta (Yahoo)	1.32	Beta (own, 5y daily GBP)	0.52
Upside beta (beta+)	0.71	Downside beta (beta-)	0.64
CAPM cost of equity Ke	5.9%	WACC	8.4%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-22.4%	Return 1y (GBP)	-30.0%
Volatility 5y	80.5%	Max drawdown 5y	-81.8%
Skew / excess kurtosis	2.15 / 21.31	Sharpe / Sortino 5y	0.34 / 0.57
Judge score (asymmetry / backing / timing)	7.54 (7.6 / 7.6 / 7.3)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 1,936k	GBP 1,000 as % of a day	0.07%
10x end-state market cap	USD 1.88bn	100x arithmetically possible?	Not inside a 5-10 year hold

Source: data/fundamentals_table.csv row VYGR; data/stats_daily.csv row VYGR; universe/unicorn_final.json; universe/dossiers/VYGR.md

Business. Voyager Therapeutics, Inc., a biotechnology company, focuses on the human genetics for the cure of neurological diseases. Its product pipeline includes VY1706, a tau silencing gene therapy and VY7523, an anti-tau antibody program for the treatment of alzheimer's disease. The company also provides research program for the treatment of Huntington's disease.

Thesis. Voyager is an AAV capsid arms dealer, not a drug developer. Its TRACER platform screens capsid libraries directly in non-human primates for blood-brain-barrier crossing after ordinary IV dosing plus liver de-targeting - the actual binding constraint on CNS gene therapy, since AAV9 crosses poorly and today's therapies need neurosurgery. Fourteen partnered TRACER-enabled programmes pay milestones and royalties regardless of which disease wins. SEC-verified cash and investments of \$148.8m at 30-Jun-2026 against a \$187.9m market cap leaves an enterprise value near \$39m.

Key backers. Novartis - capsid licence for SMA and Huntington's, \$100m upfront including a \$20m equity purchase, up to \$1.2bn milestones plus tiered royalties (Voyager IR, 2024-01-02; terms confirmed against the 8-K); Neurocrine Biosciences - multi-programme collaboration including Friedreich's ataxia, Neurocrine funding development (10-Q, 2024-11); an additional undisclosed licensee taking the partnered portfolio to 14 programmes. Insider ownership 14.8%, institutions 57.8%.

Catalysts, 12-24 months. New capsid licences, each cash plus milestones against a ~\$39m EV; partner INDs and clinical starts triggering milestones; data from Voyager's own anti-tau antibody VY7523 in Alzheimer's; read-across from uniQure's AMT-130 BLA, since any CNS gene therapy approval validates the category Voyager taxes.

Why 10x. 10x = \$1.88bn. Two or three of the 14 partnered programmes must reach late-stage development, converting contingent milestones into a visible royalty stream, plus two further large capsid deals. A royalty business with \$150-250m of annual milestone-and-royalty income trades at 8-12x revenue. Critically this requires no Voyager-owned drug to work at all. 100x = \$18.8bn implies Royalty Pharma-scale economics from a capsid estate - conceivable over 15 years, not a 5-10 year case.

What kills it. Capsid obsolescence - focused-ultrasound BBB opening, engineered exosomes or rival capsids from Capsida, Dyno or Regenxbio commoditise TRACER, and there is no regulatory moat; partner attrition, since licensees can walk and Voyager gets nothing further, with a Novartis deprioritisation removing the anchor; and burn of ~\$22.9m a quarter means the cash cushion that makes the EV look cheap is being spent on an owned pipeline worth possibly less than the cash. The company has already failed twice - this is its third business model.

What falsifies it. Novartis terminating or deprioritising the licensed capsid programmes. That single event removes the milestone anchor, the validation and the read-across in one move.

Confidence. 7/10. The royalty-stacking structure, real insider ownership and modest dilution are rare in this market cap band; the burn rate is the honest offset.

3. AML3D Limited (AL3.AX)

ASX | Industrials / Metal Fabrication | Australia | quoted in AUD | Moonshot sleeve rank 3 of 40, GBP 1,000, 14,004 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 11.1%. The headline below is shares x price.

Price (local)	0.135 AUD	Price (GBP)	0.071
52w high / low (local)	0.295 / 0.090	% below 52w high	-54.2%
Market cap (GBP)	GBP 40.5m	Market cap (USD m)	55
Enterprise value (GBP)	GBP 27.2m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	30.38	Price / book	2.17
EV / EBITDA	-10.40	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 1.6m	Cash	AUD 26.8m
Net debt	AUD -25.2m	Debt / equity (book)	0.03
Interest cover (EBITDA / interest)	-26.65	Interest cover (EBIT proxy)	-23.17
Capex, latest FY	AUD -1.8m	Capex / revenue	14.3%
Free cash flow	AUD -6.4m	Revenue (latest FY)	AUD 12.5m
Operating margin	-2.6%	Revenue growth	234.4%
Beta (Yahoo)	0.00	Beta (own, 5y daily GBP)	0.40
Upside beta (beta+)	0.16	Downside beta (beta-)	0.58
CAPM cost of equity Ke	5.4%	WACC	5.5%
Cost of debt Kd	11.9% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-17.9%	Return 1y (GBP)	-37.9%
Volatility 5y	97.0%	Max drawdown 5y	-75.5%
Skew / excess kurtosis	1.64 / 9.79	Sharpe / Sortino 5y	0.38 / 0.64
Judge score (asymmetry / backing / timing)	7.23 (6.9 / 7.6 / 7.5)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 229k	GBP 1,000 as % of a day	0.59%
10x end-state market cap	USD 0.55bn	100x arithmetically possible?	No - 2-3% tail at most

Source: data/fundamentals_table.csv row AL3.AX; data/stats_daily.csv row AL3.AX; universe/unicorn_final.json; universe/dossiers/AL3.AX.md

Business. AML3D Limited designs, manufactures, and sells industrial-scale metal 3D printing systems and contract manufacturing services to the aerospace, marine, defence, oil and gas, mining, and general manufacturing sectors in Australia, the United Kingdom, and the United States. The company produces metal components and structures through automated wire-fed 3D printing technology.

Thesis. AML3D sells ARCEMY, a containerised wire-arc additive manufacturing cell that 3D-prints large metal parts from welding wire. The mechanism is a physical bottleneck in AUKUS: the US submarine industrial base cannot hit Virginia-class build rates because castings and forgings carry 18-36 month lead times and many original manufacturers no longer exist. Through BlueForge Alliance the US Navy is installing AML3D machines across its own supply chain rather than merely buying parts - six units at Newport News Shipbuilding and three at Austal USA. FY26 revenue A\$12.5m (+70%) with the first EBITDA-positive half.

Key backers. BlueForge Alliance / US Navy Submarine Industrial Base - customer and channel, A\$2.61m (US\$1.84m) order for obsolete submarine replacement components (Metal AM, 2026-03-26); Newport News Shipbuilding (Huntington Ingalls) - six ARCEMY systems contracted; Austal USA - three systems; US Navy Maritime Industrial Base qualifying AML3D alloys for naval use. Insider ownership only 11.2% - the alignment argument is the customer, not the register.

Catalysts, 12-24 months. Follow-on ARCEMY system orders from the SIB, each material against A\$12.5m of revenue; first full year of positive EBITDA in FY2027; UK expansion under AUKUS Pillar 1 into the BAE/Rolls-Royce submarine supply chain; qualification of a safety-critical component class, which multiplies the addressable part list at a stroke.

Why 10x. 10x = US\$545m from US\$54.5m. That needs roughly A\$90-120m of revenue at 5-7x sales - about 8-10x FY2026 - achieved by going from a handful of installed systems to 100-150 across US, UK and Australian naval supply chains plus the consumables and licence tail. At A\$2-4m per system plus service, 100 systems is A\$200-400m of cumulative hardware with recurring revenue underneath. This is the most mechanically checkable 10x in the sleeve: arithmetic on machine count, not a scientific breakthrough. 100x (US\$5.5bn) would need WAAM to displace a large share of global castings and forgings with AML3D holding the standard - a 2-3% tail.

What kills it. Sovereignty risk - an Australian supplier inside a US sovereign defence programme is permanently exposed to the Navy insourcing WAAM or mandating a domestic vendor, and this is unhedgeable; qualification slippage, since additive metal has repeatedly under-delivered on timelines and the jump to safety-critical parts may never come; and contract lumpiness, where one delayed system flips the half back into loss on a A\$12.5m revenue base.

What falsifies it. The US Navy awarding a comparable WAAM installation programme to a domestic vendor, or the FY2027 order intake failing to exceed the A\$16.8m already contracted. Either shows the installed base is not compounding.

Confidence. 6/10 - the lowest here. The commercial evidence is good and the first EBITDA-positive half is real, but customer concentration is extreme and the 10x case depends on repeat orders not yet contracted.

4. ONWARD Medical N.V. (ONWD.BR)

Euronext Brussels | Healthcare / Biotechnology | Netherlands | quoted in EUR | Moonshot sleeve rank 4 of 40, GBP 1,000, 417 shares

Price (local)	2.800 EUR	Price (GBP)	2.398
52w high / low (local)	5.770 / 2.270	% below 52w high	-51.5%
Market cap (GBP)	GBP 166.7m	Market cap (USD m)	225
Enterprise value (GBP)	GBP 108.7m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-4.52	Price / book	2.75
EV / EBITDA	-2.92	Dividend yield	n/a (not reported by Yahoo)
Total debt	EUR 13.8m	Cash	EUR 81.5m
Net debt	EUR -67.7m	Debt / equity (book)	0.21
Interest cover (EBITDA / interest)	-19.61	Interest cover (EBIT proxy)	-17.72
Capex, latest FY	EUR -432.0k	Capex / revenue	5.1%
Free cash flow	EUR -29.1m	Revenue (latest FY)	EUR 8.4m
Operating margin	-556.7%	Revenue growth	253.8%
Beta (Yahoo)	0.85	Beta (own, 5y daily GBP)	0.74
Upside beta (beta+)	0.44	Downside beta (beta-)	0.86
CAPM cost of equity Ke	6.8%	WACC	7.2%
Cost of debt Kd	16.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-38.7%	Return 1y (GBP)	-45.9%
Volatility 5y	64.6%	Max drawdown 5y	-83.7%
Skew / excess kurtosis	2.85 / 24.62	Sharpe / Sortino 5y	-0.23 / -0.39
Judge score (asymmetry / backing / timing)	7.16 (7 / 7.2 / 7.5)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 361k	GBP 1,000 as % of a day	0.37%
10x end-state market cap	USD 2.32bn	100x arithmetically possible?	No - prevalence does not support it

Source: data/fundamentals_table.csv row ONWD.BR; data/stats_daily.csv row ONWD.BR; universe/unicorn_final.json; universe/dossiers/ONWD.BR.md

Business. Onward Medical N.V., a medical technology company, develops and commercializes therapies to enable functional recovery for people with spinal cord injury (SCI) in the United States, Europe, and internationally.

Thesis. ONWARD develops ARC Therapy - targeted spinal cord stimulation that amplifies surviving nerve signal below a spinal cord injury to restore movement, blood pressure and autonomic function. ARC-EX is FDA-cleared and selling (159 units in H1 2026, +430% year on year, 130+ US clinics, VA home-use access); ARC-IM is the implant; ARC-BCI couples the implant to a cortical BCI so the patient's own intention drives stimulation. It is the only listed BCI pure-play - Neuralink, Precision, Paradromics and Synchron are all private - and unlike them it already has revenue and reimbursement data.

Key backers. EQT Life Sciences - EUR 25m of the EUR 40.6m April 2026 equity raise; BlackRock managed funds and accounts - facility of up to EUR 50m with EUR 20m drawn at closing; US Department of Veterans Affairs - expanded home-use access for ARC-EX, a government payer for a device with no reimbursement precedent; CEA-Clinattec and EPFL/NeuroRestore - WIMAGINE cortical implant and the digital-bridge science behind ARC-BCI. Insider ownership 28.5%. Cash EUR 81.5m at 30-Jun-2026.

Catalysts, 12-24 months. Empower BP global pivotal study of ARC-IM for blood pressure, 22 participants enrolled across 14 active sites at H1 2026, with interim data in the window; first ARC-IM bladder feasibility study; early-feasibility implants in Parkinson's and stroke, each expanding the addressable population by an order of magnitude; continued ARC-EX unit growth; any broad CMS coverage decision.

Why 10x. 10x = US\$2.23bn. ARC-EX must reach roughly EUR 50-80m of annual revenue with permanent reimbursement, and ARC-IM must hit its pivotal blood-pressure endpoint to give ONWARD an approved implant. An approved neuromodulation implant with a BCI roadmap would be valued like an early Inspire Medical or Axonics - both exceeded \$3bn - plus the scarcity premium of being the only listed BCI pure-play. 100x (US\$22bn) is not defensible: US SCI prevalence is roughly 300,000 people, and even with Parkinson's and stroke extensions the market does not support it.

What kills it. Dilution rather than death - EUR 24.7m lost in six months on EUR 4.2m of revenue, share count already +55.8% in 18 months, so holders can be right on the science and still lose; reimbursement is the real gate, and without a permanent CMS code ARC-EX stays a cash-pay and VA niche; and a missed Empower BP endpoint removes the implant leg entirely, leaving a thinly-traded external-device company on ~US\$373k a day.

What falsifies it. Empower BP missing its primary endpoint, or a financing struck at a materially lower price than the April 2026 round. Either one breaks the "funded path to an approved implant" that the entire 10x case rests on.

Confidence. 7/10. Real revenue growth, a strong backer set for the size, high insider ownership and genuine scarcity value - against a burn rate that guarantees further dilution.

5. Lightbridge Corporation (LTBR)

NASDAQ Capital Market | Industrials / Electrical Equipment & Parts | United States | quoted in USD | Moonshot sleeve rank 5 of 40, GBP 1,000, 208 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 5.4%. The headline below is shares x price.

Price (local)	6.470 USD	Price (GBP)	4.802
52w high / low (local)	31.340 / 6.330	% below 52w high	-79.4%
Market cap (GBP)	GBP 169.2m	Market cap (USD m)	228
Enterprise value (GBP)	GBP -7.0m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	n/a (not reported by Yahoo)	Price / book	0.91
EV / EBITDA	n/a (not reported by Yahoo)	Dividend yield	n/a (not reported by Yahoo)
Total debt	n/a (not reported by Yahoo)	Cash	USD 237.5m
Net debt	USD -237.5m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	n/a (not reported by Yahoo)	Interest cover (EBIT proxy)	-4.37
Capex, latest FY	USD -10.5k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	USD -10.4m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	2.15	Beta (own, 5y daily GBP)	1.44
Upside beta (beta+)	1.52	Downside beta (beta-)	1.55
CAPM cost of equity Ke	9.8%	WACC	9.8%
Cost of debt Kd	5.8% (fallback: rf + 2% (no usable interest expense / debt))	Tax rate	25.0%
Return YTD (GBP)	-48.8%	Return 1y (GBP)	-53.8%
Volatility 5y	108.9%	Max drawdown 5y	-83.2%
Skew / excess kurtosis	1.70 / 9.35	Sharpe / Sortino 5y	0.53 / 0.91
Judge score (asymmetry / backing / timing)	7.13 (7.5 / 7.6 / 5.5)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 6,691k	GBP 1,000 as % of a day	0.02%
10x end-state market cap	USD 2.28bn	100x arithmetically possible?	No - implausible against Westinghouse/Framatome/GNF

Source: data/fundamentals_table.csv row LTBR; data/stats_daily.csv row LTBR; universe/unicorn_final.json; universe/dossiers/LTBR.md

Business. Lightbridge Corporation, together with its subsidiaries, engages in research, developing, and commercializing nuclear fuel. It develops nuclear fuel for water-cooled reactors; and metallic fuels under the Lightbridge Fuel. The company has a strategic partnership with Studsvik Scandpower, Inc. to develop software for modeling metallic nuclear fuel. The company is headquartered in Reston, Virginia.

Thesis. Lightbridge develops an all-metallic, helically-twisted multi-lobe nuclear fuel rod designed as a drop-in replacement for uranium-oxide fuel in existing PWRs and SMRs. Fuel, not reactors, is where nuclear economics change fastest: a fuel letting each of roughly 440 operating reactors produce ~10% more electricity from the same steel needs no new build, site or grid connection, only NRC qualification. The striking fact is the balance sheet - \$237.5m of cash, zero debt, against a \$228.0m market cap at the 2026-09-16 close of \$6.47, its 52-week low. Enterprise value is negative.

Key backers. US DOE Nuclear Energy Launch Pad INL / NRIC - selected 2026-08-31 for accelerated authorisation to design, build and operate a Lead Test Assembly fabrication facility at Idaho National Laboratory; Battelle Energy Alliance / INL - CRADA Project Task Statement No. 2 granting Advanced Test Reactor loop access for a ~60-month irradiation campaign under prototypic commercial LWR conditions (2026-09-09); Studsvik Scandpower (fuel-performance modelling) and Quadrant Nuclear Industries (HALEU supply). Insider ownership only 6.3%.

Catalysts, 12-24 months. DOE authorisation of the SHED pilot facility; first rodlets in-reactor at ATR; NRC pre-application for the production facility; a named utility partner for Lead Test Assemblies; further national patent grants (Japan allowed 2026-09-01). Against a stock at negative enterprise value, any one of these re-rates it.

Why 10x. 10x = \$2.28bn. It requires the ATR loop campaign to produce clean irradiation data, DOE to authorise the SHED pilot facility, and at least one US utility to commit to Lead Test Assemblies in a commercial core; the market would then price Lightbridge as a fuel licensor with line of sight to single-digit share of the ~\$10bn/yr global fabrication market plus uprate economics. Because enterprise value is negative the technology option is free, so on EV any successful qualification produces an unbounded multiple. On market cap, 100x (\$22.8bn) would require dominating Western fleet fuel supply against Westinghouse, Framatome and GNF - implausible.

What kills it. The clock rather than the balance sheet - fuel qualification is a 10-15 year regulatory marathon and the ATR loop campaign alone runs about 60 months, so the thesis can time out inside a 5-10 year hold; metallic fuel physics, since metallic fuels have historically suffered swelling and fuel-cladding chemical interaction at high burn-up and there is no fallback product; and a decade-long management record of announcing pathways rather than delivering fuel, funded throughout by dilution (+87.6% shares in 21 months), against utilities that are intensely conservative about core loading.

What falsifies it. Irradiation data from the ATR campaign showing unacceptable swelling or fuel-cladding interaction - or DOE declining to authorise SHED. Either removes the only product.

Confidence. 7/10. The set-up is unusual: a debt-free company with ~10 years of runway, two fresh state validations, trading below its own cash at its 52-week low. The downside is bounded by the cash; the upside needs patience most investors do not have.

6. Aclara Resources Inc. (ARA.TO)

Toronto Stock Exchange | Basic Materials / Other Industrial Metals & Mining | Peru | quoted in CAD | Moonshot sleeve rank 6 of 40, GBP 1,000, 512 shares

Price (local)	3.660 CAD	Price (GBP)	1.951
52w high / low (local)	5.400 / 1.780	% below 52w high	-32.2%
Market cap (GBP)	GBP 484.0m	Market cap (USD m)	652
Enterprise value (GBP)	GBP 461.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-64.02	Price / book	3.15
EV / EBITDA	-96.39	Dividend yield	n/a (not reported by Yahoo)
Total debt	CAD 546.0k	Cash	CAD 42.6m
Net debt	CAD -42.0m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-11.52	Interest cover (EBIT proxy)	-10.47
Capex, latest FY	CAD -39.2m	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	CAD -44.7m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	0.84	Beta (own, 5y daily GBP)	0.35
Upside beta (beta+)	0.63	Downside beta (beta-)	0.24
CAPM cost of equity Ke	5.2%	WACC	5.2%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	66.6%	Return 1y (GBP)	126.0%
Volatility 5y	89.2%	Max drawdown 5y	-84.3%
Skew / excess kurtosis	0.50 / 6.87	Sharpe / Sortino 5y	0.57 / 0.89
Judge score (asymmetry / backing / timing)	7.05 (5.5 / 8.8 / 8.3)	Confidence (dossier)	8 / 10
Avg daily traded value	USD 266k	GBP 1,000 as % of a day	0.50%
10x end-state market cap	USD 6.49bn	100x arithmetically possible?	No - exceeds the whole non-Chinese rare earth industry

Source: data/fundamentals_table.csv row ARA.TO; data/stats_daily.csv row ARA.TO; universe/unicorn_final.json; universe/dossiers/ARA.TO.md

Business. Aclara Resources Inc., a mining company, engages in the exploration and development of rare-earth mineral resources in Chile, Brazil and Peru. Aclara Resources Inc. was formerly known as 1303714 B.C. Ltd and changed its name to Aclara Resources Inc. in October 2021. Aclara Resources Inc. was incorporated in 2021 and is headquartered in Santiago De Surco, Peru.

Thesis. Aclara is an ionic adsorption clay rare-earth developer with the Penco Module in Chile, the flagship Carina Module in Brazil, and a planned heavy rare earth separation facility in Louisiana - intended to be the first in the United States. The US has no domestic heavy rare earth separation capability at all: dysprosium and terbium, the elements that let a magnet survive motor temperatures, are almost entirely Chinese. Ionic clays are the geological prize because the heavies desorb with a mild salt solution at ambient temperature, with no crushing, roasting or thorium waste stream, and almost every such deposit sits in China or Myanmar.

Key backers. EXIM Bank - US\$750m Letter of Interest (2026-09-04), larger than the entire market cap but explicitly non-binding; US Department of Energy - selected for federal funding (2026-07-23); US Government - direct investment (2025-09-02); JOGMEC, Japan's state resource agency - ionic-clay heavy rare earth exploration partnership in Brazil (2026-09-08); Hochschild Mining as controlling shareholder group (74.5% insider-held) and CAP S.A. as Chilean JV partner. All dates verified on aclara-re.com/news.

Catalysts, 12-24 months. Conversion of the EXIM Letter of Interest into a committed facility - the event that matters; Carina final investment decision after the feasibility study filed 2026-04-13; Louisiana construction start targeted Q1 2027 following the air permit secured 2026-09-14 and a US\$20m+ state tax exemption (2026-06-26); Penco construction after the positive Environmental Qualification Resolution of 2026-06-23; a first magnet-maker

or automaker offtake.

Why 10x. 10x = US\$6.49bn from US\$649m. That needs Carina in production, the Louisiana separation plant operating, and Aclara established as the West's primary non-Chinese dysprosium and terbium supplier with binding offtakes at strategic pricing. MP Materials reached a multi-billion valuation on light rare earths alone with a US price floor; a heavy rare earth equivalent with EXIM project financing and two producing modules supports the number. 100x = US\$65bn exceeds the plausible value of the entire rare earth industry outside China - the arithmetic does not close.

What kills it. Chile - Penco was previously rejected on environmental grounds and the company relinquished all water rights in December 2025 to get it through, so community opposition can return and has before; the EXIM Letter of Interest is explicitly non-binding and many never convert, and with ~\$42.6m of cash against a multi-hundred-million build a failure to convert forces heavy dilution; and rare earth prices are administered by Chinese policy, so a deliberate price war can render every Western project uneconomic mid-construction, with minorities in a 74.5%-controlled, ~US\$248k-a-day stock powerless to respond.

What falsifies it. EXIM declining to convert the US\$750m Letter of Interest into a committed facility, or a renewed Chilean environmental/community block on Penco. Either turns a financed national-security asset back into an ordinary junior developer.

Confidence. 8/10 - the highest here, on backer quality: three separate US federal actions and one Japanese state action all point the same way. The offset is that the market cap already reflects some of it.

7. Silex Systems Limited (SLX.AX)

ASX | Technology / Semiconductors | Australia | quoted in AUD | Moonshot sleeve rank 7 of 40, GBP 1,000, 430 shares

Price (local)	4.390 AUD	Price (GBP)	2.322
52w high / low (local)	10.850 / 4.070	% below 52w high	-59.5%
Market cap (GBP)	GBP 646.6m	Market cap (USD m)	871
Enterprise value (GBP)	GBP 551.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-88.37	Price / book	6.17
EV / EBITDA	1,211.62	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 642.2k	Cash	AUD 180.7m
Net debt	AUD -180.1m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	12.73	Interest cover (EBIT proxy)	-570.34
Capex, latest FY	AUD -107.7k	Capex / revenue	0.6%
Free cash flow	AUD 1.7m	Revenue (latest FY)	AUD 18.9m
Operating margin	-42.9%	Revenue growth	0.0%
Beta (Yahoo)	0.48	Beta (own, 5y daily GBP)	1.09
Upside beta (beta+)	1.70	Downside beta (beta-)	0.92
CAPM cost of equity Ke	8.3%	WACC	8.3%
Cost of debt Kd	10.5% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-45.9%	Return 1y (GBP)	20.2%
Volatility 5y	72.7%	Max drawdown 5y	-66.5%
Skew / excess kurtosis	-0.05 / 4.21	Sharpe / Sortino 5y	0.57 / 0.85
Judge score (asymmetry / backing / timing)	7.04 (6 / 8.6 / 7.3)	Confidence (dossier)	8 / 10
Avg daily traded value	USD 3,852k	GBP 1,000 as % of a day	0.03%
10x end-state market cap	USD 8.71bn	100x arithmetically possible?	No - larger than the whole Western enrichment industry

Source: data/fundamentals_table.csv row SLX.AX; data/stats_daily.csv row SLX.AX; universe/unicorn_final.json; universe/dossiers/SLX.AX.md

Business. Silex Systems Limited, a technology commercialization company, engages in the research and development, commercialization, and license of SILEX laser enrichment technology in Australia, the United States, and the United Kingdom. It operates through three segments: Silex Systems, Translucent, and Silex USA. The company also sells its cREO technology.

Thesis. Silex owns the SILEX laser uranium enrichment process and holds 51% of Global Laser Enrichment, a JV with Cameco (49%), building the Paducah Laser Enrichment Facility in Kentucky. Enrichment, not reactors, is the true chokepoint - every SMR, HALEU reactor and LEU+ uprate consumes SWU, and the West lost roughly 45% of supply when Russian enriched product was sanctioned. Laser separation uniquely can economically re-enrich DOE's stockpile of over 200,000 tonnes of depleted tails under a 2016 agreement running up to 30 years, feedstock incumbents cannot touch. Near-breakeven EBITDA and US\$128.5m of investments make it unusually survivable for a moonshot.

Key backers. Cameco Corporation - 49% JV partner in GLE, confirmed on silex.com.au, cameco.com and in Cameco's SEC-filed 6-K exhibit (2026-02); US DOE - supplies over 200,000 tonnes of depleted tails feedstock for up to 30 years under a 2016 agreement; Commonwealth of Kentucky and Paducah-McCracken County - state, county and federal support for PLEF; US NRC - full licence application filed July 2025, accepted for formal review 2025-08-04, first in line for a new US enrichment licence. Insider ownership 25.5%.

Catalysts, 12-24 months. NRC construction and operating licence decision for PLEF; finalisation of the DOE tails-transfer contract; GLE final investment decision and the associated capital call, which is also the principal dilution risk; commercial LEU and HALEU contracts; TRL-7 cascade demonstration; a possible DOE HALEU

award.

Why 10x. 10x = US\$8.69bn. It needs PLEF licensed, built and operating at planned scale with conversion and enrichment services attached; at mid-cycle pricing that is a multi-hundred-million-dollar annual EBITDA asset, and Silex's 51% share on Centrus-type multiples supports the number. The required conditions are specific and checkable: licence granted, JV funded without Silex diluting below 51%, enrichment pricing sustained above roughly US\$120/SWU. 100x = US\$87bn is larger than the plausible enterprise value of the entire Western enrichment industry combined.

What kills it. The capital call - a GLE final investment decision forces Silex to fund 51% of a multi-billion-dollar plant, so it either issues heavy equity at a depressed price or dilutes below 51% and loses control; this is the most probable value-destroying path and it is triggered by success, not failure. Second, NRC licensing can slip or fail, and nobody outside the classified programme can independently verify that separation factors hold at full cascade scale. Third, Cameco holds half the money and the commercial channel and could walk, while a peace settlement restoring Russian enriched product would collapse SWU pricing.

What falsifies it. An NRC licence denial or material delay, or a GLE final investment decision structured so that Silex is diluted below 51%. Either converts the owner of a Western enrichment utility back into a royalty holder.

Confidence. 8/10. A near-breakeven, debt-free company with US\$128m of investments, 51% of the only third-generation enrichment technology near licensing, a state feedstock monopoly and an investment-grade JV partner - bought 58% below its high.

8. Nyxoah SA (NYXH)

NASDAQ (also Euronext Brussels, NYXH.BR) | Healthcare / Medical Devices | Belgium | quoted in USD | Moonshot sleeve rank 8 of 40, GBP 1,000, 869 shares

Price (local)	1.550 USD	Price (GBP)	1.150
52w high / low (local)	6.875 / 1.260	% below 52w high	-77.5%
Market cap (GBP)	GBP 117.2m	Market cap (USD m)	158
Enterprise value (GBP)	GBP 89.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-1.90	Price / book	1.67
EV / EBITDA	-1.45	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 60.0m	Cash	USD 97.8m
Net debt	USD -37.8m	Debt / equity (book)	0.91
Interest cover (EBITDA / interest)	-44.77	Interest cover (EBIT proxy)	-47.20
Capex, latest FY	USD -3.8m	Capex / revenue	17.4%
Free cash flow	USD -46.1m	Revenue (latest FY)	USD 21.7m
Operating margin	-268.2%	Revenue growth	472.8%
Beta (Yahoo)	0.96	Beta (own, 5y daily GBP)	0.58
Upside beta (beta+)	-0.03	Downside beta (beta-)	0.55
CAPM cost of equity Ke	6.2%	WACC	5.1%
Cost of debt Kd	3.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-66.3%	Return 1y (GBP)	-71.9%
Volatility 5y	74.9%	Max drawdown 5y	-95.7%
Skew / excess kurtosis	-0.23 / 17.51	Sharpe / Sortino 5y	-0.44 / -0.63
Judge score (asymmetry / backing / timing)	7.01 (7.5 / 6 / 7.3)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 457k	GBP 1,000 as % of a day	0.29%
10x end-state market cap	USD 1.58bn	100x arithmetically possible?	Theoretically, but improbable after dilution

Source: data/fundamentals_table.csv row NYXH; data/stats_daily.csv row NYXH; universe/unicorn_final.json; universe/dossiers/NYXH.md

Business. Nyxoah SA, a medical technology company, focuses on the development and commercialization of solutions to treat obstructive sleep apnea. Its lead solution is the Genio system, a hypoglossal neurostimulations therapy for obstructive sleep apnea. The company was incorporated in 2009 and is headquartered in Mont-Saint-Guibert, Belgium.

Thesis. Nyxoah makes Genio, a bilateral hypoglossal nerve stimulator for obstructive sleep apnoea in CPAP-intolerant patients. Unlike the incumbent implant it has no battery and no lead - a chip under the chin powered externally by a nightly disposable patch - so there is no replacement surgery every 10-11 years, and it is MRI-compatible. Bilateral stimulation lets Genio treat complete concentric collapse, a population for which the incumbent device is contraindicated. This is not a market that must be created: Inspire Medical proved physicians implant these and payers reimburse them, so Nyxoah only has to take share.

Key backers. European Investment Bank - \$15m of non-dilutive debt alongside the 2026 package, carrying genuine third-party technical and commercial due diligence; institutional investors in the June 2026 \$110m raise that funded the US commercial build-out; 180 high-volume US implanting centres, doubled during 2026, which in this business are the asset; BREATHE post-approval study investigators, first patient implanted 2026-09-07. Insider ownership 33.2%.

Catalysts, 12-24 months. Hitting the EUR 36-40m 2026 guidance, the most testable near-term claim in this group; account count moving from 180 toward 400+; BREATHE post-approval data supporting broader payer coverage; a CMS or large commercial payer decision for the complete-concentric-collapse population; and the next financing, whose terms determine how much upside existing holders keep.

Why 10x. 10x = \$1.58bn. Nyxoah needs roughly \$250-350m of revenue - about ten times current - at 60-70% gross margins valued at 5-6x sales, implying 800-1,200 active implanting centres and 10-15% share of the US HGNS market. Inspire's own trajectory shows that revenue level is attainable in this category; the question is purely whether Nyxoah funds the commercial build without destroying the equity. 100x = \$15.8bn is roughly the whole OSA neurostimulation category at maturity - possible in theory, improbable given the dilution its burn makes near-certain.

What kills it. The financing rather than the science - a ~\$110m annual loss against ~\$45m of net cash (\$97.8m cash and investments less \$52.6m debt) means repeated raises at a \$158m market cap, and the share count is already up 172% in 18 months, so holders can be entirely right about Genio and still be diluted out of the return. Second, Inspire has a decade of physician relationships, a built salesforce and the balance sheet to compete on price while Nyxoah is capital-constrained. Third, any adverse safety finding in the BREATHE post-approval study would be fatal to a single-product company with no cash buffer.

What falsifies it. A 2027 financing struck at a deep discount with warrant coverage, or the EUR 36-40m guidance being missed. The first destroys the return arithmetic; the second destroys the commercial thesis behind it.

Confidence. 7/10. The clinical and commercial evidence is the strongest in this group - approval, met endpoints, 339% growth, doubled accounts, high insider ownership. The confidence is capped not by the product but by the certainty of dilution.

9. Aeluma, Inc. (ALMU)

NasdaqCM | Technology / Semiconductors | United States | quoted in USD | Moonshot sleeve rank 9 of 40, GBP 1,000, 100 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 14.0%. The headline below is shares x price.

Price (local)	13.430 USD	Price (GBP)	9.968
52w high / low (local)	31.790 / 10.240	% below 52w high	-57.8%
Market cap (GBP)	GBP 182.5m	Market cap (USD m)	246
Enterprise value (GBP)	GBP 155.2m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-51.22	Price / book	7.06
EV / EBITDA	-32.94	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 1.1m	Cash	USD 37.8m
Net debt	USD -36.7m	Debt / equity (book)	0.05
Interest cover (EBITDA / interest)	-8.88	Interest cover (EBIT proxy)	-3.23
Capex, latest FY	USD -161.0k	Capex / revenue	3.1%
Free cash flow	USD -443.4k	Revenue (latest FY)	USD 5.2m
Operating margin	-173.9%	Revenue growth	-2.6%
Beta (Yahoo)	-0.12	Beta (own, 5y daily GBP)	0.73
Upside beta (beta+)	-0.42	Downside beta (beta-)	0.87
CAPM cost of equity Ke	6.8%	WACC	6.8%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-21.8%	Return 1y (GBP)	-9.2%
Volatility 5y	121.7%	Max drawdown 5y	-60.5%
Skew / excess kurtosis	1.47 / 13.17	Sharpe / Sortino 5y	0.84 / 1.42
Judge score (asymmetry / backing / timing)	6.88 (7 / 6.6 / 7)	Confidence (dossier)	5 / 10
Avg daily traded value	USD 8,973k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 2.46bn	100x arithmetically possible?	Conceivable on a 15-year view only

Source: data/fundamentals_table.csv row ALMU; data/stats_daily.csv row ALMU; universe/unicorn_final.json; universe/dossiers/ALMU.md

Business. Aeluma, Inc. develops optoelectronic and electronic devices for sensing, communication, and computing applications in the United States. The company produces semiconductor materials and chips using compound semiconductors on diameter substrates that are used to manufacture mass market microelectronics.

Thesis. Aeluma grows compound-semiconductor (III-V) material directly on large-diameter silicon wafers and builds photodetectors, lasers and infrared sensors from it, targeting AI/datacentre optical interconnect, defence and aerospace sensing, mobile, automotive lidar, AR/VR and quantum. Co-packaged optics is the declared 2026-28 hyperscaler roadmap and the binding constraint is that every optical engine today imports its laser and detector as separate III-V die - Aeluma attacks exactly that bottleneck, and US industrial policy is now funding it directly. It sells development contracts and NRE today, not volume product: FY2026 revenue \$4.5m, Q4 FY2026 \$0.6m vs \$1.3m prior year, FY2026 net loss \$9.2m, gross margin 37.7%, cash \$56.0m at 30 June 2026 (>3 years of runway). All seven fact-checked claims confirmed verbatim.

Key backers. US Department of Commerce CHIPS R&D Office - letter of intent for up to \$30m for 'development and commercialization of photonics for AI and Advanced Computing' (GlobeNewswire, 16 Sep 2026); an LOI, not a funded contract. Tower Semiconductor - production-foundry flow agreement for qualification and scaling (13 May 2026 / 16 Sep 2026). Sumitomo Chemical Advanced Technologies - wafer production and fabrication support. NASA and the US Navy - among six new FY2026 contracts totalling \$5.3m.

Catalysts, 12-24 months. Conversion of the \$30m CHIPS LOI into a definitive, milestone-funded award. Management-flagged 'potential multi-million-dollar NRE negotiations in AI datacom' for FY2027 - the first such win is the thesis-changing event. Tower qualification progress. Any shift in mix from government R&D to commercial product.

Why 10x. 10x on the 16 September close is ~\$2.1bn. Aeluma must convert CHIPS money and the Tower production-foundry flow into a qualified 200/300mm III-V-on-Si process with a named datacom or defence customer shipping volume, taking revenue from \$4.5m to \$100m+. At \$2.1bn it would trade as a scarce US-domiciled photonics materials platform - below what compound-semiconductor substrate peers command, so the multiple is not the stretch; the revenue is. The \$56m cash pile buys the three-plus years that most moonshots do not have. 100x is \$21bn, roughly Coherent's position as default laser supplier: conceivable on a 15-year view, not underwritable now.

What kills it. Revenue is going backwards - FY2026 fell to \$4.5m and Q4 halved to \$0.6m; the company is being paid to do research and even that is lumpy. The physics may not scale: defect density in III-V grown on large silicon wafers is the standing objection that has defeated far better-resourced organisations. The CHIPS letter of intent is politically contingent and explicitly non-binding, and at ~\$211m on \$4.5m of revenue success is already embedded in the price. Falsifying fact: FY2027 passing without a commercial (non-government) design win in datacom optics.

What falsifies it. FY2027 passing without a commercial (non-government) design win or NRE contract in datacom optics. That would confirm this is a grant-funded research house, not a supplier.

Confidence. 5/10. The balance sheet (\$56m cash, >3 years) buys time that most moonshots do not have, and the CHIPS/Tower/Sumitomo triangle is real external validation. But the Q4 revenue collapse is a material deterioration against the record I inherited, which assumed ~\$1.2m/quarter and \$37.8m cash.

10. Weebit Nano Limited (WBT.AX)

ASX | Technology / Semiconductors | Australia | quoted in AUD | Moonshot sleeve rank 10 of 40, GBP 1,000, 619 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 6.9%. The headline below is shares x price.

Price (local)	3.050 AUD	Price (GBP)	1.613
52w high / low (local)	8.890 / 2.730	% below 52w high	-65.7%
Market cap (GBP)	GBP 388.4m	Market cap (USD m)	523
Enterprise value (GBP)	GBP 299.5m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-19.02	Price / book	8.78
EV / EBITDA	-9.94	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 373.0k	Cash	AUD 168.3m
Net debt	AUD -168.0m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-288.01	Interest cover (EBIT proxy)	-274.20
Capex, latest FY	AUD -1.1m	Capex / revenue	7.0%
Free cash flow	AUD 200.8k	Revenue (latest FY)	AUD 15.3m
Operating margin	-280.7%	Revenue growth	158.0%
Beta (Yahoo)	0.96	Beta (own, 5y daily GBP)	0.94
Upside beta (beta+)	1.04	Downside beta (beta-)	1.01
CAPM cost of equity Ke	7.7%	WACC	7.7%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-35.9%	Return 1y (GBP)	31.9%
Volatility 5y	78.4%	Max drawdown 5y	-86.8%
Skew / excess kurtosis	0.48 / 2.36	Sharpe / Sortino 5y	0.34 / 0.52
Judge score (asymmetry / backing / timing)	6.78 (6 / 7.4 / 7.8)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 6,197k	GBP 1,000 as % of a day	0.02%
10x end-state market cap	USD 5.23bn	100x arithmetically possible?	No - would approach Arm's valuation

Source: data/fundamentals_table.csv row WBT.AX; data/stats_daily.csv row WBT.AX; universe/unicorn_final.json; universe/dossiers/WBT.AX.md

Business. Weebit Nano Limited engages in the development of non-volatile memory using a resistive random access memory technology based on fab-friendly materials in South Korea and the United States. It offers its products to aerospace and defense, IoT devices, secure devices, edge artificial intelligence, automotive, data center, neuromorphic computing, data logging, and medical devices industries. The company was incorporated in 2010 and is headquartered in Melbourne, Australia.

Thesis. Weebit Nano licenses ReRAM, a non-volatile embedded memory intended to replace embedded flash below 28nm, where flash physically stops scaling. This is a pure semiconductor-IP model: licence fees and NRE now, production royalties later, at IP margins. FY2026 revenue A\$15.3m (+158% y/y), gross margin ~100%, cash A\$168.3m at 30 Jun 2026 against an A\$57m annual burn - funded to the royalty inflection. The validation is unusually hard for a pre-royalty IP company: Texas Instruments licensed the technology in December 2025 and is described as Weebit's largest customer to date; onsemi 65nm BCD demo chips are performing as expected; DB HiTek has JEDEC-qualified it in a 130nm BCD process across three wafer lots.

Key backers. Texas Instruments - ReRAM licensed December 2025 covering IP licensing, technology transfer, design and qualification in TI's advanced process nodes, described as largest customer to date (Weebit, 30 Jan 2026). onsemi - demo chips on its 65nm BCD platform confirmed performing as expected, qualification expected to conclude later in 2026. DB HiTek - Korean merchant foundry, JEDEC-qualified in 130nm BCD across three wafer

lots. Note: the automotive-grade high-temperature claim belongs to the DB HiTek qualification, not to TI.

Catalysts, 12-24 months. Completion of onsemi's 65nm BCD qualification (guided to conclude during 2026). A third tier-1 foundry/IDM agreement. First production royalties, expected 2027 - the moment the model changes from lumpy NRE to recurring revenue. Customer product launches from taped-out designs. Any placement of ReRAM into an analog in-memory-compute product.

Why 10x. 10x = ~A\$7.3bn (~US\$4.9bn). ReRAM must become the default embedded NVM below 28nm across three or more major foundries, with royalties compounding toward a few hundred million dollars a year at IP margins. That is a normal valuation for a successful semiconductor IP company - and it is among the most mechanically plausible 10x paths in the sleeve because the royalty arithmetic is already known; only the adoption rate is uncertain. 100x (~US\$49bn) would put a single memory-IP family near Arm's valuation - no.

What kills it. Royalties keep moving right - now a 2027 story, and they have slipped before; until they arrive, revenue is NRE that stops when the work stops. The burn is A\$57m a year against A\$15m of revenue: the A\$168m cash pile came from shareholders, not operations, and the register has been diluted heavily to build it. MRAM is the incumbent alternative (Everspin, GlobalFoundries, Samsung) with more design wins in some segments, and two tier-1 licences are already in the price. Insider ownership of 7% is the one alignment weakness.

What falsifies it. The onsemi 65nm qualification failing, or 2027 passing with no production royalty revenue recognised. Either would mean the licences are research budgets, not product commitments. Insider ownership of 7% is the one alignment weakness.

Confidence. 7/10. Two of the world's largest embedded IDMs and a qualified merchant foundry is about as much third-party validation as a pre-royalty IP company can produce, and the balance sheet funds it to the royalty inflection.

11. Vulcan Energy Resources Limited (VUL.AX)

ASX | Basic Materials / Other Industrial Metals & Mining | Australia | quoted in AUD | Moonshot sleeve rank 11 of 40, GBP 1,000, 843 shares

Price (local)	2.240 AUD	Price (GBP)	1.185
52w high / low (local)	7.520 / 2.175	% below 52w high	-70.2%
Market cap (GBP)	GBP 567.1m	Market cap (USD m)	764
Enterprise value (GBP)	GBP 446.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-17.85	Price / book	0.79
EV / EBITDA	-21.43	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 46.0m	Cash	AUD 273.9m
Net debt	AUD -227.9m	Debt / equity (book)	0.03
Interest cover (EBITDA / interest)	-28.00	Interest cover (EBIT proxy)	-47.10
Capex, latest FY	AUD -50.2m	Capex / revenue	914.6%
Free cash flow	AUD -246.1m	Revenue (latest FY)	AUD 5.5m
Operating margin	-1408.3%	Revenue growth	-45.2%
Beta (Yahoo)	1.60	Beta (own, 5y daily GBP)	1.17
Upside beta (beta+)	1.70	Downside beta (beta-)	1.55
CAPM cost of equity Ke	8.6%	WACC	8.4%
Cost of debt Kd	3.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-46.0%	Return 1y (GBP)	-35.1%
Volatility 5y	69.9%	Max drawdown 5y	-87.4%
Skew / excess kurtosis	0.51 / 2.75	Sharpe / Sortino 5y	-0.24 / -0.36
Judge score (asymmetry / backing / timing)	6.68 (5.8 / 8.4 / 6.3)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 4,072k	GBP 1,000 as % of a day	0.03%
10x end-state market cap	USD 7.46bn	100x arithmetically possible?	No - exceeds Albemarle and SQM at their peaks combined

Source: data/fundamentals_table.csv row VUL.AX; data/stats_daily.csv row VUL.AX; universe/unicorn_final.json; universe/dossiers/VUL.AX.md

Business. Vulcan Energy Resources Limited engages in the geothermal energy, and lithium exploration and production activities in Europe, Germany, and Australia. The company is involved in exploration and development of battery materials and geothermal lithium; and provision of engineering services. It holds interests in the Phase One Lionheart Project. The company was formerly known as Koppar Resources Limited and changed its name to Vulcan Energy Resources Limited in September 2019.

Thesis. Vulcan Energy is building the world's first integrated geothermal-plus-direct-lithium-extraction operation in Germany's Upper Rhine Valley Brine Field, producing lithium hydroxide and district heat from the same brine. The single most important fact is the financing: a EUR 2.2bn Lionheart Phase One package closed December 2025 with EIB committing EUR 250m inside a syndicate of a dozen-plus banks and export credit agencies - a dozen sophisticated lenders have done the diligence for you, which is rare at a ~A\$1bn equity value. Product already exists (geothermal heat has been flowing to Landau since April 2025); lithium production is targeted for 2028. Correction applied: the BASF permits of Feb 2025 covered 2D survey measurements only, with 3D flagged as a possible follow-on, not as already permitted.

Key backers. European Investment Bank (EUR 250m within the EUR 2.2bn Lionheart Phase One close, Dec 2025); bank and ECA syndicate of ABN AMRO, UniCredit, Natixis, ING, BNP Paribas, Kommunalkredit, OCBC plus Export Finance Australia, Export Development Canada, EIFO, BPI France and SACE; BASF (joint Vorderpfalz geothermal-heat project - 2D seismic permits only, Feb 2025, corrected); Infraserb Hoechst (77,517 m2 brownfield lease in the Frankfurt Hoechst chemical park); EnergieSuedwest AG (Landau district-heat offtake, flowing since April 2025); German federal government support. All v-er.eu/news, re-fetched 2026-09-17.

Catalysts, 12-24 months. Honest timeline - lithium production is 2028, so the next two years are construction milestones, not product. Drawdowns under the EUR 2.2bn facility; G-LEP Landau commissioning; central plant construction at Hoechst; V-LiON offtake qualification with battery makers converting into priced contracts; expansion of geothermal heat contracts; a BASF Ludwigshafen decision (2D then 3D then FID - a multi-step sequence); lithium price recovery.

Why 10x. 10x = ~US\$7.5bn (~A\$10.7bn). Requires Lionheart Phase One delivered on budget at 24,000 tpa, Phase Two sanctioned, and lithium hydroxide back above ~US\$20,000/t with a European green premium. Own arithmetic: at 24,000 tpa and a US\$8,000/t margin Phase One alone is ~US\$190m EBITDA; a 48,000 tpa business plus a 30-year geothermal annuity at 15-18x supports the target. The structural case is that Europe has no domestic lithium chemical supply and Vulcan's product is net-zero by construction. 100x (US\$75bn) exceeds Albemarle and SQM at their peaks combined - not possible.

What kills it. First-of-a-kind commissioning risk: no integrated geothermal-DLE plant of this scale exists anywhere, and Upper Rhine geothermal carries a history of induced seismicity - Basel and Landau projects were stopped by it - so a felt earthquake is a political event, not just a technical one. Lithium prices staying depressed for years: the plant gets built and the equity still earns nothing. And the EUR 2.2bn is DEBT - at a ~A\$1bn equity value the project is heavily geared, a cost overrun impairs equity first, and FCF of -A\$246m shows cash already going out of the door.

What falsifies it. A material Lionheart capex overrun or schedule slip beyond 2028 announced in a quarterly report - with this gearing, that is the difference between a 10x and a recapitalisation.

Confidence. 7/10. The financing close is the rare case where a dozen sophisticated lenders have done the diligence for you, and the product already exists. The discount is real because the commodity is hated and first production is still two years away.

12. Intellicheck, Inc. (IDN)

NASDAQ | Technology / Software - Application | United States | quoted in USD | Moonshot sleeve rank 12 of 40, GBP 1,000, 566 shares

Price (local)	2.380 USD	Price (GBP)	1.766
52w high / low (local)	9.080 / 2.350	% below 52w high	-73.8%
Market cap (GBP)	GBP 35.8m	Market cap (USD m)	48
Enterprise value (GBP)	GBP 27.0m	Trailing P/E	14.94
Forward P/E	31.87	Price / book	2.15
EV / EBITDA	10.09	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 0.00	Cash	USD 11.8m
Net debt	USD -11.8m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	721.20	Interest cover (EBIT proxy)	267.20
Capex, latest FY	USD -265.0k	Capex / revenue	1.1%
Free cash flow	USD 1.4m	Revenue (latest FY)	USD 24.1m
Operating margin	9.6%	Revenue growth	16.0%
Beta (Yahoo)	0.78	Beta (own, 5y daily GBP)	0.71
Upside beta (beta+)	0.88	Downside beta (beta-)	0.60
CAPM cost of equity Ke	6.7%	WACC	6.7%
Cost of debt Kd	5.8% (fallback: rf + 2% (no usable interest expense / debt))	Tax rate	4.4%
Return YTD (GBP)	-64.4%	Return 1y (GBP)	-56.4%
Volatility 5y	82.8%	Max drawdown 5y	-83.3%
Skew / excess kurtosis	2.63 / 44.03	Sharpe / Sortino 5y	0.04 / 0.06
Judge score (asymmetry / backing / timing)	6.61 (5.5 / 8.2 / 7)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 1,276k	GBP 1,000 as % of a day	0.11%
10x end-state market cap	USD 0.48bn	100x arithmetically possible?	No - would be acquired long before \$4.8bn

Source: data/fundamentals_table.csv row IDN; data/stats_daily.csv row IDN; universe/unicorn_final.json; universe/dossiers/IDN.md

Business. Intellicheck, Inc., a technology company, provides on-demand digital identity validation solutions for KYC, fraud, and age verification needs in North America. It validates both digital and physical identities for financial services, fintech companies, BNPL providers, e-commerce and retail commerce businesses, and law enforcement and government agencies.

Thesis. Intellicheck does authoritative-source identity validation - it reads the encoded data on a US driver's licence and checks it against exclusive state DMV data relationships rather than inspecting an image. That contractual data access, not the software, is the moat, and it cannot be reproduced by a better-funded competitor buying GPUs. TTM revenue \$24.11m (+16.7%), gross margin 91%, net income \$3.14m / EPS \$0.15, ~500 revenue-generating enterprise customers across banking, fintech, retail and government, no debt, positive cash flow - on a \$48.0m market cap after a 57% de-rating.

Key backers. US state DMVs - exclusive authoritative-source data relationships, contractual rather than technical, and the actual moat. Channel partner securing 'one of the world's largest investment management companies' (Business Wire, 19 Aug 2026). ~500 enterprise customers. Russell 2000/3000 inclusion effective 26 June 2026 (Business Wire, corrected date) for passive-ownership support. Institutions 53.5%; insiders only 4.85%, the honest weakness.

Catalysts, 12-24 months. Quarterly revenue holding mid-to-high-teens growth with the 91% margin intact. New state age-verification statutes taking effect and driving transaction counts. Further channel wins in wealth management and insurance. Any large bank displacing an incumbent vendor. Standing takeover interest from Experian, LexisNexis, Socure or Mastercard.

Why 10x. 10x = \$483m. That needs revenue of roughly \$70-90m - three to four times current - at the same 91% margin, with operating leverage taking net margin to 20-25%; on 6-8x sales or 25-30x earnings that clears \$480m. This is the most arithmetically ordinary 10x in the sleeve: it requires no new scientific category, only a small company with an unreproducible data asset compounding at 15-25% for six or seven years while state age-verification and KYC statutes keep expanding the addressable transaction count. With only 20.3m shares out, a single large contract moves EPS visibly. 100x (\$4.8bn) would need \$600m-1bn of revenue and would be acquired long before.

What kills it. Customer concentration - a handful of large financial-services accounts drive revenue; losing one shows up immediately in a quarter, which is what a 57% drawdown on a profitable company usually means the market is already pricing. It may simply be a niche vendor: 16-17% growth is respectable, not explosive, and Socure, Persona, LexisNexis and Experian are better capitalised and can bundle identity into broader contracts. Acquisition at a modest premium caps the upside - with a \$48m cap, no debt and clean cash flow, a 40% premium take-out is plausible and thesis-ending.

What falsifies it. Two consecutive quarters of revenue decline, or disclosure that a top-five customer has not renewed. Either would confirm concentration, not category leadership.

Confidence. 7/10. This is the odd one out in a moonshot sleeve - it is already profitable, cash-generative and debt-free, so the downside is not zero, which caps neither the logic nor the asymmetry given the 57% de-rating. The 4.85% insider stake is the honest weakness.

13. Gelion plc (GELN.L)

LSE (AIM) | Industrials / Electrical Equipment & Parts | United Kingdom | quoted in GBp | Moonshot sleeve rank 13 of 40, GBP 1,000, 6,896 shares

Price (local)	0.145 GBP	Price (GBP)	0.145
52w high / low (local)	0.280 / 0.120	% below 52w high	-48.2%
Market cap (GBP)	GBP 33.3m	Market cap (USD m)	45
Enterprise value (GBP)	GBP 22.8m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-5.37	Price / book	1.96
EV / EBITDA	-5.50	Dividend yield	n/a (not reported by Yahoo)
Total debt	GBP 27.0k	Cash	GBP 10.5m
Net debt	GBP -10.5m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-153.41	Interest cover (EBIT proxy)	-221.33
Capex, latest FY	GBP -483.0k	Capex / revenue	17.0%
Free cash flow	GBP -3.0m	Revenue (latest FY)	GBP 2.8m
Operating margin	-576.5%	Revenue growth	32.8%
Beta (Yahoo)	0.47	Beta (own, 5y daily GBP)	0.12
Upside beta (beta+)	-0.45	Downside beta (beta-)	0.44
CAPM cost of equity Ke	4.2%	WACC	4.3%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-20.5%	Return 1y (GBP)	-28.9%
Volatility 5y	71.9%	Max drawdown 5y	-96.4%
Skew / excess kurtosis	4.98 / 67.39	Sharpe / Sortino 5y	-0.41 / -0.73
Judge score (asymmetry / backing / timing)	6.59 (6.5 / 6.8 / 6.5)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 35k	GBP 1,000 as % of a day	3.83%
10x end-state market cap	USD 0.44bn	100x arithmetically possible?	No - cycle life rules out mass-market displacement

Source: data/fundamentals_table.csv row GELN.L; data/stats_daily.csv row GELN.L; universe/unicorn_final.json; universe/dossiers/GELN.L.md

Business. Gelion plc, together with its subsidiary, engages in the research and development, manufacture, and sale of battery systems in the United Kingdom and internationally. The company offers storage batteries, which include lithium-sulfur, and room temperature sodium sulfur, lithium-Ion batteries, recycling technology, bespoke battery energy storage systems, and zinc hybrid cell batteries.

Thesis. Gelion is a ~GBP33m AIM lithium-sulphur battery company at 14.5p, licensing a sulphur cathode rather than building gigafactories, so it does not need heavy capital. Li-S offers a step-change in gravimetric energy density with no nickel and no cobalt, which is why the first real markets are drones and defence, where weight dominates cost. Revenue GBP2.84m (+32.8%), cash GBP10.52m, no debt, insiders 39.55%. The partner list is far better than a GBP33m AIM company normally assembles. All RNS dates confirmed verbatim against Investegate.

Key backers. TDK Corporation - 'Full Collaboration Agreement with TDK', 21 Oct 2025, a global passive-component and battery major engaging directly with the sulphur cathode. Unnamed tier-one battery manufacturer - 'Collaboration with Tier-One Battery Partner', 16 Oct 2025. Nissan - 'Nissan Project, Grant Award, Investor Presentation', 2 Jun 2026: a named OEM plus third-party grant money. Prof. Rachid Yazami - Technology Advisor, 15 Jan 2026. Prof. Thomas Maschmeyer / University of Sydney - founder. Also an Exclusive Licence Agreement (16 Jun 2026) and a funded development and scale-up agreement (7 Aug 2026). All Investegate GELN RNS list, accessed 2026-09-17.

Catalysts, 12-24 months. Naming the tier-one partner and the exclusive licensee - both anonymous, and this is the single thing a sceptic should demand. Conversion of the 7 Aug 2026 funded scale-up agreement into a royalty-bearing licence. The outcome of the Nissan project. Drone customer qualification following the 26 Aug

2026 mission testing. First royalty revenue.

Why 10x. 10x = ~GBP333m (US\$444m). It needs one named, royalty-bearing licence with a global cell maker plus a defence or drone design win, taking licence and royalty revenue to roughly GBP15-25m annualised. On the 15-20x revenue multiple licensing models attract once the counterparty is named, GBP333m is reachable from GBP33m - the licensing model means the capital requirement to get there is modest, which is what makes the arithmetic work. 100x is US\$4.4bn, requiring Li-S to displace lithium-ion in a mainstream mass market inside the holding period - not credible; cycle life is the reason.

What kills it. The Oxis Energy base rate - a UK Li-S company with real IP and real partners went into administration in 2021 doing exactly this; that is the honest prior. Cycle life is unsolved physics: the polysulphide shuttle has defeated forty years of attempts, and if the cathode cannot exceed a few hundred cycles it is a niche drone product. Every major commercial relationship is unnamed - the tier-one partner, the exclusive licensee and the funded scale-up counterparty are all anonymous - and with ~GBP26k of daily turnover you can neither verify the thesis nor exit it.

What falsifies it. Independent third-party cycle-life data below ~500 cycles at commercially relevant sulphur loading, or the exclusive licence lapsing without the licensee ever being named.

Confidence. 6/10. The partner list (TDK, Nissan, Yazami, a funded scale-up agreement) is far better than a GBP 33m AIM company normally assembles, insider ownership is 39.6%, there is no debt, and the licensing model means it does not need gigafactory capital. Against that, the science is the hardest in this batch and the liquidity is the worst.

14. Editas Medicine, Inc. (EDIT)

NASDAQ | Healthcare / Biotechnology | United States | quoted in USD | Moonshot sleeve rank 14 of 40, GBP 1,000, 497 shares

Price (local)	2.710 USD	Price (GBP)	2.011
52w high / low (local)	4.537 / 1.660	% below 52w high	-40.3%
Market cap (GBP)	GBP 308.9m	Market cap (USD m)	416
Enterprise value (GBP)	GBP 163.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-3.59	Price / book	3.94
EV / EBITDA	-2.72	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 15.5m	Cash	USD 211.6m
Net debt	USD -196.1m	Debt / equity (book)	0.42
Interest cover (EBITDA / interest)	-13.13	Interest cover (EBIT proxy)	-24.94
Capex, latest FY	USD -607.0k	Capex / revenue	1.3%
Free cash flow	USD -44.3m	Revenue (latest FY)	USD 47.0m
Operating margin	-167.3%	Revenue growth	232.3%
Beta (Yahoo)	2.09	Beta (own, 5y daily GBP)	1.23
Upside beta (beta+)	1.20	Downside beta (beta-)	0.97
CAPM cost of equity Ke	8.9%	WACC	9.4%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	32.1%	Return 1y (GBP)	0.5%
Volatility 5y	93.7%	Max drawdown 5y	-98.3%
Skew / excess kurtosis	2.55 / 34.57	Sharpe / Sortino 5y	-0.26 / -0.41
Judge score (asymmetry / backing / timing)	6.59 (6.5 / 5.8 / 8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 6,975k	GBP 1,000 as % of a day	0.02%
10x end-state market cap	USD 4.16bn	100x arithmetically possible?	No - a takeout at \$3-8bn is the realistic ceiling

Source: data/fundamentals_table.csv row EDIT; data/stats_daily.csv row EDIT; universe/unicorn_final.json; universe/dossiers/EDIT.md

Business. Editas Medicine, Inc., a clinical stage genome editing company, focuses on developing transformative genomic medicines to treat a range of serious diseases. The company develops a proprietary gene editing platform based on CRISPR technology. Its lead program is EDIT-401, a one-time therapy designed to reduce LDL cholesterol through the upregulation of the LDL receptor to treat hyperlipidemia.

Thesis. Editas Medicine has pivoted to EDIT-401, an in vivo LNP-delivered gene editor that up-regulates LDL receptor expression in the liver, aiming to cut LDL-C, Lp(a) and ApoB simultaneously with a single dose. That is a different and more ambitious mechanism than the PCSK9-knockdown editors, and it is aimed at the largest chronic-disease market in medicine. The company holds exclusive Broad Institute / Harvard licences to foundational Cas9 and Cas12a patents, giving freedom to operate and leverage over other editors. Best-funded name in its batch, with runway extended into H2 2028 and 79.2% institutional ownership; a dated binary readout sits inside the holding period.

Key backers. Broad Institute / Harvard - exclusive licences to foundational Cas9 and Cas12a patents (GlobeNewswire, 5 Aug 2026). Bristol Myers Squibb - historic collaboration, but its opt-in expiry is what drove the H1 2026 revenue recognition, so this is a relationship ending. Institutional register 79.2%, with specialist funds taking up the financing that extended runway into H2 2028. Management insider ownership only 0.18%.

Catalysts, 12-24 months. Australian Clinical Trial Notification, which management said it planned to submit in August 2026, followed by first-in-human dosing in the HeFH Phase 1/2. A data update in Q1 2027 with topline results anticipated during 2027 - this is the binary: the thesis is proved or dies inside the window. A large-pharma cardiovascular partnership, since Editas cannot fund prevention trials alone.

Why 10x. 10x = \$4.2bn. EDIT-401 must reproduce even half the primate lipid effect in humans with clean safety, and Editas must sign a large-pharma partner. The comparable is priced: Lilly paid roughly \$1.3bn upfront plus CVRs for Verve on PCSK9 editing data alone, so a one-shot triple-lipid editor with human data should be worth several multiples of that; \$4.2bn is roughly where Intellia peaked in 2021 on comparable evidence. 100x is \$42bn - cardiovascular outcomes trials run to tens of thousands of patients and are far beyond this company, so the equity will almost certainly be crystallised by a takeout in the \$3-8bn range.

What kills it. Translation failure - primate lipid effects have translated poorly before, and LDLR up-regulation may prove less durable in humans than knockdown; a Q1 2027 update showing 30% rather than 90% reduction ends it. Safety: liver off-target editing or an LNP immune response, in a PREVENTION indication where tolerance for harm is near zero. Zero insider ownership and a record of pivots - management holds 0.18%, and Editas has pivoted from LCA10 to sickle cell to lipids with each pivot destroying value. Note the H1 2026 revenue line is a one-off from the Bristol Myers Squibb opt-in EXPIRING - a relationship ending, not deepening.

What falsifies it. The Q1 2027 human data update failing to show durable, clinically meaningful reductions in all three of LDL-C, Lp(a) and ApoB. Everything else in the file is secondary to that one readout.

Confidence. 6/10. Best-funded name in this batch, with a genuinely differentiated mechanism, a dated binary readout inside the holding period, and a validated M&A comparable. Against that: the revenue line is a one-off from a partner walking away, the dilution has been heavy, and management owns essentially none of it.

15. Smart Eye AB (publ) (SEYE.ST)

Nasdaq Stockholm | Technology / Software - Application | Sweden | quoted in SEK | Moonshot sleeve rank 15 of 40, GBP 1,000, 248 shares

Price (local)	53.050 SEK	Price (GBP)	4.027
52w high / low (local)	101.700 / 48.880	% below 52w high	-47.8%
Market cap (GBP)	GBP 157.9m	Market cap (USD m)	213
Enterprise value (GBP)	GBP 180.5m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-35.88	Price / book	1.52
EV / EBITDA	142.73	Dividend yield	n/a (not reported by Yahoo)
Total debt	SEK 383.9m	Cash	SEK 85.9m
Net debt	SEK 298.0m	Debt / equity (book)	0.02
Interest cover (EBITDA / interest)	0.53	Interest cover (EBIT proxy)	-5.22
Capex, latest FY	SEK -117.2m	Capex / revenue	24.0%
Free cash flow	SEK -87.9m	Revenue (latest FY)	SEK 488.8m
Operating margin	-14.6%	Revenue growth	52.9%
Beta (Yahoo)	1.35	Beta (own, 5y daily GBP)	1.22
Upside beta (beta+)	0.93	Downside beta (beta-)	1.31
CAPM cost of equity Ke	8.8%	WACC	8.4%
Cost of debt Kd	8.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-36.6%	Return 1y (GBP)	-37.6%
Volatility 5y	63.0%	Max drawdown 5y	-84.6%
Skew / excess kurtosis	-0.32 / 18.34	Sharpe / Sortino 5y	-0.12 / -0.17
Judge score (asymmetry / backing / timing)	6.54 (6 / 6.6 / 7.8)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 2,718k	GBP 1,000 as % of a day	0.05%
10x end-state market cap	USD 2.13bn	100x arithmetically possible?	No - not defensible for an automotive software supplier

Source: data/fundamentals_table.csv row SEYE.ST; data/stats_daily.csv row SEYE.ST; universe/unicorn_final.json; universe/dossiers/SEYE.ST.md

Business. Smart Eye AB (publ) develops human insight artificial intelligence (AI) technology solutions that understand, support, and predict human behavior in the Nordics countries, the rest of Europe, North America, Asia, and internationally. The company provides driver monitoring systems and interior sensing solutions for automotive market; and AIS, an end-to-end driver monitoring hardware and software system for fleet and aftermarket.

Thesis. Smart Eye supplies driver monitoring systems and interior sensing to automotive OEMs, a category that stopped being optional: EU General Safety Regulation driver-drowsiness and attention-warning requirements became binding during 2026, making DMS a regulatory line item on new type approvals rather than a premium feature. The setup is unusually concrete for a moonshot: 379 cumulative design wins across 24 OEMs, >SEK 7.6bn of estimated remaining lifetime order value (roughly 3.6x the market cap), 20 customers already in production across 175 models, 128% automotive organic growth, EBITDA turned positive, and the stock at a 52-week low.

Key backers. 24 automotive OEMs with 379 cumulative design wins and >SEK 7.6bn estimated remaining lifetime order value, 20 already in production across 175 models (StockTitan, 26 Aug 2026). An existing European premium OEM - follow-on DMS design win announced 27 August 2026, one new model, ~SEK 15m lifetime value, start of production mid-2027. The dossier explicitly corrects the inherited record here, which had described this as a 17 July 2026 Interior Sensing order across three DMS programmes. A leading unnamed global car manufacturer - four-model win, ~SEK 50m, production 2028-29 (Yahoo Finance, 1 Jun 2026).

Catalysts, 12-24 months. Q3 and Q4 2026 results showing whether 128% automotive organic growth holds and whether EBITDA positivity converts into positive free cash flow. Further Interior Sensing upsells onto existing DMS programmes - the margin story. Design wins with the four OEM customers not yet in production. The mid-2027 and 2028-29 production starts entering the backlog conversion curve. Acquisition interest: a supplier with 379 design wins is strategically attractive to a tier-one.

Why 10x. 10x = ~SEK 21bn (US\$2.1bn). It requires the SEK 7.6bn order book to convert plus continued win accumulation, taking revenue to roughly SEK 1.5-2bn at software gross margins with real operating profit - an ordinary valuation for a profitable automotive software supplier with that backlog. The margin kicker is Interior Sensing upsold onto existing DMS programmes: follow-on orders require no change to the software, so incremental wins drop through at near-full margin. 100x (US\$21bn) is not defensible for an automotive software supplier and should not be underwritten.

What kills it. Net debt of ~SEK 300m against SEK 86m of cash with negative free cash flow - the company has grown revenue for years without converting it to cash, so another dilutive raise before breakeven is the live risk, and it would be priced off a 52-week low. Competition that can bundle: Seeing Machines competes directly while Bosch, Continental and Aptiv can give DMS away inside a cockpit domain controller, and Chinese OEMs increasingly use domestic DMS. Design-win value is an estimate of future deliveries, not a contracted minimum - it can be cancelled with a model programme.

What falsifies it. An equity raise on dilutive terms, or automotive organic growth falling back below ~30% in Q3/Q4 2026, which would mean the July 2026 regulation did not produce the volume step-up the whole thesis rests on.

Confidence. 7/10. The regulation is dated and binding, the backlog is 3.6x the market cap, EBITDA has turned positive, and the stock is at a 52-week low - that is an unusually concrete setup. The balance sheet is the reason it is not higher.

16. Quantum-Si Incorporated (QSI)

NASDAQ | Healthcare / Medical Devices | United States | quoted in USD | Moonshot sleeve rank 16 of 40, GBP 1,000, 1,845 shares

Price (local)	0.730 USD	Price (GBP)	0.542
52w high / low (local)	3.100 / 0.690	% below 52w high	-76.5%
Market cap (GBP)	GBP 107.8m	Market cap (USD m)	145
Enterprise value (GBP)	GBP 27.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-1.98	Price / book	0.84
EV / EBITDA	-0.42	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 3.3m	Cash	USD 111.7m
Net debt	USD -108.5m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-11.24	Interest cover (EBIT proxy)	-12.03
Capex, latest FY	USD -2.5m	Capex / revenue	156.8%
Free cash flow	USD -50.1m	Revenue (latest FY)	USD 1.6m
Operating margin	-7450.3%	Revenue growth	-41.8%
Beta (Yahoo)	3.25	Beta (own, 5y daily GBP)	1.77
Upside beta (beta+)	2.40	Downside beta (beta-)	1.23
CAPM cost of equity Ke	11.1%	WACC	11.4%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-33.7%	Return 1y (GBP)	-38.9%
Volatility 5y	141.3%	Max drawdown 5y	-92.6%
Skew / excess kurtosis	8.67 / 150.88	Sharpe / Sortino 5y	0.16 / 0.35
Judge score (asymmetry / backing / timing)	6.49 (7.5 / 5.8 / 5)	Confidence (dossier)	5 / 10
Avg daily traded value	USD 2,732k	GBP 1,000 as % of a day	0.05%
10x end-state market cap	USD 1.45bn	100x arithmetically possible?	Arithmetically conceivable, a tail not a plan

Source: data/fundamentals_table.csv row QSI; data/stats_daily.csv row QSI; universe/unicorn_final.json; universe/dossiers/QSI.md

Business. Quantum-Si incorporated, a life sciences company, develops proprietary, universal, and single-molecule detection platform applied to proteomics to enable next-gen protein sequencing (NGPS).

Thesis. Semiconductor-chip single-molecule protein sequencing from the founder of 454 Life Sciences and Ion Torrent. Platinum ships today; next-gen Proteus is the platform bet. Market cap US\$145-160m at \$0.73 against \$169.9m of cash and marketable securities at 30-Jun-2026 - the equity trades below net cash because the Proteus launch slipped to Q2 2027.

Key backers. Jonathan Rothberg (founder/chairman; 454 Life Sciences sold to Roche, Ion Torrent sold ~\$725m; holds the controlling Class B stock). The balance sheet itself - \$169.9m funds the company through launch without returning to market. NOTABLE GAP: no disclosed strategic corporate investor, the weakest part of the case.

Catalysts, 12-24 months. Proteus commercial launch Q2 2027 (the event). Early-access placements converting to pre-orders with consumable pull-through. Peer-reviewed accuracy data vs mass spec. Evidence the ~\$12m annualised savings from the 20% workforce reduction land without damaging development. FY26 guidance reiterated: revenue ~\$1.0m, adj opex <=\$98.0m, cash usage <=\$93.0m.

Why 10x. 10x = \$1.60bn. Proteus must ship on time in Q2 2027 and convert to instrument-plus-consumable revenue; at \$80-120m of revenue growing 50%+, tools platforms trade at 12-20x sales. Detection on CMOS means the cost curve should follow Moore's law, as Ion Torrent's did, not a mass spectrometer's. A newly disclosed >\$4bn non-human proteomics TAM (agri, pathogen typing, viral surveillance) widens the frame beyond the original clinical pitch. 100x = \$16.0bn (~45% of Illumina) is arithmetically conceivable but a tail, not a plan.

What kills it. Proteus read length, accuracy and throughput may not beat mass spec on real biological questions - the extra design cycle and 20% reduction in force say management is not yet confident in manufacturing readiness. A second slip at \$0.73 invites Nasdaq minimum-bid trouble and a reverse split. Class B super-voting stock (19.94m shares x 20 votes = ~66.7% of votes, own calc from the 10-Q cover, 2026-08-11) leaves minorities no recourse.

What falsifies it. Any further delay to the Q2 2027 Proteus launch, or a launch whose first datasets do not show sequencing accuracy competitive with mass spectrometry. Either ends the thesis regardless of the cash pile.

Confidence. 5/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

17. Caribou Biosciences, Inc. (CRBU)

NASDAQ | Healthcare / Biotechnology | United States | quoted in USD | Moonshot sleeve rank 17 of 40, GBP 1,000, 1,077 shares

Price (local)	1.250 USD	Price (GBP)	0.928
52w high / low (local)	3.535 / 1.250	% below 52w high	-64.6%
Market cap (GBP)	GBP 99.5m	Market cap (USD m)	134
Enterprise value (GBP)	GBP 35.9m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-0.94	Price / book	1.29
EV / EBITDA	-0.46	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 26.7m	Cash	USD 112.3m
Net debt	USD -85.7m	Debt / equity (book)	0.16
Interest cover (EBITDA / interest)	-13,076.12	Interest cover (EBIT proxy)	-18,583.38
Capex, latest FY	USD -1.4m	Capex / revenue	13.5%
Free cash flow	USD -59.6m	Revenue (latest FY)	USD 10.0m
Operating margin	-1690.4%	Revenue growth	-43.8%
Beta (Yahoo)	2.28	Beta (own, 5y daily GBP)	1.01
Upside beta (beta+)	0.52	Downside beta (beta-)	0.78
CAPM cost of equity Ke	8.0%	WACC	6.7%
Cost of debt Kd	0.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-21.4%	Return 1y (GBP)	-34.3%
Volatility 5y	84.5%	Max drawdown 5y	-97.2%
Skew / excess kurtosis	0.56 / 7.07	Sharpe / Sortino 5y	-0.35 / -0.51
Judge score (asymmetry / backing / timing)	6.48 (7 / 5.4 / 6.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 2,699k	GBP 1,000 as % of a day	0.05%
10x end-state market cap	USD 1.34bn	100x arithmetically possible?	No - the equity ceiling is nearer \$4-7bn

Source: data/fundamentals_table.csv row CRBU; data/stats_daily.csv row CRBU; universe/unicorn_final.json; universe/dossiers/CRBU.md

Business. Caribou Biosciences, Inc. operates as a clinical-stage clustered regularly interspaced short palindromic repeats (CRISPR) genome-editing biopharmaceutical company. It offers a genome-editing platform based on novel CRISPR hybrid RNA-DNA genome-editing technology that enables precise genome editing of allogeneic cell therapies.

Thesis. Off-the-shelf allogeneic CAR-T using proprietary chRDNA (hybrid DNA-RNA guide) editing, co-founded by Jennifer Doudna. Market cap US\$134m at \$1.25 against \$113.8m of cash and marketable securities and \$26.7m of debt, i.e. an enterprise value of roughly \$45m - the market ascribes almost nothing to a platform that has FDA alignment on a pivotal Phase 3 design.

Key backers. The FDA itself - alignment reached on the ANTLER-3 pivotal Phase 3 design (verified, Q2 2026 release); vispa-cel holds RMAT, Fast Track and Orphan Drug designations. Jennifer Doudna (Nobel laureate co-founder, origin of the chRDNA IP; widely documented, not primary-verified this session). Blood Cancer United (LLS) Therapy Acceleration Program, non-dilutive funding. CAVEAT: the Pfizer \$25m equity investment (2023) could NOT be re-verified from 2026 filings - treat as unconfirmed.

Catalysts, 12-24 months. CB-011 dose-expansion data H2 2026 (>15 BCMA-naive and BCMA-exposed patients, >=3 months follow-up, plus longer follow-up on the 12-patient escalation cohort). ANTLER-3 initiation and, critically, how it gets funded - partnership, licence or dilutive raise. Longer vispa-cel follow-up. A takeout: at ~\$45m EV with pivotal alignment in hand this is a cheap option on someone else concluding allogeneic works.

Why 10x. 10x = \$1.33bn. The evidence is far better than the price: at EHA 2026, optimised vispa-cel (80m cells, donor <30, >=2 matched HLA alleles, n=27) in 2L LBCL gave 82% ORR, 67% CR and 17.1-month median PFS; CB-011 at the 450m-cell expansion dose in 12 BCMA-naïve myeloma patients gave 92% ORR, 83% >=CR, 91% MRD-negativity and 50% still in >=CR at 15 months (Q2 release, 2026-08-13). FDA has agreed ANTLE-3: ~250 CD19-naïve 2L LBCL patients who cannot access autologous CAR-T - a population incumbents structurally cannot serve. From a \$45m EV the first 3x is just the market ceasing to price liquidation. 100x = \$13.3bn improbable: \$113.8m cannot fund two registrational programmes, so a partner takes half the economics.

What kills it. Allogeneic durability - the classic failure is host rejection, responses looking excellent at 3-6 months and collapsing by 18; 17.1-month median PFS is Phase 1, n=27, single-arm. The funding gap: management is explicitly 'exploring multiple options to fully fund' ANTLE-3, so a 250-patient Phase 3 means handing away the asset or destroying the share count. In-vivo CAR-T (Capstan/AbbVie, Umoja) could make ex-vivo allogeneic obsolete before approval.

What falsifies it. CB-011 dose-expansion data in H2 2026 showing responses materially below the 92% ORR / 83% >=CR of dose escalation, or vispa-cel follow-up showing relapses clustering after 18 months. Either would say the allogeneic durability problem is unsolved.

Confidence. 6/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

18. Collectis S.A. (ALCLS.PA)

Euronext Paris | Healthcare / Biotechnology | France | quoted in EUR | Moonshot sleeve rank 18 of 40, GBP 1,000, 1,014 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 23.3%. The headline below is shares x price.

Price (local)	1.151 EUR	Price (GBP)	0.986
52w high / low (local)	4.840 / 1.151	% below 52w high	-76.2%
Market cap (GBP)	GBP 71.7m	Market cap (USD m)	97
Enterprise value (GBP)	GBP -2.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-0.69	Price / book	2.21
EV / EBITDA	0.06	Dividend yield	n/a (not reported by Yahoo)
Total debt	EUR 80.3m	Cash	EUR 166.7m
Net debt	EUR -86.3m	Debt / equity (book)	0.91
Interest cover (EBITDA / interest)	-5.76	Interest cover (EBIT proxy)	-7.39
Capex, latest FY	EUR -3.5m	Capex / revenue	5.5%
Free cash flow	EUR 107.7m	Revenue (latest FY)	EUR 63.8m
Operating margin	-340.7%	Revenue growth	-62.1%
Beta (Yahoo)	2.77	Beta (own, 5y daily GBP)	1.84
Upside beta (beta+)	2.09	Downside beta (beta-)	1.34
CAPM cost of equity Ke	11.4%	WACC	9.5%
Cost of debt Kd	10.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-73.1%	Return 1y (GBP)	-59.1%
Volatility 5y	114.0%	Max drawdown 5y	-92.3%
Skew / excess kurtosis	14.99 / 400.33	Sharpe / Sortino 5y	-0.03 / -0.08
Judge score (asymmetry / backing / timing)	6.48 (6 / 7.4 / 6.3)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 581k	GBP 1,000 as % of a day	0.23%
10x end-state market cap	USD 1.42bn	100x arithmetically possible?	No - AZ's 44% caps minority capture

Source: data/fundamentals_table.csv row ALCLS.PA; data/stats_daily.csv row ALCLS.PA; universe/unicorn_final.json; universe/dossiers/ALCLS.PA.md

Business. Collectis S.A., a clinical stage biotechnological company, develops products based on gene-editing with a portfolio of allogeneic chimeric antigen receptor T-cells product candidates in the field of immuno-oncology and gene therapy product candidates in other therapeutic indications.

Thesis. TALEN gene-editing pioneer and allogeneic CAR-T developer, now roughly 44% owned by AstraZeneca. Market cap ~US\$142m at EUR1.151 against \$188m of cash and fixed-term deposits at 31-Mar-2026 with runway into Q4 2027. Own clinical programmes lasme-cel (r/r B-ALL) and eti-cel (r/r NHL) both read out in Q4 2026.

Key backers. AstraZeneca plc - JRCA from Nov 2023: \$25m upfront plus \$22m of milestones received by end-2024; a \$105m November 2023 tranche (which INCLUDES that \$25m upfront plus an \$80m equity investment - corrected at fact-check) and \$140m (May 2024) of preferred equity at \$5.00/share taking AZ to ~44% fully converted and ~30% of voting rights; exclusive rights to 25 genetic targets with an option on up to 10 candidate products; three active programmes. European Investment Bank - \$45.48m facility, final \$5.68m tranche drawn Dec 2024, non-dilutive.

Catalysts, 12-24 months. lasme-cel (r/r B-ALL) first interim Phase 2 analysis expected Q4 2026. eti-cel (r/r NHL) full Phase 1 dataset expected Q4 2026. Development-candidate nomination or IND on any AZ programme (each triggers milestones). AZ taking further targets under the 25-target framework, or bidding for the remainder - at a

\$142m market cap, buying out the minority is a rounding error for AZ.

Why 10x. 10x = \$1.42bn; Cellectis traded above \$1.5bn in 2021 on weaker evidence. The arithmetic that makes this defensible: AstraZeneca paid \$5.00 per preferred share for its stake, while the ordinary line closed at EUR1.151 (~\$1.32) on 2026-09-16 - AZ is roughly 74% underwater on equity it bought after diligence, and you are offered the same asset at about a quarter of its price. With cash above market cap at the last report, the first ~1.5x needs nothing but the market ceasing to price the equity below net cash; the remaining ~7x needs one Q4 2026 clinical success or one AZ programme entering the clinic. 100x = \$14.2bn is the wrong bet - AZ's 44% means minorities capture a fraction.

What kills it. The 44% stake cuts both ways - AZ can block, dictate terms or low-ball the minority, and preferred structures can rank ahead of ordinary shareholders. A decade in the clinic without a breakout dataset; persistence and host rejection remain unsolved, and Q4 2026 is where that gets tested. Thin liquidity (~EUR340k/day median, the thinnest of its group) and a weakening balance sheet - cash and equivalents down to \$35.6m against \$84.6m of debt at 30-Jun-2026.

What falsifies it. The Q4 2026 lasme-cel interim Phase 2 and eti-cel Phase 1 datasets failing to show durable responses, or AstraZeneca declining to advance a collaboration programme. Either removes the only two reasons to hold this rather than cash.

Confidence. 6/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

19. Fractyl Health, Inc. (GUTS)

NASDAQ | Healthcare / Biotechnology | United States | quoted in USD | Moonshot sleeve rank 19 of 40, GBP 1,000, 2,393 shares

Insufficient history. 662 daily bars in the five-year window (first bar 2024-02-02). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Price (local)	0.563 USD	Price (GBP)	0.418
52w high / low (local)	2.445 / 0.377	% below 52w high	-77.0%
Market cap (GBP)	GBP 66.5m	Market cap (USD m)	90
Enterprise value (GBP)	GBP 76.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-1.22	Price / book	-53.65
EV / EBITDA	-1.22	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 60.7m	Cash	USD 47.1m
Net debt	USD 13.6m	Debt / equity (book)	4.77
Interest cover (EBITDA / interest)	-58.78	Interest cover (EBIT proxy)	-96.75
Capex, latest FY	USD -557.0k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	USD -48.4m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	2.33	Beta (own, 5y daily GBP)	1.18
Upside beta (beta+)	1.06	Downside beta (beta-)	1.59
CAPM cost of equity Ke	8.7%	WACC	5.9%
Cost of debt Kd	2.4% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-74.4%	Return 1y (GBP)	-44.6%
Volatility 5y	110.9%	Max drawdown 5y	-97.1%
Skew / excess kurtosis	-0.97 / 20.12	Sharpe / Sortino 5y	-0.49 / -0.67
Judge score (asymmetry / backing / timing)	6.47 (7.5 / 3.2 / 8.8)	Confidence (dossier)	5 / 10
Avg daily traded value	USD 1,202k	GBP 1,000 as % of a day	0.11%
10x end-state market cap	USD 0.90bn	100x arithmetically possible?	Arithmetically live but defeated by dilution

Source: data/fundamentals_table.csv row GUTS; data/stats_daily.csv row GUTS; universe/unicorn_final.json; universe/dossiers/GUTS.md

Business. Fractyl Health, Inc., a metabolic therapeutics company, develops therapies for the treatment of type 2 diabetes (T2D) and obesity. It develops Revita DMR System (Revita), an outpatient procedural therapy designed to durably modify duodenal dysfunction, a pathologic consequence of a high fat and high sugar diet, which can initiate T2D and obesity in humans.

Thesis. Two assets attacking metabolic disease at the organ, not the hormone: Revita, a 45-minute endoscopic duodenal resurfacing procedure to stop weight regain after GLP-1 discontinuation, and Rejuva, a pancreatic gene-therapy platform making islet cells produce incretins endogenously. Market cap US\$89.6m at \$0.563. A genuine binary deliberately funded only to the pivotal readout. MATERIAL: on 2026-09-10 Nasdaq notified Fractyl its securities are subject to delisting after it failed to regain \$1.00 minimum bid compliance by 2026-09-09; a Hearings Panel request stays suspension (8-K filed 2026-09-11), and this is material to whether the position stays holdable on ii.

Key backers. Dutch CCMO - CTA authorisation for RJVA-001 granted April 2026, enabling first-in-human pancreatic gene-therapy dosing. Australian ethics committee approval July 2026 (second jurisdiction). Mike Zumdahl appointed SVP Market Access and Commercial Strategy June 2026, previously built reimbursement infrastructure for procedural therapies at Inari Medical. NOTABLE GAP: no strategic pharma or device investor, the weakest part of the case.

Catalysts, 12-24 months. Early Q4 2026: topline 6-month randomised REMAIN-1 Pivotal Cohort data (randomisation complete) - this is the event. Late Q4 2026: potential FDA De Novo marketing application in post-GLP-1 weight maintenance. H2 2026: first-in-human dosing of RJVA-001 with preliminary data, the first pancreatic gene therapy for metabolic disease ever dosed. Q1 2027: one-year pivotal data. Plus the financing that must follow and the Nasdaq hearing.

Why 10x. 10x = \$0.90bn. The randomised evidence is stronger than the price implies: in the REMAIN-1 Midpoint Cohort at one year, in the pre-specified optimised population (complete ablation >14cm plus run-in loss >=17.5%), Revita patients maintained ~84% of prior GLP-1-induced weight loss versus 46% for sham - LSM regain 4.1% vs 13.5% (n=10 vs 8) - with no device- or procedure-related SAEs through one year; REVEAL-1 open-label at one year (n=15) maintained ~78%. The dose-response concentration of benefit where ablation was complete is mechanistic, not post-hoc luck. Durability is the largest unmet need in medicine's largest market. 100x = \$9.0bn is below where single-asset obesity companies have traded pre-approval, so it is arithmetically live - but dilution defeats it.

What kills it. REMAIN-1 Pivotal misses - sham-controlled device trials fail often and the Midpoint result came from an optimised subgroup of 18 patients. The capital structure: \$47.1m cash against \$60.7m of debt and stockholders' equity of MINUS \$1.7m at 30-Jun-2026, runway only into early 2027 - lenders sit ahead of shareholders, and a miss means equity holders are wiped out. Delisting: if the Hearings Panel is not persuaded the shares leave Nasdaq, and OTC is outside the ii mandate entirely.

What falsifies it. The early-Q4-2026 REMAIN-1 Pivotal Cohort 6-month topline failing to show a statistically significant weight-maintenance benefit versus sham in the full randomised population. Nothing else about this company matters until that number prints.

Confidence. 5/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

20. Prime Medicine, Inc. (PRME)

NasdaqGM | Healthcare / Biotechnology | United States | quoted in USD | Moonshot sleeve rank 20 of 40, GBP 1,000, 449 shares

Price (local)	3.000 USD	Price (GBP)	2.227
52w high / low (local)	6.940 / 2.670	% below 52w high	-56.8%
Market cap (GBP)	GBP 403.9m	Market cap (USD m)	544
Enterprise value (GBP)	GBP 416.8m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-3.25	Price / book	12.74
EV / EBITDA	-2.97	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 112.5m	Cash	USD 95.1m
Net debt	USD 17.4m	Debt / equity (book)	0.69
Interest cover (EBITDA / interest)	-45.58	Interest cover (EBIT proxy)	-47.48
Capex, latest FY	USD -4.5m	Capex / revenue	111.2%
Free cash flow	USD -75.8m	Revenue (latest FY)	USD 4.1m
Operating margin	-3750.9%	Revenue growth	3.5%
Beta (Yahoo)	2.35	Beta (own, 5y daily GBP)	1.29
Upside beta (beta+)	0.14	Downside beta (beta-)	2.18
CAPM cost of equity Ke	9.1%	WACC	8.0%
Cost of debt Kd	3.7% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-13.6%	Return 1y (GBP)	-26.1%
Volatility 5y	90.6%	Max drawdown 5y	-95.3%
Skew / excess kurtosis	0.74 / 2.47	Sharpe / Sortino 5y	-0.11 / -0.17
Judge score (asymmetry / backing / timing)	6.47 (6.5 / 6.4 / 6.5)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 11,289k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 5.44bn	100x arithmetically possible?	No - a US\$5-15bn acquisition is the realistic ceiling

Source: data/fundamentals_table.csv row PRME; data/stats_daily.csv row PRME; universe/unicorn_final.json; universe/dossiers/PRME.md

Business. Prime Medicine, Inc., a biotechnology company, engages in delivering genetic medicines to address the spectrum of diseases by deploying gene editing technology in the United States.

Thesis. Prime Medicine develops prime editing - a search-and-replace genome edit that nicks one DNA strand and writes a new sequence from an attached template, with no double-strand break and no donor DNA. In principle it can correct the large majority of known pathogenic human variants rather than the narrow subset addressable by base editing or by knocking a gene out, which is where the volume of human genetic disease actually lives. This is no longer a preclinical claim: PM359 produced clinical proof of prime editing in humans, the FDA granted it RMAT designation on 2026-06-22, and a BLA is targeted for the first half of 2027. Cash and investments of US\$108.8m at 30-Jun-2026 against a ~US\$42m quarterly loss make a financing inside the holding period close to certain.

Key backers. Bristol Myers Squibb - partnership developing prime-edited CAR-T products, the source of the US\$1.2m Q2 2026 collaboration revenue (Q2 2026 results, 2026-08-06). Cystic Fibrosis Foundation - supporting the in vivo cystic fibrosis programme (same release); the CFF is the disease-foundation funder with the best track record in medicine. US FDA - RMAT designation for PM359 (2026-06-22), IND cleared for PM577a in Wilson disease (2026-07-23), CTA approved in New Zealand (2026-06-18). Arbitration with Beam Therapeutics resolved in Prime's favour on 2026-07-08, confirming rights to PM647. Insiders 23.7%, institutions 71.2%. Prime editing originates in David Liu's laboratory at the Broad Institute (widely documented; not primary-verified this session).

Catalysts, 12-24 months. PM359 BLA submission targeted 1H 2027 - potentially the first prime-edited medicine filed anywhere. PM647 (AATD) IND/CTA filing in Q3 2026 with initial data 2027. PM577a Wilson disease initial

clinical data in 2027 - the in vivo proof point that decides whether this is a platform or a cell-therapy company. The financing, a scheduled hazard whose terms set how much upside holders keep. A large-pharma partnership on an in vivo programme.

Why 10x. 10x = US\$5.44bn. The required chain is specific: PM359 approved or clearly approvable, PM577a showing durable in vivo correction in Wilson disease with clean safety, and at least one large-pharma deal on an in vivo programme. That combination would make Prime the only company with a clinically validated, correction-capable editor - and editing platforms have carried US\$5-10bn valuations on thinner evidence (Intellia peaked above US\$10bn in 2021 on a single interim). The multiple is not the stretch; surviving the financing with the share count intact is. 100x = US\$54.4bn is Vertex-scale and requires commercial infrastructure Prime cannot fund; the realistic terminal event is an acquisition in the US\$5-15bn band. Note dilution cuts the per-share multiple: a US\$5.4bn enterprise reached after two raises might deliver 6-7x on today's price.

What kills it. The financing, at this price - US\$108.8m against a ~US\$42m quarterly loss and an 'into 2027' guide means raising from a share price 57% off its high. This is the most likely way to be right and still lose money. Translation: ex vivo success in CGD says little about whether a prime editor delivered to the liver corrects enough hepatocytes to change a Wilson disease phenotype, and the 2027 PM577a readout is a genuine binary. Category risk - gene-editing equities have de-rated as a group and even good data has repeatedly failed to re-rate them.

What falsifies it. PM359's BLA slipping out of 2027, or the PM577a 2027 Wilson data failing to show durable correction in the liver. Either breaks the platform claim, and the *ex vivo* CGD asset alone does not support US\$544m.

Confidence. 6/10. The clearest "why this could be the next big thing" argument in the sleeve's biology bucket - correction rather than disruption, already in humans, with FDA RMAT, a pharma partner and the CF Foundation behind it - married to the sleeve's most predictable balance-sheet hazard.

21. ProQR Therapeutics N.V. (PRQR)

NASDAQ | Healthcare / Biotechnology | Netherlands | quoted in USD | Moonshot sleeve rank 21 of 40, GBP 1,000, 724 shares

Price (local)	1.860 USD	Price (GBP)	1.380
52w high / low (local)	3.100 / 1.330	% below 52w high	-40.0%
Market cap (GBP)	GBP 194.8m	Market cap (USD m)	262
Enterprise value (GBP)	GBP 119.2m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-3.87	Price / book	2.17
EV / EBITDA	-3.83	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 15.3m	Cash	USD 117.1m
Net debt	USD -101.8m	Debt / equity (book)	0.23
Interest cover (EBITDA / interest)	-45.10	Interest cover (EBIT proxy)	-44.39
Capex, latest FY	USD -1.0m	Capex / revenue	5.5%
Free cash flow	USD -19.1m	Revenue (latest FY)	USD 18.4m
Operating margin	-100.7%	Revenue growth	120.4%
Beta (Yahoo)	0.06	Beta (own, 5y daily GBP)	0.96
Upside beta (beta+)	0.58	Downside beta (beta-)	0.88
CAPM cost of equity Ke	7.8%	WACC	7.6%
Cost of debt Kd	6.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-8.0%	Return 1y (GBP)	-14.7%
Volatility 5y	101.9%	Max drawdown 5y	-92.7%
Skew / excess kurtosis	4.95 / 126.66	Sharpe / Sortino 5y	0.18 / 0.31
Judge score (asymmetry / backing / timing)	6.46 (6 / 7 / 6.8)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 2,080k	GBP 1,000 as % of a day	0.06%
10x end-state market cap	USD 2.62bn	100x arithmetically possible?	No - royalty-taker economics cap it

Source: data/fundamentals_table.csv row PRQR; data/stats_daily.csv row PRQR; universe/unicorn_final.json; universe/dossiers/PRQR.md

Business. ProQR Therapeutics N.V., a clinical-stage biotechnology company, focuses on the discovery and development of novel therapeutic medicines. The company develops an RNA editing platform through its Axiomer RNA editing technology for genetic disorders with unmet needs. Its lead product candidate, AX-0810, targets na-taurocholate cotransporting polypeptide (NTCP) for the treatment of cholestatic diseases.

Thesis. RNA-editing company whose Axiomer platform uses chemically modified oligonucleotides to recruit the body's own ADAR enzymes for a single A-to-I edit in a target mRNA. Market cap US\$262.5m at \$1.86 against EUR117.1m (~\$134m) of cash at 30-Jun-2026 and runway into mid-2028 - roughly half the market value is cash, so the platform, two clinical assets and the Lilly package are valued at about \$144m.

Key backers. Eli Lilly and Company - Axiomer licensing and research collaboration plus repeated direct equity: on 2026-06-25 Lilly signed a share purchase agreement for 5,100,780 ordinary shares at \$1.81 (~\$9.2m), concurrent with the public offering and subject to a lock-up (verified, 6-K 2026-06-26). A partner buying alongside the market, at the market price, with a lock-up, is a real signal. BofA Securities, Evercore and Cantor Fitzgerald underwrote the concurrent 27,624,310-share registered direct offering at \$1.81, net ~\$46.5m. EMA accepted the AX-0811 CTA in July 2026.

Catalysts, 12-24 months. Full AX-0810 Phase 1 dataset at a medical/scientific conference later in 2026 including Cohort 3. AX-0810 9 mg/kg cohort data and 12-week follow-up by year-end 2026. AX-0811 initial healthy-volunteer data by year-end 2026. H1 2027: biliary atresia IIT in China initial data, AX-0422 (IDUA) first-in-patient data in MPS I Hurler, AX-2911 (PNPLA3) first-in-human. Mid-2027: potentially registration-enabling Phase 2 starts. Further Lilly option exercises and milestone payments.

Why 10x. 10x = \$2.63bn. The key upgrade versus the candidate record: the first human editing signal is no longer a future catalyst, it has already happened. In June 2026 AX-0810 Phase 1 delivered up to an 8-fold increase in total serum bile acids at 6 mg/kg against a pre-defined 2-fold threshold - clearing its own bar by 4x - with dose-dependent conjugated bile acid changes, no serious adverse events, no pruritus and an eight-week half-life, which the company calls the first clinical validation of Axiomer. Axiomer is the pragmatic middle ground: re-dosable, reversible, no viral vector or LNP, using GalNAc delivery that already works commercially. Wave traded above \$1bn on ADAR proof-of-concept alone. 100x = \$26.6bn: NO - ProQR has partnered its biggest opportunities to Lilly and takes royalties.

What kills it. Target engagement is not efficacy - bile-acid movement in healthy volunteers is a biomarker, and Korro's 2026 KRRO-110 failure is the direct precedent for a similar mechanism disappointing at the patient stage. Liver-only delivery: GalNAc works and nothing else does yet, capping indications until extrahepatic delivery is solved. Lilly can walk, and ProQR has a long failure history - sepofarsen and ultevursen both failed and the company has already pivoted itself once.

What falsifies it. The biliary atresia or other patient data in H1 2027 showing no clinical benefit despite confirmed editing, which would mean ADAR recruitment works biochemically but not therapeutically.

Confidence. 7/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

22. Median Technologies SA (ALMDT.PA)

Euronext Growth Paris | Healthcare / Medical Devices | France | quoted in EUR | Moonshot sleeve rank 22 of 40, GBP 1,000, 308 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 8.2%. The headline below is shares x price.

Price (local)	3.785 EUR	Price (GBP)	3.242
52w high / low (local)	7.270 / 2.625	% below 52w high	-47.9%
Market cap (GBP)	GBP 158.8m	Market cap (USD m)	214
Enterprise value (GBP)	GBP 164.8m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-9.29	Price / book	-3.66
EV / EBITDA	-13.64	Dividend yield	n/a (not reported by Yahoo)
Total debt	EUR 25.3m	Cash	EUR 18.4m
Net debt	EUR 6.9m	Debt / equity (book)	-0.62
Interest cover (EBITDA / interest)	-2.50	Interest cover (EBIT proxy)	-5.00
Capex, latest FY	EUR -747.0k	Capex / revenue	3.2%
Free cash flow	EUR 8.3m	Revenue (latest FY)	EUR 23.5m
Operating margin	-57.8%	Revenue growth	0.7%
Beta (Yahoo)	0.52	Beta (own, 5y daily GBP)	1.18
Upside beta (beta+)	0.89	Downside beta (beta-)	1.46
CAPM cost of equity Ke	8.7%	WACC	9.6%
Cost of debt Kd	22.3% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-21.5%	Return 1y (GBP)	11.9%
Volatility 5y	87.0%	Max drawdown 5y	-93.2%
Skew / excess kurtosis	2.82 / 27.41	Sharpe / Sortino 5y	0.04 / 0.07
Judge score (asymmetry / backing / timing)	6.44 (7 / 4.6 / 7.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 868k	GBP 1,000 as % of a day	0.15%
10x end-state market cap	USD 2.31bn	100x arithmetically possible?	No - exceeds the plausible total US screening market

Source: data/fundamentals_table.csv row ALMDT.PA; data/stats_daily.csv row ALMDT.PA; universe/unicorn_final.json; universe/dossiers/ALMDT.PA.md

Business. Median Technologies SA develops and markets software solutions and services for the use of medical images for the diagnosis and monitoring of patients with cancer in Europe, North America, and Asia. The company offers iCRO, an imaging service for oncology clinical trials; and eyonis, an end-to-end CAdE/CADx software as medical devices. The company was founded in 2002 and is based in Valbonne, France.

Thesis. AI Software-as-a-Medical-Device for lung cancer screening. eyonis LCS detects AND characterises lung nodules on low-dose CT - a combined CAdE/CADx device, not just nodule-finding. Market cap ~US\$231m at EUR3.785. Two caveats up front: this is Euronext GROWTH (a junior MTF, not main Euronext Paris) and ii's coverage of that venue was NOT confirmed - check on ii.co.uk before committing capital; and the balance sheet is the weakest in its group.

Key backers. US FDA (510(k) clearance 2026-02-09) and CMS (New Tech APC, ~\$601-700 per exam for 2026) - the single most valuable thing Median owns. Tempus AI - distribution partnership announced 2026-02-12, the fact that most changes the case: a partner with an existing oncology footprint gets eyonis into radiology workflows without a sales force. Oran Muduroglu, President of Median eyonis Inc. (Feb 2026): Cemax sold to 3M/Imation 1995, co-founded Stentor sold to Philips 2005, CEO of Medicalis sold to Siemens Healthineers 2017. CAVEAT: the quoted 93.3% sensitivity / 92.4% specificity / 99.9% NPV figures could NOT be verified from the primary press

releases fetched - treat as company-claimed.

Catalysts, 12-24 months. First US sites operational in Q3 2026 - i.e. now - and the first reported per-exam revenue, proving the reimbursement code converts to cash. Signed contracts with US screening networks and radiology groups, and evidence of Tempus AI pull-through. European rollout as national screening programmes start. A financing, which the balance sheet requires - a scheduled hazard rather than a catalyst, and its terms are the key risk to watch.

Why 10x. $10x = \$2.32bn$, and this is the cleanest arithmetic in the sleeve because CMS has already set the price per unit. FDA 510(k) clearance was granted 2026-02-09 and CMS New Tech APC reimbursement pays approximately \$601-700 per exam for 2026, against ~14.5 million Americans meeting USPSTF screening criteria. At ~\$600/exam with software gross margins, one million exams is ~\$600m of revenue; even 500,000 exams a year is ~\$300m of very high-margin revenue, which at 7-8x sales is ~\$2.3bn. A cleared device that already has a payment code and a per-exam price is a different asset class from an AI company hoping for one. $100x = \$23bn$: NO, that implies \$2-3bn of annual revenue from a single screening indication.

What kills it. Adoption, not technology - radiology procurement cycles are long, and Aidence/RadNet, Optellum, Coreline and the Siemens/GE in-house tools are already inside the workflow. The balance sheet, which is the likeliest way to lose money here: stockholders' equity of MINUS EUR35.2m at 31-Dec-2025, more debt (EUR21.8-25.3m) than cash (EUR18.2-18.4m), and shares up from 18.42m (Dec-2024) to 49.00m now - +166% in ~21 months. Venue and reimbursement risk - Euronext Growth means lighter disclosure and ~EUR550k/day liquidity, and New Tech APC rates are reviewed annually and can fall.

What falsifies it. The first half-year of US commercial reporting showing negligible reimbursed exam volume, which would mean the reimbursement code exists but nobody is using the product.

Confidence. 6/10. (Marked down from the candidate record's 7 because the performance statistics could not be verified, the balance sheet is worse than described, and ii tradeability on Euronext Growth is unconfirmed.)

23. Clarity Pharmaceuticals Ltd (CU6.AX)

ASX | Healthcare / Biotechnology | Australia | quoted in AUD | Moonshot sleeve rank 23 of 40, GBP 1,000, 904 shares

Price (local)	2.090 AUD	Price (GBP)	1.105
52w high / low (local)	5.870 / 1.800	% below 52w high	-64.4%
Market cap (GBP)	GBP 412.4m	Market cap (USD m)	556
Enterprise value (GBP)	GBP 318.2m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-7.81	Price / book	4.17
EV / EBITDA	-5.56	Dividend yield	n/a (not reported by Yahoo)
Total debt	n/a (not reported by Yahoo)	Cash	AUD 178.3m
Net debt	AUD -178.3m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-22.70	Interest cover (EBIT proxy)	-12.41
Capex, latest FY	AUD -182.7k	Capex / revenue	1.7%
Free cash flow	AUD -59.4m	Revenue (latest FY)	AUD 10.5m
Operating margin	-1095.6%	Revenue growth	-1.1%
Beta (Yahoo)	1.18	Beta (own, 5y daily GBP)	0.80
Upside beta (beta+)	0.80	Downside beta (beta-)	1.35
CAPM cost of equity Ke	7.1%	WACC	7.1%
Cost of debt Kd	5.8% (fallback: rf + 2% (no usable interest expense / debt))	Tax rate	25.0%
Return YTD (GBP)	-34.2%	Return 1y (GBP)	-27.8%
Volatility 5y	72.1%	Max drawdown 5y	-85.0%
Skew / excess kurtosis	0.57 / 1.96	Sharpe / Sortino 5y	0.42 / 0.65
Judge score (asymmetry / backing / timing)	6.43 (5.5 / 6.6 / 8.5)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 4,388k	GBP 1,000 as % of a day	0.03%
10x end-state market cap	USD 5.58bn	100x arithmetically possible?	No - exceeds the whole radiopharmaceutical industry

Source: data/fundamentals_table.csv row CU6.AX; data/stats_daily.csv row CU6.AX; universe/unicorn_final.json; universe/dossiers/CU6.AX.md

Business. Clarity Pharmaceuticals Ltd, a clinical stage radiopharmaceutical company, engages in research and development of radiopharmaceutical products for oncology in Australia and the United States. The company's lead product is SAR-bisPSMA, a theranostic radiopharmaceutical for the treatment of Prostate Specific Membrane Antigen (PSMA) cancer.

Thesis. Copper-64/copper-67 radiopharmaceutical theranostics using the proprietary SAR sarcophagine chelator. Lead ⁶⁴Cu-SAR-bisPSMA is a prostate-cancer PET imaging agent in registrational Phase 3 (CLARIFY); the matched therapeutic ⁶⁷Cu-SAR-bisPSMA uses the same targeting molecule. Market cap ~US\$558m at A\$2.09, debt-free with A\$178.3m of reported cash. SCEPTIC'S FIRST QUESTION, FLAGGED HONESTLY: the shares are down roughly 63% from a 52-week high of A\$5.70 and sit ~15% above the 52-week low; the cause of that de-rating could NOT be established from the sources reachable this session - read Clarity's 2026 ASX announcements before acting.

Key backers. US FDA - after a successful end-of-phase meeting the FDA gave positive guidance on the prospective, non-randomised, single-arm, open-label, multicentre Phase 3 design, with results intended to support an approval application (verified, Clarity 2023-07-04). CAVEAT: the frequently cited Fast Track designation was NOT re-verified this session. The US/Australian registrational network - CLARIFY recruits 383 participants across US and Australian sites, first patient dosed 2023-12-23 at the Urology Cancer Center/XCancer, Omaha. The Australian isotope ecosystem (ANSTO) underpins copper-64/67 supply. Insiders hold 21.1%; debt-free with only ~16% dilution in 15 months.

Catalysts, 12-24 months. CLARIFY topline against surgical histopathology - the dated binary event, inside the window. Readout of the second pivotal study in biochemical recurrence. NDA submission for ⁶⁴Cu-SAR-bisPSMA. ⁶⁷Cu-SAR-bisPSMA therapeutic data - the imaging agent is the wedge, the therapy is where the value is. A bid: given the M&A backdrop an approach is plausible at any point.

Why 10x. 10x = US\$5.58bn. The mechanism is a manufacturing and logistics advantage, not just chemistry: gallium-68's 68-minute half-life forces production beside the patient, whereas copper-64's 12.7-hour half-life and Clarity's stated 48-hour shelf life allow central manufacture and overnight shipping, and next-day imaging when background has cleared - which is why the CLARIFY protocol images at Day 1 and Day 2. The SAR cage holds copper tightly enough not to leak in vivo, which is what previously made copper unusable. And the same molecule carries copper-67 for therapy, giving identical biodistribution a gallium/lutetium pair can never guarantee. Precedents: BMS paid ~\$4.1bn for RayzeBio and AstraZeneca ~\$2.4bn for Fusion, both pre-approval. 100x = US\$55.8bn: NO - that exceeds the entire current radiopharmaceutical industry.

What kills it. The comparator, not the agent - CLARIFY's endpoint is sensitivity and specificity for pelvic nodal disease against surgical pathology in 383 high-risk patients going to radical prostatectomy with lymph node dissection, a brutal standard on which imaging trials fail even when the agent works. Radiochemistry at scale - copper-64 supply and manufacturing scale-up is a genuine risk, and the 48-hour shelf-life advantage only exists if production is reliable. Entrenched competition (Novartis, Lantheus, Telix) plus a demanding starting price: at ~\$558m the shares already discount considerable success, and the unexplained 63% drawdown means the market may know something this dossier does not.

What falsifies it. CLARIFY missing its co-primary sensitivity/specificity endpoints against surgical pathology. Everything else - the therapeutic, the M&A optionality, the supply chain - is downstream of that one result.

Confidence. 6/10. (Marked down from the candidate record's 7 solely because of the unexplained 12-month drawdown and the unverified Fast Track and trial-timing claims.)

24. SEALSQ Corp (LAES)

NasdaqGS | Technology / Semiconductors | Switzerland | quoted in USD | Moonshot sleeve rank 24 of 40, GBP 1,000, 615 shares

Price (local)	2.190 USD	Price (GBP)	1.625
52w high / low (local)	8.710 / 1.990	% below 52w high	-74.9%
Market cap (GBP)	GBP 363.2m	Market cap (USD m)	489
Enterprise value (GBP)	GBP 10.5m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-7.72	Price / book	0.90
EV / EBITDA	-0.31	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 7.1m	Cash	USD 482.3m
Net debt	USD -475.1m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-206.61	Interest cover (EBIT proxy)	-152.40
Capex, latest FY	USD -1.2m	Capex / revenue	5.1%
Free cash flow	USD -32.4m	Revenue (latest FY)	USD 24.6m
Operating margin	-288.6%	Revenue growth	131.1%
Beta (Yahoo)	-7.71	Beta (own, 5y daily GBP)	1.01
Upside beta (beta+)	0.65	Downside beta (beta-)	0.60
CAPM cost of equity Ke	8.0%	WACC	7.9%
Cost of debt Kd	3.1% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-42.1%	Return 1y (GBP)	-23.3%
Volatility 5y	172.6%	Max drawdown 5y	-98.5%
Skew / excess kurtosis	3.62 / 26.77	Sharpe / Sortino 5y	0.41 / 0.83
Judge score (asymmetry / backing / timing)	6.42 (6 / 6.2 / 7.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 32,650k	GBP 1,000 as % of a day	0.00%
10x end-state market cap	USD 4.89bn	100x arithmetically possible?	No - that is Infineon scale

Source: data/fundamentals_table.csv row LAES; data/stats_daily.csv row LAES; universe/unicorn_final.json; universe/dossiers/LAES.md

Business. SEALSQ Corp, together with its subsidiaries, designs, develops, and markets semiconductors in North America, Europe, the Middle East, Africa, the Asia Pacific, and Latin America. It offers semiconductors, such as VaultIC secure elements, quantum-resistant RISC-V secure chips, secure ARM platform, and smart card reader chips; custom chip design services; and other PKI services, including personal digital identity services.

Thesis. Swiss designer of secure microcontrollers and hardware security modules converting its franchise to post-quantum: the QS7001 Quantum Shield secure element and QVault TPM implement NIST post-quantum algorithms in hardware. The central fact: market cap US\$489m at \$2.19 against \$486.1m of cash at 30-Jun-2026 and \$7.1m of debt - an enterprise value of roughly \$14m for a business guided to \$27-36m of FY2026 revenue growing 131% at ~48% gross margin with a >\$225m pipeline. Either the market believes the cash will be destroyed, or this is mispriced; both readings deserve respect.

Key backers. NIST - QS7001 achieved SP 800-90B Entropy Source Validation and completed Common Criteria testing (verified directly in the H1 2026 release, 2026-09-11). WISeKey International Holding (Nasdaq: WKEY) - parent participating directly in R&D and commercial development, holding non-traded Class F shares (1.5m outstanding at 30-Jun-2026, confirmed); the widely reported 52% voting control was NOT re-confirmed in that release and rests on WISeKey's SEC/6-K filings. Wecan Group, Quobly and Miraex - strategic stakes, and also the related-party transactions that make the governance risk concrete.

Catalysts, 12-24 months. Initial commercial revenue from QS7001 and QVault TPM toward the end of H2 2026 - i.e. within one quarter; this has been 'about to start' for several quarters, so delivery or another slip is the near-term test. Completion of EAL5+ certification. Conversion of the >\$225m pipeline (as at 2026-09-09, >\$100m

post-quantum) into booked orders, and FY2026 revenue landing inside the reaffirmed \$27-36m guidance. Deployment of the ~\$486m cash - the biggest swing factor in either direction.

Why 10x. 10x = \$4.89bn, but with \$486.1m of cash in hand the operating business only has to be worth ~\$4.4bn - and it does not need to raise again to get there. Post-quantum cryptography cannot live only in software: firmware signing, secure boot and device identity must be rooted in hardware, so every one of the tens of billions of connected devices shipped this decade needs a root of trust running algorithms that did not exist when today's secure elements were designed. That is a forced, dated replacement cycle driven by regulation (EU Cyber Resilience Act, US federal migration timetables). Hardware root-of-trust is a certification moat: QS7001 has NIST SP 800-90B Entropy Source Validation and has completed Common Criteria testing. Revenue must go from ~\$30m to \$300-400m. 100x = \$49bn: NO, that is Infineon scale.

What kills it. The dilution record is the thesis risk - that \$486m was raised by issuing shares, with the count going from 101.54m (31-Dec-2024) to 224.93m (30-Jun-2026), +121.5% in 18 months, and nothing stops it continuing; cash per share, not cash, is what you own. Governance - WISEKey's voting control through non-traded Class F shares means minorities cannot object to related-party transactions, and the group has a history of them (Wecan, Quobly, Miraex). Infineon, NXP and STMicroelectronics will all ship post-quantum secure elements and own the automotive design-in relationships.

What falsifies it. QS7001 commercial revenue failing to appear by the end of H1 2027, or a large cash deployment into a related party. The first would mean the certified part cannot win sockets; the second would mean the balance sheet is not really the shareholders'.

Confidence. 6/10. The dossier records the score in its heading without a separate note; its falsifier is the line above.

25. Terrestrial Energy Inc. (IMSR)

NASDAQ Global Select | Utilities / Utilities - Regulated Electric | United States | quoted in USD | Moonshot sleeve rank 25 of 40, GBP 1,000, 317 shares

Insufficient history. 219 daily bars in the five-year window (first bar 2025-11-03). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 33.9%. The headline below is shares x price.

Price (local)	4.250 USD	Price (GBP)	3.154
52w high / low (local)	16.380 / 4.140	% below 52w high	-74.1%
Market cap (GBP)	GBP 260.9m	Market cap (USD m)	352
Enterprise value (GBP)	GBP 60.8m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-5.21	Price / book	1.55
EV / EBITDA	-2.17	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 3.9m	Cash	USD 273.5m
Net debt	USD -269.6m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-8.70	Interest cover (EBIT proxy)	-5.45
Capex, latest FY	USD -1.1m	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	USD -41.7m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	n/a (not reported by Yahoo)	Beta (own, 5y daily GBP)	3.59
Upside beta (beta+)	2.88	Downside beta (beta-)	2.72
CAPM cost of equity Ke	18.7%	WACC	18.7%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-30.5%	Return 1y (GBP)	n/a (not reported by Yahoo)
Volatility 5y	109.2%	Max drawdown 5y	-73.6%
Skew / excess kurtosis	0.84 / 1.97	Sharpe / Sortino 5y	-0.89 / -1.33
Judge score (asymmetry / backing / timing)	6.38 (6 / 7.4 / 5.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 9,979k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 3.52bn	100x arithmetically possible?	No - not conceivable in the hold period

Source: data/fundamentals_table.csv row IMSR; data/stats_daily.csv row IMSR; universe/unicorn_final.json; universe/dossiers/IMSR.md

Business. Terrestrial Energy Inc. engages in the development of small modular nuclear plants using its generation IV integral molten salt reactor (IMSR). The IMSR captures the transformative operating benefits of molten salt reactor technology in a plant design of nuclear energy supply. The company uses a plant design together with molten salt reactor technology and uranium in its fuel for a nuclear plant.

Thesis. Terrestrial Energy develops the Integral Molten Salt Reactor - 390 MWt / ~195 MWe delivering 585C process heat, with the whole primary loop inside a sealed, replaceable seven-year Core-unit. Nasdaq-listed via the HCM II de-SPAC, development site at Texas A&M-RELLIS. Trading at \$4.25 against a \$31.50 52-week high while licensing evidence accumulates. Not flagged by the fact-checker.

Key backers. Texas A&M University System (ground lease, ~77 acres at RELLIS, 2026-06-18); Zachry Group (Engineering Services Agreement 2026-07-28, a top-tier US nuclear EPC); Riot Platforms (up-to-4 GW datacentre MOU, 2026-05-06); US NRC (Postulated Initiating Events topical approved 2026-05-12, graphite topical submitted 2026-09-10); Pamela Cowan (EVP Engineering, 30+ years nuclear) and Kathryn McCarthy (board). Cash and investments US\$283.4m.

Catalysts, 12-24 months. Third 2026 topical submission and NRC acceptance of the graphite topical; a RELLIS construction permit application being docketed - the single re-rating event; conversion of the Riot MOU into a binding PPA; a DOE cost-share or loan award; appointment of a fuel-salt supplier.

Why 10x. 10x on ~US\$352m is US\$3.5bn, requiring a construction permit application docketed and accepted, one binding multi-GW offtake, and a demonstrated LEU fuel-salt supply chain. On the company's stated ~US\$2.7bn lifetime revenue per plant, an 8-12 plant pipeline implies a US\$22-32bn backlog; the market paid US\$5-8bn for less in this sector before the 2026 de-rating. Two structural choices matter: the sealed Core-unit sidesteps 60-year component corrosion, and IMSR runs on standard-assay LEU rather than HALEU, avoiding the enrichment bottleneck gating Oklo and X-energy. 100x is not conceivable in the hold period.

What kills it. Molten-salt chemistry - tritium permeation, fission-product management and salt corrosion defeated every programme since Oak Ridge's MSRE, and no molten-salt reactor has ever been NRC-licensed, so the review timetable is open-ended. The Riot Platforms deal is an MOU, not a contract. Construction-stage dilution: US\$283m does not build a reactor, and the raise that funds first concrete comes from a register nursing a 93% drawdown.

What falsifies it. An NRC rejection or indefinite deferral of a topical on graphite or salt-chemistry grounds, or an NRC position that Core-unit replacement needs a new licensing framework - either pushes first concrete beyond the hold period.

Confidence. 6/10. Balance sheet, the LEU choice and an accumulating NRC record make this the best-evidenced name in its sector here; unsolved materials science caps it.

26. EMVision Medical Devices Ltd (EMV.AX)

ASX | Healthcare / Medical Devices | Australia | quoted in AUD | Moonshot sleeve rank 26 of 40, GBP 1,000, 1,089 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 5.2%. The headline below is shares x price.

Price (local)	1.735 AUD	Price (GBP)	0.918
52w high / low (local)	2.360 / 1.470	% below 52w high	-26.5%
Market cap (GBP)	GBP 85.3m	Market cap (USD m)	115
Enterprise value (GBP)	GBP 78.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-14.08	Price / book	10.76
EV / EBITDA	-17.43	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 3.4m	Cash	AUD 17.1m
Net debt	AUD -13.7m	Debt / equity (book)	0.11
Interest cover (EBITDA / interest)	-41.63	Interest cover (EBIT proxy)	-39.72
Capex, latest FY	AUD -124.2k	Capex / revenue	1.5%
Free cash flow	AUD -3.6m	Revenue (latest FY)	AUD 8.4m
Operating margin	-100.8%	Revenue growth	187.5%
Beta (Yahoo)	1.06	Beta (own, 5y daily GBP)	0.30
Upside beta (beta+)	0.12	Downside beta (beta-)	0.66
CAPM cost of equity Ke	5.0%	WACC	5.0%
Cost of debt Kd	6.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	3.7%	Return 1y (GBP)	-18.1%
Volatility 5y	55.0%	Max drawdown 5y	-66.7%
Skew / excess kurtosis	0.71 / 4.23	Sharpe / Sortino 5y	0.01 / 0.02
Judge score (asymmetry / backing / timing)	6.36 (7 / 5 / 6.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 40k	GBP 1,000 as % of a day	3.34%
10x end-state market cap	USD 1.09bn	100x arithmetically possible?	Arithmetically possible but acquisition would pre-empt it

Source: data/fundamentals_table.csv row EMV.AX; data/stats_daily.csv row EMV.AX; universe/unicorn_final.json; universe/dossiers/EMV.AX.md

Business. EMVision Medical Devices Ltd engages in the research, development, and commercialization of neurodiagnostic technology for stroke diagnosis and monitoring, and other medical imaging needs in Australia. It offers non invasive brain scanners, including emu Brain Scanner, a bedside device; and First Responder Brain Scanner, a device embedded in standard road and air ambulances to deliver pre-hospital stroke diagnosis and care to patients.

Thesis. EMVision is developing emu, a portable electromagnetic brain-imaging device for pre-hospital and bedside stroke triage - distinguishing haemorrhagic from ischaemic stroke at the point of care, which currently requires a CT scanner. A pivotal trial is running inside live code-stroke workflows at eight leading academic centres. Not flagged by the fact-checker.

Key backers. Recruiting hospitals named in the 4C (2026-07-15): Mayo Clinic Jacksonville, Mount Sinai and Mount Sinai West, Ronald Reagan UCLA, Memorial Hermann Texas Medical Center, Royal Melbourne, Liverpool Hospital NSW - with no device-related adverse events reported to date. Australian Government: A\$1.17m Industry Growth Program payment plus A\$0.4m CRC-P instalment, A\$4.6m non-dilutive funding remaining. Australian Stroke Alliance, Titan Pre-Hospital Innovation, SA Rural Support Service. Royal Flying Doctor Service aeromedical feasibility study. Directors and founders hold 33.2%.

Catalysts, 12-24 months. Full pivotal enrolment from late CY2026 / early CY2027 with sequential cohort readouts; primary endpoint is haemorrhage detection sensitivity and specificity above 80%; an FDA pre-submission meeting to add an acute ischaemia detection endpoint; launch of the Regional Benefit Study in 2H CY26; then the De Novo submission itself.

Why 10x. 10x is ~US\$1.09bn. The pivotal trial must clear 80% sensitivity and specificity, the FDA must grant De Novo clearance, and emu must sell into US stroke centres and ambulance fleets. Roughly 1,800 certified US stroke centres and tens of thousands of ambulances at a US\$100-200k device price plus service means a few thousand units is US\$300-500m of revenue; a cleared, category-creating device with that installed base supports US\$1bn+ (Inspire Medical and Shockwave both passed it shortly after clearance). 100x (US\$10.9bn) is arithmetically possible on the Shockwave path but assumes EMVision is not acquired long before.

What kills it. Binary trial risk - miss 80% sensitivity or specificity and the product is not clinically usable; the company is blinded until recruitment completes, so there is no early warning. No regulatory precedent: electromagnetic brain imaging has never been FDA-cleared, so De Novo timing and evidentiary demands are unknown. Liquidity - at roughly A\$56k of average daily traded value this is the thinnest name in the sleeve after HyTerra and Gelion; the price will gap in both directions and cannot be exited quickly.

What falsifies it. The pivotal readout showing haemorrhage-detection sensitivity or specificity below 80%. One number decides this company.

Confidence. 6/10. Nature-portfolio peer review, eight named tertiary centres, 33% insider ownership and A\$4.6m of non-dilutive funding still to come make this defensible; a single blinded endpoint and near-zero liquidity cap it.

27. Solid Power, Inc. (SLDP)

NASDAQ Global Select | Consumer Cyclical / Auto Parts | United States | quoted in USD | Moonshot sleeve rank 27 of 40, GBP 1,000, 563 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 6.5%. The headline below is shares x price.

Price (local)	2.390 USD	Price (GBP)	1.774
52w high / low (local)	8.860 / 1.980	% below 52w high	-73.0%
Market cap (GBP)	GBP 404.8m	Market cap (USD m)	545
Enterprise value (GBP)	GBP 230.5m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-4.99	Price / book	1.02
EV / EBITDA	-3.53	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 7.7m	Cash	USD 242.4m
Net debt	USD -234.7m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-3,521.92	Interest cover (EBIT proxy)	-3,738.32
Capex, latest FY	USD -11.1m	Capex / revenue	149.9%
Free cash flow	USD -44.8m	Revenue (latest FY)	USD 7.4m
Operating margin	2978.3%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	1.95	Beta (own, 5y daily GBP)	1.28
Upside beta (beta+)	1.08	Downside beta (beta-)	1.37
CAPM cost of equity Ke	9.1%	WACC	9.0%
Cost of debt Kd	0.3% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-43.8%	Return 1y (GBP)	-34.8%
Volatility 5y	88.3%	Max drawdown 5y	-93.2%
Skew / excess kurtosis	1.54 / 11.19	Sharpe / Sortino 5y	0.06 / 0.10
Judge score (asymmetry / backing / timing)	6.25 (5.5 / 7.8 / 5.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 8,607k	GBP 1,000 as % of a day	0.02%
10x end-state market cap	USD 5.83bn	100x arithmetically possible?	No - larger than any non-Chinese battery-materials company

Source: data/fundamentals_table.csv row SLDP; data/stats_daily.csv row SLDP; universe/unicorn_final.json; universe/dossiers/SLDP.md

Business. Solid Power, Inc., a research and development-stage company, develops solid-state battery technologies for the electric vehicles markets in the United States. It engages in the development of sulfide-based solid electrolytes. The company was founded in 2011 and is headquartered in Louisville, Colorado.

Thesis. Solid Power is pivoting from being a cell maker to being the merchant supplier of sulphide solid electrolyte - the scarce input every solid-state battery programme needs - with BMW building Solid Power cells on its own line and SK On having purchased and installed a Solid Power-built electrolyte pilot line. About 71% of the market cap is cash, so the market values the technology near zero. Not flagged by the fact-checker.

Key backers. BMW Group - long-standing development partner, and on 2026-07-01 BMW's Uwe Breitweg (VP Powertrain, Emission and Battery Strategy) joined the Solid Power board, the strongest single governance signal in this group. SK On - purchased and installed a Solid Power-built electrolyte pilot line, commissioning on track for end of 2026. Samsung SDI - Joint Evaluation Agreement with Samsung SDI and BMW. US Department of Energy - prior awards supporting electrolyte scale-up.

Catalysts, 12-24 months. Commissioning of the continuous sulphide electrolyte pilot line, targeted end-2026 - the key operational milestone; a first merchant electrolyte supply agreement with disclosed pricing; the outcome of the Samsung SDI/BMW joint evaluation; BMW's decision on whether solid-state enters a production platform; A-2 and

B-sample cell delivery.

Why 10x. 10x is US\$5.83bn. It requires the electrolyte business to convert from pilot to merchant supply with at least two named offtakes, and one automaker to commit solid-state to a production vehicle programme. At scale sulphide electrolyte is plausibly a US\$500m-1bn annual revenue line at chemical-specialty margins, and US\$5.8bn is 6-10x that - what sole-source battery-materials suppliers command. 100x (US\$58bn) is not credible: more than the market value of any battery-materials company outside China.

What kills it. Revenue that goes backwards - Q2 2026 revenue was negative US\$268k on changed milestones, which says the customer relationships are not yet contractual annuities. Being designed out: automakers have a long record of internalising battery chemistry once it works, and the Joint Evaluation Agreement structure makes that easy for BMW, SK On or Samsung SDI. The market may simply be right - solid-state timelines have slipped across the industry before.

What falsifies it. BMW or SK On announcing an in-house or alternative sulphide electrolyte source. That removes the merchant-supply thesis entirely and leaves a cash shell with a cell-design licence.

Confidence. 6/10. The best backer evidence in this group (a BMW VP on the board, a line installed at SK On) and a balance sheet that removes financing risk for four years, against a business that has yet to prove it can bill anyone reliably.

28. HyTerra Ltd (HYT.AX)

ASX | Energy / Oil & Gas E&P | Australia | quoted in AUD | Moonshot sleeve rank 28 of 40, GBP 1,000, 315,091 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 16.7%. The headline below is shares x price.

Price (local)	0.006 AUD	Price (GBP)	0.003
52w high / low (local)	0.027 / 0.005	% below 52w high	-77.8%
Market cap (GBP)	GBP 6.8m	Market cap (USD m)	9
Enterprise value (GBP)	GBP 4.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	n/a (not reported by Yahoo)	Price / book	0.42
EV / EBITDA	-1.93	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 39.0k	Cash	AUD 4.2m
Net debt	AUD -4.1m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-2,367.53	Interest cover (EBIT proxy)	-2,270.88
Capex, latest FY	AUD -15.8m	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	AUD -11.8m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	0.22	Beta (own, 5y daily GBP)	0.54
Upside beta (beta+)	-0.10	Downside beta (beta-)	-0.01
CAPM cost of equity Ke	6.0%	WACC	6.0%
Cost of debt Kd	4.9% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-71.0%	Return 1y (GBP)	-65.8%
Volatility 5y	100.2%	Max drawdown 5y	-98.4%
Skew / excess kurtosis	-1.39 / 43.15	Sharpe / Sortino 5y	-0.19 / -0.25
Judge score (asymmetry / backing / timing)	6.24 (8.6 / 2.6 / 5.8)	Confidence (dossier)	3 / 10
Avg daily traded value	USD 19k	GBP 1,000 as % of a day	7.09%
10x end-state market cap	USD 0.08bn	100x arithmetically possible?	Yes - the clearest 100x arithmetic in the sleeve, on the lowest probability

Source: data/fundamentals_table.csv row HYT.AX; data/stats_daily.csv row HYT.AX; universe/unicorn_final.json; universe/dossiers/HYT.AX.md

Business. Hyterra Ltd focuses on the exploration, development, and production of natural hydrogen in the United States. The company explores for hydrogen and helium resources. The company's flagship project includes the 100% owned Nemaha project covers an area of approximately 72,500 acres located in Kansas. Hyterra Ltd was incorporated in 2010 and is based in Subiaco, Australia.

Thesis. HyTerra is a sub-cent ASX natural-hydrogen explorer on the Nemaha Ridge in Kansas with ~80,000 acres, plus a 16% earn-in on the Geneva Project in Nebraska. The dossier already withdrew two backer claims the candidate record carried: Fortescue could not be found anywhere on HyTerra's site or news archive (checked 2026-09-17) and is treated as unconfirmed, and the widely circulated Mitsubishi Gas Chemical / Toyota / Eneos story concerns Gold Hydrogen, a different ASX company, not HyTerra. This is the smallest position in the sleeve at US\$7.6m and the purest lottery ticket in it.

Key backers. Thin, and this is the biggest weakness of the case: GeoKiln Energy Innovation (binding 12-month technology agreement, 2025-11-26); Natural Hydrogen Energy LLC (joint development partner, Geneva Project, operator of Hoarty NE3); insiders 53.8%. No strategic corporate, no government contract, no institutional holder above 1%. Fortescue UNCONFIRMED; Mitsubishi Gas Chemical / Toyota / Eneos belong to a different company.

Catalysts, 12-24 months. Further Nemaha drilling results; the GeoKiln controlled wellbore test - a world first and the most differentiated catalyst here; any sustained flow test demonstrating a producible rate rather than a concentration; a maiden resource estimate; helium offtake from co-produced gas; and the near-certain equity raise.

Why 10x. 10x is only ~US\$76m (about A\$107m) - in an exploration play that is one well demonstrating sustained commercial-rate hydrogen flow in Kansas, re-rating this from pre-discovery explorer to holder of a proven North American natural hydrogen field. That is a routine magnitude of move for a microcap explorer on discovery (Helix Exploration and Pulsar Helium both did comparable moves on helium). The requirement is flow rate, not concentration. 100x (US\$760m) is arithmetically the clearest case in the sleeve: a proven, developable geologic hydrogen field with 80,000 surrounding acres in a US industrial corridor plus helium credits is plausibly worth that - Koloma was privately valued in that region without a proven producing field.

What kills it. The play type may simply not work - no one has ever commercially produced natural hydrogen at scale, and concentration measurements are not flow rates. The raise: roughly twelve months of cash on 2.14bn shares at A\$0.005 means the next placement will be brutally dilutive, and sub-cent ASX microcaps have a very high base rate of permanent capital loss. The best result is not theirs - the 44% hydrogen / 13% helium Hoarty NE3 figure comes from a well HyTerra holds only 16% of, not from its 100%-owned flagship.

What falsifies it. A Nemaha well flowing hydrogen at sub-commercial rates, or not flowing at all. Concentration without rate is worth nothing.

Confidence. 3/10. Cut from the candidate's 4 because the backer set collapsed under checking: no Fortescue, and the Mitsubishi/Toyota money belongs to a competitor. What remains is a real acreage position, a genuinely novel GeoKiln test, and the best 100x arithmetic in the group - on twelve months of cash and almost no liquidity. Own it as a lottery ticket, sized as one.

29. Satellos Bioscience Inc. (MSCL.TO)

Toronto Stock Exchange | Healthcare / Biotechnology | Canada | quoted in CAD | Moonshot sleeve rank 29 of 40, GBP 1,000, 144 shares

Price (local)	13.000 CAD	Price (GBP)	6.931
52w high / low (local)	197.280 / 7.670	% below 52w high	-93.4%
Market cap (GBP)	GBP 147.1m	Market cap (USD m)	198
Enterprise value (GBP)	GBP 114.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-7.01	Price / book	3.19
EV / EBITDA	-5.97	Dividend yield	n/a (not reported by Yahoo)
Total debt	n/a (not reported by Yahoo)	Cash	CAD 61.8m
Net debt	CAD -61.8m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	n/a (not reported by Yahoo)	Interest cover (EBIT proxy)	n/a (not reported by Yahoo)
Capex, latest FY	CAD -6.0k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	CAD -16.9m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	1.14	Beta (own, 5y daily GBP)	0.17
Upside beta (beta+)	0.35	Downside beta (beta-)	0.50
CAPM cost of equity Ke	4.4%	WACC	4.4%
Cost of debt Kd	5.8% (fallback: rf + 2% (no usable interest expense / debt))	Tax rate	25.0%
Return YTD (GBP)	-86.6%	Return 1y (GBP)	-87.1%
Volatility 5y	100.0%	Max drawdown 5y	-97.0%
Skew / excess kurtosis	-1.60 / 38.48	Sharpe / Sortino 5y	0.11 / 0.16
Judge score (asymmetry / backing / timing)	6.22 (6 / 5.2 / 8.3)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 203k	GBP 1,000 as % of a day	0.66%
10x end-state market cap	USD 2.03bn	100x arithmetically possible?	Only via a sarcopenia read-across it cannot currently fund

Source: data/fundamentals_table.csv row MSCL.TO; data/stats_daily.csv row MSCL.TO; universe/unicorn_final.json; universe/dossiers/MSCL.TO.md

Business. Satellos Bioscience Inc., a clinical-stage drug development company, focuses on restoring natural muscle repair and regeneration in degenerative muscle diseases in Canada, the United States, and Australia. The company uses its proprietary discovery platform MyoReGenX, an automated microscopy system that recapitulates the muscle stem cell environment ex-vivo. It develops SAT-3247, an oral, small molecule drug candidate for the treatment of Duchenne muscular dystrophy disease.

Thesis. Satellos Bioscience is developing SAT-3247, an oral, mutation-agnostic small molecule that restores satellite-cell polarity to regenerate muscle in Duchenne muscular dystrophy - addressing all DMD genotypes rather than the subsets served by exon-skipping and gene therapy. Dual-listed TSX and Nasdaq, funded to the pivotal readout with C\$61.8m. Not flagged by the fact-checker.

Key backers. US FDA - Fast Track Designation for SAT-3247 in DMD, and previously cleared initiation of the paediatric Phase 2 BASECAMP study. February 2026 cross-border underwriting syndicate - a Canada/US public offering with full exercise of the underwriters' option, a US\$50m-plus raise showing real institutional appetite and funding the company to the Phase 2 readouts. Nasdaq listing broadening the register beyond the TSX. Muscular Dystrophy Association - interim data presented at the 2026 MDA Clinical & Scientific Conference.

Catalysts, 12-24 months. BASECAMP paediatric clinical data, on track for Q4 2026 - the placebo-controlled test of the hypothesis and by far the most important event; continued TRAILHEAD adult data (interim already shows reduced muscle fat fraction, increased total effort and improved quality of life); regulatory interactions on a

registrational path; and a possible in-licensing partnership with a large rare-disease company.

Why 10x. 10x is ~US\$2.03bn. BASECAMP must show functional benefit - not just biomarker movement - in placebo-controlled paediatric patients. A mutation-agnostic oral therapy for all DMD patients addresses roughly 20,000 people in the US and EU at orphan pricing: a US\$1-2bn peak-sales asset. Sarepta exceeded US\$10bn on a franchise serving only a fraction of genotypes, so US\$2bn for an oral serving all of them is conservative if the drug works. 100x (US\$20bn) needs SAT-3247 to succeed in DMD and translate into sarcopenia or GLP-1-associated muscle loss; the mechanism supports the read-across but Satellos cannot currently fund that second indication.

What kills it. The interim data are the weakest form of evidence - TRAILHEAD is open-label, and the blinded Q4 2026 paediatric readout can easily fail; DMD trials are notoriously hard to power on functional endpoints in small paediatric cohorts. The hypothesis has never produced an approved drug - satellite-cell polarity restoration has zero regulatory precedent. Funded to the data, not through it: C\$61.8m against a C\$11.7m quarterly loss funds the readout and about a year beyond, so a negative or ambiguous result means the next raise is at a fraction of today's price.

What falsifies it. BASECAMP missing its primary functional endpoint in Q4 2026. Everything else here is commentary.

Confidence. 6/10. Fast Track, a differentiated and genuinely mutation-agnostic mechanism, zero debt and 50% institutional ownership against one binary readout in roughly three months.

30. Robosense Technology Co., Ltd (2498.HK)

HKEX | Technology / Computer Hardware | China | quoted in HKD | Moonshot sleeve rank 30 of 40, GBP 1,000, 656 shares

Insufficient history. 682 daily bars in the five-year window (first bar 2024-01-05). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Price (local)	16.090 HKD	Price (GBP)	1.523
52w high / low (local)	46.500 / 15.600	% below 52w high	-65.4%
Market cap (GBP)	GBP 701.9m	Market cap (USD m)	946
Enterprise value (GBP)	GBP 513.7m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	19.49	Price / book	1.80
EV / EBITDA	-21.28	Dividend yield	n/a (not reported by Yahoo)
Total debt	HKD 669.5m	Cash	HKD 2.66bn
Net debt	HKD -1.99bn	Debt / equity (book)	0.02
Interest cover (EBITDA / interest)	-25.30	Interest cover (EBIT proxy)	-11.46
Capex, latest FY	HKD -96.8m	Capex / revenue	4.4%
Free cash flow	n/a (not reported by Yahoo)	Revenue (latest FY)	HKD 2.18bn
Operating margin	-17.0%	Revenue growth	23.2%
Beta (Yahoo)	0.04	Beta (own, 5y daily GBP)	0.98
Upside beta (beta+)	0.57	Downside beta (beta-)	0.62
CAPM cost of equity Ke	7.9%	WACC	7.3%
Cost of debt Kd	1.5% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-56.4%	Return 1y (GBP)	-61.6%
Volatility 5y	85.7%	Max drawdown 5y	-88.2%
Skew / excess kurtosis	-1.80 / 46.17	Sharpe / Sortino 5y	0.02 / 0.03
Judge score (asymmetry / backing / timing)	6.18 (4.8 / 7.4 / 7.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 14,143k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 9.46bn	100x arithmetically possible?	No - geopolitically constrained

Source: data/fundamentals_table.csv row 2498.HK; data/stats_daily.csv row 2498.HK; universe/unicorn_final.json; universe/dossiers/2498.HK.md

Business. Robosense Technology Co., Ltd, an investment holding company, provides LiDAR and perception solutions in the People's Republic of China and internationally. The company offers LiDAR products for advanced driver assistance systems, robotics, cleaning, industrial, logistics, public services, inspection, and other applications. It also provides AI active camera series and tool kit for robotic manipulations; and technology development services.

Thesis. RoboSense is China's largest lidar maker by volume, deliberately re-pointing from cars to robots and selling the whole humanoid sensing-and-motion bill of materials: eyes (lidar, Space Camera), skin (visuo-tactile and MEMS tactile sensors) and muscles (joint modules, actuators). The conversion is already in the numbers: H1 2026 revenue RMB 1.02bn (+30.2% YoY), total lidar shipments 719,200 units (+169.6%) and robotics lidar shipments 282,600 units (+510.4% YoY) (2026 interim results, PR Newswire, 2026-08-26), against a stock at HK\$16.09 versus a HK\$46.50 52-week high. Fact-check correction: Cainiao is described as a top-two pre-IPO shareholder, the previously claimed 'largest, 10.46%' having been withdrawn as uncorroborated.

Key backers. Cainiao (Alibaba's logistics arm) - a top-two pre-IPO shareholder, described by EqualOcean as second-largest; exact stake UNVERIFIED pending the HKEX prospectus (the prior 'largest, 10.46%' claim has been withdrawn). Shenzhen Nanshan Strategic Emerging Industry Investment (state-owned) - IPO cornerstone, USD 100m / HK\$780m. Strategic and venture investors across twelve rounds totalling RMB 3.628bn: BYD (plus a framework supply agreement), Geely, GAC, BAIC, Yutong, Xiaomi, China Mobile, TCL, Fosun RZ Capital,

Yunfeng Fund. Luxshare Precision manufacturing partnership. Insiders 21.6%.

Catalysts, 12-24 months. Commercial delivery of the MEMS tactile skin by September 2026 with SOP in Q4 2026 - the named, dated milestone; joint-module volume delivery Q4 2026; Space Camera customer deliveries by end-Q3 2026; a design win with a named humanoid OEM; robotics passing 50% of revenue with a gross-margin step-up; FY2026 results showing the loss narrowing.

Why 10x. 10x on ~US\$946m is ~US\$9.5bn: roughly RMB 8-12bn of revenue at 5-8x sales (four to six times the current run-rate) with robotics carrying 40%+ gross margins. The mechanism is concrete - one to two million humanoid and service robots a year, each taking US\$1,500-3,000 of RoboSense sensing and actuation content - and RoboSense is the only listed company selling all three layers. The tactile skin matters more than the lidar: higher ASP, far less commoditised, and a MEMS tactile sensor that does not drift is the unsolved problem in dexterous manipulation. 100x (US\$95bn) is not possible; the constraint is geopolitical, not arithmetic.

What kills it. The ADAS lidar price war eats the funding base - ASPs around RMB 1,000 and falling against a 24.5% gross margin; if automotive losses widen faster than robotics margin arrives, the IPO cash is consumed before the transition completes. Export controls or entity listing close the Western market, and China alone will not support a 10x multiple; plus HKEX settlement and geopolitical risk for a UK holder. Humanoid demand may be state subsidy rather than demand, and vision-based software may solve tactile sensing more cheaply than hardware, stranding the skin investment.

What falsifies it. Robotics revenue growth decelerating sharply in the FY2026 results while ADAS gross margin stays below 25%. That would mean the transition is not outrunning the price war, and there is no thesis left.

Confidence. 6/10. The only name in this group with real revenue, real customers (3,400+), a dated product roadmap and a top-tier strategic register - and the only one where the 10x case needs no scientific breakthrough, just execution on a visible S-curve. The geopolitical ceiling and the automotive price war are what hold it at 6.

31. General Fusion Group Ltd. (GFUZ)

NASDAQ Global Market | Industrials / Specialty Industrial Machinery | Canada | quoted in USD | Moonshot sleeve rank 31 of 40, GBP 1,000, 171 shares

Insufficient history. 243 daily bars in the five-year window (first bar 2025-09-30). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Price (local)	7.840 USD	Price (GBP)	5.819
52w high / low (local)	14.850 / 6.080	% below 52w high	-47.2%
Market cap (GBP)	GBP 469.7m	Market cap (USD m)	633
Enterprise value (GBP)	GBP 449.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	n/a (not reported by Yahoo)	Price / book	-0.60
EV / EBITDA	-12.86	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 4.3m	Cash	USD 32.0m
Net debt	USD -27.7m	Debt / equity (book)	-0.02
Interest cover (EBITDA / interest)	-9.77	Interest cover (EBIT proxy)	-5.49
Capex, latest FY	USD -442.0k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	n/a (not reported by Yahoo)	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	n/a (not reported by Yahoo)	Beta (own, 5y daily GBP)	0.79
Upside beta (beta+)	-0.61	Downside beta (beta-)	1.10
CAPM cost of equity Ke	7.1%	WACC	7.2%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-22.8%	Return 1y (GBP)	n/a (not reported by Yahoo)
Volatility 5y	72.0%	Max drawdown 5y	-51.0%
Skew / excess kurtosis	0.69 / 11.54	Sharpe / Sortino 5y	-0.06 / -0.10
Judge score (asymmetry / backing / timing)	6.15 (7.1 / 6 / 4)	Confidence (dossier)	4 / 10
Avg daily traded value	USD 1,350k	GBP 1,000 as % of a day	0.10%
10x end-state market cap	USD 9.40bn	100x arithmetically possible?	Only on a who-cracks-fusion-first outcome; low single-digit percent

Source: data/fundamentals_table.csv row GFUZ; data/stats_daily.csv row GFUZ; universe/unicorn_final.json; universe/dossiers/GFUZ.md

Business. General Fusion Group Ltd., together with its subsidiaries, engages in the research and development of commercial fusion machines in Canada. The company's core technology is based on magnetized target fusion (MTF), which combines magnetic confinement with rapid mechanical compression to achieve fusion conditions.

Thesis. General Fusion is the only listed pure-play fusion company (Magnetised Target Fusion; LM26 machine in Richmond BC), de-SPACed 2026-07-13 and trading at roughly half its debut price. CORRECTED AT FACT-CHECK: the ~US\$135m post-close cash and ~3-year runway stand, but the market cap did not. The F-1 contradicts itself (53,132,058 SVS on the cover vs 63,548,721 SVS plus 10,416,663 MVS in the ownership section, same date); yfinance's 80,723,040 shares / US\$631m matches no filing and has been removed as a Yahoo data trap. Corrected market cap ~US\$498m at the \$7.84 close, corroborated by stockanalysis.com. ADV also corrected to a 20-day ~US\$1.35m/day.

Key backers. BDC Capital (Business Development Bank of Canada, federal Crown corporation) 5.83% plus 19.14% of warrants; Government of Canada Strategic Innovation Fund (2019 agreement, amended to Jan 2025); Segra New Energy 11.86%; Pender Capital 11.70%; Chrysalix Energy III 3.55%; Cenovus Environmental Opportunity Fund; Braemar Energy. NOT verified and excluded: Bezos Expeditions, Temasek, GIC, Tobias Lutke - none appear in the 2026 F-1 ownership tables.

Catalysts, 12-24 months. LM26 crossing 1 keV (reported 0.72 keV / 8.4m degrees C in June 2026); a compression shot with measurable neutron yield; a government pilot-plant award in Canada, the US or UK; the identity and pricing of the next raise given the ~1.7:1 overhang.

Why 10x. Scarcity of listed access plus documented Canadian state backing: BDC Capital (a wholly owned subsidiary of the Business Development Bank of Canada, a federal Crown corporation) holds 3,287,483 shares and 9,076,980 warrants, and a Strategic Innovation Fund contribution agreement with the Minister of Industry has been amended through January 2025. The 10x has been recomputed honestly and is harder than the previous draft claimed: a 10x in the SHARE PRICE (\$7.84 to \$78.40) converts the Multiple Voting Shares (up to 44,810,873 SVS) and the Earnout Shares (11,978,950) and puts every warrant in the money, implying a US\$9.4bn company on that base and ~US\$13.6bn fully diluted (173.5m shares) - not US\$6.3bn. That is roughly twice the largest private fusion comparable, so a pure re-rating toward private peers delivers perhaps 3-5x; the 10x needs a genuine scientific or state-award event.

What kills it. The physics - MTF has never reached breakeven, and the company concedes liquid-liner compression has not been demonstrated at power-plant-relevant scale. Dilution: the F-1 offers 109,928,827 SVS issuable by the company and registers 102,262,208 SVS issuable on exercise/conversion for resale - ~173% of shares outstanding - with PIPE MVS converting at an assumed \$5.00 plus maximum accrued-value increase through the fifth anniversary of closing. Being second: first-of-a-kind plant not before the mid-2030s; if CFS or Helion reaches net energy first, MTF is a stranded science project.

What falsifies it. LM26 failing to reach 1 keV, or the programme being redirected away from the Lawson-criterion ladder. That is the entire near-term thesis.

Confidence. 4/10. Still raised above the candidate's implied distress case - the balance sheet is four times better than recorded (~US\$135m, not US\$32m) and the Canadian state backing is documented in the F-1's own tables. Cut one point because the correction work changed two things that matter: the true share count sits above the headline (and the filing cannot state it consistently), and a 10x on the share price implies a US\$9.4bn-13.6bn company rather than US\$6.3bn - roughly double the largest private fusion comparable. Physics risk and a ~1.7:1 dilution overhang do the rest.

32. Archer Materials Limited (AXE.AX)

ASX | Technology / Semiconductors | Australia | quoted in AUD | Moonshot sleeve rank 32 of 40, GBP 1,000, 10,219 shares

Price (local)	0.185 AUD	Price (GBP)	0.098
52w high / low (local)	0.505 / 0.175	% below 52w high	-63.4%
Market cap (GBP)	GBP 27.8m	Market cap (USD m)	37
Enterprise value (GBP)	GBP 23.3m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	n/a (not reported by Yahoo)	Price / book	4.53
EV / EBITDA	-6.72	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 15.0k	Cash	AUD 8.4m
Net debt	AUD -8.4m	Debt / equity (book)	0.00
Interest cover (EBITDA / interest)	-13.69	Interest cover (EBIT proxy)	-12.28
Capex, latest FY	AUD -86.7k	Capex / revenue	3.7%
Free cash flow	AUD -3.2m	Revenue (latest FY)	AUD 2.4m
Operating margin	-359.2%	Revenue growth	15.3%
Beta (Yahoo)	-0.16	Beta (own, 5y daily GBP)	0.44
Upside beta (beta+)	0.63	Downside beta (beta-)	0.74
CAPM cost of equity Ke	5.6%	WACC	5.6%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-42.1%	Return 1y (GBP)	-24.5%
Volatility 5y	82.5%	Max drawdown 5y	-91.2%
Skew / excess kurtosis	1.51 / 6.35	Sharpe / Sortino 5y	-0.17 / -0.28
Judge score (asymmetry / backing / timing)	6.14 (7.4 / 3.8 / 6.5)	Confidence (dossier)	4 / 10
Avg daily traded value	USD 111k	GBP 1,000 as % of a day	1.21%
10x end-state market cap	USD 0.37bn	100x arithmetically possible?	Not underwritten, but not absurd from a US\$37m base

Source: data/fundamentals_table.csv row AXE.AX; data/stats_daily.csv row AXE.AX; universe/unicorn_final.json; universe/dossiers/AXE.AX.md

Business. Archer Materials Limited operates as a quantum technologies company in Australia. The company develops semiconductor devices, including chips relevant to quantum computing, sensing, and medical diagnostics. It also engages in the technology research and development of TMR sensor project, lab-on-a-chip biosensing chip, and carbon nano onion quantum material.

Thesis. Archer Materials runs two independent semiconductor programmes on the same carbon/graphene device physics: 12CQ, a qubit processor chip designed to hold and read an electron spin at ROOM TEMPERATURE, and a graphene field-effect biochip for point-of-care blood chemistry. The quantum claim attacks cost-per-qubit rather than qubit count - the cryogenic bill of materials, not the qubit count, is what makes quantum computers cost tens of millions each. At US\$37.3m it is the cheapest position in the sleeve in absolute dollars, with A\$10.3m of cash at March 2026 and no debt. On 2026-06-30 it signed a three-year Quantum Compute Agreement with IonQ - but note the direction of the money: Archer PAYS IonQ US\$1.5m over three years for Forte/Tempo-class access and reselling rights, so this is commercial intent, not a backer.

Key backers. THE WEAKEST BACKER SET IN THE SLEEVE, stated honestly. IonQ Inc. - three-year Quantum Compute Agreement, 2026-06-30, Forte-class and forthcoming Tempo-class access plus advisory services, with Archer paying US\$250k on signing and US\$250k every six months (archerx.com.au newsroom, fetched 2026-09-17); validation of Archer as a commercial partner in-region, NOT financial backing. EPFL (Swiss Federal Institute of Technology Lausanne) - quantum magnetometer sensing collaboration. Emergence Quantum - graphene collaboration. NSW Health Commercialisation Training Showcase Best Innovation Award, 2026-06-17. No strategic corporate investor, no government contract of scale; insiders 14.6%, institutions 1.1%.

Catalysts, 12-24 months. Full 12CQ qubit demonstrator device targeted end-2026 - the binary event. Biochip beta-prototype later in 2026 (alpha already complete for blood potassium at ± 0.3 mmol/L), then CLIA validation. Quantum machine-learning fraud-detection prototype targeted end-2026, built on 280,000+ bank transaction records - the first invoiceable product. First revenue under the IonQ agreement. The near-certain 2027 equity raise and the price at which it is struck.

Why 10x. 10x = ~US\$373m (A\$525m), reachable by either of two independent routes. Quantum route: a working room-temperature spin readout at wafer scale turns 12CQ into licensable IP rather than a hardware competitor, and listed quantum names with no revenue and one credible physics claim have carried A\$400m-1bn valuations this cycle. Biochip route: a CLIA-validated point-of-care potassium chip is a real medical-device business (dialysis, cardiology, emergency medicine) and such companies routinely reach A\$300-600m with a cleared product and a partner. 100x = ~US\$3.7bn is possible only if 12CQ becomes standard IP inside other people's machines - not underwritten, but not absurd from US\$37m.

What kills it. The physics simply does not close - room-temperature spin coherence in carbon has been 'close' for years, and a 2026 demonstrator that fails to read out a single spin removes the entire quantum leg. Dilution is the base case, not a risk: A\$10.3m against a two-programme R&D cost base plus US\$500k a year payable to IonQ means raising into whatever price the demonstrator produces. And nobody is aligned - 1.1% institutional, no strategic investor and no anchor shareholder, which for an extraordinary physics claim is itself evidence a sceptic may weigh. Liquidity of ~US\$111k/day means the position cannot be exited into bad news.

What falsifies it. End-2026 passing without a 12CQ qubit demonstrator device, or the 2027 raise being struck at a discount to A\$0.185. Either outcome means the sleeve should replace this name at the annual review.

Confidence. 4/10. The asymmetry is genuine and the price is genuinely small; the backing is genuinely poor, and I have deliberately not dressed the IonQ agreement up as something it is not.

33. Molecular Partners AG (MOLN.SW)

SIX Swiss Exchange | Healthcare / Biotechnology | Switzerland | quoted in CHF | Moonshot sleeve rank 33 of 40, GBP 1,000, 343 shares

Price (local)	3.210 CHF	Price (GBP)	2.909
52w high / low (local)	3.960 / 2.660	% below 52w high	-18.9%
Market cap (GBP)	GBP 111.4m	Market cap (USD m)	150
Enterprise value (GBP)	GBP 52.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	13.73	Price / book	2.07
EV / EBITDA	-1.14	Dividend yield	n/a (not reported by Yahoo)
Total debt	CHF 3.0m	Cash	CHF 67.9m
Net debt	CHF -64.9m	Debt / equity (book)	0.03
Interest cover (EBITDA / interest)	-2,825.00	Interest cover (EBIT proxy)	-3,423.94
Capex, latest FY	CHF -714.0k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	CHF -27.0m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	0.69	Beta (own, 5y daily GBP)	0.52
Upside beta (beta+)	0.67	Downside beta (beta-)	0.80
CAPM cost of equity Ke	5.9%	WACC	5.8%
Cost of debt Kd	0.6% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-8.4%	Return 1y (GBP)	11.7%
Volatility 5y	66.7%	Max drawdown 5y	-88.5%
Skew / excess kurtosis	-0.14 / 12.37	Sharpe / Sortino 5y	-0.20 / -0.29
Judge score (asymmetry / backing / timing)	6.14 (6 / 5.8 / 7)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 65k	GBP 1,000 as % of a day	2.07%
10x end-state market cap	USD 1.52bn	100x arithmetically possible?	No - Orano co-owns the lead economics

Source: data/fundamentals_table.csv row MOLN.SW; data/stats_daily.csv row MOLN.SW; universe/unicorn_final.json; universe/dossiers/MOLN.SW.md

Business. Molecular Partners AG, a clinical-stage biotechnology company, designs and develops designed ankyrin repeat proteins therapeutics for the treatment of oncology diseases in Switzerland.

Thesis. DARPins are near-ideal vectors for short-lived alpha emitters: a tenth the mass of an antibody, antibody-like affinity, thermostable, with a tunable half-life long enough to reach tumour and short enough to clear before marrow takes the dose. Orano Med, the principal industrial producer of lead-212, chose DARPins and now co-develops TEN programmes (MP0712 212Pb/DLL3 in clinic, MP0714 225Ac, MP0726 MSLN), an endorsement from the party controlling isotope supply. Market cap CHF124.5m against CHF67.9m cash leaves an EV of roughly CHF60m for three clinical assets plus secured isotope access.

Key backers. Orano Med (10-programme Radio-DARPin co-development plus 212Pb/225Ac isotope supply, pipeline page confirmed 2026-09-17); Eckert & Ziegler (non-exclusive 225Ac access, FY25 results 2026-03); NuMeRI South Africa (first-in-human MP0712 imaging/dosimetry, 2026-01); MP0317 Phase 1 published in Nature Cancer 2026-05-01. Cash CHF 67.9m at 30-Jun-2026, H1 2026 net loss CHF 26.7m, guided into late 2027 (confirmed at fact-check).

Catalysts, 12-24 months. MP0712 cohort 2 at 105 MBq after cohort 1 completed with NO dose-limiting toxicities and only transient grade 1-2 events (2026-08-25); initial efficacy data guided within months, broader readout 2027. MP0317 TACTIC randomised Ph2, 75 patients, nine sites activated (2026-07-16). MP0533 AML path update. New platform partnership.

Why 10x. 10x = CHF1.25bn (~\$1.5bn). From a CHF60m EV, most of the return is the platform being priced at all. Requires MP0712 tumour responses in SCLC with tolerable marrow toxicity. Comparables acquired pre-approval on early data: Point \$1.4bn, Fusion \$2.4bn, RayzeBio \$4.1bn. 100x (CHF12.5bn) is not credible - the declared strategy is partnering and Orano co-owns the lead economics.

What kills it. Management destroyed most of the company's value twice already (abicipar failed at FDA with Novartis; ensovibep abandoned). Alpha-emitter renal/marrow toxicity appears only at higher doses - cohort 1 cleanliness at 75 MBq says nothing about 105 MBq+. Lead-212 supply concentrated in one partner: both moat and single point of failure. Plus ~\$65k/day SIX liquidity.

What falsifies it. A dose-limiting marrow or renal toxicity in MP0712 cohort 2 or 3 that caps dosing below a therapeutic level - that ends the Radio-DARPin thesis outright.

Confidence. 6/10. The asymmetry is real (CHF 60m EV, three clinical assets, secured 212Pb) but the execution history is genuinely bad.

34. Li-S Energy Limited (LIS.AX)

ASX | Industrials / Electrical Equipment & Parts | Australia | quoted in AUD | Moonshot sleeve rank 34 of 40, GBP 1,000, 12,603 shares

Price (local)	0.150 AUD	Price (GBP)	0.079
52w high / low (local)	0.195 / 0.115	% below 52w high	-23.1%
Market cap (GBP)	GBP 50.8m	Market cap (USD m)	68
Enterprise value (GBP)	GBP 42.9m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	n/a (not reported by Yahoo)	Price / book	3.06
EV / EBITDA	-13.64	Dividend yield	n/a (not reported by Yahoo)
Total debt	AUD 685.0k	Cash	AUD 15.6m
Net debt	AUD -14.9m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-88.70	Interest cover (EBIT proxy)	-62.54
Capex, latest FY	AUD -4.8m	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	AUD -7.2m	Revenue (latest FY)	n/a (not reported by Yahoo)
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	0.79	Beta (own, 5y daily GBP)	0.67
Upside beta (beta+)	0.81	Downside beta (beta-)	0.79
CAPM cost of equity Ke	6.5%	WACC	6.5%
Cost of debt Kd	9.8% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	2.9%	Return 1y (GBP)	4.7%
Volatility 5y	86.9%	Max drawdown 5y	-96.9%
Skew / excess kurtosis	1.40 / 10.56	Sharpe / Sortino 5y	-0.26 / -0.40
Judge score (asymmetry / backing / timing)	6.12 (6 / 6.4 / 6)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 18k	GBP 1,000 as % of a day	7.40%
10x end-state market cap	USD 0.68bn	100x arithmetically possible?	No - dilution to scale destroys the multiple

Source: data/fundamentals_table.csv row LIS.AX; data/stats_daily.csv row LIS.AX; universe/unicorn_final.json; universe/dossiers/LIS.AX.md

Business. Li-S Energy Limited engages in the development and commercialization of lithium sulphur and metal batteries in Australia. It serves electric vehicle and e-mobility; aviation and drone technology; and consumer devices and internet of things. The company was incorporated in 2019 and is headquartered in Brisbane, Australia.

Thesis. Lithium-sulphur/lithium-metal cell maker in Brisbane using boron nitride nanotubes to attack dendrite formation and polysulphide shuttling simultaneously, with captive BNNT supply via PPK. It has stopped being a science project: it shipped its first cells to US Army DEVCOM in June 2026. Li-S wins first where mass is the binding constraint - high-altitude UAVs, undersea autonomy - not where cost per kWh is. Now vertically integrated into lithium foil, a genuine Western supply gap.

Key backers. US Army DEVCOM (first cells shipped, Jun 2026; headline confirmed on lis.energy/news at fact-check, exact date not independently pinned); ARENA A\$7.8m grant (2025); Australian Industry Growth Program A\$1.7m grant - CORRECTION: dates to 2024 at the Phase 3 opening, not 2026; Kea Aerospace (high-altitude UAV), Praetorian Aeronautics (sovereign defence), MSubs (undersea endurance); PPK Group (BNNT supply). All via lis.energy/news.

Catalysts, 12-24 months. DEVCOM follow-on or programme order (the binary event); production contract from Kea, Praetorian or MSubs; semi-solid-state cell qualification after nail-penetration testing; third-party lithium foil sales; an AUKUS/ADF sovereign-capability award. ISO 9001 and international shipping approvals already in hand.

Why 10x. 10x = US\$683m (~A\$960m). Requires one defence relationship converting to a multi-year supply programme worth A\$50-100m annually plus foil revenue; ASX battery-materials names with defence offtakes have been paid 8-12x revenue, implying A\$80-120m of revenue visibility. 100x (US\$6.8bn) is not possible: a single Phase 3 line with a ~A\$6m cost base cannot become a global cell maker without raises that destroy the multiple.

What kills it. Cycle life - every Li-S programme in history has died here and nothing public proves the BNNT cell survives at defence-qualification standard. Evaluation orders usually go nowhere; DEVCOM shipments are cheap for the Army and existential for the company. The 69.5% insider block is also an overhang and governance concentration (the PPK related-party structure has drawn scrutiny), with ~US\$18k/day turnover making the position effectively unsellable in stress.

What falsifies it. Published third-party cycle-life data showing the BNNT cell degrading below ~300 cycles at useful depth of discharge would end it.

Confidence. 6/10. Defensible because the cash discipline is real and the customer is the hardest one in the world.

35. GSI Technology, Inc. (GSIT)

NasdaqGS | Technology / Semiconductors | United States | quoted in USD | Moonshot sleeve rank 35 of 40, GBP 1,000, 280 shares

Share-class note. Own-calculated market cap (shares outstanding x price) differs from Yahoo's all-share-class figure by 8.5%. The headline below is shares x price.

Price (local)	4.810 USD	Price (GBP)	3.570
52w high / low (local)	18.150 / 3.230	% below 52w high	-73.5%
Market cap (GBP)	GBP 137.2m	Market cap (USD m)	185
Enterprise value (GBP)	GBP 86.1m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-10.55	Price / book	1.85
EV / EBITDA	-5.79	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 8.2m	Cash	USD 77.0m
Net debt	USD -68.9m	Debt / equity (book)	0.07
Interest cover (EBITDA / interest)	-21.63	Interest cover (EBIT proxy)	-13.45
Capex, latest FY	USD -486.0k	Capex / revenue	1.9%
Free cash flow	USD -9.5m	Revenue (latest FY)	USD 25.2m
Operating margin	-85.3%	Revenue growth	0.5%
Beta (Yahoo)	2.10	Beta (own, 5y daily GBP)	1.75
Upside beta (beta+)	1.79	Downside beta (beta-)	1.06
CAPM cost of equity Ke	11.1%	WACC	11.0%
Cost of debt Kd	11.3% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-22.6%	Return 1y (GBP)	45.5%
Volatility 5y	152.1%	Max drawdown 5y	-81.0%
Skew / excess kurtosis	12.38 / 248.53	Sharpe / Sortino 5y	0.46 / 1.15
Judge score (asymmetry / backing / timing)	6.11 (5.5 / 7.2 / 6)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 4,856k	GBP 1,000 as % of a day	0.03%
10x end-state market cap	USD 1.85bn	100x arithmetically possible?	No - 20x is the tail case

Source: data/fundamentals_table.csv row GSIT; data/stats_daily.csv row GSIT; universe/unicorn_final.json; universe/dossiers/GSIT.md

Business. GSI Technology, Inc. designs, develops, and markets semiconductor memory solutions for networking, industrial, test equipment, medical, aerospace, and military customers in the United States, China, Singapore, Germany, the Netherlands, and internationally. It offers associative processing unit products, which focus on applications using similarity search in visual search queries for e-commerce, computer vision, drug discovery, cybersecurity, and service markets.

Thesis. Thirty-year SRAM maker funding the Associative Processing Unit, a compute-in-memory chip that does search and inference inside the memory array. The mechanism is the memory wall: inference cost is dominated by moving weights and vectors, not arithmetic, and computing in-array removes the movement rather than accelerating it - largest advantage exactly for retrieval, similarity and vector search, now the default enterprise AI pattern. Unlike compute-in-memory startups, GSI already owns the memory process and holds radiation-tolerant defence variants; Gemini-II passed radiation testing July 2026.

Key backers. US Army xTech/SBIR Phase II ~\$2.0m for a ruggedised Gemini-II edge AI platform (GlobeNewswire 2026-04-28); Hsinchu County Taiwan Smart City Phase I using Gemini-II for video analytics (2026-05-18) - first disclosed commercial Gemini-II win; Cornell research group independent peer-reviewed validation of Gemini-I; Lee-Lean Shu founder-CEO since 1995; Russell 2000 inclusion effective 2026-06-29.

Catalysts, 12-24 months. AI SDK release targeted September 2026 - software is the actual gating item, so this matters more than any silicon milestone. Hsinchu Phase II and further municipal wins. Ruggedised Army platform delivery and a sole-source-able SBIR Phase III. Hyperscaler or vector-DB evaluation. SRAM stability funding the programme without another raise.

Why 10x. 10x = \$1.85bn. Gemini-II must win a repeatable design - a programme of record or a datacentre retrieval accelerator - taking revenue from \$25m to \$200-300m at APU gross margins. At ~\$1.8bn it would be a credible compute-in-memory challenger, still below where private peers d-Matrix and EnCharge have raised. 100x (\$18.5bn) is not credible: a declining SRAM annuity plus one architecture does not become a platform vendor; 20x is the tail case.

What kills it. Gemini-I is the precedent and it failed commercially - the barrier was never silicon but the difficulty of programming a non-von-Neumann architecture; if the September 2026 AI SDK doesn't change that, nothing will. The legacy SRAM base is structurally shrinking and Q1 FY2027 revenue was already flat y/y. d-Matrix and EnCharge attack the same wall with far more capital, and further dilution is likely after the +31.9% share-count increase already absorbed.

What falsifies it. The AI SDK shipping and still producing no named, paying Gemini-II design win within 12 months.

Confidence. 6/10. The downside is unusually well protected - at \$131.7m EV you are paying ~5x sales for a business with \$68.9m net cash, so the floor is cash plus a declining annuity.

36. BioAge Labs, Inc. (BIOA)

NASDAQ | Healthcare / Drug Manufacturers - Specialty & Generic | United States | quoted in USD | Moonshot sleeve rank 36 of 40, GBP 1,000, 173 shares

Insufficient history. 498 daily bars in the five-year window (first bar 2024-09-26). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Price (local)	7.750 USD	Price (GBP)	5.752
52w high / low (local)	25.955 / 4.680	% below 52w high	-70.1%
Market cap (GBP)	GBP 264.0m	Market cap (USD m)	356
Enterprise value (GBP)	GBP 29.5m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-3.08	Price / book	0.91
EV / EBITDA	-0.38	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 2.8m	Cash	USD 318.7m
Net debt	USD -315.9m	Debt / equity (book)	0.01
Interest cover (EBITDA / interest)	-152.11	Interest cover (EBIT proxy)	-114.65
Capex, latest FY	USD -719.0k	Capex / revenue	6.9%
Free cash flow	USD -58.1m	Revenue (latest FY)	USD 10.4m
Operating margin	-1196.8%	Revenue growth	1.6%
Beta (Yahoo)	n/a (not reported by Yahoo)	Beta (own, 5y daily GBP)	0.85
Upside beta (beta+)	1.86	Downside beta (beta-)	0.03
CAPM cost of equity Ke	7.3%	WACC	7.4%
Cost of debt Kd	24.6% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-41.4%	Return 1y (GBP)	58.6%
Volatility 5y	106.1%	Max drawdown 5y	-87.9%
Skew / excess kurtosis	-4.21 / 52.88	Sharpe / Sortino 5y	0.34 / 0.43
Judge score (asymmetry / backing / timing)	6.10 (6 / 6 / 6.5)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 14,076k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 3.56bn	100x arithmetically possible?	No - it has no commercial infrastructure and partners out

Source: data/fundamentals_table.csv row BIOA; data/stats_daily.csv row BIOA; universe/unicorn_final.json; universe/dossiers/BIOA.md

Business. BioAge Labs, Inc., a clinical-stage biopharmaceutical company, develops therapeutic product candidates for metabolic diseases. The company's technology platform and differentiated human datasets that allows users to identify targets based on insights into molecular changes that drive aging.

Thesis. BioAge mines 30-year prospective human longevity cohorts - samples banked before disease onset, outcomes recorded decades later - to back-solve causal ageing biology into drug targets. You cannot buy or build such a cohort in under 30 years, which is a temporal moat. Lead asset BGE-102 is an oral, brain-penetrant NLRP3 inhibitor that produced median ~86% hsCRP reductions at 60mg and 120mg in Phase 1 with consistent IL-6 and fibrinogen falls. \$381.3m of cash and securities at 30-Jun-2026 against a \$355.8m market cap means the market values the entire pipeline below zero.

Key backers. Novartis (multi-year discovery collaboration, confirmed ongoing in the Q2 2026 results summary, 2026-08-05; the specific ageing-biology/exercise-physiology framing comes from a BioAge release that could not be re-fetched at fact-check and is flagged as partially confirmed); Eli Lilly ExploR&D (antibodies against novel metabolic ageing targets from BioAge cohorts, same release); 90.5% institutional register of specialist healthcare funds.

Catalysts, 12-24 months. QUELL-CV Phase 2 in obesity with inflammation - data anticipated LATE 2026, the near-term binary. QUELL-DME Phase 2 in diabetic macular edema - first participant dosed 2026-09-08, ~180

participants, topline H2 2027. New target nominations under Novartis/Lilly. With cash approximately equal to market cap, a strategic or activist approach is live.

Why 10x. 10x = \$3.56bn. BGE-102 must convert an 86% hsCRP reduction into a clinical endpoint - visual acuity in DME or weight/inflammation benefit adjunctive to GLP-1s - and be partnered with Novartis or Lilly at billion-dollar economics. Large pharma has repeatedly paid up for oral NLRP3; a best-in-class brain-penetrant version with Ph2 data supports \$3-5bn, and with a negative starting EV essentially all of it is pipeline recognition. 100x (\$35.6bn) would need statin-scale chronic use with BioAge owning the economics - it has no commercial infrastructure and a declared partnering strategy.

What kills it. It has already lost a lead asset - azelaprag discontinued on liver transaminitis in STRIDES Ph2. NLRP3 has a poor clinical record: the class repeatedly produces beautiful biomarker data then fails on hepatotoxicity or efficacy; hsCRP reduction is not clinical benefit. 2.4% insider ownership with a 90.5% institutional register means no internal skin in the game and violent moves on fund exits - the \$4.90-\$25.10 52-week range proves it.

What falsifies it. A clinically significant ALT/AST signal in QUELL-CV or QUELL-DME - hepatotoxicity is what has killed this class twice and killed BioAge's last lead asset.

Confidence. 6/10. A free option on a well-funded Phase 2 programme, with two shots on goal inside 18 months.

37. Airship AI Holdings, Inc. (AISP)

NASDAQ | Technology / Software - Infrastructure | United States | quoted in USD | Moonshot sleeve rank 37 of 40, GBP 1,000, 707 shares

Price (local)	1.905 USD	Price (GBP)	1.414
52w high / low (local)	7.200 / 1.685	% below 52w high	-73.5%
Market cap (GBP)	GBP 48.7m	Market cap (USD m)	66
Enterprise value (GBP)	GBP 40.0m	Trailing P/E	2.77
Forward P/E	46.41	Price / book	-7.43
EV / EBITDA	n/a (not reported by Yahoo)	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 650.7k	Cash	USD 12.4m
Net debt	USD -11.7m	Debt / equity (book)	-0.07
Interest cover (EBITDA / interest)	n/a (not reported by Yahoo)	Interest cover (EBIT proxy)	30.23
Capex, latest FY	n/a (not reported by Yahoo)	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	USD -3.4m	Revenue (latest FY)	USD 18.1m
Operating margin	-36.1%	Revenue growth	92.1%
Beta (Yahoo)	0.61	Beta (own, 5y daily GBP)	0.31
Upside beta (beta+)	0.21	Downside beta (beta-)	0.03
CAPM cost of equity Ke	5.0%	WACC	5.3%
Cost of debt Kd	30.0% (interest expense (FY) / total debt)	Tax rate	0.0%
Return YTD (GBP)	-34.1%	Return 1y (GBP)	-58.1%
Volatility 5y	129.9%	Max drawdown 5y	-88.0%
Skew / excess kurtosis	11.83 / 281.78	Sharpe / Sortino 5y	0.20 / 0.43
Judge score (asymmetry / backing / timing)	6.08 (6 / 5.4 / 7.3)	Confidence (dossier)	5 / 10
Avg daily traded value	USD 918k	GBP 1,000 as % of a day	0.15%
10x end-state market cap	USD 0.66bn	100x arithmetically possible?	Conceivable but not underwriteable

Source: data/fundamentals_table.csv row AISP; data/stats_daily.csv row AISP; universe/unicorn_final.json; universe/dossiers/AISP.md

Business. Airship AI Holdings, Inc. provides artificial intelligence (AI)-driven data management platforms in the United States. The company offers Outpost AI, which structures and analyzes data at the source using Airship AI trained models; Acropolis, an enterprise management software; and Airship Command, a suite of visualization tools that allow customers to interact with their data and devices.

Thesis. AI video and sensor data management (Outpost AI edge devices, Acropolis, Airship Command) already deployed inside DoJ and DHS environments. The moat is accreditation, not algorithms: Airship has cleared the security, procurement-vehicle and integration hurdles that keep better-funded commercial AI vendors out for years, and edge processing is the architecturally correct answer for agencies that cannot send video to a commercial cloud. A 75% gross margin on a hardware-inclusive product is the evidence the software does the work. At ~3.1x EV/sales with a \$206m validated pipeline, one programme of record changes the company.

Key backers. US DHS \$2.1m firm fixed-price BRAND-NAME-ONLY award for AI video/sensor intelligence supporting border operations (2026-04-16) - brand-name-only means DHS specified Airship rather than competing it; Fortune 100 transportation/e-commerce customer \$1.9m maintenance plus \$1.4m additional orders (2026-06-23); David V. Aguilar, former Acting Commissioner of US CBP, first Advisory Board member (2026-04-29); Victor Huang founder-CEO since 2006. NOTE: the Roth Capital Buy/\$6.50 target could NOT be re-confirmed - stockanalysis.com shows no analyst rating for AISP as at 2026-09-17.

Catalysts, 12-24 months. Conversion of the \$206m validated pipeline (up from \$165.3m at Q1) into a programme of record rather than task orders - the binary event. Delivery on the stated target of cash-flow-positive operations by end-2026. Further DHS/DoJ awards as enforcement budgets deploy. Commercial expansion beyond the single

Fortune 100 anchor. Clearing the warrant overhang so earnings become interpretable.

Why 10x. 10x = \$656m. Needs revenue of \$90-120m - five to seven times current - at 70%+ gross margin on a 6-8x sales multiple, realistically via one large multi-year federal programme (~\$50m+) plus a commercial channel. Airship's whole value is being pre-positioned for exactly that award. 100x (\$6.6bn) is arithmetically conceivable but would put it alongside the largest public-safety software vendors with decades of installed base; I would not underwrite it.

What kills it. Revenue is small, lumpy and government-dependent - Q1 FY26 \$6.35m fell to \$4.12m in Q2, a 35% sequential drop; a continuing resolution or lost recompetes removes a large share. Palantir, Axon and Motorola Solutions compete in exactly this segment with the relationships, accreditations and far more capital. SPAC legacy: warrant/PIPE overhang, reported GAAP profit is a derivative fair-value artefact (operating income negative in all last five quarters), and the stock is down 63% in a year.

What falsifies it. Two consecutive quarters of declining revenue with backlog falling - that would show the federal pipeline is conversation, not procurement.

Confidence. 5/10. The lowest confidence in this group, justified only by the price: ~3x sales for a 75%-gross-margin business inside DHS and DoJ with a near-breakeven cost base.

38. Filtronic plc (FTC.L)

LSE AIM | Technology / Communication Equipment | United Kingdom | quoted in GBp | Moonshot sleeve rank 38 of 40, GBP 1,000, 429 shares

Price (local)	2.330 GBP	Price (GBP)	2.330
52w high / low (local)	4.800 / 1.140	% below 52w high	-51.5%
Market cap (GBP)	GBP 512.5m	Market cap (USD m)	690
Enterprise value (GBP)	GBP 505.2m	Trailing P/E	115.51
Forward P/E	59.65	Price / book	12.42
EV / EBITDA	101.45	Dividend yield	n/a (not reported by Yahoo)
Total debt	GBP 5.6m	Cash	GBP 12.9m
Net debt	GBP -7.3m	Debt / equity (book)	0.14
Interest cover (EBITDA / interest)	10.46	Interest cover (EBIT proxy)	8.90
Capex, latest FY	GBP -12.8m	Capex / revenue	23.1%
Free cash flow	GBP -3.8m	Revenue (latest FY)	GBP 55.5m
Operating margin	4.5%	Revenue growth	-1.5%
Beta (Yahoo)	1.02	Beta (own, 5y daily GBP)	0.75
Upside beta (beta+)	0.34	Downside beta (beta-)	1.60
CAPM cost of equity Ke	6.9%	WACC	6.9%
Cost of debt Kd	8.4% (interest expense (FY) / total debt)	Tax rate	0.0%
Return YTD (GBP)	32.4%	Return 1y (GBP)	70.1%
Volatility 5y	63.8%	Max drawdown 5y	-55.8%
Skew / excess kurtosis	2.24 / 21.87	Sharpe / Sortino 5y	1.20 / 2.13
Judge score (asymmetry / backing / timing)	6.05 (4.3 / 8 / 7.5)	Confidence (dossier)	7 / 10
Avg daily traded value	USD 6,520k	GBP 1,000 as % of a day	0.02%
10x end-state market cap	USD 6.91bn	100x arithmetically possible?	No - impossible on the addressable revenue

Source: data/fundamentals_table.csv row FTC.L; data/stats_daily.csv row FTC.L; universe/unicorn_final.json; universe/dossiers/FTC.L.md

Business. Filtronic plc designs, develops, manufactures, and sells radio frequency (RF) technology in the United Kingdom, Europe, the Americas, and internationally. It provides Morpheus II and X2, an E-band transceiver module; Cerus, an E-band power amplifier for long range communications; Hercules II, a high power E-band transceiver; Taurus, an E-band power amplifier; Hades X2, an E-band active diplexer; tower top amplifiers; and transmit and receive modules.

Thesis. Filtronic is the sole qualified external supplier of E-band solid-state power amplifiers into SpaceX's Starlink ground gateways, plus mmWave transceivers for European defence EW and space primes, manufactured ITAR-free in the UK. The moat is qualification: SpaceX does not re-qualify a flight-critical RF part casually, and its revealed preference was to take warrants rather than vertically integrate. Order flow proves the cycle - \$62.5m SpaceX (Aug 2025), GBP13.4m defence prime (Jul 2025), EUR7m constellation (Nov 2025), \$8.0m US (Mar 2026), \$8m follow-on as recently as 2026-09-16. The only profitable, cash-generative, net-cash name in the sleeve.

Key backers. SpaceX (strategic customer since Apr 2024, \$62.5m largest-ever order 2025-08-26, warrant holder - but the exact warrant mechanics of 10,949,079 at 92.8p vesting to 15% could NOT be re-verified this session: Proactive returned HTTP 403 and Investigate shows no warrant RNS); Leonardo (named defence prime contract 2025-05-29); Airbus (constellation satellites contract 2025-05-15); undisclosed European defence prime GBP13.4m (2025-07-14) plus further contract 2026-04-01; UK Space Agency GBP1.2m NSIP award for a 550W Ka-band SSPA.

Catalysts, 12-24 months. A SpaceX IPO in 2026 and the gateway capex cycle it funds. Further SSPA follow-on orders (one landed 2026-09-16). V-band and Ka-band product launches in calendar 2026. Conversion of Leonardo and the European prime into repeat volume. FY2027 interims showing margin recovery on the new Sedgefield facility. ~90% of FY2026 revenue already covered by contracted orders.

Why 10x. 10x = ~\$6.9bn (GBP5.2bn). On GBP55.5m of revenue that needs GBP400-600m at an 8-12x sales multiple, roughly a ten-fold volume increase, achieved by becoming the mmWave front-end supplier to multiple constellations (Starlink, Kuiper, IRIS2) AND European EW simultaneously - several fabs' worth of capacity, so expect capex and dilution en route. 100x (\$69bn) is impossible: even at 100% share of every Western gateway amplifier the revenue does not support it.

What kills it. Total customer concentration - SpaceX can insource E-band amplifiers at any point and the warrants would be all shareholders got. The customer appears to hold the pricing power: EBITDA fell from GBP8.7m to GBP5.1m on flat revenue while SpaceX volumes expanded; management calls it deliberate investment, a sceptic calls it margin transfer. Warrant dilution of up to 15%, plus lumpy gateway build-out where a single pause shows in the next half immediately.

What falsifies it. A second consecutive period of falling EBITDA on flat or rising revenue, which would confirm Filtronic is a price-taking subcontractor rather than a bottleneck supplier.

Confidence. 7/10. The highest confidence here - a real business with a record order book, 90% revenue visibility and net cash - but correspondingly the lowest asymmetry, since 10x from \$700m is a genuine stretch.

39. BTQ Technologies Corp. (BTQ)

NasdaqGM | Technology / Software - Infrastructure | Canada | quoted in USD | Moonshot sleeve rank 39 of 40, GBP 1,000, 512 shares

Insufficient history. 245 daily bars in the five-year window (first bar 2025-09-26). Every statistic on this row is arithmetic on that sample, not a statistical estimate.

Price (local)	2.630 USD	Price (GBP)	1.952
52w high / low (local)	16.000 / 2.090	% below 52w high	-83.6%
Market cap (GBP)	GBP 287.1m	Market cap (USD m)	387
Enterprise value (GBP)	GBP 280.6m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	-23.78	Price / book	28.20
EV / EBITDA	-6.64	Dividend yield	n/a (not reported by Yahoo)
Total debt	USD 1.1m	Cash	USD 9.8m
Net debt	USD -8.7m	Debt / equity (book)	0.03
Interest cover (EBITDA / interest)	-934.79	Interest cover (EBIT proxy)	-403.16
Capex, latest FY	USD -21.3k	Capex / revenue	n/a (not reported by Yahoo)
Free cash flow	USD -3.0m	Revenue (latest FY)	USD 0.00
Operating margin	0.0%	Revenue growth	n/a (not reported by Yahoo)
Beta (Yahoo)	-15.50	Beta (own, 5y daily GBP)	2.52
Upside beta (beta+)	2.76	Downside beta (beta-)	2.30
CAPM cost of equity Ke	14.2%	WACC	14.2%
Cost of debt Kd	5.6% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-48.7%	Return 1y (GBP)	n/a (not reported by Yahoo)
Volatility 5y	143.5%	Max drawdown 5y	-84.7%
Skew / excess kurtosis	1.68 / 5.30	Sharpe / Sortino 5y	0.13 / 0.24
Judge score (asymmetry / backing / timing)	6.02 (5.6 / 5.4 / 8)	Confidence (dossier)	5 / 10
Avg daily traded value	USD 11,963k	GBP 1,000 as % of a day	0.01%
10x end-state market cap	USD 3.87bn	100x arithmetically possible?	No - larger than the listed quantum complex combined

Source: data/fundamentals_table.csv row BTQ; data/stats_daily.csv row BTQ; universe/unicorn_final.json; universe/dossiers/BTQ.md

Business. BTQ Technologies Corp. engages in the development of computer-based technology related to post-quantum cryptography for applications in blockchain and related technologies in Canada, Taiwan, Australia, and the United States.

Thesis. Full-stack 'trusted quantum': neutral-atom emulation/digital-twin/control via the completed QPerfect acquisition, the QSSN post-quantum security stack, and Bitcoin Quantum - a quantum-safe Bitcoin fork using NIST-standardised signatures. The mechanism is a forced migration with a deadline: 6-7m BTC sit in address types with public keys already exposed on-chain, NIST has finalised FIPS 203/204/205, and national timetables run to 2030-2035. BTQ is the only listed company that has shipped a working quantum-safe Bitcoin testnet - 75+ miners, 300,000+ blocks, 150+ open-source contributors - rather than a whitepaper.

Key backers. QPerfect SA Strasbourg acquisition COMPLETED 2026-07-08, now wholly owned (PRNewswire) - though the EUR18.59m consideration and EUR5.67m earnout in the candidate record are NOT in the completion release and remain unconfirmed; South Korean institutions iM Bank, Finger, Danal, Daou Data and Keypair, with QSSN selected for one of Korea's first bank-led KRW stablecoin PoCs (Q2 2026 update) - the most commercially meaningful relationships BTQ has; SDT Inc., qBraid and Scaleway as QPerfect distribution partners; Quadrilium Ventures and A.G.P. Canada in ~\$39m of funding; French government foreign-investment clearance (2026-07-01). Testnet milestone figures confirmed verbatim at fact-check, though they were reached in Q1 and

reported retrospectively.

Catalysts, 12-24 months. Bitcoin Quantum mainnet targeted late 2026 pending security audits and infrastructure readiness - the dominant catalyst and dominant risk. Conversion of a South Korean PoC, particularly the KRW stablecoin work, into a paid production contract - the first real revenue. First QPerfect emulation/control revenue from other quantum hardware builders. Any exchange, custodian or central bank adopting the fork. A financing, which is near-certain and will likely arrive before any of the above.

Why 10x. $10x = \sim \$3.87bn$. Bitcoin Quantum must launch and be adopted by a meaningful share of miners plus at least one large custodian or exchange, with QSSN converting a flagship enterprise contract to recurring revenue. At $\sim \$3.9bn$ the market would value BTQ as critical infrastructure for the post-quantum migration of digital assets - an adoption-and-narrative re-rate, not an earnings one, since revenue would still be small. $100x (\$38.7bn)$ is not credible: larger than almost every listed quantum and crypto-infrastructure company combined, on a share count that must keep rising.

What kills it. The Bitcoin community rejects the fork - forks without social consensus die (BCH, BSV) and Bitcoin governance is structurally hostile to outside commercial actors; 75 testnet miners is not social consensus. FINANCING RISK IS ACUTE - C\$9.73m cash at 30-Jun-2026 against a C\$5.94m quarterly burn is ~ 1.6 quarters; BTQ must issue equity into a stock already 81% below its high, and each raise compounds the share count against any 10x. Priced entirely on narrative - \$387m market cap on \$0 revenue means it de-rates with the quantum complex regardless of execution, with founder control at 35%.

What falsifies it. Bitcoin Quantum mainnet slipping past 2026 while cash falls below one quarter of burn - that combination forces a deeply dilutive rescue financing before the thesis can be tested.

Confidence. 5/10. Held for the genuine uniqueness of a live quantum-safe Bitcoin testnet and the named Korean financial relationships, not for the balance sheet. This is the most fragile name in the group and should be sized as a true lottery ticket.

40. Bango PLC (BGO.L)

LSE (AIM) | Technology / Software - Infrastructure | United Kingdom | quoted in GBp | Moonshot sleeve rank 40 of 40, GBP 1,000, 1,600 shares

Price (local)	0.625 GBP	Price (GBP)	0.625
52w high / low (local)	1.290 / 0.550	% below 52w high	-51.6%
Market cap (GBP)	GBP 48.2m	Market cap (USD m)	65
Enterprise value (GBP)	GBP 64.4m	Trailing P/E	n/a (not reported by Yahoo)
Forward P/E	18.46	Price / book	2.94
EV / EBITDA	29.01	Dividend yield	n/a (not reported by Yahoo)
Total debt	GBP 21.6m	Cash	GBP 5.3m
Net debt	GBP 16.3m	Debt / equity (book)	0.98
Interest cover (EBITDA / interest)	1.30	Interest cover (EBIT proxy)	-3.52
Capex, latest FY	GBP -15.3m	Capex / revenue	29.3%
Free cash flow	GBP -593.9k	Revenue (latest FY)	GBP 52.2m
Operating margin	13.2%	Revenue growth	-7.9%
Beta (Yahoo)	0.67	Beta (own, 5y daily GBP)	0.55
Upside beta (beta+)	0.60	Downside beta (beta-)	0.95
CAPM cost of equity Ke	6.0%	WACC	6.0%
Cost of debt Kd	7.9% (interest expense (FY) / total debt)	Tax rate	25.0%
Return YTD (GBP)	-28.6%	Return 1y (GBP)	-35.9%
Volatility 5y	49.0%	Max drawdown 5y	-78.5%
Skew / excess kurtosis	-0.55 / 32.19	Sharpe / Sortino 5y	-0.34 / -0.49
Judge score (asymmetry / backing / timing)	5.97 (5.5 / 6.2 / 6.8)	Confidence (dossier)	6 / 10
Avg daily traded value	USD 146k	GBP 1,000 as % of a day	0.92%
10x end-state market cap	USD 0.64bn	100x arithmetically possible?	No - the take rate does not support it

Source: data/fundamentals_table.csv row BGO.L; data/stats_daily.csv row BGO.L; universe/unicorn_final.json; universe/dossiers/BGO.L.md

Business. Bango PLC, together with its subsidiaries, develops, markets, and sells technology that enables the marketing and sale of products and services to mobile phone users. It operates through two segments, Payment and Subscription. The company offers Digital Vending Machine, a lifecycle subscription bundling platform that connects subscription providers with brands that distribute digital services to consumers, such as Telcos, retailers, banks, etc.

Thesis. Bango's Digital Vending Machine is the entitlement, provisioning, revenue-share, tax and cancellation plumbing under subscription bundling, plus the direct carrier billing rails. It is a genuine two-sided network - content owners integrate once to reach every operator, operators integrate once to reach every content owner - and network effects at a GBP48m market cap are rare. Four independent blue-chip validations in six months of 2026: KDDI, Telin/LMT, LinkedIn and Turkcell. It trades at 1.24x EV/sales on an 84.5% gross margin.

Key backers. All four 2026 wins verified exactly against Investigate: LinkedIn (Microsoft) partners with the DVM 2026-05-07; KDDI Japan selects the DVM 2026-01-28; Turkcell selects Bango for subscription bundling 2026-06-25; Telin and LMT select Bango to power DCB 2026-04-21. Founder Ray Anderson with ~26% insider holding. New Chair appointed 2026-05-12 plus further directorate change 2026-06-26.

Catalysts, 12-24 months. Interim results (notice given 2026-09-03) - the test of whether the four 2026 wins convert to ARR and of the debt trajectory. An AI-subscription bundling deal would be the category-defining win, since AI subscriptions are exactly the category now needing bundled distribution. US telco adoption, the largest unpenetrated market. Net-debt reduction. New Chair bedding in.

Why 10x. 10x = ~\$643m (GBP480m). Needs annualised recurring revenue of roughly \$120-150m at 30%+ EBITDA margins on a 4-5x sales multiple, with the DVM becoming the default bundling layer for 30-40 large operators rather than a dozen, and at least one hyperscaler AI subscription distributed through it. The network effect makes adoption non-linear - the last twenty operators are far easier than the first twenty. 100x (\$6.4bn) is not possible: Bango earns a share of a share of a subscription and the take rate does not support a top-tier payments multiple.

What kills it. THE BALANCE SHEET - gross debt tripled from GBP6.89m (Dec-2024) to GBP21.59m (Dec-2025) against GBP5.31m cash, giving net debt equal to 34% of market cap; a covenant issue or rescue rights issue at these prices is the realistic path to permanent loss and would arrive before any good outcome. Disintermediation - Amazon, Google, Apple or a telco consortium could build the plumbing themselves. The gap between 'selected by' and cash can be a year or more; demand the ARR bridge, not the logo slide.

What falsifies it. Interim results showing net debt rising again with flat recurring revenue - that would prove the 2026 wins are not converting and the balance sheet is running out of time.

Confidence. 6/10. Genuinely cheap against revenue and margin, with verifiable blue-chip adoption; the leverage is what stops this being an 8.

8. Implementing this on interactive investor

The rates below were checked against ii's own pages, twice. The 11 September build carried every ii figure with a warning that the commission and FX numbers came from third-party summaries. The 16 September refresh opened **ii's own charges page** directly ([ii.co.uk/our-charges](https://www.ii.co.uk/our-charges), accessed 2026-09-16), which changed the answer: international (non-US) dealing is **not** GBP 3.99, and the Core plan cannot hold a GBP 200,000 portfolio. The moonshot rebuild then checked five specific venue questions raised by the new sleeve against ii's own market pages on 2026-09-17 (section 6.8), which added the telephone-dealing line below. Stamp-duty rules and the exemptions list are carried over from the earlier pass and were not re-checked; verify everything against ii's current published rates before placing an order.

8.1 Dealing costs

Item	ii figure as recorded	Source	Verified?
Platform fee	Core GBP 5.99/month but capped at a GBP 100,000 portfolio ; Plus GBP 14.99/month, unlimited size; Premium GBP 39.99/month	https://www.ii.co.uk/our-charges	Yes - ii's own charges page
UK and US shares, ETFs, investment trusts	GBP 3.99 per online trade (Core and Plus); GBP 2.99 (Premium)	https://www.ii.co.uk/our-charges	Yes - ii's own charges page
Other international shares (Canada, Europe, ASX, HKEX, SGX and the rest)	GBP 9.99 (Core), GBP 7.99 (Plus), GBP 5.99 (Premium)	https://www.ii.co.uk/our-charges	Yes - corrects the previous build, which had used GBP 3.99 for all international dealing
FX conversion charge	0.75% (Core); 0.75% on the first GBP 50,000 then 0.25% above (Plus); 0.25% (Premium)	https://www.ii.co.uk/our-charges	Yes - more granular than the previous build's flat ~0.75%
Gilt dealing, secondary market	GBP 3.99 per online trade	https://www.ii.co.uk/analysis-commentary/everything-you-need-know-about-investing-gilts-ii528437	ii's own guide
Gilt dealing, new auction	No commission and no bid/offer spread when applying via ii for a new gilt auction	https://www.ii.co.uk/analysis-commentary/everything-you-need-know-about-investing-gilts-ii528437	ii's own guide
Stamp duty (SDRT)	0.5% on purchases of most UK-listed company shares, auto-applied on CREST settlement; AIM shares, gilts, corporate bonds and ETFs exempt	https://www.ii.co.uk/learn/tax/stamp-duty	Carried over, not re-checked this pass

Source: universe/market_params.json -> ii_dealing_costs, as at 2026-09-16

A GBP 200,000 portfolio cannot sit on the Core plan. ii's Core plan states a GBP 100,000 portfolio ceiling, so the realistic assumption for this model is the **Plus plan at GBP 14.99 a month** (GBP 179.88 a year, 0.09% of capital), or splitting the money across account types. That is a new finding in this refresh and it was not in the previous build's cost table.

What that costs on this portfolio, on the Plus plan.

Cost	Basis	Amount	% of GBP 200,000
Dealing commission	37 UK/US lines at GBP 3.99 + 17 other international lines at GBP 7.99	GBP 283.46	0.142%
Telephone-dealing surcharge	GBP 49 each on the two lines ii deals by phone only (SEYE.ST, MOLN.SW)	GBP 98.00	0.049%

Cost	Basis	Amount	% of GBP 200,000
FX conversion	0.75% on the first GBP 50,000 and 0.25% above, on GBP 58,922 of non-sterling-quoted lines	GBP 397.31	0.199%
Stamp duty (SDRT)	0.5% on GBP 25,661 of UK main-market shares (RR.L, AZN.L, SHEL.L, HSBA.L only - all three UK moonshot lines are AIM and exempt)	GBP 128.31	0.064%
Total, day one		GBP 907.08	0.45%
Platform fee	Plus plan, GBP 14.99 a month	GBP 179.88 a year	0.09% a year
Ongoing fund charges	weighted TER 0.20% on the core funds	GBP 315 a year	0.16% a year

Source: own calculation from data/combined_positions.csv invested_gbp and the ii rates in universe/market_params.json -> ii_dealing_costs | arithmetic is ours; the rate assumptions are ii's own published figures as at 2026-09-16

On the Core plan the same build would cost about GBP 986 (GBP 317 of commission, GBP 98 of telephone fees and GBP 442 of FX), but the Core plan's GBP 100,000 ceiling rules it out. So roughly **0.45% of capital to build it and about 0.25% a year to hold it** including the platform fee, before bid-offer spread, which is not modelled anywhere in this report and on the smallest moonshot lines will be the largest single cost of all.

Three things about this table are new in the moonshot build. First, the **GBP 49 telephone fee** appears at all: Smart Eye (Nasdaq Stockholm) and Molecular Partners (SIX) can only be dealt on ii by phone, so each round trip on those two lines costs GBP 56.98 on the Plus plan against GBP 7.99 for every other international line - five to ten times as much. Second, the **stamp-duty bill fell** from GBP 138 to GBP 128, because all three UK moonshot lines (GELN.L, FTC.L, BGO.L) are AIM and therefore exempt, where the previous sleeve held two UK main-market shares. Third, the **FX bill rose** from GBP 389 to GBP 397, because more of the sleeve is now quoted outside sterling. Net, the build costs about GBP 92 more than the sleeve it replaces, almost all of it the two telephone trades.

And one cost is still not in the table at all. The bid-offer spread on a sub-cent ASX line, or on a name turning over USD 19k a day, can exceed everything above put together on that position. Nine of the forty trade under USD 250k a day. Use limit orders.

8.2 The ISA and SIPP wrapper

Nothing in this report is tax advice, and the right wrapper depends on circumstances this analysis knows nothing about. Three facts are worth stating plainly:

- Inside a stocks-and-shares **ISA** or a **SIPP**, there is no UK capital gains tax and no UK dividend tax on anything held. The annual ISA subscription limit is well below GBP 200,000, so a portfolio this size cannot be wrapped in one year. A SIPP has its own annual and lifetime rules and locks the money up until pension age.
- **Stamp duty is not avoided by a wrapper.** The 0.5% SDRT on UK main-market share purchases applies inside an ISA exactly as it does outside one.
- **Withholding tax on foreign dividends is not avoided either**, and it is not modelled in any return figure in this report. The Irish-domiciled UCITS ETFs here reclaim some of it at fund level; a directly-held US stock in an ISA does not have the same treatment as one in a SIPP.

The practical consequence for sequencing is that if only part of the GBP 200,000 can be wrapped, the lines with the highest expected income are the ones that benefit most from being inside it.

8.3 Gilt dealing

ii's own gilts guide states that the most popular gilts can be traded online at GBP 3.99 per online trade, while less liquid issues have to be dealt by telephone on 0345 607 6001. Applying for a new gilt at auction through ii carries no commission and no bid-offer spread. **Neither gilt is held in the headline portfolio**, so this matters only if variant A (section 4.6) is adopted, in which case T36N would be a 16.4% position - GBP 26,270 of nominal, 271 units at the 2026-09-16 clean price of 96.75. Nothing in the moonshot rebuild touches this.

Two unresolved items were recorded and are still unresolved: ii's **minimum online deal size for gilts** was not found on any page retrieved, and T56's online tradability was confirmed only via a third-party gilt-data site rather

than ii's own order screen.

One live development sits directly on top of this section. The Bank of England was reported to be about to stop actively selling 20- and 30-year gilts from the Asset Purchase Facility, with the announcement expected alongside the **17 September MPC decision** - the day after this report's as-of date - and the DMO has already cut long conventional gilt sales to about GBP 23bn, roughly 9% of its GBP 252.1bn remit, with a pilot switch auction on 24 September. If that supply relief is confirmed it re-prices T56 directly, and none of it is in the model.

8.4 Order sizes and residual cash

Shares are whole numbers, so the model floors every position and leaves the remainder in cash.

	Allocated	Invested	Residual cash	Residual as %
Core sleeve	GBP 160,000	GBP 159,287	GBP 713	0.45%
Moonshot sleeve	GBP 40,000	GBP 39,962	GBP 38	0.10%
Total	GBP 200,000	GBP 199,249	GBP 751	0.38%

Source: data/combined_positions.csv

The largest individual rounding residuals are:

Ticker	Price (GBP)	Allocated	Shares	Invested	Left over
BRK-B	385.80	2,276	5	1,929	347
PLTR	129.40	8,000	61	7,893	107
SPCX	111.98	4,000	35	3,919	81
BRNT.L	73.38	8,274	112	8,219	55
SGLN.L	62.70	23,726	378	23,701	25
AZN.L	121.36	1,724	14	1,699	25

Source: data/combined_positions.csv

Berkshire is the awkward one: at GBP 386 a share, a GBP 2,276 target buys 5 shares and leaves GBP 347 unspent. **The moonshot sleeve rounds far better than the sleeve it replaces** - GBP 38 of unspent cash across all forty lines against GBP 182 before - precisely because the names are smaller and cheaper: the most expensive line is Aeluma at GBP 9.97 a share and the cheapest is HyTerra at GBP 0.003, where a GBP 1,000 ticket buys 315,091 shares. That is a rounding convenience and a liquidity warning in the same number. None of this is material at the portfolio level, but it is worth knowing that the model's weights and the executable weights are not identical.

8.5 Rebalancing

The sleeve rule changed with the mandate, and the change is deliberate. The original brief's rule was 20/80 annually: once a year, restore the sleeve split to 20% moonshot and 80% core, and inside the sleeve restore every name to GBP 1,000. The moonshot mandate replaces that for Sleeve A with **let winners run, and trim only on size**.

- **Inside the moonshot sleeve there is no annual rebalance to GBP 1,000.** A position is trimmed only when it exceeds **3% of the total GBP 200,000 portfolio** - GBP 6,000, about 6x from cost - and then only by half. The reasoning is in section 6.6: rebalancing a moonshot sleeve is the mechanism by which you sell the one position that was supposed to pay for the other thirty-nine. Selling half at 6x banks GBP 3,000, three times the original stake, and leaves GBP 3,000 compounding. There are no stop losses; a 70% drawdown in an individual name is an expected event in this sleeve, not a signal.
- **The sleeve is reviewed annually on thesis, not on price.** Each of the forty dossiers names its own falsifier, reproduced in the per-name blocks in section 7.2, so the annual review is a test rather than an opinion: if the named event has happened, the name is replaced; if it has not, a bad year is not a reason to sell.
- **Note the mismatch between that rule and the statistics in this report.** The sleeve index used in sections 5 and 6 is equal-weight and **daily rebalanced**, which is the opposite of the rule adopted. It is used because the combined-book covariance needs one return series, and it flatters the sleeve relative to the buy-and-hold basket that would actually be held. Nothing in this report models the let-winners-run rule's actual path.

- **Restoring the 20/80 split between sleeves is a separate question and is still open.** With the moonshot sleeve at 26.5% volatility against the core's 12.1%, the sleeve is the one that will drift. Restoring the split in a good year means selling the sleeve that just worked, which sits awkwardly beside the let-winners-run rule inside it. This report does not choose; it notes that the 3%-of-portfolio trim already provides a size discipline, and that applying a second, calendar-based one on top would partly undo it.
- **Within the core sleeve**, drift back to the optimised weights is the natural rule, but the optimiser's own weights move a long way with the expected-return input (section 4.6), so re-solving every year and trading to the new answer would generate turnover far in excess of the value of the information. The 16 September refresh is the cleanest available evidence for that: five days of extremely eventful news moved the headline portfolio by **1.5% one-way turnover**, which on the cost table above is about GBP 12 of commission and would be swamped by spread. The moonshot rebuild is cleaner still - replacing 34 of 40 sleeve names moved the core by GBP 0.0034. A tolerance band, rebalancing only when a line is more than, say, 5 percentage points from target, costs far less than a calendar rule.
- **What a rebalance costs.** 54 lines is up to GBP 567 of commission if every line is traded on both sides, plus GBP 196 if the two telephone-only lines are touched, plus the FX charge on every non-sterling line, plus 0.5% stamp duty on any UK main-market share re-bought. A full annual rebalance of everything would cost more than the initial build. That argues for tolerance bands rather than calendar dates - and, in the moonshot sleeve, for the trim rule rather than either.

8.6 Position caps

The caps that shaped this portfolio, and which would have to be re-imposed on any rebalance:

Cap	Limit	Where it binds today
Single equity/bond/commodity/gilt fund line	20%	Not binding (SMGB.L largest at 16.6%)
Single trend line	15%	Not binding (DBMG.L at 9.5%)
Single stock	5%	Binding on RR.L, PLTR, SHELL, HSBA.L
All single stocks	25%	Binding
Commodities	5-20%	Binding at 20%
Bonds (bond ETFs + gilts)	10-40%	Binding at the 10% floor
Equity ETFs	30% minimum	Not binding (35.5%)
SPCX policy weight	2.5%, fixed	By decision, not by the optimiser
Moonshot sleeve, per name	GBP 1,000 at cost, trimmed by half above 3% of the total portfolio	No position is near it today; every line is 0.5% of the book

Source: data/analysis_results.json -> config.constraints, portfolios.Moderate12.weight_by_bucket

Six of the eight constraints are binding or at a boundary, and the 12% volatility target binds at exactly 12.00% on top of them. That is a portfolio being shaped by its constraints rather than by its inputs, which is worth knowing: it means the caps are doing more of the work than the covariance matrix is, and it is also why three days of news moved so little.

9. Limitations and disclaimers

This is educational analysis, not personalised investment advice. Nobody who worked on this project is a licensed adviser. This is not a personal recommendation and takes no account of anyone's circumstances, tax position, time horizon or capacity for loss. Every number in it is a model output, not a forecast. **The moonshot sleeve is designed on the explicit expectation that most of its forty positions lose money and a meaningful number go to zero;** it is sized at 20% of the book so that total failure costs 20% of the portfolio, and it should not be held by anyone who would not accept writing the whole GBP 40,000 off.

The full limitation list carried in the analysis output, restated onto this build's figures (data/analysis_results.json -> limitations stores the same list; its strings were repatched on 2026-09-17 to this build's figures - 67 bars for SPCX, a 2026-09-17 .info capture date against the 2026-09-16 close, and the 16 September gilt yields - so the stored list and the list below now agree). Items 12 to 16, 18 and 19 are specific to this build and are not in the stored list:

1. Educational analysis. Not personalised investment advice. Nobody on this project is a licensed adviser, and every number below is a model output, not a forecast.
2. Estimation error dominates. The 5y sample means are extreme (several core lines annualise above 20%) because 2021-2026 contained a gold bull market, an AI capex boom and an oil spike. The 50/50 blend with CAPM halves that but does not remove it: the headline blended expected return of 15.4% for the core sits far above the CAPM-only estimate of 7.2%, and the truth is unlikely to be at the top of that range.
3. The optimiser is a maximiser of estimation error (Michaud 1989). Small changes in the expected-return vector move the weights a long way; the frontier and the four portfolios should be read as a map, not as a recommendation of one point on it.
4. The 12% volatility constraint binds on the ex-ante Ledoit-Wolf volatility of the optimised 97.5% of the core. Including the 2.5% SPCX policy weight at its 67-bar pairwise covariance raises the estimate to about 12.8%; SPCX's row and column of that matrix are not reliable.
5. SPCX has 67 daily bars in the window. Its beta (2.95), volatility (90%) and every other statistic are reported for completeness and are not estimates. Adding three bars in this refresh moved its beta by 0.12, which is what a non-converged statistic looks like.
6. No true 10-year backtest exists for this universe (see the log): the headline is 5 years of monthly rebalancing with no dealing costs, no 0.5% stamp duty on the UK single stocks, no ii FX charge on the non-GBP lines and no bid-offer spread. Real-world returns would be lower.
7. Backward-looking betas and correlations are computed against VWRP.L in GBP. ^FTAS stops on 2026-07-22 in Yahoo's feed, so the All-Share betas use a shorter sample - and in this refresh ^FTAS could not be refetched at all (Yahoo returned a single row), so its column was reinstated from the 11 September build's pristine cache. ^FTAS is reporting-only: it is not in the optimiser universe, the covariance matrix or any expected return.
8. The moonshot sleeve's 5y statistics are survivorship-biased, and more so than the sleeve they replace: the 40 names were selected in September 2026 from companies that still exist and are still listed on an ii-dealt exchange. A sleeve chosen in 2021 on the same mandate would contain names that have since failed or delisted, and these names are smaller and earlier-stage than the previous forty, i.e. drawn from exactly the part of the distribution where non-survival is most common. The bias cannot be quantified from this data.
9. The two individual gilts and the long-duration bond ETF get zero weight because the 50/50 blend averages a forward-looking redemption yield (T36N 5.30%, T56 5.87%) with a 5-year sample return of -4.0% and -11.1% a year - the once-in-a-generation repricing from ~1% to ~5.3% yields that has ALREADY happened. For a gilt held to maturity the yield is the better estimate, and the expected_return_sensitivity block shows what the portfolio looks like when the optimiser is fed the yield/CAPM vector alone. This is a judgement call the Inputs tab should expose, not a settled answer.
10. Beta asymmetry is not robust: it changes sign with the measurement frequency AND with the treatment of SPCX. Measured on the optimised 97.5% alone (daily), the headline core has beta+ 0.66 below beta- 0.75, i.e. negative asymmetry; including the 2.5% SPCX policy weight at its own 67-bar betas - the portfolio as actually held, and the convention the beta fields use - gives beta+ 0.85 against beta- 0.77, asymmetry +0.08. SPCX's 67-bar beta+ of 8.34 is not an estimate, so that positive figure rests on 67 days of one 2.5% line. On

MONTHLY data the same weights give beta+ above beta- for the optimised part as well. Read the positive-skew claim as unproven either way rather than as established in either direction.

11. Concentration: SMGB.L (VanEck Semiconductor UCITS ETF) takes 16.6% of the headline core weight and contributes 35.8% of its ex-ante risk. The 20% single-ETF cap is what stops the largest ETF line going further; a tighter cap is a reasonable Inputs-tab experiment, priced as variant B in section 4.6.
12. The moonshot sleeve's five-year statistics are close to meaningless as a forecast and are reported because the combined-book covariance needs a single return series, not because they predict anything. A payoff distribution designed as "most go to zero, one or two go up 10x-30x over five to ten years" is not described by an annualised mean and a standard deviation. 27 of the 40 names are live on the sleeve index's first date and 13 start later, so the early index is a smaller and different portfolio from the late one; 6 of the 40 have fewer than three years of daily history in the window.
13. The sleeve index used throughout this report is equal-weight and DAILY REBALANCED. The rule actually adopted is the opposite - let winners run, trim half only above 3% of the total portfolio - so the index is not the sleeve's realised path and flatters it relative to a buy-and-hold basket. Nothing in this report models the adopted rule.
14. Liquidity is not modelled anywhere. The sleeve's median name trades about USD 2.4m a day, nine names trade under USD 250k a day, and the thinnest (LIS.AX, HYT.AX) would have seen a GBP 1,000 ticket exceed 30% of that day's entire turnover on roughly a quarter to a third of sessions in the last three months. Positions are fillable; exit at size is assumed nowhere and should not be assumed by a reader.
15. One tradeability item is open: GUTS (Fractyl Health) received a Nasdaq delisting determination on 2026-09-10, stayed by a Hearings Panel request on 2026-09-11. If the panel is not persuaded the line moves to OTC, which is outside the ii mandate entirely. A second, ALMDT.PA on Euronext Growth Paris, is carried on a best-evidence reading of ii's country-level market disclosure that could not be confirmed either way.
16. Six dossiers carry load-bearing claims explicitly labelled UNVERIFIED (Caribou's 2023 Pfizer investment, Filtronic's SpaceX warrant mechanics, Clarity's Fast Track designation, Median Technologies' sensitivity and specificity figures, BTQ's QPerfect consideration, RoboSense's Cainiao stake). They are labelled in place rather than deleted.
17. Reference portfolios are not constrained. EqualWeight breaches the 25% all-stocks cap (10 stocks x 2.71% plus SPCX at 2.50% = 29.6%) and is kept only as a comparison.
18. A level-base defect in the total-return level of every return series was found by blind verification and fixed on 2026-09-17: the level was compounded from 1.0 on the date of the first return rather than the day before it, so total_return and cagr omitted the first period. Mean, volatility, Sharpe, Sortino, skew, kurtosis, VaR, CVaR, the betas and the correlations are unaffected. Every total return, CAGR and drawdown-path figure in this report is on the corrected basis, and the archived comparison figures in section 6.8 are restated onto it before any old-versus-new difference is formed.
19. The 77x77 combined covariance is made positive semi-definite by eigenvalue clipping applied to the whole matrix, so replacing 34 sleeve rows also perturbs the core block (largest cell change 0.0115, on the SPCX row; largest implied core volatility change 0.85 pp, on SGLN.L). The OPTIMISER's covariance is untouched to 3.1e-16, so no core weight moves, but the combined book's ex-ante volatility, betas and risk split rest on a matrix whose core block has shifted marginally.
20. Listing-region weights record the LISTING VENUE, not economic exposure. Most core lines are LSE-listed UCITS wrappers holding global assets.
21. yfinance's .info was captured live on 2026-09-17, not at the 2026-09-16 close used everywhere else. Every price-linked field taken from it is restated onto the as-of close: trailing and forward P/E, P/B and EV/EBITDA are rescaled by close(as-of)/price-at-fetch, and market cap is rebuilt as shares outstanding x the GBP price per share (Yahoo's own figure is in the company's reporting currency, which made the HKD and AUD lines out by 19x and 1.9x). The unrestated Yahoo values are kept beside them as *_yahoo_live / market_cap_yahoo_raw_local_ccy. What cannot be restated is the NON-price side of .info - shares outstanding, EPS, book value, debt, cash, margins and growth rates are whatever Yahoo held on 2026-09-17, and the statement-derived fields are the latest reported fiscal year or four quarters.
22. Three price-data faults (DBMG.L quote-scale and placeholder bars, SMEA.L quote-convention break, WLDS.L placeholder bar) are repaired in the pipeline and the repaired panel is written back over

data/prices_gbp_daily.csv, _monthly, _weekly and data/prices_local_daily.csv. The unrepaired fetch output is preserved as *_raw.csv. Anything that reads a *_raw.csv file, or a copy of the price CSVs taken before this run, will reproduce the fabricated DBMG.L bars and the two placeholder prints.

23. Twenty-five LSE lines had their 2026-09-16 bar reconstructed because Yahoo's daily history omits it (section 2.2). Twenty-three take their close from the official figure recorded in the universe files; two - the gilt proxies IGLT.L and GLTL.L - fall back to the last intraday 60-minute bar, which is typically within 0.1% of the official close but is not it.
24. The common calendar is the union of VWRP.L's trading days and the ^FTSE/^FTAS sessions VWRP.L is missing. That is wider than the SPEC's literal wording and is adopted deliberately, because it recovers real LSE sessions. One consequence: VWRP.L has no own bar on 2025-10-24 or 2026-03-06, so the benchmark return there is a forward-filled 0.00% and every beta regression contains those two manufactured observations.
25. The expected returns are five-year estimates. They respond to news only through the risk-free rate, the ERP and the yield inputs, and in the 16 September refresh the first two did not move at all. Nothing in the 11-16 September news flow (section 1.4) enters the optimiser directly, and nothing at all about the market changed in the 17 September moonshot rebuild - the only input that moved was the 40-name Sleeve A list.

Source: data/analysis_results.json -> limitations; data/REFRESH_DIFF.md sections 1 and 7

And the limitations specific to the two variants in section 4.6:

1. Educational analysis, not personalised advice; every figure is a model output.
2. Both variants are single-change sensitivities on one already noisy problem. A mean-variance optimiser maximises estimation error (Michaud 1989), so the weight differences are larger than the confidence one should attach to them.
3. Variant A replaces the two gilts' blended mu with their redemption yield but leaves every other line (including the HY/EM credit ETFs and IBTM.L, which already use a yield-based estimate #2) on the 50/50 blend, so it is a sensitivity, not a coherent all-forward-looking expected-return vector.
4. A gilt's redemption yield is the forward return only if it is HELD TO MATURITY and coupons are reinvested at that yield; the covariance matrix still uses the ETF-proxy history (IGLT.L for T36N, GLTL.L for T56), whose duration and volatility are those of a rolling fund, not of a single bond pulling to par.
5. Variant B's 10% single-fund cap is applied to every fund bucket including the trend fund DBMG.L, whose separate 15% group limit is left in place but is no longer binding. With four lines pinned at the cap the residual is a near-degenerate choice between two near-substitute high-yield ETFs, and a tiny input move flips it.
6. The 12% target is the ex-ante Ledoit-Wolf volatility of the optimised 97.5%; adding the 2.5% SPCX policy weight at its 67-bar pairwise covariance raises the full-core figure (reported alongside as ex_ante_vol_full_core_incl_spcx).
7. Backtests are 5 years of monthly rebalancing with no dealing costs, no stamp duty, no FX charge and no spread; they apply the variant weights to past returns and are not a track record.
8. Beta/beta+/beta- are as-held sums including SPCX at its own 67-bar betas, which are not estimates; the _excl_spcx fields are given for every portfolio.
9. Turnover is one-way (0.5 x sum of absolute weight changes) against the PUBLISHED headline weights in data/weights.csv.

Source: data/analysis_variants.json -> limitations

9.1 The five that matter most

If only five things survive from this report, make them these.

1. **Estimation error dominates everything.** The gap between the historical-only and the structural expected return for this core sleeve is 23.6% against 7.2%. The headline blend sits in between. The optimiser is a maximiser of that error, and switching estimator moves 83% of the portfolio.
2. **The backtest is not a track record.** Weights chosen in 2026 applied to 2021-2026 returns, with no costs, no stamp duty, no FX charge and no spread. Real returns would be lower, and the window contained no crisis.

3. **The moonshot sleeve is a bet on a mechanism this report cannot test.** Its history is survivorship-biased and cannot be de-biased from this data; its five-year statistics describe the wrong horizon; most of its forty positions are expected to lose money; and nothing here establishes that a dossier-defended forty is more likely to contain a 10x than any other forty. What the report can show is the arithmetic: a total loss of the sleeve costs 20% of the book, and a single 30x pays for three-quarters of it.
4. **The positive-skew objective was not met** in the core sleeve on daily data, the headline reading of the beta asymmetry depends on a 2.5% line with 67 days of history whose beta moved 0.12 when three bars were added, and the moonshot sleeve's own beta asymmetry is now clearly negative at -0.13 against +0.003 for the sleeve it replaced.
5. **Three instruments rest on proxy history** (DBMG.L on DBMF, T36N on IGLT.L, T56 on GLTL.L), three price series had to be repaired before they were usable, 25 lines needed their final bar reconstructed because Yahoo omitted it, and a level-base defect in every series' total return was found and fixed on 2026-09-17. The model is only as good as those repairs, which are documented in section 2.2 and at the end of section 4, and reproducible from the shipped files.

Appendix

A1. Full per-asset statistics, all 80 series

5y daily, GBP total return, 2021-09-16 to 2026-09-16 (or the instrument's own shorter history). n is the number of daily observations; rows with fewer than three years are marked *.

Ticker	Sleeve	n	Mean (ann.)	CAGR	Vol	Sharpe	Sortino	Skew	Exc. kurt.
IITU.L	core	1,261	22.1%	21.6%	22.6%	0.81	1.17	-0.16	1.46
VMID.L	core	1,261	4.6%	3.5%	15.2%	0.06	0.08	0.07	2.47
IWQU.L	core	1,261	11.0%	10.1%	16.7%	0.44	0.63	0.06	2.39
VWRP.L	core	1,261	11.6%	11.4%	13.0%	0.60	0.85	-0.38	3.05
CSP1.L	core	1,261	13.4%	13.2%	14.5%	0.67	0.95	-0.28	2.72
ISF.L	core	1,261	12.6%	12.6%	12.5%	0.71	0.99	-0.60	4.70
EMIM.L	core	1,261	9.9%	8.9%	16.5%	0.37	0.52	-0.21	2.79
WLDS.L	core	1,259	8.2%	7.5%	15.6%	0.28	0.39	-0.38	2.29
SMEA.L	core	1,155	9.3%	9.9%	14.0%	0.40	0.56	-0.11	3.48
IJPA.L	core	1,261	10.6%	9.2%	19.1%	0.36	0.52	0.09	1.96
SMGB.L	core	1,261	33.2%	32.2%	32.4%	0.91	1.31	-0.16	1.21
XDWH.L	core	1,261	6.3%	5.2%	15.7%	0.16	0.23	-0.03	2.04
SGLN.L	core	1,261	20.0%	20.3%	17.1%	0.95	1.37	-0.32	4.84
SSLN.L	core	1,261	22.0%	15.1%	39.4%	0.46	0.64	-0.69	18.10
BRNT.L	core	1,261	30.1%	26.2%	37.0%	0.71	1.01	-0.32	3.01
COPA.L	core	1,261	11.7%	8.4%	26.6%	0.30	0.41	-1.37	21.07
ICOM.L	core	1,261	13.6%	12.7%	18.0%	0.55	0.77	-0.28	2.25
YCA.L	core	1,261	15.6%	9.4%	36.4%	0.33	0.48	0.25	2.28
RR.L	core	1,261	60.5%	68.3%	41.6%	1.37	2.20	1.19	13.13
PLTR	core	1,261	58.6%	44.3%	67.2%	0.82	1.32	1.17	8.49
SPCX *	core	67	13.7%	-23.1%	90.3%	0.11	0.17	0.51	2.52
AZN.L	core	1,261	13.1%	10.6%	24.7%	0.38	0.53	-0.24	5.75
SHEL.L	core	1,261	24.7%	24.3%	24.4%	0.86	1.21	-0.47	3.39
HSBA.L	core	1,261	37.3%	40.8%	24.7%	1.35	1.93	-0.51	4.35
ULVR.L	core	1,261	5.2%	3.3%	19.7%	0.07	0.10	-0.15	13.75
MSFT	core	1,261	15.1%	11.5%	29.2%	0.39	0.58	0.52	6.20
ASML.AS	core	1,261	21.6%	14.4%	40.2%	0.44	0.64	-0.18	2.84
SIE.DE	core	1,261	18.4%	14.7%	30.8%	0.48	0.69	0.06	3.20
BRK-B	core	1,261	14.9%	14.0%	18.8%	0.59	0.85	-0.10	3.27
IHYU.L	core	1,261	4.5%	4.1%	10.0%	0.08	0.12	0.33	2.90
IHYG.L	core	1,261	2.7%	2.4%	7.4%	-0.15	-0.22	0.46	3.38
GHYS.L	core	1,244	3.3%	3.0%	6.1%	-0.08	-0.11	-0.11	3.81
IEMB.L	core	1,261	1.9%	1.3%	11.5%	-0.16	-0.23	0.36	3.81
IBTM.L	core	1,261	-0.9%	-1.3%	9.4%	-0.50	-0.70	0.28	2.38
T36N	core	1,261	-4.0%	-4.5%	10.3%	-0.76	-1.10	1.25	16.12
T56	core	1,261	-11.1%	-12.2%	19.7%	-0.75	-1.12	2.16	31.97
DBMG.L	core	1,258	10.3%	9.7%	15.3%	0.43	0.61	-0.16	2.76
IPWR	moonshot	1,261	9.7%	-23.6%	87.3%	0.07	0.11	1.48	8.22
VYGR	moonshot	1,261	31.4%	0.7%	80.5%	0.34	0.57	2.15	21.31
AL3.AX	moonshot	1,261	40.6%	-4.0%	97.0%	0.38	0.64	1.64	9.79
ONWD.BR	moonshot	1,236	-10.9%	-26.3%	64.6%	-0.23	-0.39	2.85	24.62
LTBR	moonshot	1,261	61.2%	5.6%	108.9%	0.53	0.91	1.70	9.35
ARA.TO	moonshot	1,200	54.6%	16.2%	89.2%	0.57	0.89	0.50	6.87
SLX.AX	moonshot	1,261	45.5%	20.8%	72.7%	0.57	0.85	-0.05	4.21

Ticker	Sleeve	n	Mean (ann.)	CAGR	Vol	Sharpe	Sortino	Skew	Exc. kurt.
NYXH	moonshot	1,261	-29.5%	-44.4%	74.9%	-0.44	-0.63	-0.23	17.51
ALMU	moonshot	896	105.8%	41.6%	121.7%	0.84	1.42	1.47	13.17
WBT.AX	moonshot	1,261	30.4%	0.0%	78.4%	0.34	0.52	0.48	2.36
VUL.AX	moonshot	1,261	-13.3%	-31.3%	69.9%	-0.24	-0.36	0.51	2.75
IDN	moonshot	1,261	6.8%	-22.9%	82.8%	0.04	0.06	2.63	44.03
GELN.L	moonshot	1,208	-25.9%	-38.8%	71.9%	-0.41	-0.73	4.98	67.39
EDIT	moonshot	1,261	-20.3%	-46.1%	93.7%	-0.26	-0.41	2.55	34.57
SEYE.ST	moonshot	1,261	-3.9%	-21.6%	63.0%	-0.12	-0.17	-0.32	18.34
QSI	moonshot	1,261	26.7%	-38.6%	141.3%	0.16	0.35	8.67	150.88
CRBU	moonshot	1,261	-25.8%	-45.9%	84.5%	-0.35	-0.51	0.56	7.07
ALCLS.PA	moonshot	1,261	-0.2%	-36.9%	114.0%	-0.03	-0.08	14.99	400.33
GUTS *	moonshot	662	-50.4%	-70.3%	110.9%	-0.49	-0.67	-0.97	20.12
PRME	moonshot	986	-6.2%	-37.2%	90.6%	-0.11	-0.17	0.74	2.47
PRQR	moonshot	1,261	22.5%	-25.3%	101.9%	0.18	0.31	4.95	126.66
ALMDT.PA	moonshot	1,261	7.2%	-24.3%	87.0%	0.04	0.07	2.82	27.41
CU6.AX	moonshot	1,261	34.1%	8.9%	72.1%	0.42	0.65	0.57	1.96
LAES	moonshot	840	74.9%	-39.8%	172.6%	0.41	0.83	3.62	26.77
IMSR *	moonshot	219	-93.2%	-78.1%	109.2%	-0.89	-1.33	0.84	1.97
EMV.AX	moonshot	1,261	4.4%	-10.0%	55.0%	0.01	0.02	0.71	4.23
SLDP	moonshot	1,261	9.4%	-24.5%	88.3%	0.06	0.10	1.54	11.19
HYT.AX	moonshot	1,261	-15.0%	-53.6%	100.2%	-0.19	-0.25	-1.39	43.15
MSCL.TO	moonshot	1,261	14.9%	-43.0%	100.0%	0.11	0.16	-1.60	38.48
2498.HK *	moonshot	682	5.5%	-32.2%	85.7%	0.02	0.03	-1.80	46.17
GFUZ *	moonshot	243	-0.9%	-23.4%	72.0%	-0.06	-0.10	0.69	11.54
AXE.AX	moonshot	1,261	-10.4%	-35.0%	82.5%	-0.17	-0.28	1.51	6.35
MOLN.SW	moonshot	1,261	-9.5%	-27.5%	66.7%	-0.20	-0.29	-0.14	12.37
LIS.AX	moonshot	1,253	-18.8%	-42.5%	86.9%	-0.26	-0.40	1.40	10.56
GSIT	moonshot	1,261	74.2%	-2.6%	152.1%	0.46	1.15	12.38	248.53
BIOA *	moonshot	498	39.8%	-35.7%	106.1%	0.34	0.43	-4.21	52.88
AISP	moonshot	1,261	30.1%	-27.3%	129.9%	0.20	0.43	11.83	281.78
FTC.L	moonshot	1,261	80.0%	83.3%	63.8%	1.20	2.13	2.24	21.87
BTQ *	moonshot	245	22.4%	-51.5%	143.5%	0.13	0.24	1.68	5.30
BGO.L	moonshot	1,261	-12.8%	-22.3%	49.0%	-0.34	-0.49	-0.55	32.19
^FTAS	benchmark	1,218	8.1%	7.3%	12.6%	0.34	0.47	-0.59	4.62
^GSPC	benchmark	1,261	12.8%	11.7%	18.9%	0.48	0.69	0.05	5.38
ACWI	benchmark	1,261	12.3%	11.2%	18.1%	0.47	0.68	0.12	4.87

Ticker	Max DD	Beta VWRP	Beta FTAS	beta+	beta-	Up capt.
IITU.L	-28.0%	1.48	0.77	1.47	1.35	1.72
VMID.L	-28.8%	0.82	1.01	0.88	0.84	0.65
IWQU.L	-18.7%	0.98	0.88	0.98	0.91	0.97
VWRP.L	-17.6%	1.00	0.74	1.00	1.00	1.00
CSP1.L	-20.8%	1.06	0.69	1.08	1.04	1.13
ISF.L	-12.7%	0.65	0.99	0.71	0.74	0.59
EMIM.L	-21.0%	0.93	0.76	0.89	0.93	0.97
WLDS.L	-21.6%	1.07	0.90	1.09	1.13	0.97
SMEA.L	-15.8%	0.84	0.98	0.92	0.86	0.62
IJPA.L	-19.5%	0.80	0.80	0.78	0.76	0.70
SMGB.L	-36.2%	1.91	1.10	1.80	1.71	2.69
XDWH.L	-18.3%	0.55	0.68	0.58	0.53	0.47

Ticker	Max DD	Beta VWRP	Beta FTAS	beta+	beta-	Up capt.
SGLN.L	-24.9%	0.05	0.03	-0.03	0.07	0.63
SSLN.L	-64.9%	0.42	0.49	0.29	0.53	1.16
BRNT.L	-33.4%	0.11	0.27	-0.41	0.58	0.36
COPA.L	-26.4%	0.48	0.68	0.32	0.61	0.53
ICOM.L	-28.5%	0.09	0.20	-0.21	0.27	0.24
YCA.L	-48.2%	0.99	1.00	0.95	1.06	0.79
RR.L	-55.1%	1.31	1.69	1.41	1.45	2.36
PLTR	-76.1%	1.44	0.67	1.42	1.00	2.59
SPCX *	-48.9%	2.95	-1.43	8.34	1.40	n/a
AZN.L	-26.7%	0.43	0.92	0.53	0.49	0.17
SHEL.L	-20.6%	0.47	0.91	0.20	0.87	0.30
HSBA.L	-22.0%	0.85	1.28	0.89	1.09	1.19
ULVR.L	-28.4%	0.11	0.46	0.27	0.01	0.21
MSFT	-34.1%	0.67	0.30	0.65	0.47	1.02
ASML.AS	-46.4%	1.92	1.42	1.82	1.65	2.38
SIE.DE	-37.0%	1.41	1.43	1.43	1.58	1.52
BRK-B	-20.6%	0.34	0.42	0.34	0.34	0.50
IHYU.L	-10.5%	0.28	0.26	0.27	0.23	0.19
IHYG.L	-15.5%	0.24	0.25	0.27	0.21	0.16
GHYS.L	-14.7%	0.27	0.27	0.29	0.28	0.23
IEMB.L	-15.2%	0.30	0.33	0.34	0.21	0.33
IBTM.L	-16.3%	-0.03	-0.18	-0.03	-0.14	-0.02
T36N	-33.5%	0.04	0.08	0.05	-0.05	0.15
T56	-53.0%	0.07	0.15	0.02	-0.06	0.25
DBMG.L	-31.6%	0.13	0.05	-0.10	0.20	0.10
IPWR	-84.5%	1.08	0.69	0.68	1.41	1.69
VYGR	-81.8%	0.52	0.55	0.71	0.64	2.18
AL3.AX	-75.5%	0.40	0.10	0.16	0.58	1.82
ONWD.BR	-83.7%	0.74	0.67	0.44	0.86	-0.27
LTBR	-83.2%	1.44	1.16	1.52	1.55	5.70
ARA.TO	-84.3%	0.35	0.19	0.63	0.24	2.60
SLX.AX	-66.5%	1.09	0.74	1.70	0.92	4.10
NYXH	-95.7%	0.58	0.55	-0.03	0.55	1.05
ALMU	-60.5%	0.73	0.35	-0.42	0.87	1.61
WBT.AX	-86.8%	0.94	0.64	1.04	1.01	2.93
VUL.AX	-87.4%	1.17	1.08	1.70	1.55	2.10
IDN	-83.3%	0.71	0.70	0.88	0.60	0.82
GELN.L	-96.4%	0.12	0.13	-0.45	0.44	-0.17
EDIT	-98.3%	1.23	0.88	1.20	0.97	0.68
SEYE.ST	-84.6%	1.22	0.99	0.93	1.31	1.27
QSI	-92.6%	1.77	1.38	2.40	1.23	3.49
CRBU	-97.2%	1.01	0.68	0.52	0.78	2.12
ALCLS.PA	-92.3%	1.84	1.61	2.09	1.34	2.51
GUTS *	-97.1%	1.18	1.19	1.06	1.59	-0.26
PRME	-95.3%	1.29	1.26	0.14	2.18	2.49
PRQR	-92.7%	0.96	0.89	0.58	0.88	1.68
ALMDT.PA	-93.2%	1.18	1.13	0.89	1.46	-0.23

Ticker	Max DD	Beta VWRP	Beta FTAS	beta+	beta-	Up capt.
CU6.AX	-85.0%	0.80	0.78	0.80	1.35	3.29
LAES	-98.5%	1.01	0.26	0.65	0.60	1.37
IMSR *	-73.6%	3.59	2.51	2.88	2.72	n/a
EMV.AX	-66.7%	0.30	0.29	0.12	0.66	0.78
SLDP	-93.2%	1.28	1.01	1.08	1.37	2.00
HYT.AX	-98.4%	0.54	0.78	-0.10	-0.01	-1.03
MSCL.TO	-97.0%	0.17	0.33	0.35	0.50	1.21
2498.HK *	-88.2%	0.98	0.90	0.57	0.62	0.16
GFUZ *	-51.0%	0.79	0.06	-0.61	1.10	-0.97
AXE.AX	-91.2%	0.44	0.38	0.63	0.74	0.53
MOLN.SW	-88.5%	0.52	0.47	0.67	0.80	0.96
LIS.AX	-96.9%	0.67	0.65	0.81	0.79	0.37
GSIT	-81.0%	1.75	1.32	1.79	1.06	3.40
BIOA *	-87.9%	0.85	0.78	1.86	0.03	2.68
AISP	-88.0%	0.31	0.23	0.21	0.03	2.26
FTC.L	-55.8%	0.75	0.72	0.34	1.60	3.00
BTQ *	-84.7%	2.52	1.92	2.76	2.30	2.36
BGO.L	-78.5%	0.55	0.60	0.60	0.95	0.07
^FTAS	-13.6%	0.69	1.00	0.74	0.78	0.50
^GSPC	-22.0%	0.62	0.46	0.52	0.49	1.09
ACWI	-18.4%	0.60	0.55	0.51	0.49	1.00

Ticker	Down capt.	YTD	1y	3y	5y
IITU.L	1.31	21.3%	30.1%	114.9%	167.9%
VMID.L	0.94	9.9%	15.1%	43.0%	18.2%
IWQU.L	1.03	10.3%	16.9%	49.8%	61.0%
VWRP.L	1.00	12.6%	19.3%	61.8%	71.2%
CSP1.L	1.08	11.3%	17.8%	63.7%	86.1%
ISF.L	0.27	10.2%	18.4%	57.6%	79.7%
EMIM.L	1.12	20.5%	29.1%	67.8%	50.9%
WLDS.L	1.22	14.3%	19.1%	49.9%	43.2%
SMEA.L	0.48	8.5%	16.9%	51.3%	59.7%
IJPA.L	0.65	22.0%	26.0%	60.6%	54.8%
SMGB.L	1.92	66.1%	104.5%	246.6%	308.3%
XDWH.L	0.61	5.0%	16.2%	16.5%	27.3%
SGLN.L	-0.24	0.9%	20.5%	109.9%	148.4%
SSLN.L	0.25	-33.4%	16.6%	106.4%	101.4%
BRNT.L	-1.33	109.1%	100.2%	73.6%	225.4%
COPA.L	0.37	11.4%	36.0%	51.9%	46.8%
ICOM.L	-0.30	35.9%	47.8%	45.1%	82.8%
YCA.L	0.42	-5.5%	7.2%	15.1%	60.3%
RR.L	-0.22	26.3%	29.3%	557.8%	1214.5%
PLTR	-0.65	-2.0%	6.6%	923.5%	593.4%
SPCX *	n/a	n/a	n/a	n/a	n/a
AZN.L	-0.27	-10.5%	3.3%	22.7%	66.4%
SHEL.L	-0.99	34.1%	40.2%	60.4%	199.2%
HSBA.L	-0.68	33.4%	55.8%	213.0%	435.8%
ULVR.L	0.26	-2.1%	-13.9%	6.0%	16.0%

Ticker	Down capt.	YTD	1y	3y	5y
MSFT	0.76	2.0%	-0.9%	37.6%	76.7%
ASML.AS	2.31	49.5%	102.9%	147.4%	99.8%
SIE.DE	1.32	9.3%	15.0%	103.9%	102.8%
BRK-B	-0.04	3.4%	5.1%	31.8%	90.8%
IHYU.L	0.30	0.9%	3.1%	15.9%	22.5%
IHYG.L	0.36	-1.5%	0.7%	17.6%	12.9%
GHYS.L	0.42	0.4%	1.7%	22.1%	16.0%
IEMB.L	0.73	-0.1%	3.2%	18.6%	6.2%
IBTM.L	0.37	-2.9%	-1.6%	2.1%	-6.1%
T36N	0.83	-1.7%	0.5%	9.8%	-21.0%
T56	1.58	-6.2%	-1.9%	-2.6%	-48.3%
DBMG.L	-0.39	14.1%	25.8%	22.0%	60.3%
IPWR	3.01	25.8%	-25.1%	-64.5%	-73.3%
VYGR	1.75	-22.4%	-30.0%	-68.2%	-4.5%
AL3.AX	1.26	-17.9%	-37.9%	88.3%	-23.3%
ONWD.BR	1.43	-38.7%	-45.9%	-33.8%	n/a
LTBR	4.02	-48.8%	-53.8%	34.8%	14.0%
ARA.TO	0.90	66.6%	126.0%	762.6%	n/a
SLX.AX	3.02	-45.9%	20.2%	36.9%	124.9%
NYXH	4.58	-66.3%	-71.9%	-82.5%	-94.5%
ALMU	-3.14	-21.8%	-9.2%	270.5%	n/a
WBT.AX	1.72	-35.9%	31.9%	-23.0%	-8.1%
VUL.AX	4.65	-46.0%	-35.1%	-24.2%	-86.0%
IDN	1.22	-64.4%	-56.4%	-11.4%	-71.3%
GELN.L	2.68	-20.5%	-28.9%	-40.8%	n/a
EDIT	3.56	32.1%	0.5%	-72.6%	-95.6%
SEYE.ST	2.64	-36.6%	-37.6%	13.8%	-70.3%
QSI	5.81	-33.7%	-38.9%	-70.6%	-91.6%
CRBU	5.74	-21.4%	-34.3%	-81.9%	-95.4%
ALCLS.PA	4.27	-73.1%	-59.1%	-32.3%	-90.5%
GUTS *	3.46	-74.4%	-44.6%	n/a	n/a
PRME	7.09	-13.6%	-26.1%	-78.1%	n/a
PRQR	0.74	-8.0%	-14.7%	7.8%	-77.2%
ALMDT.PA	-0.04	-21.5%	11.9%	-17.0%	-77.4%
CU6.AX	2.67	-34.2%	-27.8%	85.2%	53.8%
LAES	-29.66	-42.1%	-23.3%	-57.7%	n/a
IMSR *	n/a	-30.5%	n/a	n/a	n/a
EMV.AX	1.81	3.7%	-18.1%	15.2%	-38.6%
SLDP	3.32	-43.8%	-34.8%	4.5%	-75.4%
HYT.AX	1.74	-71.0%	-65.8%	-70.5%	-97.8%
MSCL.TO	1.73	-86.6%	-87.1%	-84.2%	-94.2%
2498.HK *	0.21	-56.4%	-61.6%	n/a	n/a
GFUZ *	n/a	-22.8%	n/a	n/a	n/a
AXE.AX	2.38	-42.1%	-24.5%	-57.6%	-88.7%
MOLN.SW	2.73	-8.4%	11.7%	-29.5%	-79.8%
LIS.AX	3.24	2.9%	4.7%	-29.6%	n/a
GSIT	-0.03	-22.6%	45.5%	67.0%	-6.9%

Ticker	Down capt.	YTD	1y	3y	5y
BIOA *	4.50	-41.4%	58.6%	n/a	n/a
AISP	0.21	-34.1%	-58.1%	-83.5%	-79.8%
FTC.L	-1.31	32.4%	70.1%	1356.2%	1971.1%
BTQ *	n/a	-48.7%	n/a	n/a	n/a
BGO.L	1.46	-28.6%	-35.9%	-62.7%	-72.7%
^FTAS	0.52	6.5%	13.4%	39.6%	39.9%
^GSPC	1.11	10.3%	15.2%	56.0%	73.6%
ACWI	1.01	12.1%	17.7%	59.7%	69.8%

Source: data/stats_daily.csv (37 core + 40 moonshot + 3 benchmark series; the CSV's own bucket field still reads 'unicorn' for the sleeve rows) | upside/downside capture are monthly; the final monthly observation is a 10-trading-day stub (2026-08-31 to 2026-09-16) counted as a month, which matters for the capture ratios and monthly VaR only

A2. Expected returns, all three estimates

Ticker	Historical 5y	Estimate 2	Blend (optimiser input)	Estimate-2 kind	Beta used	Note
IITU.L	22.1%	9.9%	16.0%	capm	1.48	rf + beta_own * ERP
VMID.L	4.6%	7.2%	5.9%	capm	0.82	rf + beta_own * ERP
IWQU.L	11.0%	7.8%	9.4%	capm	0.98	rf + beta_own * ERP
VWRP.L	11.6%	7.9%	9.8%	capm	1.00	rf + beta_own * ERP
CSP1.L	13.4%	8.2%	10.8%	capm	1.06	rf + beta_own * ERP
ISF.L	12.6%	6.5%	9.5%	capm	0.65	rf + beta_own * ERP
EMIM.L	9.9%	7.6%	8.7%	capm	0.93	rf + beta_own * ERP
WLDS.L	8.2%	8.2%	8.2%	capm	1.07	rf + beta_own * ERP
SMEA.L	9.3%	7.2%	8.3%	capm	0.84	rf + beta_own * ERP
IJPA.L	10.6%	7.1%	8.9%	capm	0.80	rf + beta_own * ERP
SMGB.L	33.2%	11.7%	22.4%	capm	1.91	rf + beta_own * ERP
XDWH.L	6.3%	6.0%	6.2%	capm	0.55	rf + beta_own * ERP
SGLN.L	20.0%	4.0%	12.0%	capm	0.05	rf + beta_own * ERP
SSLN.L	22.0%	5.5%	13.7%	capm	0.42	rf + beta_own * ERP
BRNT.L	30.1%	4.2%	17.2%	capm	0.11	rf + beta_own * ERP
COPA.L	11.7%	5.8%	8.7%	capm	0.48	rf + beta_own * ERP
ICOM.L	13.6%	4.1%	8.9%	capm	0.09	rf + beta_own * ERP
YCA.L	15.6%	7.9%	11.7%	capm	0.99	rf + beta_own * ERP
RR.L	60.5%	9.2%	34.9%	capm	1.31	rf + beta_own * ERP
PLTR	58.6%	9.8%	34.2%	capm	1.44	rf + beta_own * ERP
SPCX	13.7%	16.0%	14.9%	capm	2.95	rf + beta_own * ERP
AZN.L	13.1%	5.5%	9.3%	capm	0.43	rf + beta_own * ERP
SHEL.L	24.7%	5.7%	15.2%	capm	0.47	rf + beta_own * ERP
HSBA.L	37.3%	7.3%	22.3%	capm	0.85	rf + beta_own * ERP
ULVR.L	5.2%	4.2%	4.7%	capm	0.11	rf + beta_own * ERP
MSFT	15.1%	6.5%	10.8%	capm	0.67	rf + beta_own * ERP
ASML.AS	21.6%	11.8%	16.7%	capm	1.92	rf + beta_own * ERP
SIE.DE	18.4%	9.6%	14.0%	capm	1.41	rf + beta_own * ERP
BRK-B	14.9%	5.2%	10.0%	capm	0.34	rf + beta_own * ERP
IHYU.L	4.5%	6.8%	5.7%	credit_yield	0.28	USD high yield: distribution yield less 1% expected default loss
IHYG.L	2.7%	5.5%	4.1%	credit_yield	0.24	EUR high yield: distribution yield less 1% expected default loss
GHYS.L	3.3%	6.2%	4.7%	credit_yield	0.27	GBP-hedged global high yield: distribution yield less 1% default loss
IEMB.L	1.9%	4.8%	3.4%	credit_yield	0.30	EM USD sovereign: distribution yield less 1% expected default loss
IBTM.L	-0.9%	4.9%	2.0%	govt_yield	-0.03	iShares \$ Treasury Bond 7-10yr UCITS ETF, GBP-quoted but UNHEDGED. Expected GBP return = US 10y Treasury yi...
T36N	-4.0%	5.3%	0.6%	gilt_ytm	0.04	Treasury 4 7/8% 2036 redemption yield at the as-of clean price (market_params benchmark_gilt_10y; core_univ...
T56	-11.1%	5.9%	-2.6%	gilt_ytm	0.07	Treasury 5 3/8% 2056 redemption yield at the as-of clean price (market_params benchmark_gilt_30y; core_univ...
DBMG.L	10.3%	4.3%	7.3%	capm	0.13	rf + beta_own * ERP

Source: data/expected_returns.csv | rf = 3.75%, ERP = 4.17%; SPCX is priced on CAPM only because its 67-bar historical mean is not usable, and it is outside the optimiser anyway

A3. Sources and lookups

Item	Value	As at	Source
boe bank rate	3.75%	2026-09-16 (held since 2026-07-30)	https://www.bankofengland.co.uk/monetary-policy/the-interest-rate-bank-rate
sonia or 3m tbill	3.73% (3m compounded 3.74818%)	2026-09-09 / 2026-09-14	https://www.bluegamma.io/compounded-rates/sonia
gilt yield 2y	4.63%	2026-09-16	https://tradingeconomics.com/uk/gilt-yield/2-year-note-yield
gilt yield 10y	5.30%	2026-09-16	https://tradingeconomics.com/uk/gilt-yield/government-bond-yield
gilt yield 30y	5.89%	2026-09-16	https://tradingeconomics.com/uk/gilt-yield/30-year-bond-yield
us 10y yield	5.01%	2026-09-16	yfinance ^TNX (CBOE 10-Year Treasury Note Yield Index), pulled 2026-09-16
us 2y yield	4.74%	2026-09-16	https://tradingeconomics.com/usa/2-year-note-yield
us fed funds range	3.75%-4.00% (25bp hike, unanimous 12-0)	2026-09-16	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm
FOMC projections (median funds rate end-2026 4.1%)	one further hike implied	2026-09-16	https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm
damodaran erp (the figure the model uses)	4.17%	2026-07-01	https://aswathdamodaran.substack.com/p/data-update-2-for-2026-a-testing
damodaran US S&P 500 monthly implied ERP (reference only, NOT used)	4.09%	2026-09-01	https://pages.stern.nyu.edu/~adamodar/pc/implprem/ERPbymonth.xlsx
gbpusd	1.347346	2026-09-16	yfinance GBPUSD=X (Yahoo Finance)
gbpeur	1.167620	2026-09-16	yfinance GBPEUR=X (Yahoo Finance)
usdjpy	155.266006	2026-09-16	yfinance JPY=X (Yahoo Finance); derived GBP/JPY ~ = 209.19
vix	17.71 (from 15.84 on 2026-09-11)	2026-09-16	yfinance ^VIX
brent (front month)	USD 105.44	2026-09-16	yfinance BZ=F
gold (front month)	USD 4,323.00	2026-09-16	yfinance GC=F
uk cpi latest	3.1% headline (core 2.6%, services 3.4%)	2026-08 data, released 2026-09-16	https://www.ons.gov.uk/economy/inflationandpriceindices/bulletins/consumerpriceinflation/august2026
us cpi latest	3.4% headline (core 2.4%)	2026-08 data, released 2026-09-11	https://www.usinflationcalculator.com/inflation/us-cpi-august-2026/100073342/
T36N clean price / YTM / duration (the figures actually used)	96.75 / 5.30% / 7.46	2026-09-16	https://www.hl.co.uk/shares/corporate-bonds-gilts/bond-prices/uk-gilts , recorded in universe/market_params.json -> benchmark_gilt_10y

Item	Value	As at	Source
T56 clean price / YTM / duration (the figures actually used)	93.155 / 5.87% / 14.06	2026-09-16	https://www.hl.co.uk/shares/corporate-bonds-gilts/bond-prices/uk-gilts , recorded in universe/market_params.json -> benchmark_gilt_30y
Alternative 10y gilt reference (T4Q) - recorded but NOT used	Treasury Stock 4 1/4% 2036 (T4Q)	2026-09-14 page pull	https://www.hl.co.uk/shares/corporate-bonds-gilts/bond-prices/uk-gilts
ii supported exchanges	15 exchanges listed	2026-09-14 (not re-verified this pass)	https://www.ii.co.uk/investing-with-ii/international-investing/markets
ii US-ETF restriction	US-domiciled ETFs cannot be bought by ii's UK retail clients because US ETF issuers have not produced a PRIIPs-compliant Key Information Document	2026-09-14	https://www.bankeronwheels.com/kid-etf-registration-priips-ucits/
ii dealing costs and platform fees	see section 8.1	2026-09-16	https://www.ii.co.uk/our-charges
ii telephone-trade fee	GBP 49 flat, on top of the plan's international commission	2026-09-17	https://www.ii.co.uk/our-charges
ii Sweden (Nasdaq Stockholm) - phone only	"Orders on Swedish shares can only be placed over the phone on 0345 607 6001"	2026-09-17	https://www.ii.co.uk/investing-with-ii/international-investing/markets/sweden
ii Switzerland (SIX) - phone only	"orders on Swiss shares can only be placed over the phone"	2026-09-17	https://www.ii.co.uk/investing-with-ii/international-investing/markets/switzerland
ii Belgium (Euronext Brussels) - online	listed without the phone-only asterisk; standard international dealing flow	2026-09-17	https://www.ii.co.uk/investing-with-ii/international-investing/markets/belgium
ii Hong Kong (HKEX) - online	listed without the phone-only asterisk; no VIE-structure exclusion stated	2026-09-17	https://www.ii.co.uk/investing-with-ii/international-investing/markets/hong-kong
ii France - country-level listing only, no exchange-tier detail	best-evidence basis for ALMDT.PA on Euronext GROWTH Paris; help-centre search for "Euronext Growth" returns 0 results	2026-09-17	https://www.ii.co.uk/investing-with-ii/international-investing/markets/france
ii "trading international shares" help page	dealing channels and fees by market	2026-09-17	https://www.ii.co.uk/help/trading/international-trading/trading-international-shares
Bessembinder base rate (section 6.6)	57.8% of 25,332 US stocks 1926-2016 returned less than one-month T-bills over their lifetime; 4.3% of firms account for all net wealth creation; the modal lifetime return is -100%	published 2018, read 2026-09-17	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2900447
GUTS Nasdaq delisting determination, stayed by Hearings Panel request	determination 2026-09-10, stay requested 2026-09-11 (8-K)	2026-09-17	universe/dossiers/GUTS.md and the 8-K cited there
Moonshot sleeve prices, market caps and traded values	2026-09-16 closes; GELN.L, FTC.L and BGO.L closes from Ticker.info['previousClose'] during the 2026-09-17 LSE session; GFUZ market cap and ADV are own calculations from the SEC F-1 share count	2026-09-17	universe/unicorn_final.json -> price_source
Price and fundamentals data	Yahoo Finance via yfinance	fetches 2026-09-17, priced to the 2026-09-16 close	finance.yahoo.com

Items recorded as unresolved in the source file, and still unresolved:

- The actual 2026-09-17 MPC decision is not known to this report: the file is dated the evening before, and the model uses Bank Rate 3.75% as it stood at the 2026-09-16 close.
- Damodaran UK country risk premium not confirmed numerically (recalled convention only).
- ii minimum gilt deal size not found.
- ii international exchange list not re-verified this pass (Nordics ex-Sweden, Vienna and Tokyo Stock Exchange status all unconfirmed).
- T56 ii-online-tradability confirmed only via a third-party gilt-data site, not ii's own order screen.
- The US 2-year Treasury yield is inferred from a TradingEconomics 'previous close' rather than a primary source.
- ii's Core plan GBP 100,000 portfolio ceiling against this project's GBP 200,000 total: discovered in the 16 September refresh and addressed in section 8.1 by assuming the Plus plan, but not confirmed with ii directly.
- ii's coverage of Euronext **Growth** Paris (the venue ALMDT.PA trades on) could not be confirmed either way: ii's market list is country-level and its help centre returns no results for "Euronext Growth". Carried on a best-evidence basis; verify by instrument search on ISIN FR0011049824 or by telephone before dealing.
- The GUTS (Fractyl Health) Nasdaq listing outcome is unknown: a delisting determination of 2026-09-10 was stayed by a Hearings Panel request on 2026-09-11 and the panel had not ruled at the as-of date.
- Six moonshot dossiers carry load-bearing claims that could not be re-verified against a primary source and are labelled UNVERIFIED in place (listed in section 9, item 16).

Source: universe/market_params.json -> unresolved_for_followup

A4. Data corrections applied

Correction	What was wrong	Effect	Where
DBMG.L quote-scale glitches	33 bars in DBMG.L's own-history segment (2025-04-23 to 2025-08-11) were quoted 100x too low, producing fabricated -99% falls and +7,500% to +9,900% recoveries; two further opening prints (2025-04-22/23) are flat, zero-volume placeholders and are masked, with the DBMF proxy segment re-anchored to the first clean bar (2025-04-24).	DBMG.L 5y annualised volatility falls from 7,795% to 15.3%; its beta to VWRP.L from 2.04 to 0.13. Because it feeds the shared Ledoit-Wolf covariance, this changes every optimised weight.	data/prices_gbp_daily.csv, _monthly, _weekly, data/prices_local_daily.csv (repaired in place; unrepaired originals kept as *_raw.csv)
SMEA.L quote-convention break	The feed is quoted in pounds to 2023-11-24 and in pence from 2024-04-25; the earlier segment is rescaled onto the current convention and the mis-quoted stretch in between (which is mostly a Yahoo data hole) is masked.	SMEA.L 5y annualised volatility falls from 5,245% to 14.0%.	same four files
WLDS.L placeholder bar (found in the 16 September refresh)	The 2025-10-24 raw bar is Open = High = Low = Close = 8.9258 with Volume = 0, against 6.6480 on the 23rd and 6.7285 on the 27th: +34.26% then -24.62%. Same signature as the DBMG.L placeholders.	WLDS.L 5y volatility 15.61% instead of 24.48%; skew -0.386 instead of +5.335; max drawdown -21.55% instead of -27.57%. No weight moves by more than 0.1 pp.	same four files
Twenty-five 2026-09-16 bars reconstructed (16 September refresh; re-applied identically here)	Yahoo's daily history omits the 2026-09-16 bar for most LSE-listed ETFs and ETCs; the series runs 09-15 to 09-17. Forward-filling would have mis-stated fourteen core instruments.	Each bar rebuilt from 60-minute intraday data, with the close taken from the official figure recorded in the universe files (23 lines) or the last intraday bar (IGLT.L, GLTL.L). Cross-checked against 18 LSE lines that did carry a direct print; all 18 matched.	data/fetch_log.json -> patched_bars_2026-09-16
Gilt clean price and yield taken from the same record (16 September refresh)	The first pass of this refresh paired a 2026-09-16 redemption yield with a 2026-09-14 clean price.	T36N 96.06 -> 96.75, T56 92.49 -> 93.155. MinVariance's T36N position becomes 119 units rather than 120; Variant A's becomes 271 rather than 273. Nil effect on the headline, which holds no gilts.	universe/market_params.json -> benchmark_gilt_10y / benchmark_gilt_30y
ERP series reverted (16 September refresh)	An interim pass had put Damodaran's US S&P 500 monthly implied ERP (4.09%) into the field the model reads for CAPM.	Reverted to the mature-market figure, 4.17%, the same value the 11 September build used. The ERP therefore contributes 0.00 pp of this refresh's change; the US series is kept, labelled, as a sensitivity worth at most 0.035 pp.	universe/market_params.json -> damodaran_erp
Valuation multiples repriced to the as-of date	yfinance .info was captured live on 2026-09-17. Trailing/forward P/E, P/B and EV/EBITDA are rescaled by close(2026-09-16)/price-at-fetch.	Median factor 0.9846, range 0.9000-1.2000 - a wider spread than the previous build's, because the moonshot names are small and moved further between the as-of close and the fetch; Yahoo's unscaled values kept as *_yahoo_live.	data/fundamentals_table.csv

Correction	What was wrong	Effect	Where
Market cap rebuilt in GBP	info['marketCap'] is in the company's own reporting currency. Rebuilt as shares outstanding x GBP price per share at the 2026-09-16 close; market_cap_usd_m added for every row.	Read as GBP, Yahoo's raw figure would be 13.2x the correct number on SEYE.ST (SEK), 10.6x on 2498.HK (HKD) and 1.9x on SLX.AX (AUD).	data/fundamentals_table.csv
One 52-week-high convention	High/low and pct_below_52w_high are all computed from the intraday High/Low columns over the trailing 365 calendar days, split-adjusted.	The two figures in a row now agree exactly; close-basis figures kept alongside.	data/fundamentals_table.csv, data/stats_daily.csv
Portfolio beta convention	Portfolio beta / beta+ / beta- are the weighted sums over all 37 core lines including the 2.5% SPCX policy weight (beta is linear in the weights); the optimised-97.5%-only figures are reported as *_excl_spcx.	Headline beta+ 0.85 as held against 0.66 excluding SPCX; the sign of the beta asymmetry depends on which convention is used.	data/analysis_results.json portfolios.*
Total-return level base corrected (new in the moonshot rebuild)	Every return series' total-return level was compounded from 1.0 ON the date of its first return instead of the day before it, so the first return was counted in every distributional statistic but lost from the compounded level. Found by blind verification, reproduced to 10 decimal places.	total_return, cagr and the drawdown path of every portfolio, the sleeve index and the combined book. Headline core realised total return 219% to 218%, CAGR 26.13% to 26.07%; EqualWeight and NaiveReference monthly backtest CAGRs up about 0.2 pp; archived quality sleeve restates 182.37% to 183.71%. No mean, volatility, Sharpe, Sortino, skew, kurtosis, VaR, CVaR, beta or correlation is affected.	scripts/analysis.py -> series_stats; data/analysis_results.json (level_base_date, first_period_return)
Stale *_raw.csv panels cleared (new in the moonshot rebuild)	analysis.py reads data/prices_gbp_daily_raw.csv in preference to the SPEC-named deliverable so its repairs stay idempotent. The raw panels left by the previous build still carried the OLD 40 Sleeve A names; leaving them would have silently re-analysed the old sleeve.	Cleared before the moonshot fetch and rebuilt from it. Verified: the current raw panel carries 40/40 moonshot tickers and 0 of the 34 dropped ones; the archived copy carries 40/40 quality-sleeve tickers and 0 of the 34 added ones.	data/REFRESH_DIFF.md note 10.1
^FTAS column restored again	Yahoo still does not serve ^FTAS history; the 2026-09-17 re-fetch again returned a single post-as-of row, so the freshly fetched panel had no ^FTAS column.	The column was reinstated from the previous build's pristine cache and is byte-identical on every overlapping date. ^FTAS is the SECONDARY benchmark (beta_ftas / alpha_ftas / r2_ftas only) and enters no optimiser universe, covariance, expected return or weight.	data/fetch_log.json -> tickers['^FTAS']; data/REFRESH_DIFF.md note 10.2
One core price column differs in level, not in returns	data/prices_gbp_daily_clean.csv: of the 44 shared core columns only SIE.DE differs, and only in the base level of its total-return index (constant ratio 0.99999897).	Its daily returns are identical to 8.9e-16 and no statistic changes. It is a rebuilt total-return index base from the re-fetch.	data/REFRESH_DIFF.md note 10.4

Correction	What was wrong	Effect	Where
unicorn_final.json schema changed	The sleeve file now carries weighted_score / judge_scores / confidence_1_10 / key_backers / catalysts instead of the old judge_mean. analysis.py reads judge_mean with .get(), so analysis_results.json's judge_mean field is null for all 40 names.	Metadata only - it enters no statistic, weight or covariance. Section 6 and the Excel Unicorns tab read weighted_score / judge_scores instead, which is what this report prints.	data/REFRESH_DIFF.md note 10.3

Source: data/analysis_results.json -> sources_assumptions_corrections; data/fetch_log.json; data/REFRESH_DIFF.md section 7 | the stored JSON strings for the valuation-repricing and market-cap rows still quote the 2026-09-11/14 dates; the dates above are this build's

A5. Analysis log

The run log, verbatim, so every decision above can be traced:

- data override [GBP TR panel]: SMEA.L -- 1 bars masked as bad prints (see evidence in analysis_results.json)
- proxy splice re-anchored: DBMG.L proxy segment x 1.000000 so it joins the first clean own bar (2025-04-24) along DBMF's own path
- data override [GBP TR panel]: DBMG.L -- 2 bars masked as bad prints (see evidence in analysis_results.json)
- data override [GBP TR panel]: WLDS.L -- 1 bars masked as bad prints (see evidence in analysis_results.json)
- surviving large daily moves flagged for review (not altered): 9 across 3 core instruments -- all inspected and judged genuine market moves (silver and Brent spikes, the 2022-09-28 BoE long-gilt intervention, PLTR earnings gaps); full list in analysis_results.json
- data override [local price panel]: SMEA.L -- 1 bars masked as bad prints (see evidence in analysis_results.json)
- data override [local price panel]: DBMG.L -- 2 bars masked as bad prints (see evidence in analysis_results.json)
- data override [local price panel]: WLDS.L -- 1 bars masked as bad prints (see evidence in analysis_results.json)
- GBP unit prices built for 37 core + 40 unicorn lines
- ^FTAS price history ends 2026-07-22 (before the 2026-09-16 as-of date) -- its betas use the shorter overlapping sample
- per-asset statistics computed for 80 series on the 5y daily window; 7 flagged INSUFFICIENT HISTORY (< 3 years of daily bars in the window), including SPCX with 67 bars -- every statistic for those names is reported but must not be read as an estimate
- optimiser universe = 36 instruments (excluded: SPCX)
- complete-case daily return matrix for the optimiser: 1135 x 36 rows/assets (1135 of 1261 window days retained). The dropped days are the SMEA.L stretch masked above (2023-11-27..2024-04-24, which contains Yahoo's own 2024-01-02..2024-04-19 data hole) and the GHYS.L 2026-01-02..2026-01-23 hole. Complete-case keeps the covariance internally consistent; the pairwise-complete matrices are reported separately.
- Ledoit-Wolf constant-correlation shrinkage: delta = 0.035840 (r_bar = 0.2324, pi = 0.000191459, rho = 6.9588e-05, gamma = 2.99594e-06, kappa = 40.6787, T = 1135, N = 36)
- SPCX expected return uses CAPM only (rf + beta*ERP = 0.1604); its 65-bar historical mean (0.1372 annualised) is NOT a usable estimate and is excluded from the blend.
- analytic Sortino gradient verified against central finite differences: max relative error 7.43e-10
- optimiser: 96 random starts per non-convex objective plus 4 rounds of cross-seeding (every solved portfolio used as a start for every other objective); final dominance check -- no portfolio is beaten on its own objective by any other feasible portfolio in the set
- EqualWeight is a comparison reference, not an optimiser output, and breaches: group stocks 0.29583333 > max 0.25000000
- every OPTIMISED portfolio satisfies every bound, the budget and all five group constraints to 1e-06
- efficient frontier: 30 feasible points from 6.06% to 18.47% expected return

- core 37x37 pairwise-complete covariance (SPCX row/col from 67 overlapping bars): min eigenvalue 1.099e-04 before PSD repair
- a true 10-year fixed-weight backtest is IMPOSSIBLE for this universe: the last core line to start is SMGB.L on 2020-12-10 (VWRP.L 2019-07-25, DBMF/DBMG.L 2019-05-08, PLTR 2020-09-30 are the other late starters). The 'all assets live' backtest therefore runs from 2020-12-31; the 'max available' variant runs the full 10 years but renormalises the weights over whichever names exist, so its early years are a DIFFERENT portfolio and are not comparable.
- reported-Sortino check: MaxSortino 2.564159 is the highest of the 6 portfolios (next best MaxSharpe 2.546806). The optimiser's objective and the published statistic are now the same quantity, computed on the same 1261-day pairwise-renormalised series.
- headline core portfolio = Moderate12 (max expected return subject to ex-ante vol <= 12%)
- unicorn sleeve index: 40 names, 1261 usable days (2021-09-17 .. 2026-09-16), 27-40 live names
- combined 77x77 pairwise-complete daily covariance: min eigenvalue -1.032e-01 before PSD repair
- expected-return sensitivity: switching the optimiser input from the blend to the historical-only vector moves 8% of the Moderate12 portfolio; switching to the CAPM/yield-only vector moves 83%.
- per-stock fundamentals / WACC table built for 51 names (11 core stocks + 40 unicorns)
- valuation repricing: yfinance .info was captured live on 2026-09-17, so trailing/forward P/E, P/B and EV/EBITDA are rescaled onto the 2026-09-16 close by close/price-at-fetch (median factor 0.9846, range 0.9000-1.2000); market cap and EV are rebuilt in GBP. The unscaled Yahoo values are kept as *_yahoo_live / market_cap_yahoo_raw_local_ccy.
- info['previousClose'] is NOT the as-of close for 3 lines (QSI, GUTS, GSIT) -- all ASX, where a further session had already traded when .info was captured. The rescale uses the raw 2026-09-16 bar directly, so these rows are still priced at the as-of date.
- market cap: shares-outstanding x price differs from Yahoo's own (all share classes, restated to the as-of close in GBP) by more than 5% on 13 names (SPCX 40%, BRK-B 36%, AL3.AX 11%, LTBR 5%, ALMU 14%, WBT.AX 7%, ALCLS.PA 23%, ALMDT.PA 8%, IMSR 34%, EMV.AX 5%, SLDP 6%, HYT.AX 17%, GSIT 9%) -- multiple share classes. Both figures are in the table; the headline is shares x price.
- repaired price panels written OVER the SPEC-named deliverables (prices_gbp_daily.csv, prices_gbp_monthly.csv, prices_gbp_weekly.csv, prices_local_daily.csv); the pristine fetch_data.py output is preserved as *_raw.csv and is the file this script reads on every subsequent run. Only DBMG.L and SMEA.L differ between the raw and repaired panels; every other column is byte-identical.

Source: data/analysis_results.json -> log

Three lines in that log need a reader's caution. The info['previousClose'] line names three tickers (QSI, GUTS, GSIT) and then calls them "all ASX"; all three are in fact NASDAQ lines, and the ASX phrasing is carried over from an earlier build - the substantive point, that the rescale uses the raw 2026-09-16 bar directly so those rows are still priced at the as-of date, is correct. The SPCX bar count appears as both 65 and 67 in different log lines; data/stats_daily.csv gives **67** observations in the window, and 67 is the figure used throughout this report. And the log's "7 flagged INSUFFICIENT HISTORY" counts SPCX plus the six moonshot names listed in section 6.4 (GUTS, IMSR, 2498.HK, GFUZ, BIOA, BTQ). One line that was in the previous build's log is absent from this one: the SMEA.L segment rescale no longer prints as a separate "scale repair" entry, because the re-fetch records it inside the SMEA.L override rather than as a standalone permanent break. The repair itself is unchanged and is still applied - SMEA.L's five-year volatility reads 14.0%, not 5,245% - and the evidence string is in data/analysis_results.json -> data_repairs.overrides["SMEA.L"].

A6. Dossier digest

Every moonshot dossier ends with a confidence score and a named falsifier. Both are reproduced verbatim below, in sleeve rank order, so the sleeve's own stated weak points can be read in one place without opening forty files. The falsifier is the test the annual thesis review applies (section 6.3 and 8.5); the confidence score is the dossier author's own, on a 1-10 scale, and is not a probability.

#	Ticker	Conf	Confidence statement	Falsifier
1	IPWR	7	Unusually clean asymmetry: debt-free, ~4-year runway, technology priced at ~\$22m against a named Nvidia-architecture programme.	The Q4 2026 hyperscaler prototype slipping materially or being dropped. If that milestone passes without converting to a named commercial agreement during 2027, the thesis is broken regardless of the balance sheet.
2	VYGR	7	The royalty-stacking structure, real insider ownership and modest dilution are rare in this market cap band; the burn rate is the honest offset.	Novartis terminating or deprioritising the licensed capsid programmes. That single event removes the milestone anchor, the validation and the read-across in one move.
3	AL3.AX	6	The lowest here. The commercial evidence is good and the first EBITDA-positive half is real, but customer concentration is extreme and the 10x case depends on repeat orders not yet contracted.	The US Navy awarding a comparable WAAM installation programme to a domestic vendor, or the FY2027 order intake failing to exceed the A\$16.8m already contracted. Either shows the installed base is not compounding.
4	ONWD.BR	7	Real revenue growth, a strong backer set for the size, high insider ownership and genuine scarcity value - against a burn rate that guarantees further dilution.	Empower BP missing its primary endpoint, or a financing struck at a materially lower price than the April 2026 round. Either one breaks the "funded path to an approved implant" that the entire 10x case rests on.
5	LTBR	7	The set-up is unusual: a debt-free company with ~10 years of runway, two fresh state validations, trading below its own cash at its 52-week low. The downside is bounded by the cash; the upside needs patience most investors do not have.	Irradiation data from the ATR campaign showing unacceptable swelling or fuel-cladding interaction - or DOE declining to authorise SHED. Either removes the only product.
6	ARA.TO	8	The highest here, on backer quality: three separate US federal actions and one Japanese state action all point the same way. The offset is that the market cap already reflects some of it.	EXIM declining to convert the US\$750m Letter of Interest into a committed facility, or a renewed Chilean environmental/community block on Penco. Either turns a financed national-security asset back into an ordinary junior developer.
7	SLX.AX	8	A near-breakeven, debt-free company with US\$128m of investments, 51% of the only third-generation enrichment technology near licensing, a state feedstock monopoly and an investment-grade JV partner - bought 58% below its high.	An NRC licence denial or material delay, or a GLE final investment decision structured so that Silex is diluted below 51%. Either converts the owner of a Western enrichment utility back into a royalty holder.
8	NYXH	7	The clinical and commercial evidence is the strongest in this group - approval, met endpoints, 339% growth, doubled accounts, high insider ownership. The confidence is capped not by the product but by the certainty of dilution.	A 2027 financing struck at a deep discount with warrant coverage, or the EUR 36-40m guidance being missed. The first destroys the return arithmetic; the second destroys the commercial thesis behind it.
9	ALMU	5	The balance sheet (\$56m cash, >3 years) buys time that most moonshots do not have, and the CHIPS/Tower/Sumitomo triangle is real external validation. But the Q4 revenue collapse is a material deterioration against the record I inherited, which assumed ~\$1.2m/quarter and \$37.8m cash.	FY2027 passing without a commercial (non-government) design win or NRE contract in datacom optics. That would confirm this is a grant-funded research house, not a supplier.
10	WBT.AX	7	Two of the world's largest embedded IDMs and a qualified merchant foundry is about as much third-party validation as a pre-royalty IP company can produce, and the balance sheet funds it to the royalty inflection.	The onsemi 65nm qualification failing, or 2027 passing with no production royalty revenue recognised. Either would mean the licences are research budgets, not product commitments. Insider ownership of 7% is the one alignment weakness.
11	VUL.AX	7	The financing close is the rare case where a dozen sophisticated lenders have done the diligence for you, and the product already exists. The discount is real because the commodity is hated and first production is still two years away.	A material Lionheart capex overrun or schedule slip beyond 2028 announced in a quarterly report - with this gearing, that is the difference between a 10x and a recapitalisation.
12	IDN	7	This is the odd one out in a moonshot sleeve - it is already profitable, cash-generative and debt-free, so the downside is not zero, which caps neither the logic nor the asymmetry given the 57% de-rating. The 4.85% insider stake is the honest weakness.	Two consecutive quarters of revenue decline, or disclosure that a top-five customer has not renewed. Either would confirm concentration, not category leadership.

#	Ticker	Conf	Confidence statement	Falsifier
13	GELN.L	6	The partner list (TDK, Nissan, Yazami, a funded scale-up agreement) is far better than a GBP 33m AIM company normally assembles, insider ownership is 39.6%, there is no debt, and the licensing model means it does not need gigafactory capital. Against that, the science is the hardest in this batch and the liquidity is the worst.	Independent third-party cycle-life data below ~500 cycles at commercially relevant sulphur loading, or the exclusive licence lapsing without the licensee ever being named.
14	EDIT	6	Best-funded name in this batch, with a genuinely differentiated mechanism, a dated binary readout inside the holding period, and a validated M&A comparable. Against that: the revenue line is a one-off from a partner walking away, the dilution has been heavy, and management owns essentially none of it.	The Q1 2027 human data update failing to show durable, clinically meaningful reductions in all three of LDL-C, Lp(a) and ApoB. Everything else in the file is secondary to that one readout.
15	SEYE.ST	7	The regulation is dated and binding, the backlog is 3.6x the market cap, EBITDA has turned positive, and the stock is at a 52-week low - that is an unusually concrete setup. The balance sheet is the reason it is not higher.	An equity raise on dilutive terms, or automotive organic growth falling back below ~30% in Q3/Q4 2026, which would mean the July 2026 regulation did not produce the volume step-up the whole thesis rests on.
16	QSI	5	No separate note in the dossier; the score is stated in its heading.	Any further delay to the Q2 2027 Proteus launch, or a launch whose first datasets do not show sequencing accuracy competitive with mass spectrometry. Either ends the thesis regardless of the cash pile.
17	CRBU	6	No separate note in the dossier; the score is stated in its heading.	CB-011 dose-expansion data in H2 2026 showing responses materially below the 92% ORR / 83% >=CR of dose escalation, or vispa-cel follow-up showing relapses clustering after 18 months. Either would say the allogeneic durability problem is unsolved.
18	ALCLS.PA	6	No separate note in the dossier; the score is stated in its heading.	The Q4 2026 lasme-cel interim Phase 2 and eti-cel Phase 1 datasets failing to show durable responses, or AstraZeneca declining to advance a collaboration programme. Either removes the only two reasons to hold this rather than cash.
19	GUTS	5	No separate note in the dossier; the score is stated in its heading.	The early-Q4-2026 REMAIN-1 Pivotal Cohort 6-month topline failing to show a statistically significant weight-maintenance benefit versus sham in the full randomised population. Nothing else about this company matters until that number prints.
20	PRME	6	The clearest "why this could be the next big thing" argument in the sleeve's biology bucket - correction rather than disruption, already in humans, with FDA RMAT, a pharma partner and the CF Foundation behind it - married to the sleeve's most predictable balance-sheet hazard.	PM359's BLA slipping out of 2027, or the PM577a 2027 Wilson data failing to show durable correction in the liver. Either breaks the platform claim, and the *ex vivo* CGD asset alone does not support US\$544m.
21	PRQR	7	No separate note in the dossier; the score is stated in its heading.	The biliary atresia or other patient data in H1 2027 showing no clinical benefit despite confirmed editing, which would mean ADAR recruitment works biochemically but not therapeutically.
22	ALMDT.PA	6	(Marked down from the candidate record's 7 because the performance statistics could not be verified, the balance sheet is worse than described, and ii tradeability on Euronext Growth is unconfirmed.)	The first half-year of US commercial reporting showing negligible reimbursed exam volume, which would mean the reimbursement code exists but nobody is using the product.
23	CU6.AX	6	(Marked down from the candidate record's 7 solely because of the unexplained 12-month drawdown and the unverified Fast Track and trial-timing claims.)	CLARIFY missing its co-primary sensitivity/specificity endpoints against surgical pathology. Everything else - the therapeutic, the M&A optionality, the supply chain - is downstream of that one result.

#	Ticker	Conf	Confidence statement	Falsifier
24	LAES	6	No separate note in the dossier; the score is stated in its heading.	QS7001 commercial revenue failing to appear by the end of H1 2027, or a large cash deployment into a related party. The first would mean the certified part cannot win sockets; the second would mean the balance sheet is not really the shareholders'.
25	IMSR	6	Balance sheet, the LEU choice and an accumulating NRC record make this the best-evidenced name in its sector here; unsolved materials science caps it.	An NRC rejection or indefinite deferral of a topical on graphite or salt-chemistry grounds, or an NRC position that Core-unit replacement needs a new licensing framework - either pushes first concrete beyond the hold period.
26	EMV.AX	6	Nature-portfolio peer review, eight named tertiary centres, 33% insider ownership and A\$4.6m of non-dilutive funding still to come make this defensible; a single blinded endpoint and near-zero liquidity cap it.	The pivotal readout showing haemorrhage-detection sensitivity or specificity below 80%. One number decides this company.
27	SLDP	6	The best backer evidence in this group (a BMW VP on the board, a line installed at SK On) and a balance sheet that removes financing risk for four years, against a business that has yet to prove it can bill anyone reliably.	BMW or SK On announcing an in-house or alternative sulphide electrolyte source. That removes the merchant-supply thesis entirely and leaves a cash shell with a cell-design licence.
28	HYT.AX	3	Cut from the candidate's 4 because the backer set collapsed under checking: no Fortescue, and the Mitsubishi/Toyota money belongs to a competitor. What remains is a real acreage position, a genuinely novel GeoKiln test, and the best 100x arithmetic in the group - on twelve months of cash and almost no liquidity. Own it as a lottery ticket, sized as one.	A Nemaha well flowing hydrogen at sub-commercial rates, or not flowing at all. Concentration without rate is worth nothing.
29	MSCL.TO	6	Fast Track, a differentiated and genuinely mutation-agnostic mechanism, zero debt and 50% institutional ownership against one binary readout in roughly three months.	BASECAMP missing its primary functional endpoint in Q4 2026. Everything else here is commentary.
30	2498.HK	6	The only name in this group with real revenue, real customers (3,400+), a dated product roadmap and a top-tier strategic register - and the only one where the 10x case needs no scientific breakthrough, just execution on a visible S-curve. The geopolitical ceiling and the automotive price war are what hold it at 6.	Robotics revenue growth decelerating sharply in the FY2026 results while ADAS gross margin stays below 25%. That would mean the transition is not outrunning the price war, and there is no thesis left.
31	GFUZ	4	Still raised above the candidate's implied distress case - the balance sheet is four times better than recorded (~US\$135m, not US\$32m) and the Canadian state backing is documented in the F-1's own tables. Cut one point because the correction work changed two things that matter: the true share count sits above the headline (and the filing cannot state it consistently), and a 10x on the share price implies a US\$9.4bn-13.6bn company rather than US\$6.3bn - roughly double the largest private fusion comparable. Physics risk and a ~1.7:1 dilution overhang do the rest.	LM26 failing to reach 1 keV, or the programme being redirected away from the Lawson-criterion ladder. That is the entire near-term thesis.
32	AXE.AX	4	The asymmetry is genuine and the price is genuinely small; the backing is genuinely poor, and I have deliberately not dressed the IonQ agreement up as something it is not.	End-2026 passing without a 12CQ qubit demonstrator device, or the 2027 raise being struck at a discount to A\$0.185. Either outcome means the sleeve should replace this name at the annual review.
33	MOLN.SW	6	The asymmetry is real (CHF 60m EV, three clinical assets, secured 212Pb) but the execution history is genuinely bad.	A dose-limiting marrow or renal toxicity in MP0712 cohort 2 or 3 that caps dosing below a therapeutic level - that ends the Radio-DARPin thesis outright.
34	LIS.AX	6	Defensible because the cash discipline is real and the customer is the hardest one in the world.	Published third-party cycle-life data showing the BNNT cell degrading below ~300 cycles at useful depth of discharge would end it.

#	Ticker	Conf	Confidence statement	Falsifier
35	GSIT	6	The downside is unusually well protected - at \$131.7m EV you are paying ~5x sales for a business with \$68.9m net cash, so the floor is cash plus a declining annuity.	The AI SDK shipping and still producing no named, paying Gemini-II design win within 12 months.
36	BIOA	6	A free option on a well-funded Phase 2 programme, with two shots on goal inside 18 months.	A clinically significant ALT/AST signal in QUELL-CV or QUELL-DME - hepatotoxicity is what has killed this class twice and killed BioAge's last lead asset.
37	AISP	5	The lowest confidence in this group, justified only by the price: ~3x sales for a 75%-gross-margin business inside DHS and DoJ with a near-breakeven cost base.	Two consecutive quarters of declining revenue with backlog falling - that would show the federal pipeline is conversation, not procurement.
38	FTC.L	7	The highest confidence here - a real business with a record order book, 90% revenue visibility and net cash - but correspondingly the lowest asymmetry, since 10x from \$700m is a genuine stretch.	A second consecutive period of falling EBITDA on flat or rising revenue, which would confirm Filtronic is a price-taking subcontractor rather than a bottleneck supplier.
39	BTQ	5	Held for the genuine uniqueness of a live quantum-safe Bitcoin testnet and the named Korean financial relationships, not for the balance sheet. This is the most fragile name in the group and should be sized as a true lottery ticket.	Bitcoin Quantum mainnet slipping past 2026 while cash falls below one quarter of burn - that combination forces a deeply dilutive rescue financing before the thesis can be tested.
40	BGO.L	6	Genuinely cheap against revenue and margin, with verifiable blue-chip adoption; the leverage is what stops this being an 8.	Interim results showing net debt rising again with flat recurring revenue - that would prove the 2026 wins are not converting and the balance sheet is running out of time.

Source: `universe/dossiers/<TICKER>.md` section 9 (Confidence and falsifier), verbatim; confidence score cross-checked against `universe/unicorn_final.json -> names[.confidence_1_10]`

Read the spread rather than the level. The median confidence is **6 of 10** and the range runs from **3** (HYT.AX, described in its own dossier as the purest lottery ticket in the sleeve) to **8** (ARA.TO and SLX.AX). Thirteen names score 7 or above and eight score 5 or below. A sleeve whose median name is a 6 out of 10 is not a sleeve anybody should be confident about; it is a sleeve of bets that have been written down clearly enough to be checked later.

A7. Files this report is built from

File	What it holds
universe/core_universe.json	the 37 core instruments, their metadata, ii evidence, proxies and the resolutions in section 3.2
universe/unicorn_final.json	the 40 moonshot names, weighted judge scores, theses, key backers, catalysts, 10x end caps and the selection method (the filename is unchanged; sleeve_type reads moonshot)
universe/MOONSHOT_SLEEVE_DESIGN.md	the design note behind section 6: mandate, funnel, judging weights, sleeve rules, profile, payoff arithmetic, ii dealing channels and what changed against the quality sleeve
universe/dossiers/*.md	42 per-name dossiers (40 selected plus the 2 removed at fact-check), each with sourced backers, catalysts, a 10x path, a kill list, a confidence score and a named falsifier
universe/moonshot/	the moonshot audit trail: ideation, 10 verification batches, three judging lenses and 5 adversarial fact-check groups
universe/unicorn_final_quality_2026-09-16.json	the quality-tilted 40 this sleeve replaced
universe/unicorn_candidates.json	the previous sleeve's audit trail including all 47 rejected candidates
universe/market_params.json	every looked-up market number with its URL and date
universe/NEWS_AND_IMPACT_2026-09-16.md	the 11-16 September news synthesis behind section 1.4, with sources
universe/refresh_changelog.json	what the universe refresh changed, line by line
universe/reverify_2026-09-16_batch_1..5.json	the re-verification of the 37 core instruments and the previous sleeve at the 2026-09-16 close
data/analysis_results.json	config, definitions, per-asset statistics, all six portfolios, frontier, sensitivity, limitations, log
data/analysis_variants.json	variants A and B of section 4.6
data/REFRESH_DIFF.md, data/refresh_diff.json	the full 16 September quality-sleeve vs 17 September moonshot comparison behind section 4's closing subsection and section 6.8
archive/2026-09-16-quality-sleeve/data/REFRESH_DIFF.md	the earlier 11 September vs 16 September comparison
data/stats_daily.csv	the per-asset statistics table (appendix A1)
data/expected_returns.csv	the three expected-return estimates (appendix A2)
data/weights.csv, data/weights_variants.csv	portfolio weights
data/cov_annual.csv, data/cov_lw.csv, data/corr.csv, data/cov_daily_full.csv	covariance and correlation matrices
data/frontier.csv	the 30 frontier points
data/risk_contribution.csv	the Euler risk decomposition of the headline portfolio
data/backtest_monthly.csv	monthly backtest returns for every portfolio and both benchmarks
data/combined_positions.csv	the executable GBP 200,000 position list
data/fundamentals_table.csv	every field in section 7
data/unicorn_stats.csv	per-name moonshot statistics and sizing
data/prices_gbp_daily.csv / _monthly / _weekly	the repaired GBP total-return panels (unrepaired originals kept as *_raw.csv)
data/fetch_log.json	per-ticker fetch record, the GBp dividend evidence, the 2026-09-16 patched bars and the proxy splice log
data/verify_analysis_*.md, data/verify_data_*.md	the independent verification passes and their findings

File	What it holds
archive/2026-09-16-quality-sleeve/	the complete quality-sleeve build this one replaces, for anyone who wants to reproduce the comparison
archive/2026-09-11/	the 11 September build

Prepared 2026-09-17 on data as at the 2026-09-16 close. Educational analysis, not personalised investment advice. Model outputs are estimates, not certainties.